



UNIVERSITÀ DI PAVIA
Dipartimento di
Ingegneria Industriale
e dell'Informazione

Course of Robot Control

Prof. Antonella Ferrara



Fifth Part

Outline of the Course

Basic Terminology and Concepts

Reference Frames and Rotation Matrices

Robot Direct Kinematics: the Geometric Approach

Robot Direct Kinematics: the Denavit-Hartenberg Convention Based Approach

Robot Inverse Kinematics

Robot Differential Kinematics

Robot Dynamic Model

Robot Control



UNIVERSITÀ DI PAVIA
Dipartimento di
Ingegneria Industriale
e dell'Informazione

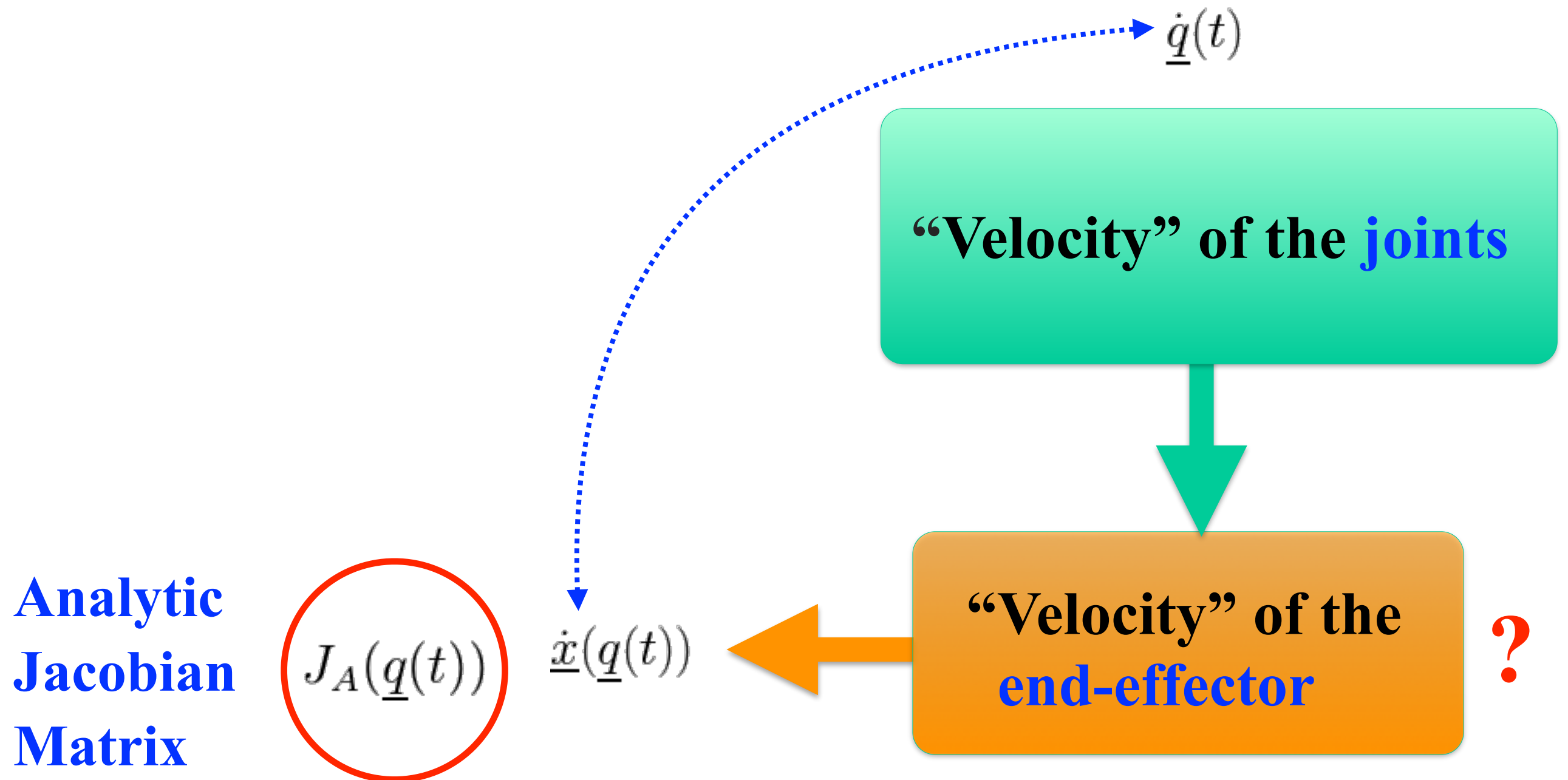
Course of Robot Control

Prof. Antonella Ferrara

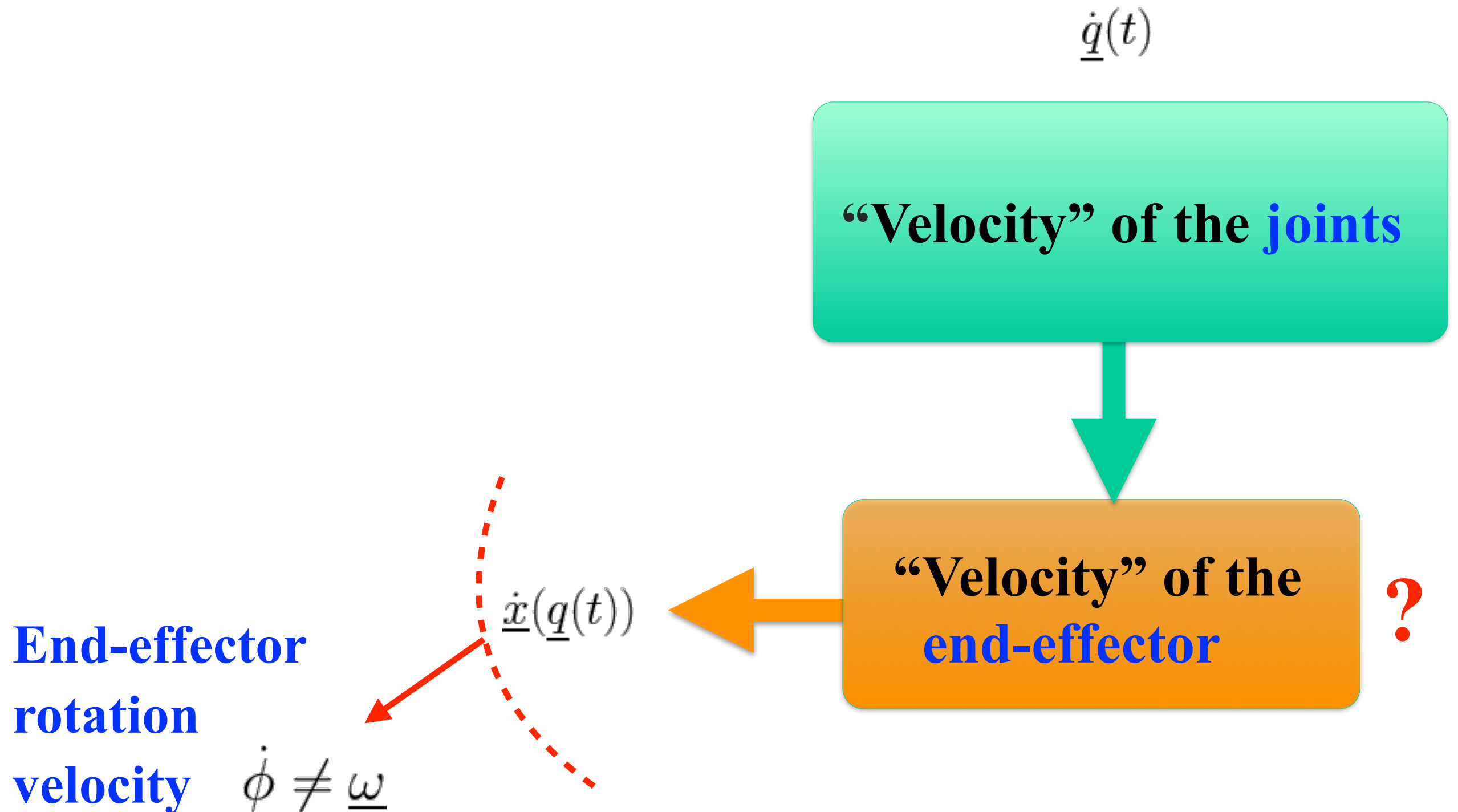


Robot Differential Kinematics: Geometric Approach

Differential Kinematics



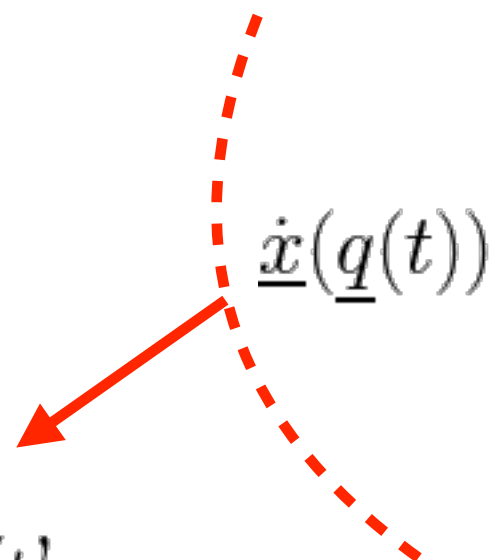
Differential Kinematics



Differential Kinematics

There is a different way to define the end-effector velocity in which the rotation velocity is the angular velocity

End-effector
rotation
velocity $\dot{\phi} \neq \underline{\omega}$


$$\dot{x}(q(t))$$

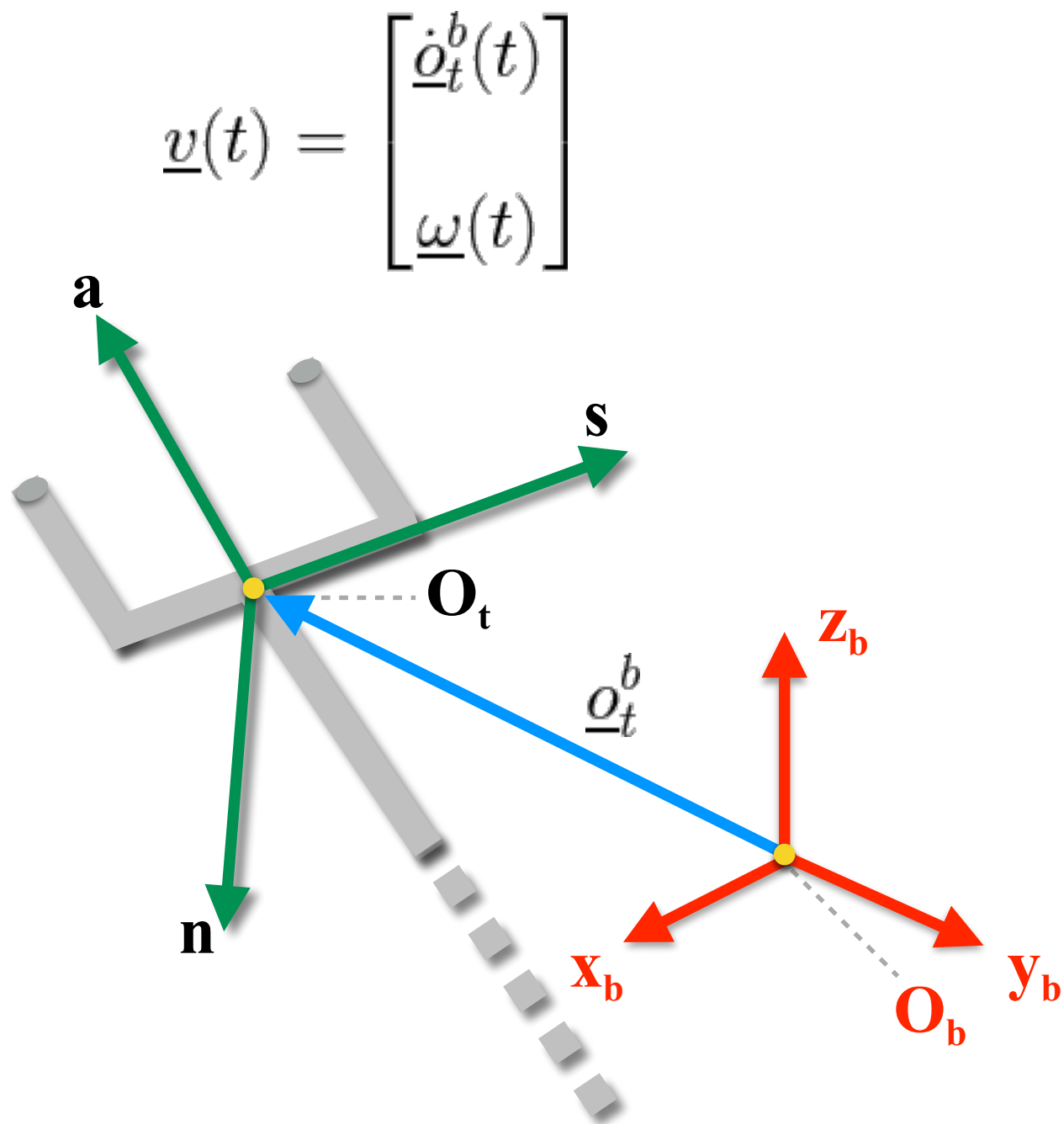
$$\dot{q}(t)$$

“Velocity” of the joints

“Velocity” of the end-effector

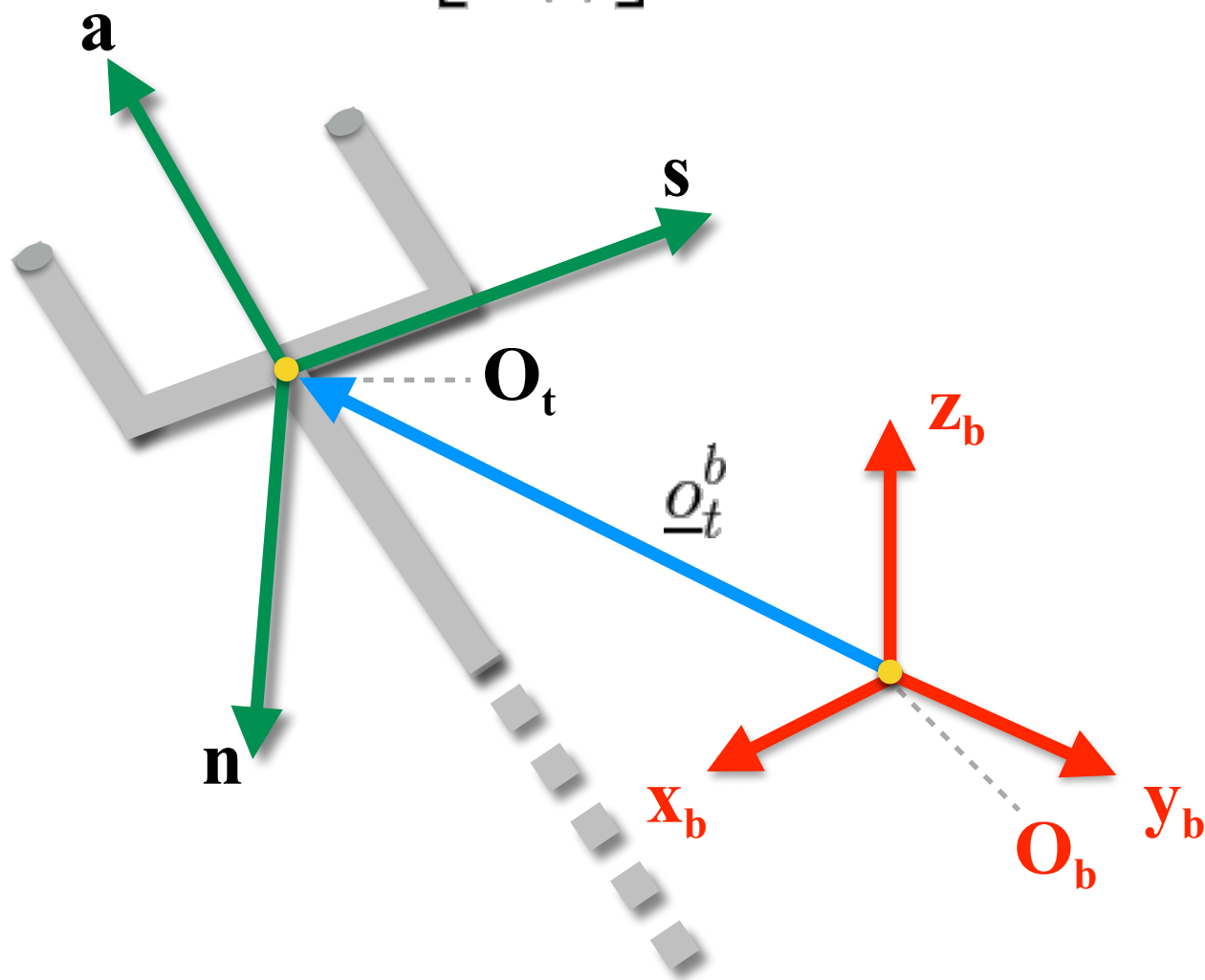
?

A Different Description of the End-Effector Velocity



A Different Description of the End-Effector Velocity

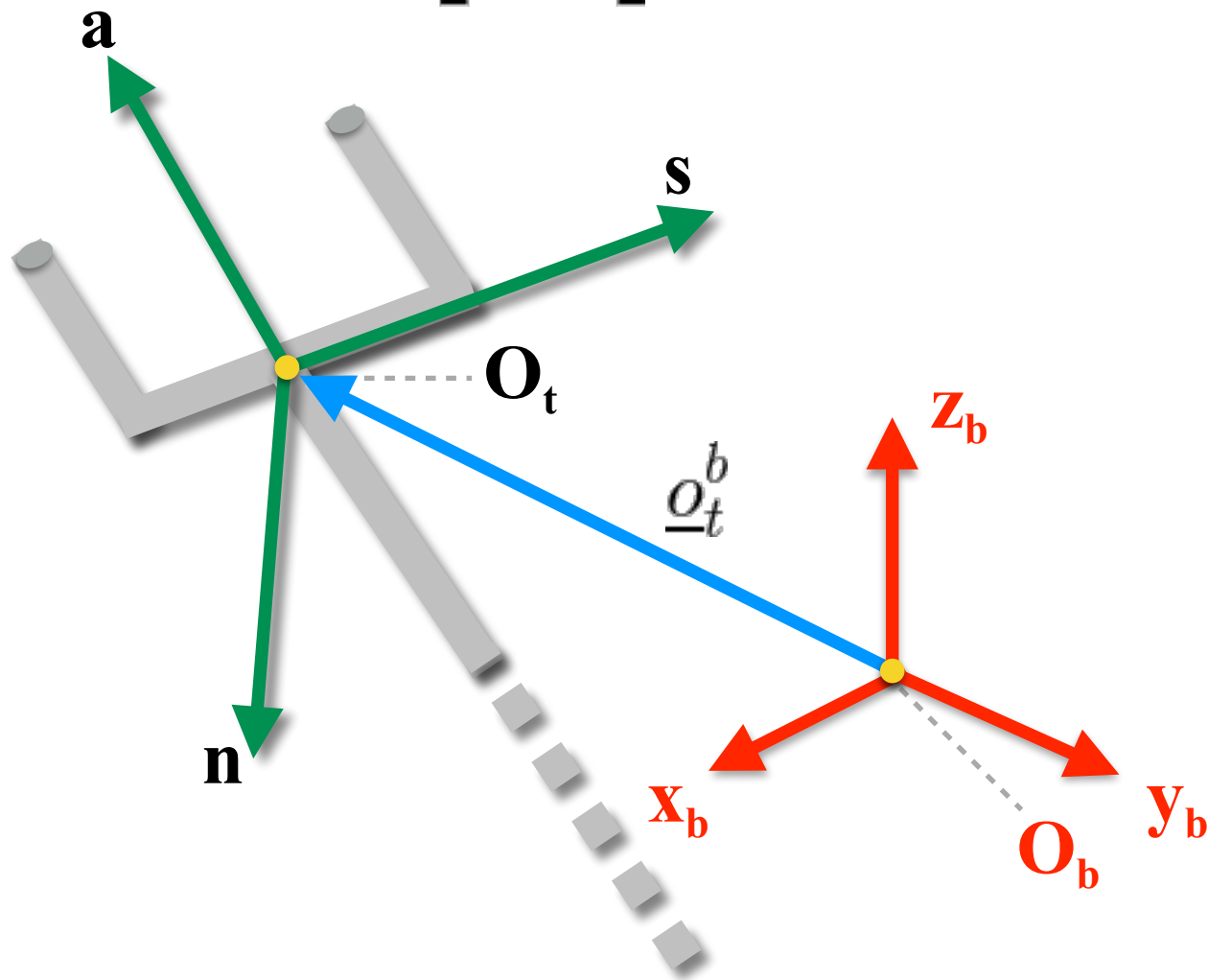
$$\underline{v}(t) = \begin{bmatrix} \dot{\underline{O}}_t^b(t) \\ \underline{\omega}(t) \end{bmatrix} \quad \longrightarrow \quad \underline{v}(t) = \begin{bmatrix} J_P(\underline{q}(t)) \\ J_O(\underline{q}(t)) \end{bmatrix} \dot{\underline{q}}(t)$$



A Different Description of the End-Effector Velocity

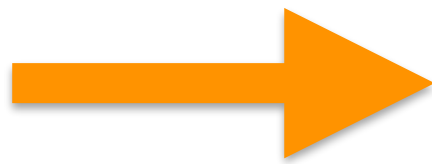
$$\underline{v}(t) = \begin{bmatrix} \dot{\underline{o}}_t^b(t) \\ \underline{\omega}(t) \end{bmatrix} \quad \longrightarrow \quad \underline{v}(t) = \begin{bmatrix} J_P(\underline{q}(t)) \\ J_O(\underline{q}(t)) \end{bmatrix} \dot{\underline{q}}(t)$$

$\frac{\partial \underline{o}_t^b(\underline{q}(t))}{\partial \underline{q}(t)}$



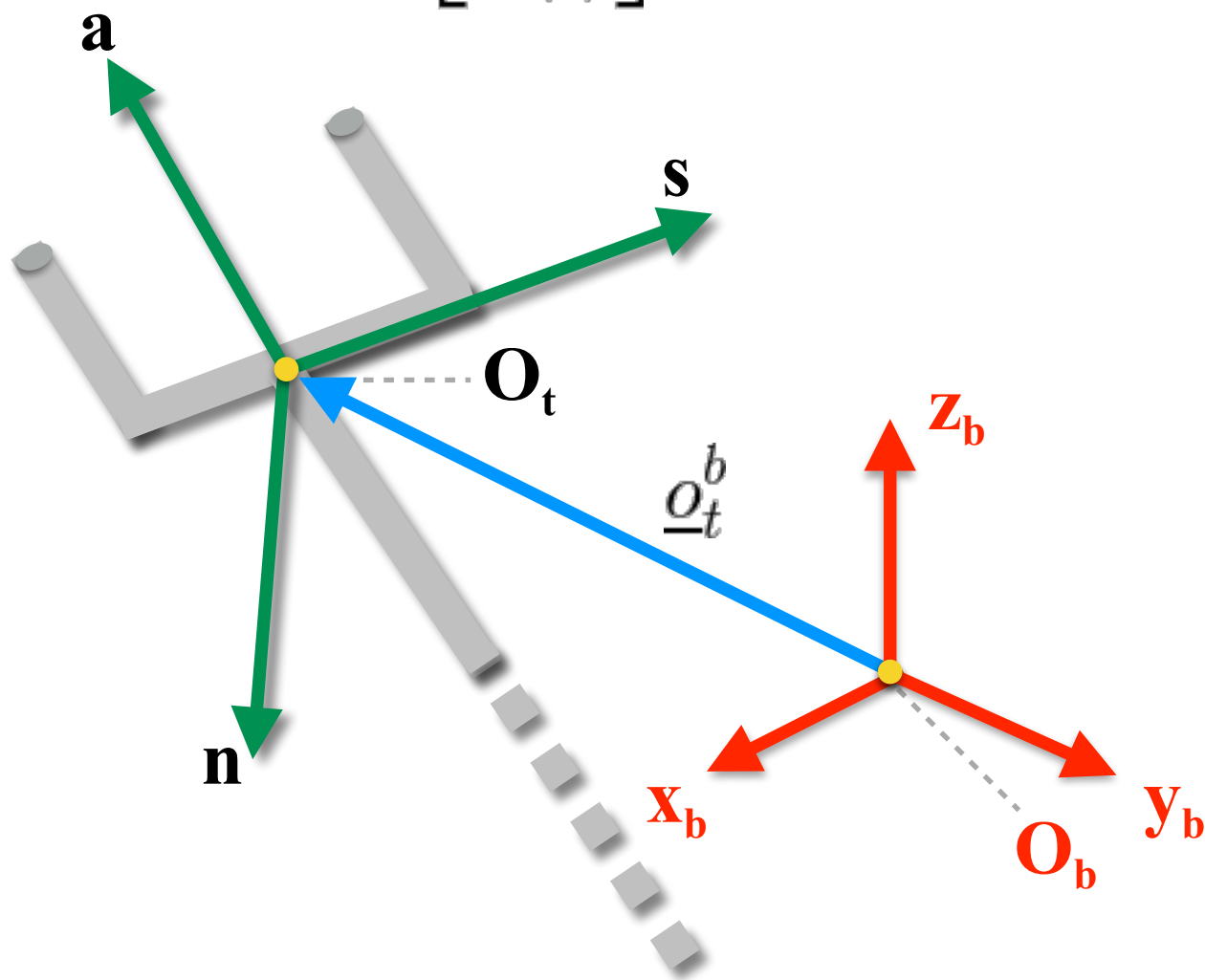
A Different Description of the End-Effector Velocity

$$\underline{v}(t) = \begin{bmatrix} \dot{\underline{O}}_t^b(t) \\ \underline{\omega}(t) \end{bmatrix}$$

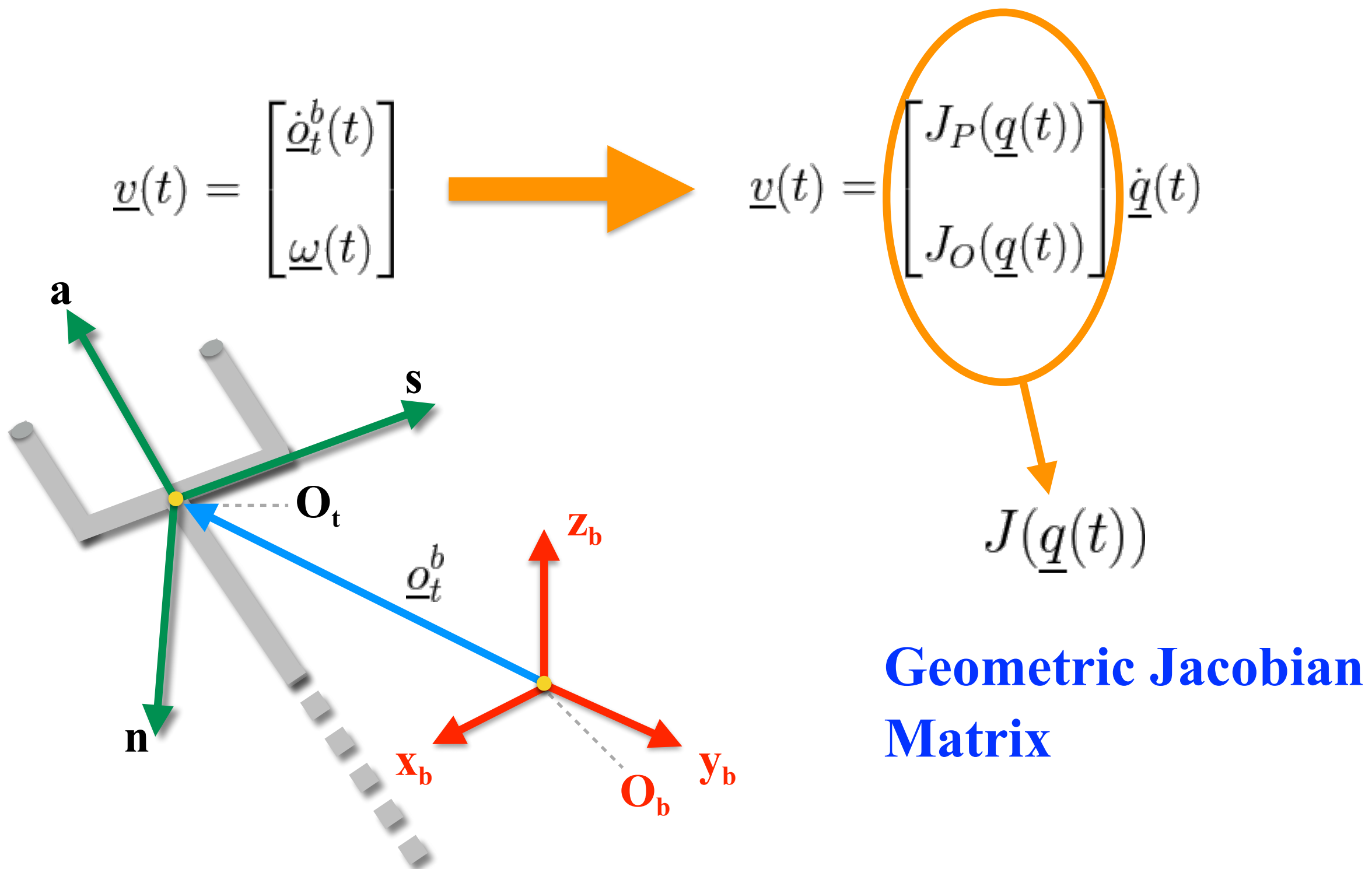


$$\underline{v}(t) = \begin{bmatrix} J_P(\underline{q}(t)) \\ J_O(\underline{q}(t)) \end{bmatrix} \dot{\underline{q}}(t)$$

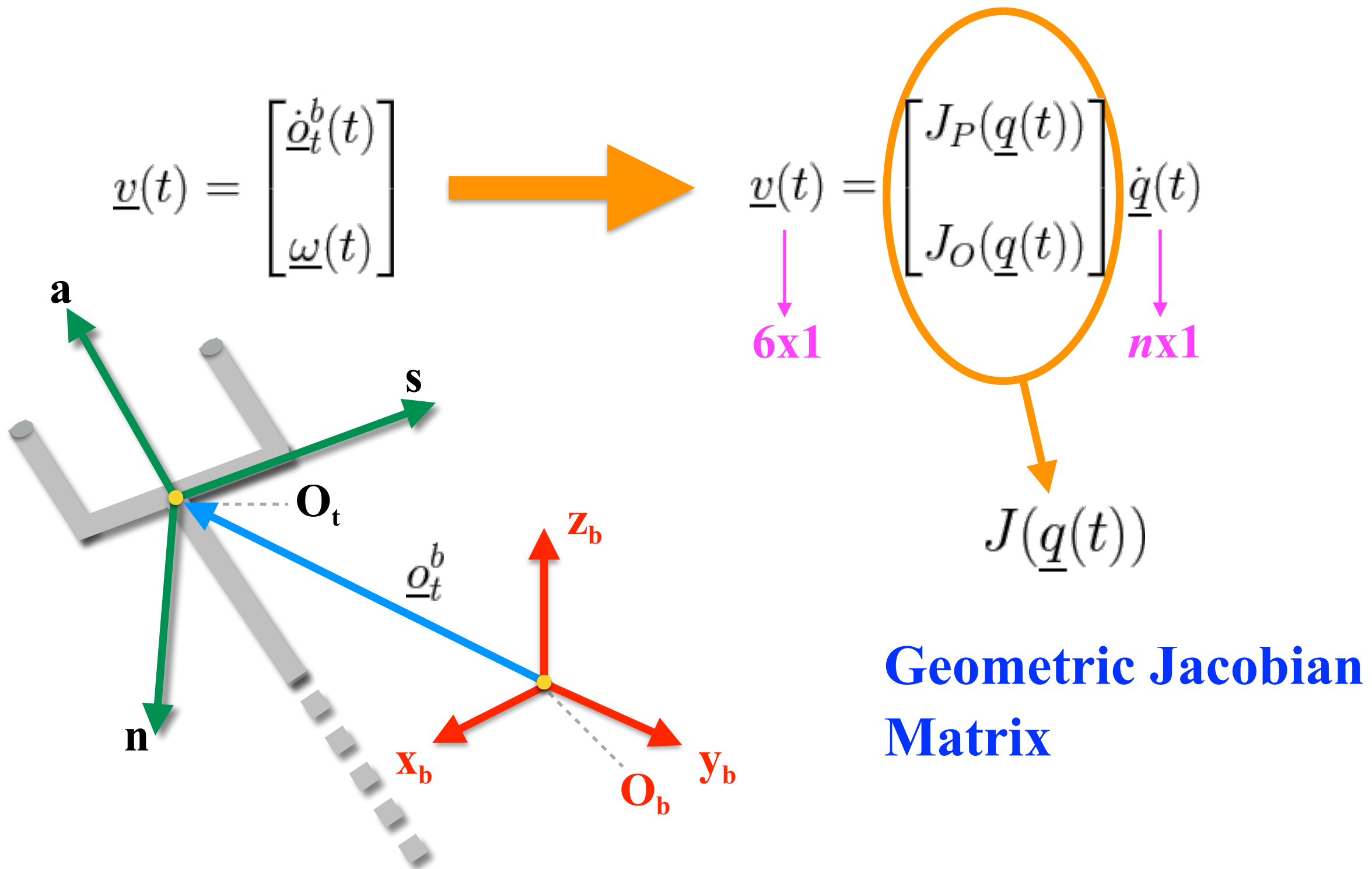
$$\underline{\omega}(t)$$



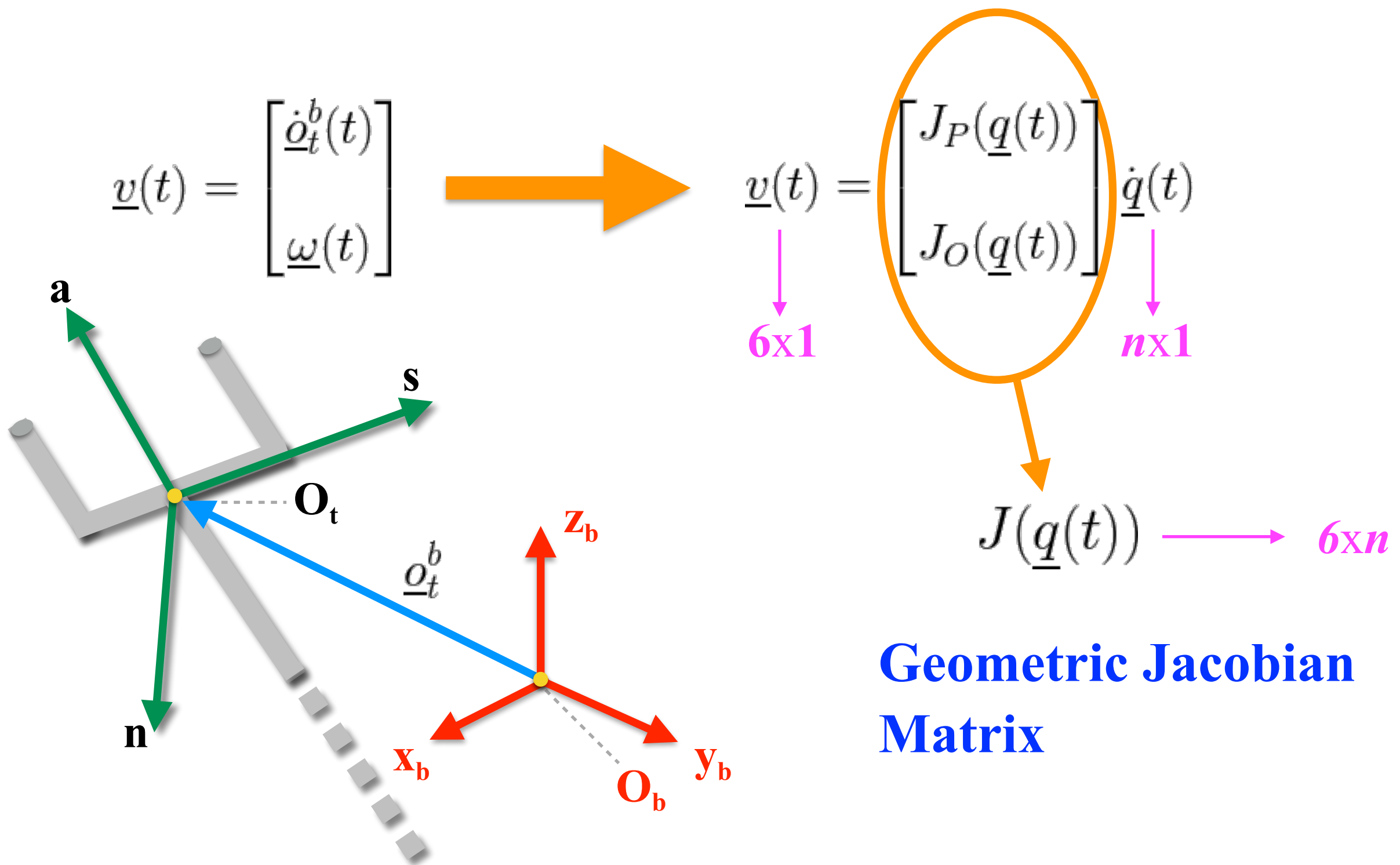
A Useful Definition: Geometric Jacobian Matrix



A Useful Definition: Geometric Jacobian Matrix



A Useful Definition: Geometric Jacobian Matrix





UNIVERSITÀ DI PAVIA
Dipartimento di
Ingegneria Industriale
e dell'Informazione

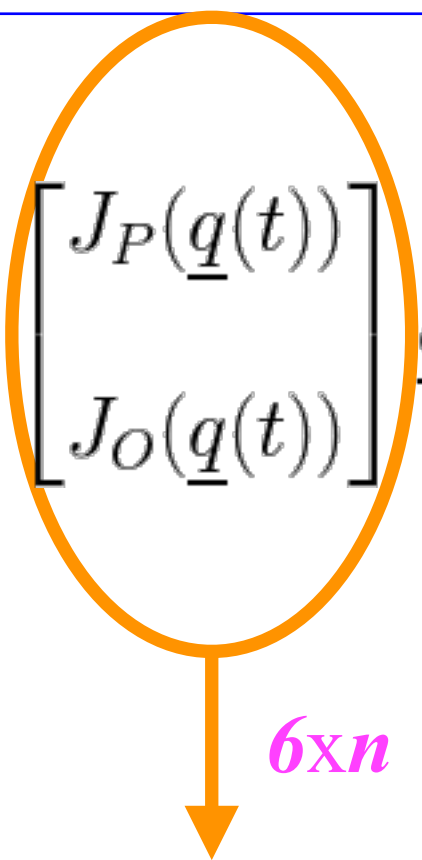
Course of Robot Control

Prof. Antonella Ferrara



Geometric Jacobian Matrix

How to Determine the Geometric Jacobian Matrix of a Robot Manipulator

$$\underline{v}(t) = \begin{bmatrix} J_P(\underline{q}(t)) \\ J_O(\underline{q}(t)) \end{bmatrix} \dot{\underline{q}}(t)$$


$6 \times n$

$$J(\underline{q}(t))$$

**Geometric Jacobian
Matrix**

How to Determine the Geometric Jacobian Matrix of a Robot Manipulator

$$\underline{v}(t) = \begin{bmatrix} J_P(\underline{q}(t)) \\ J_O(\underline{q}(t)) \end{bmatrix} \dot{\underline{q}}(t)$$

$J(\underline{q}(t)) \longrightarrow J(\underline{q}) = \begin{bmatrix} J_{P_1}(\underline{q}) & \cdots & J_{P_i}(\underline{q}) & \cdots & J_{P_n}(\underline{q}) \\ J_{O_1}(\underline{q}) & \cdots & J_{O_i}(\underline{q}) & \cdots & J_{O_n}(\underline{q}) \end{bmatrix}$

$6 \times n$

Geometric Jacobian Matrix

How to Determine the Geometric Jacobian Matrix of a Robot Manipulator

$$\underline{v}(t) = \begin{bmatrix} J_P(\underline{q}(t)) \\ J_O(\underline{q}(t)) \end{bmatrix} \dot{\underline{q}}(t)$$

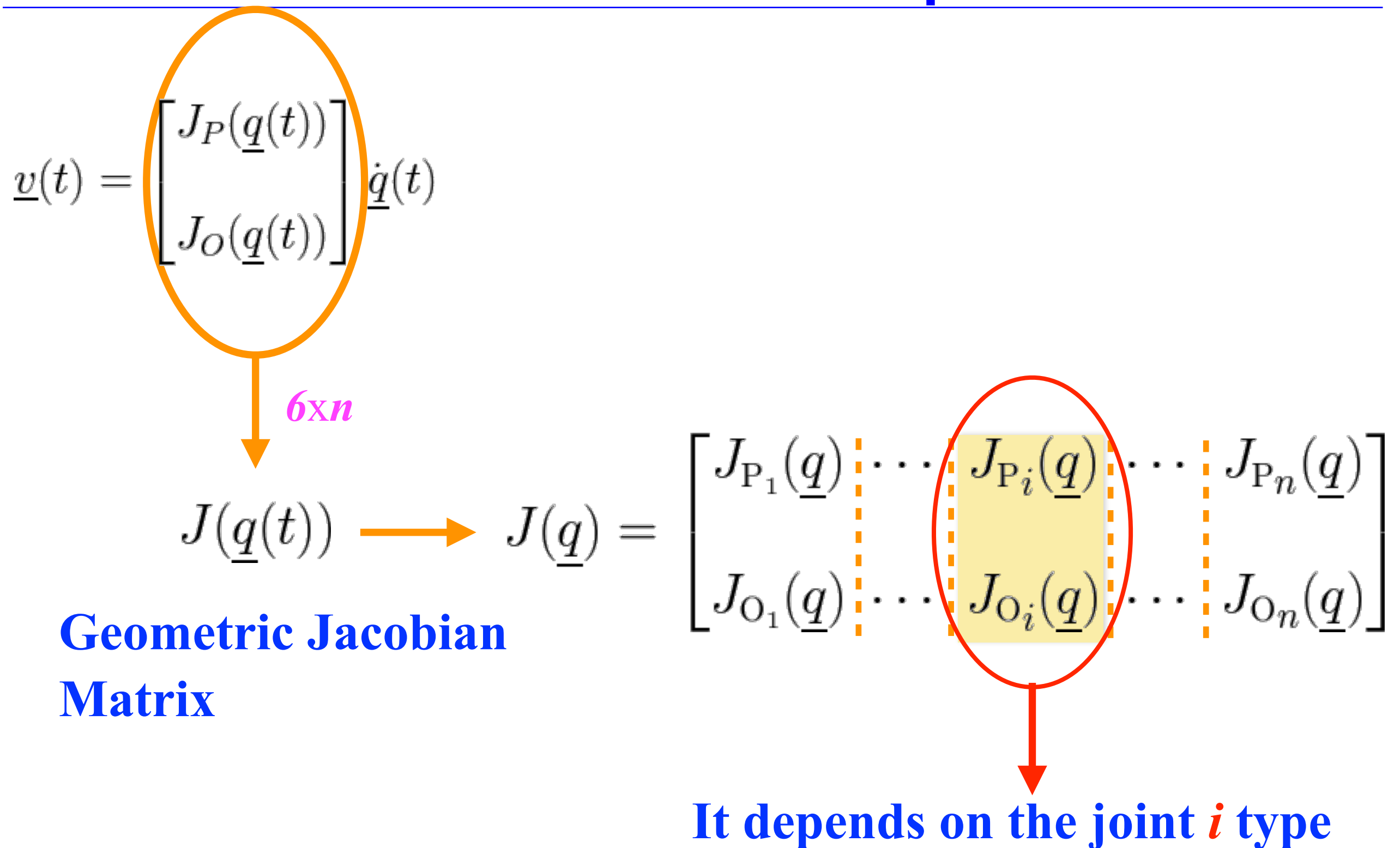
$\downarrow$ $6 \times n$
 $J(\underline{q}(t)) \longrightarrow J(\underline{q}) =$

$$\begin{bmatrix} J_{P_1}(\underline{q}) & \cdots & J_{P_i}(\underline{q}) & \cdots & J_{P_n}(\underline{q}) \\ J_{O_1}(\underline{q}) & \cdots & J_{O_i}(\underline{q}) & \cdots & J_{O_n}(\underline{q}) \end{bmatrix}$$

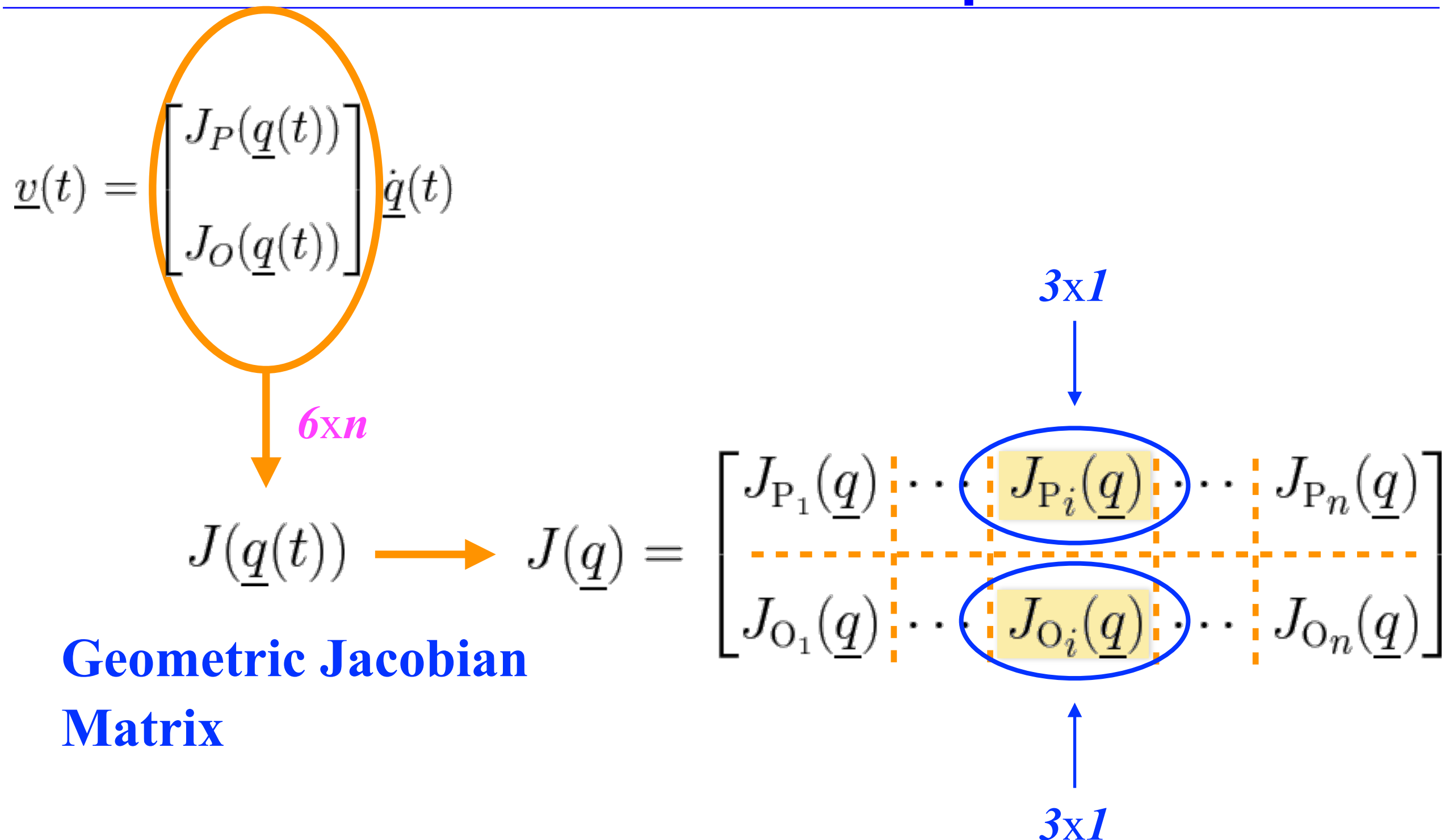
$\begin{matrix} 6 \times 1 & & 6 \times 1 & & 6 \times 1 \\ \downarrow & & \downarrow & & \downarrow \end{matrix}$

Geometric Jacobian Matrix

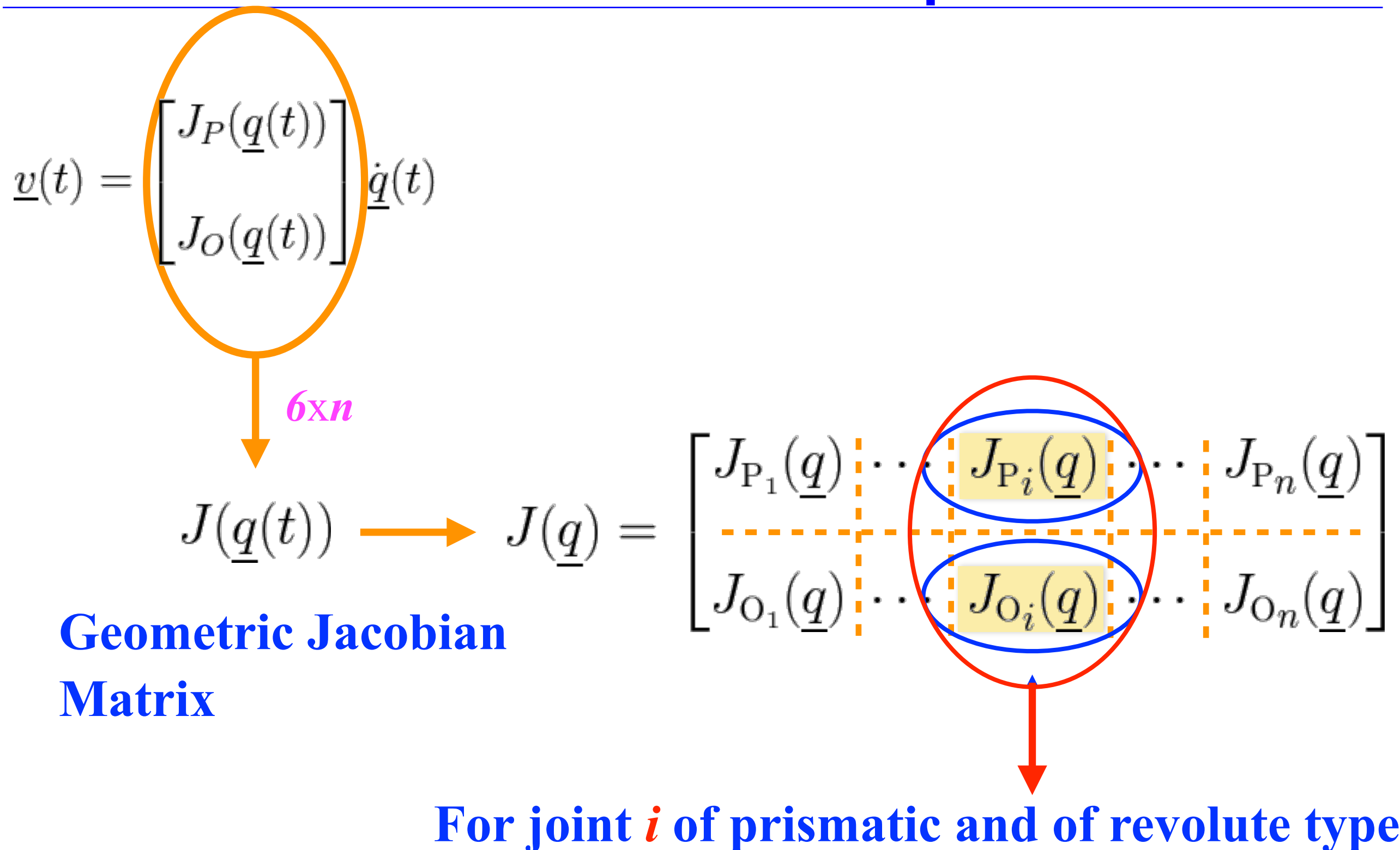
How to Determine the Geometric Jacobian Matrix of a Robot Manipulator



How to Determine the Geometric Jacobian Matrix of a Robot Manipulator



How to Determine the Geometric Jacobian Matrix of a Robot Manipulator





UNIVERSITÀ DI PAVIA
Dipartimento di
Ingegneria Industriale
e dell'Informazione

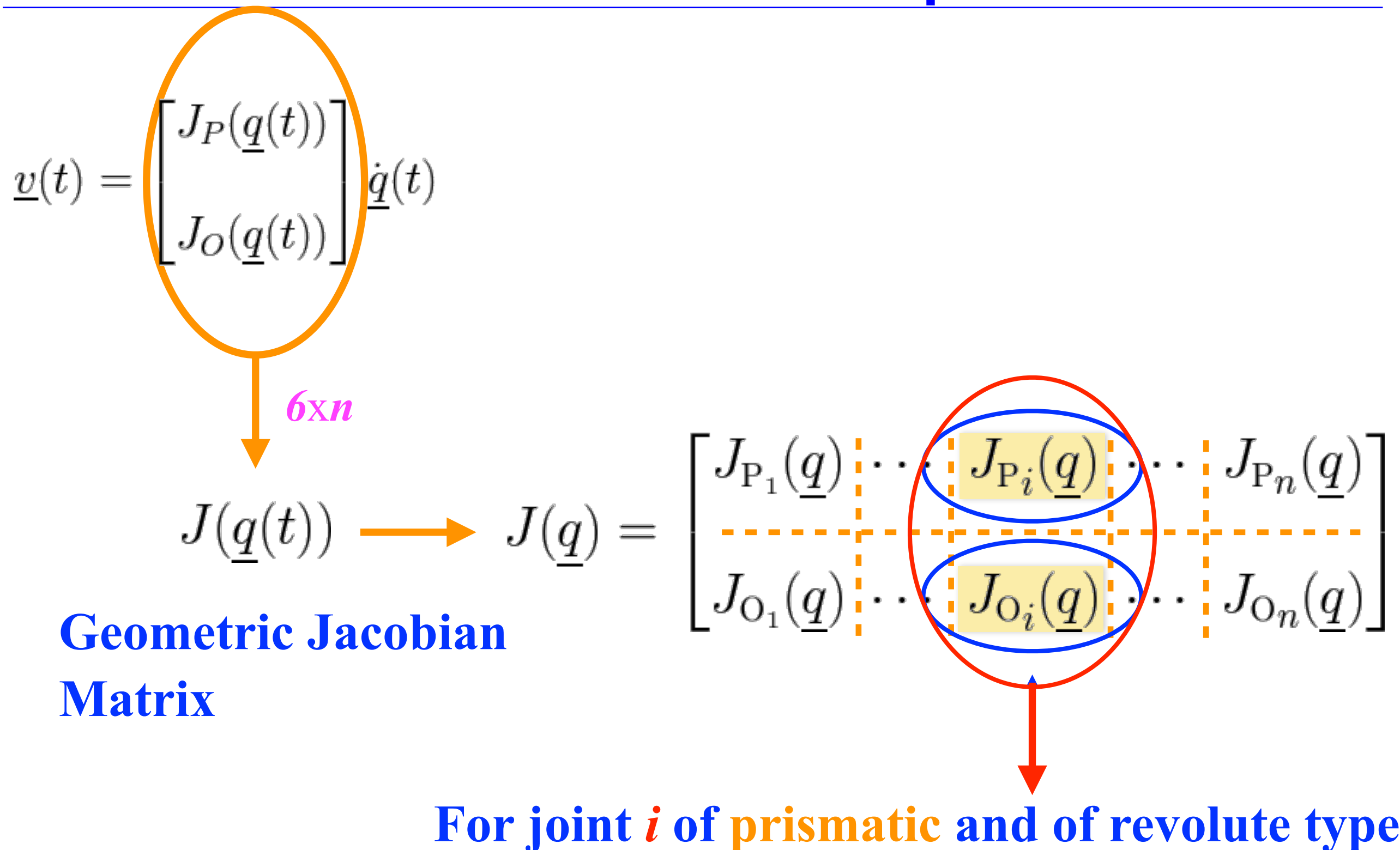
Course of Robot Control

Prof. Antonella Ferrara

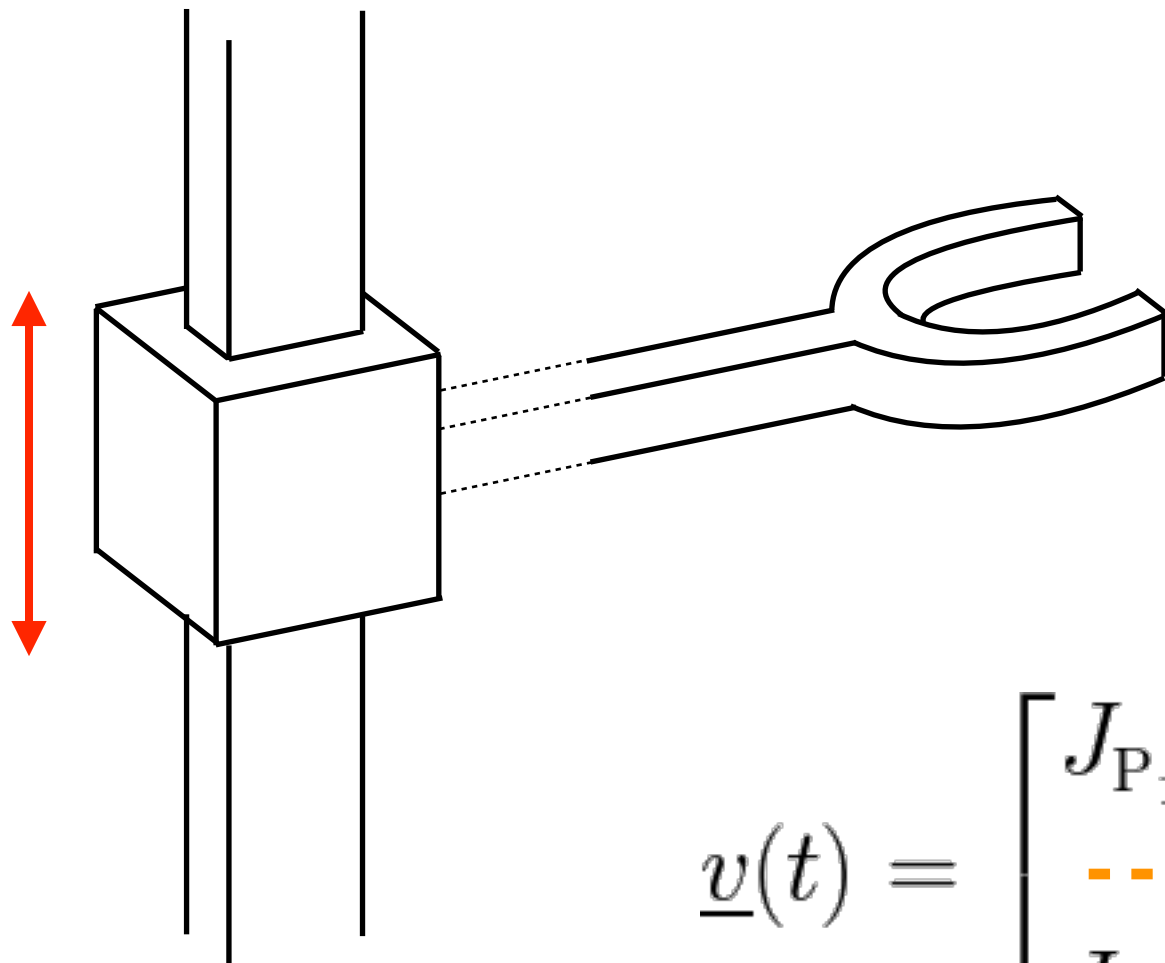


Geometric Jacobian Matrix: Prismatic Joint Case

How to Determine the Geometric Jacobian Matrix of a Robot Manipulator



How to Determine the Geometric Jacobian Matrix Column i with i prismatic joint

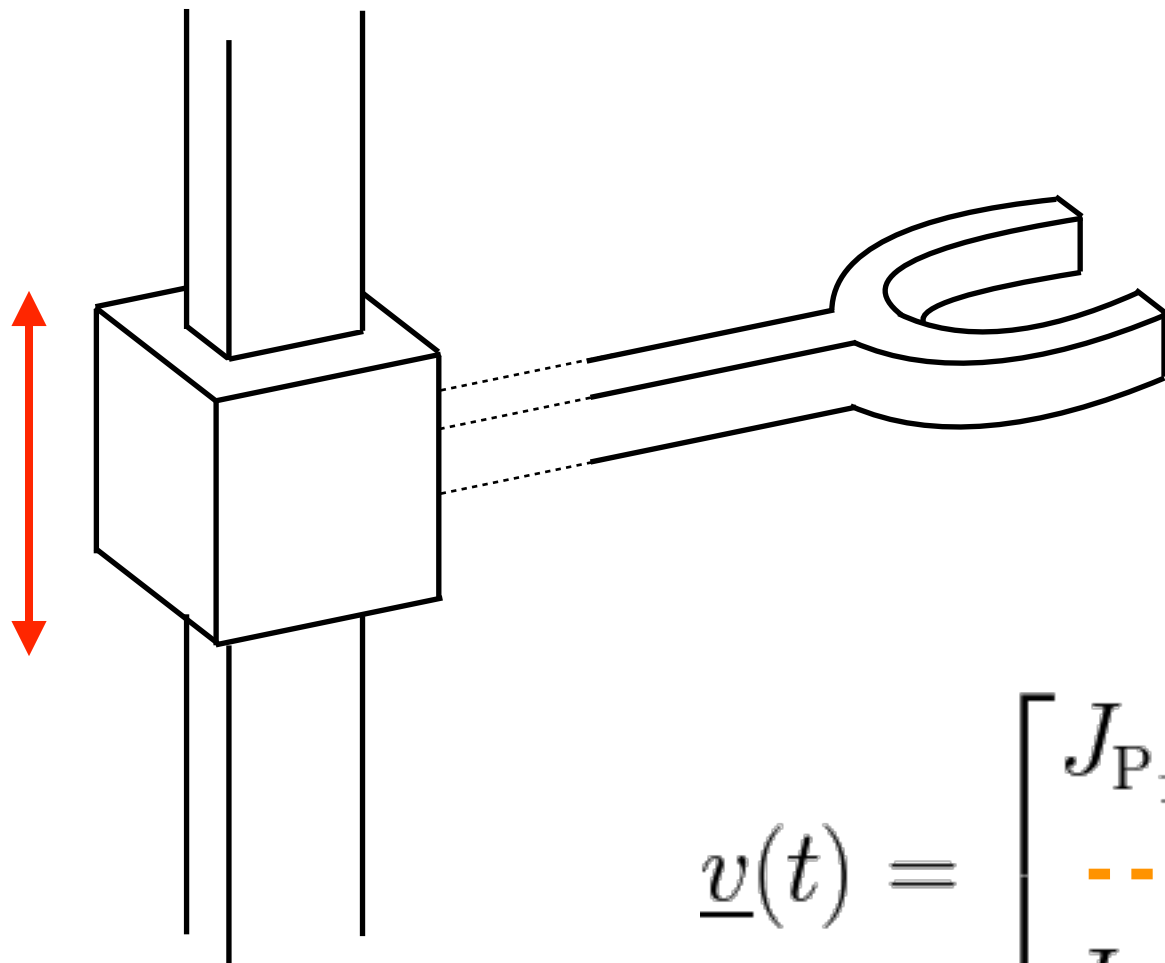


Assume joint i of prismatic type

$$\underline{v}(t) = \begin{bmatrix} J_{P_1}(\underline{q}) & \cdots & J_{P_i}(\underline{q}) & \cdots & J_{P_n}(\underline{q}) \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ J_{O_1}(\underline{q}) & \cdots & J_{O_i}(\underline{q}) & \cdots & J_{O_n}(\underline{q}) \end{bmatrix} \underline{\dot{q}}(t)$$

Geometric Jacobian Matrix

How to Determine the Geometric Jacobian Matrix Column i with i prismatic joint

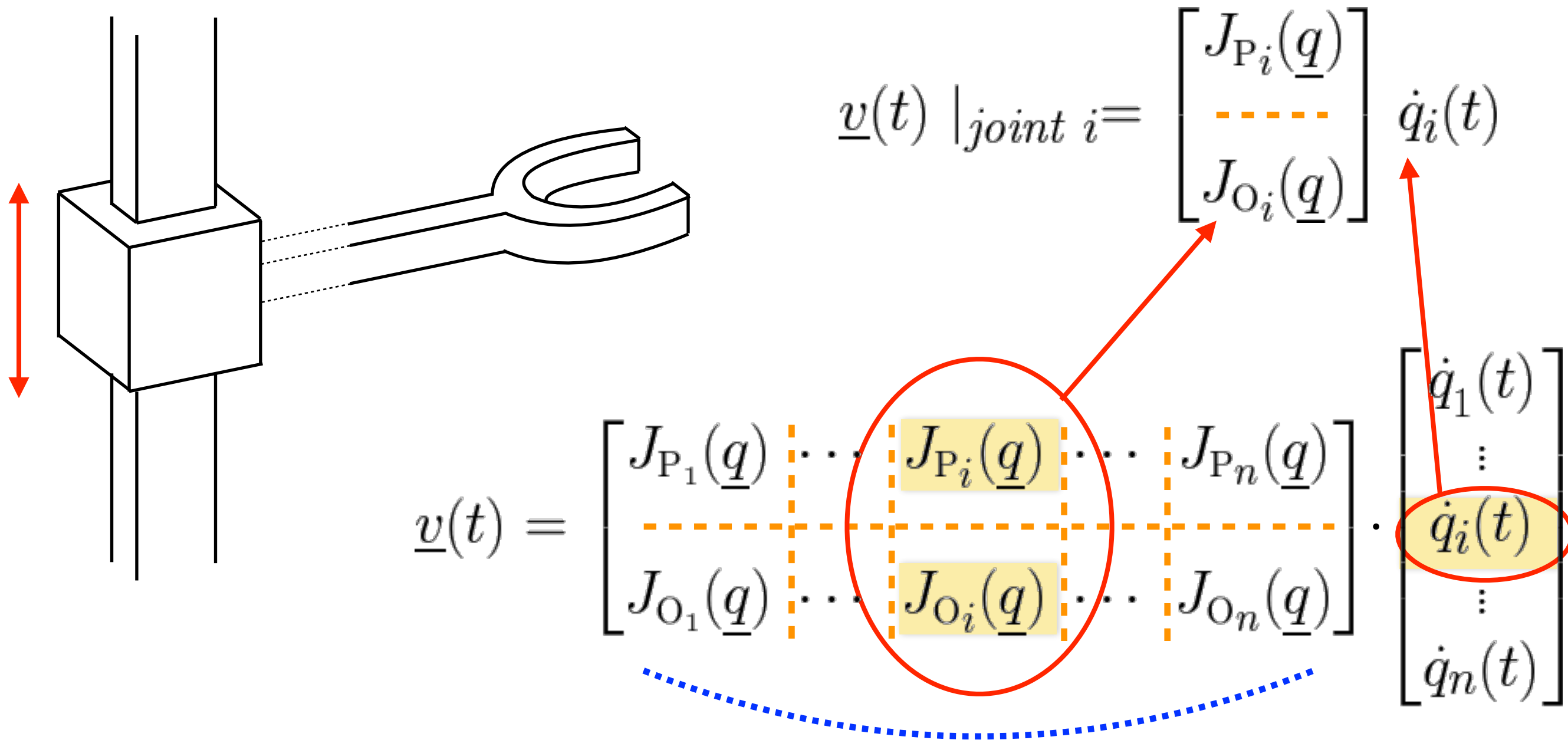


Assume joint i of prismatic type

$$\underline{v}(t) = \begin{bmatrix} J_{P_1}(\underline{q}) & \cdots & J_{P_i}(\underline{q}) & \cdots & J_{P_n}(\underline{q}) \\ \vdots & & \vdots & & \vdots \\ J_{O_1}(\underline{q}) & \cdots & J_{O_i}(\underline{q}) & \cdots & J_{O_n}(\underline{q}) \end{bmatrix} \cdot \begin{bmatrix} \dot{q}_1(t) \\ \vdots \\ \dot{q}_i(t) \\ \vdots \\ \dot{q}_n(t) \end{bmatrix}$$

Geometric Jacobian Matrix

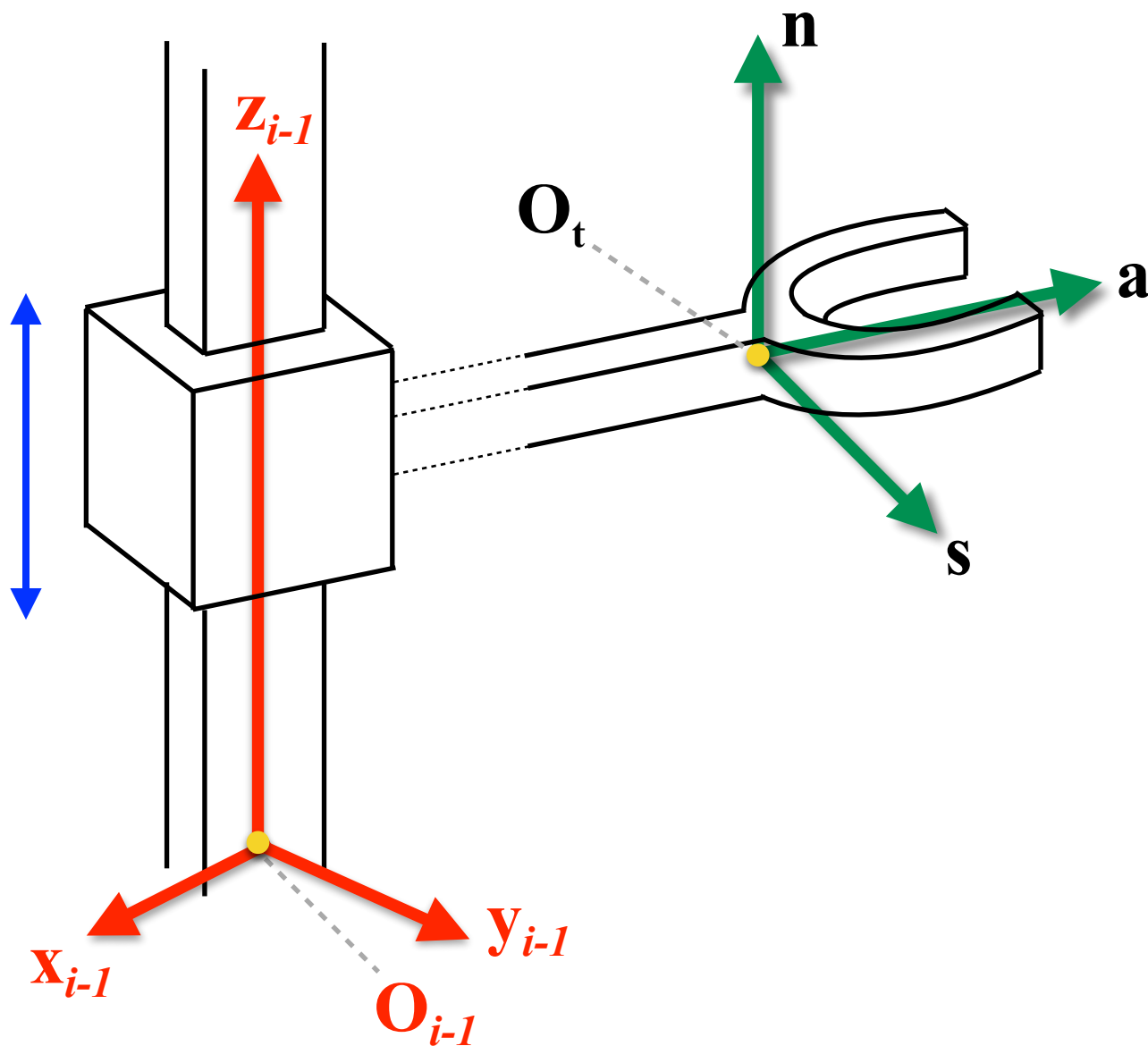
How to Determine the Geometric Jacobian Matrix Column i with i prismatic joint



Geometric Jacobian Matrix

How to Determine the Geometric Jacobian

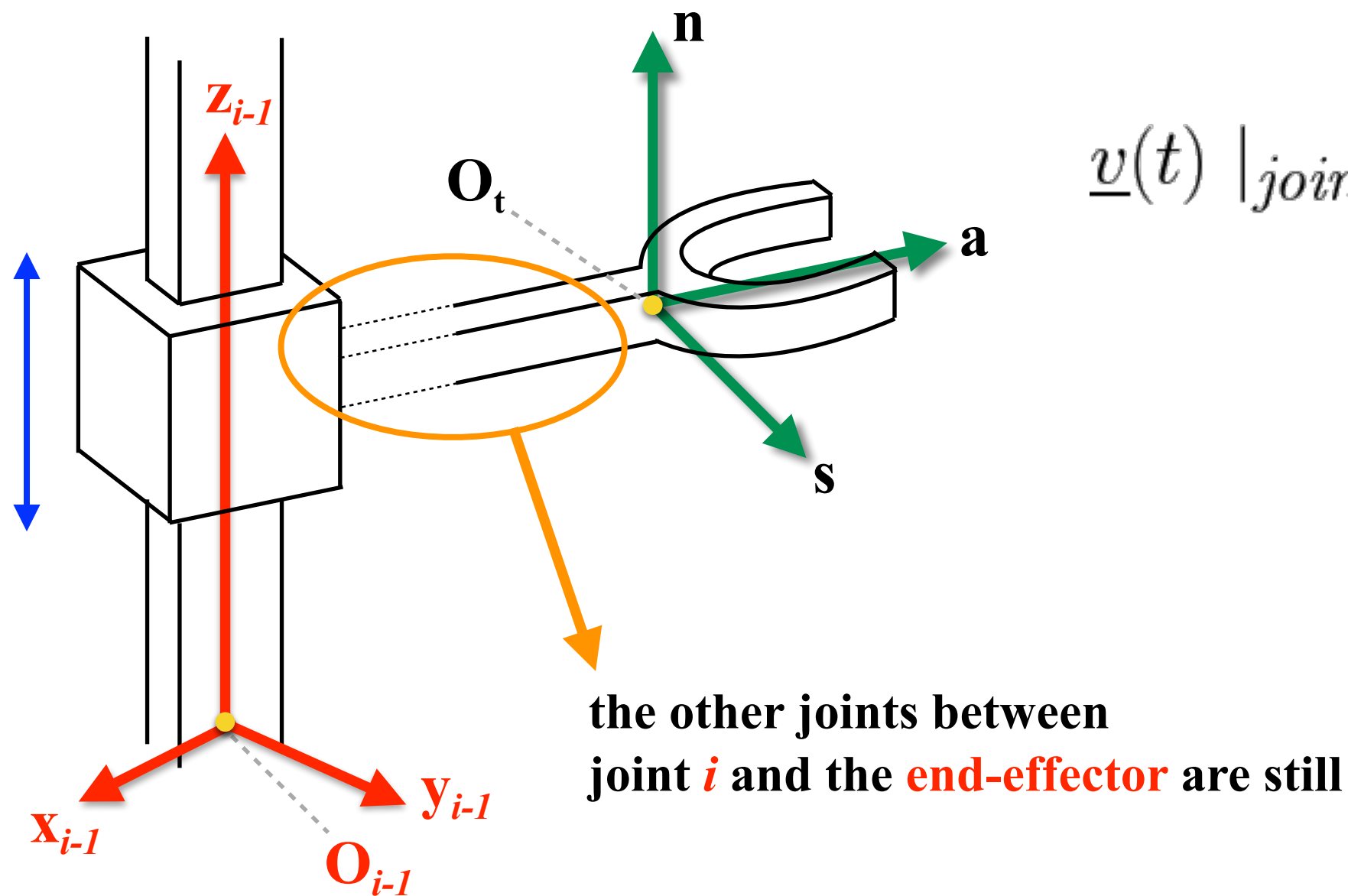
Matrix **Column i with i prismatic joint**



$$\underline{v}(t) \mid_{joint\ i} = \begin{bmatrix} J_{P_i}(\underline{q}) \\ \text{---} \\ J_{O_i}(\underline{q}) \end{bmatrix} \dot{q}_i(t)$$

joint i of prismatic type

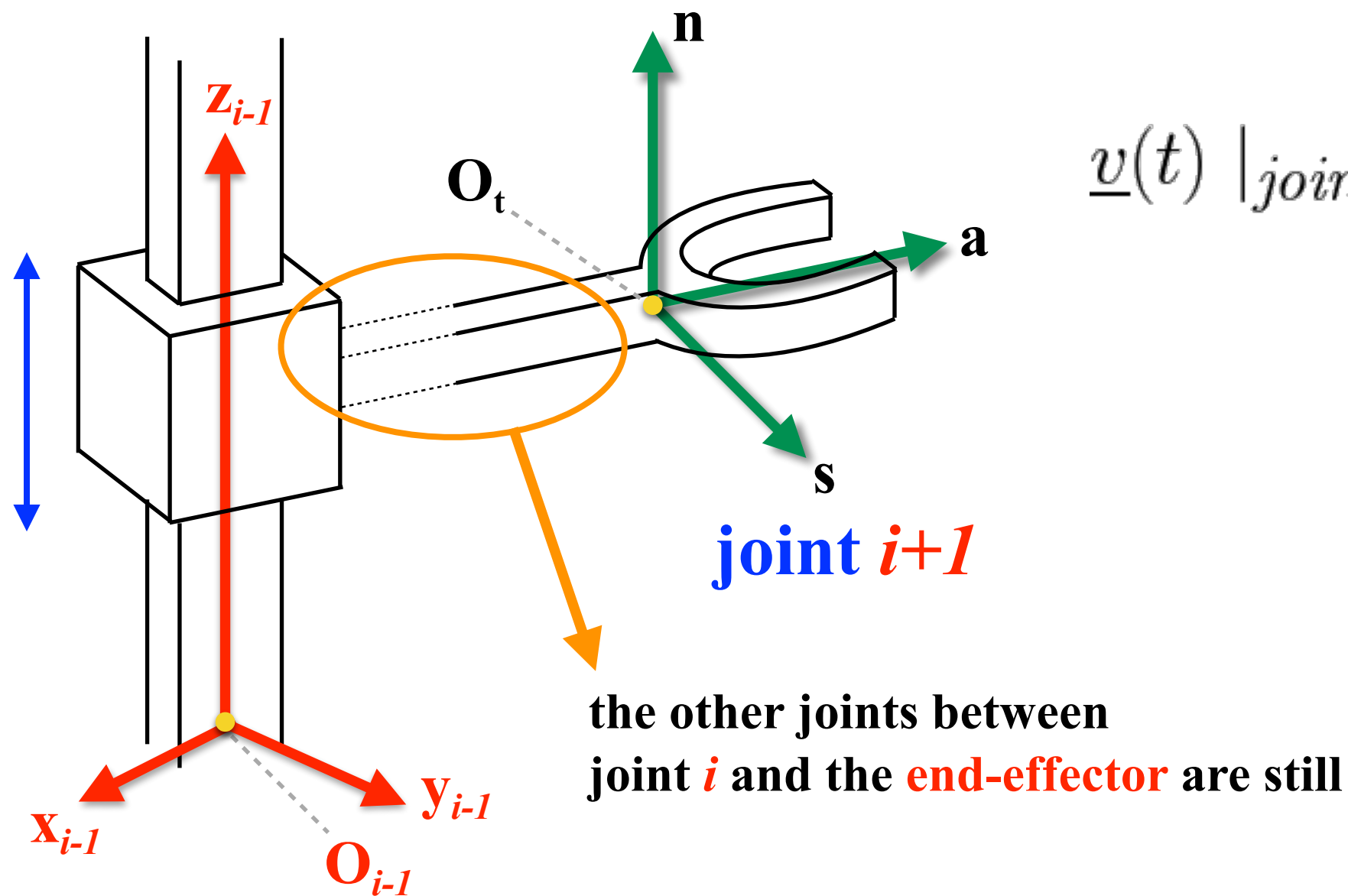
How to Determine the Geometric Jacobian Matrix Column i with i prismatic joint



$$\underline{v}(t) \mid_{joint\ i} = \begin{bmatrix} J_{P_i}(\underline{q}) \\ \text{---} \\ J_{O_i}(\underline{q}) \end{bmatrix} \dot{q}_i(t)$$

joint i of prismatic type

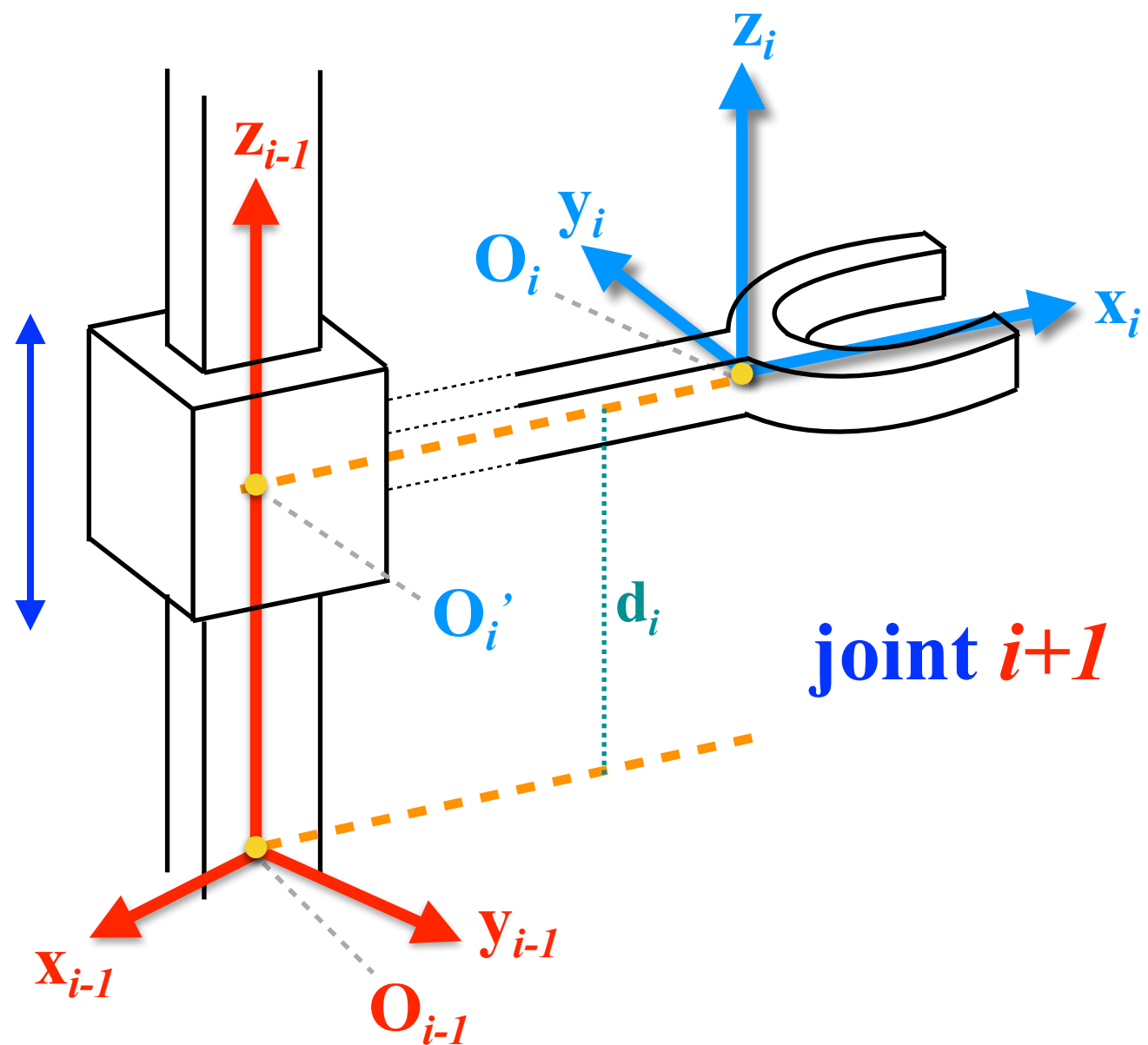
How to Determine the Geometric Jacobian Matrix Column i with i prismatic joint



$$\underline{v}(t) \mid_{joint\ i} = \begin{bmatrix} J_{P_i}(\underline{q}) \\ \text{---} \\ J_{O_i}(\underline{q}) \end{bmatrix} \dot{q}_i(t)$$

joint i of prismatic type

How to Determine the Geometric Jacobian Matrix **Column i with i prismatic joint**

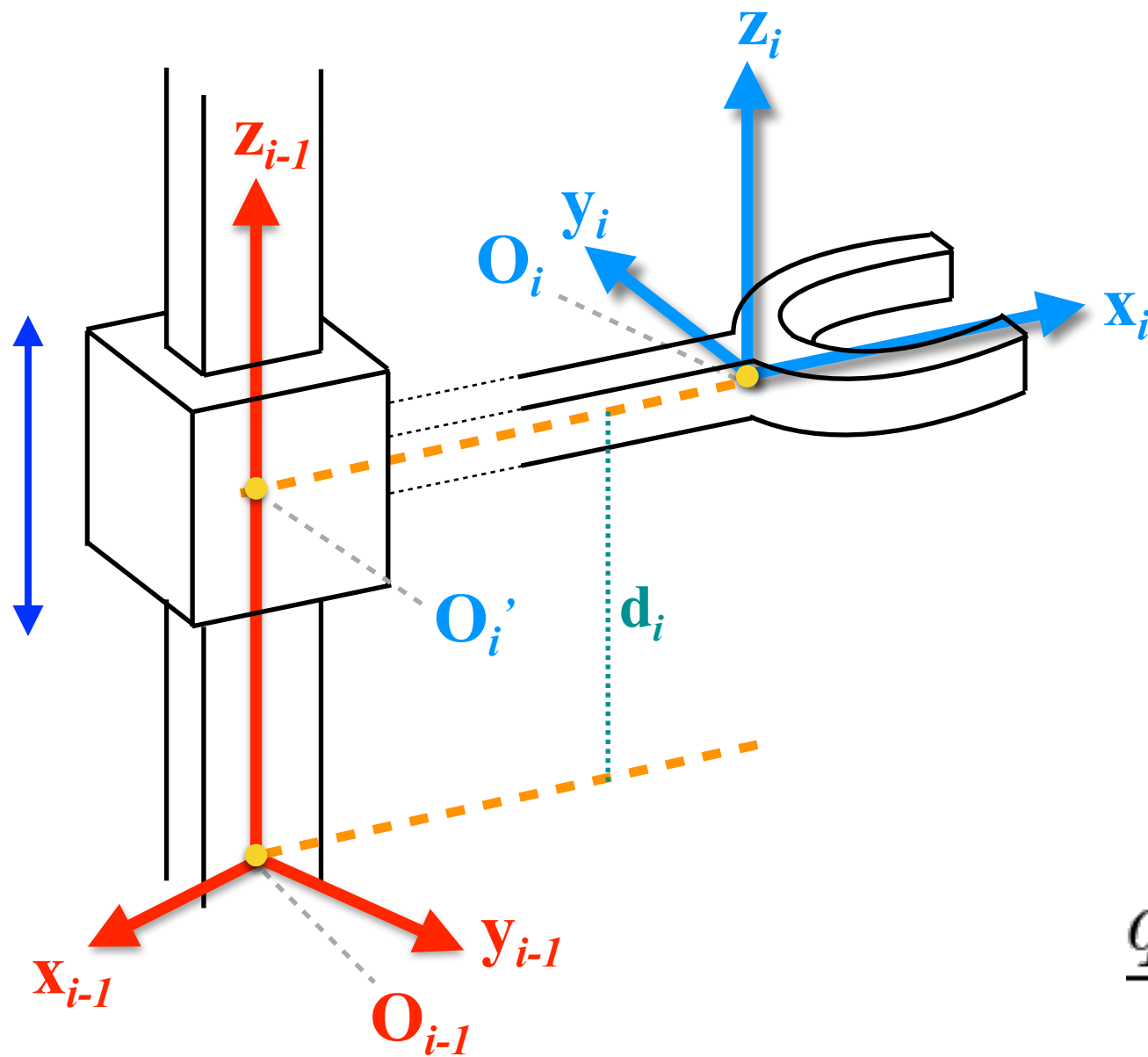


$$\underline{v}(t) \mid_{joint\ i} = \begin{bmatrix} J_{P_i}(\underline{q}) \\ \text{---} \\ J_{O_i}(\underline{q}) \end{bmatrix} \dot{q}_i(t)$$

joint i of prismatic type

How to Determine the Geometric Jacobian

Matrix **Column i with i prismatic joint**

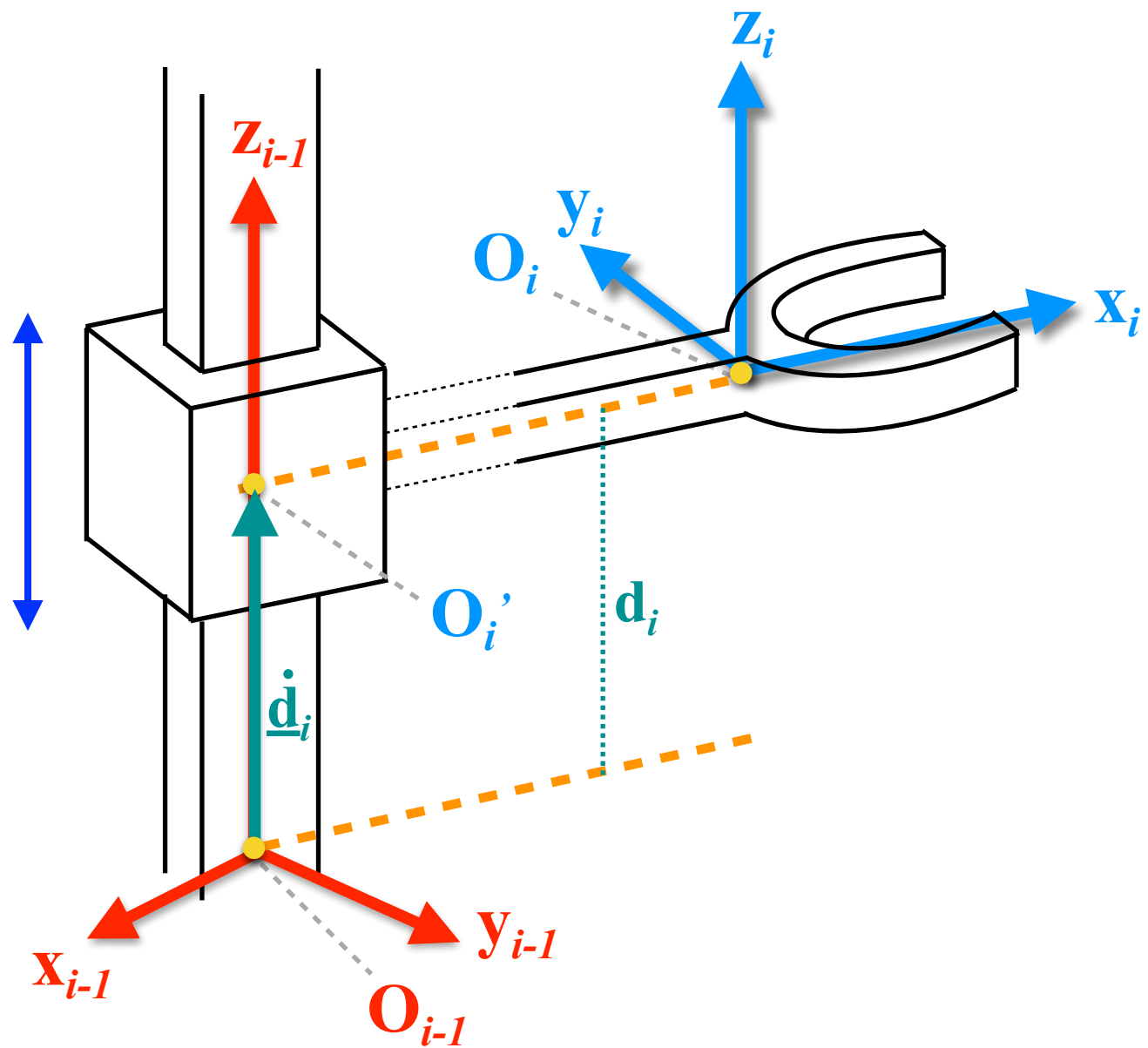


$$\underline{v}(t) \mid_{joint\ i} = \begin{bmatrix} J_{P_i}(\underline{q}) \\ \text{---} \\ J_{O_i}(\underline{q}) \end{bmatrix} \dot{q}_i(t)$$

$$\underline{q}(t) = \begin{bmatrix} q_1(t) \\ \vdots \\ q_i(t) \\ \vdots \\ q_n(t) \end{bmatrix} = \begin{bmatrix} q_1(t) \\ \vdots \\ d_i(t) \\ \vdots \\ q_n(t) \end{bmatrix}$$

joint i of prismatic type

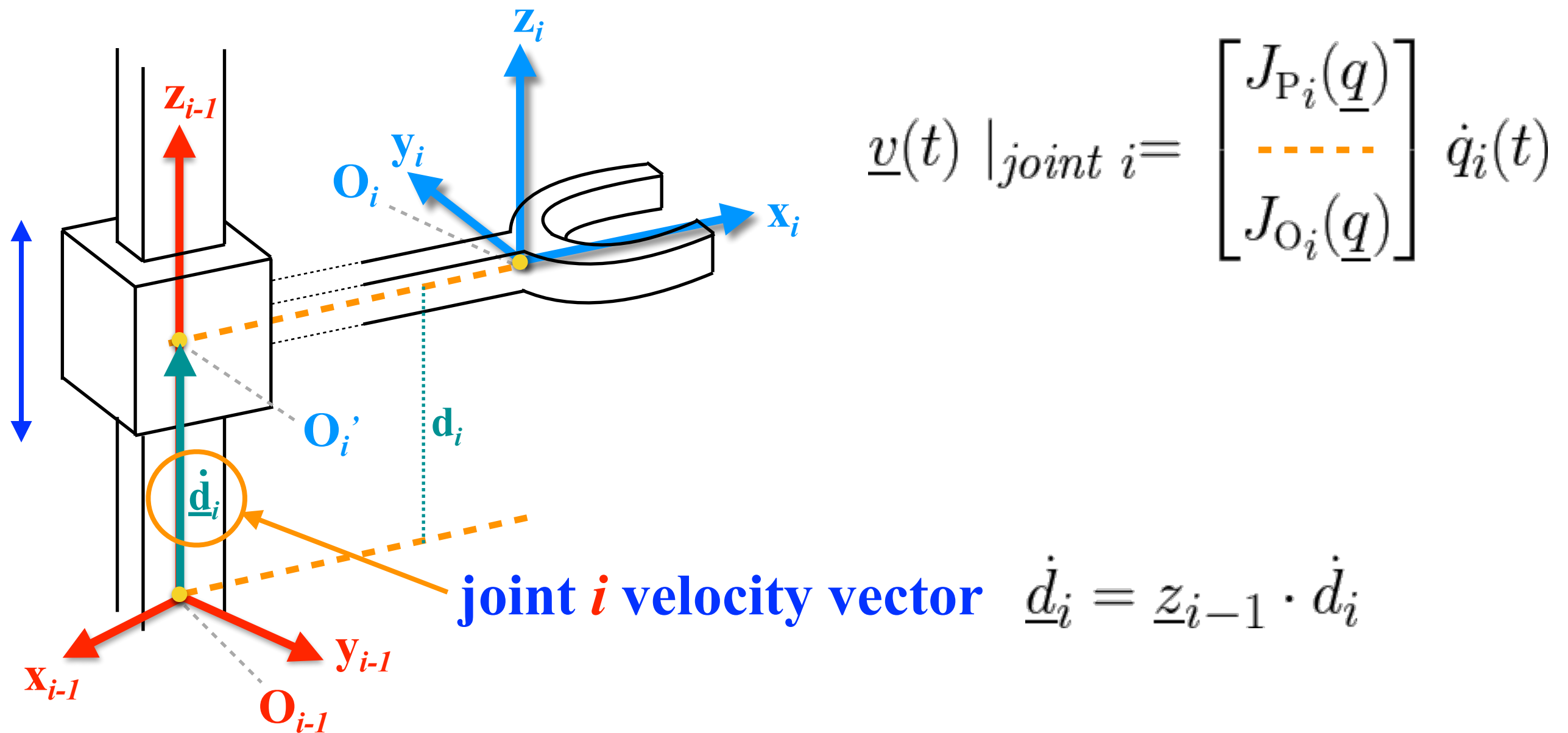
How to Determine the Geometric Jacobian Matrix Column i with i prismatic joint



$$\underline{v}(t) \mid_{joint \ i} = \begin{bmatrix} J_{P_i}(\underline{q}) \\ \text{---} \\ J_{O_i}(\underline{q}) \end{bmatrix} \dot{q}_i(t)$$

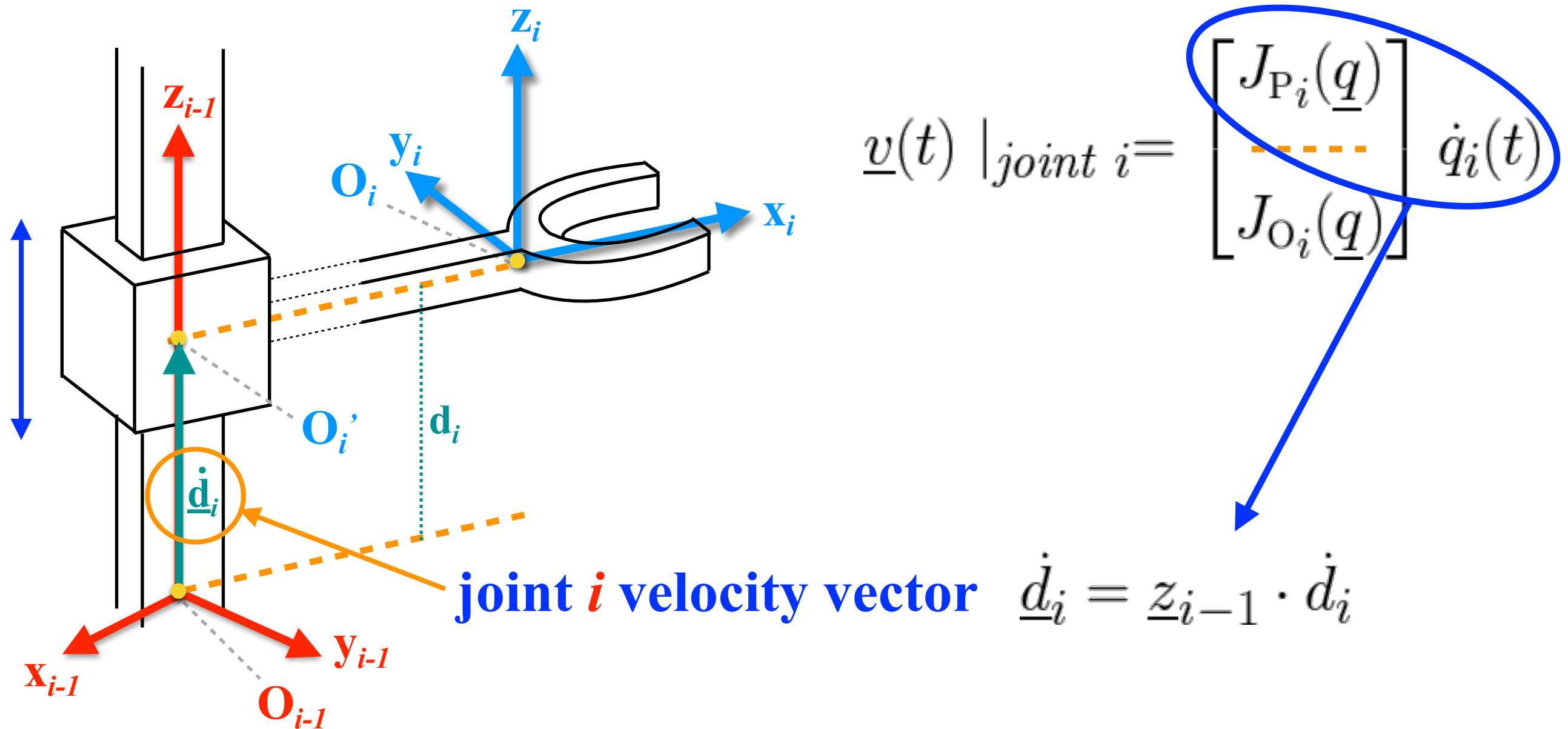
joint i of prismatic type

How to Determine the Geometric Jacobian Matrix Column i with i prismatic joint



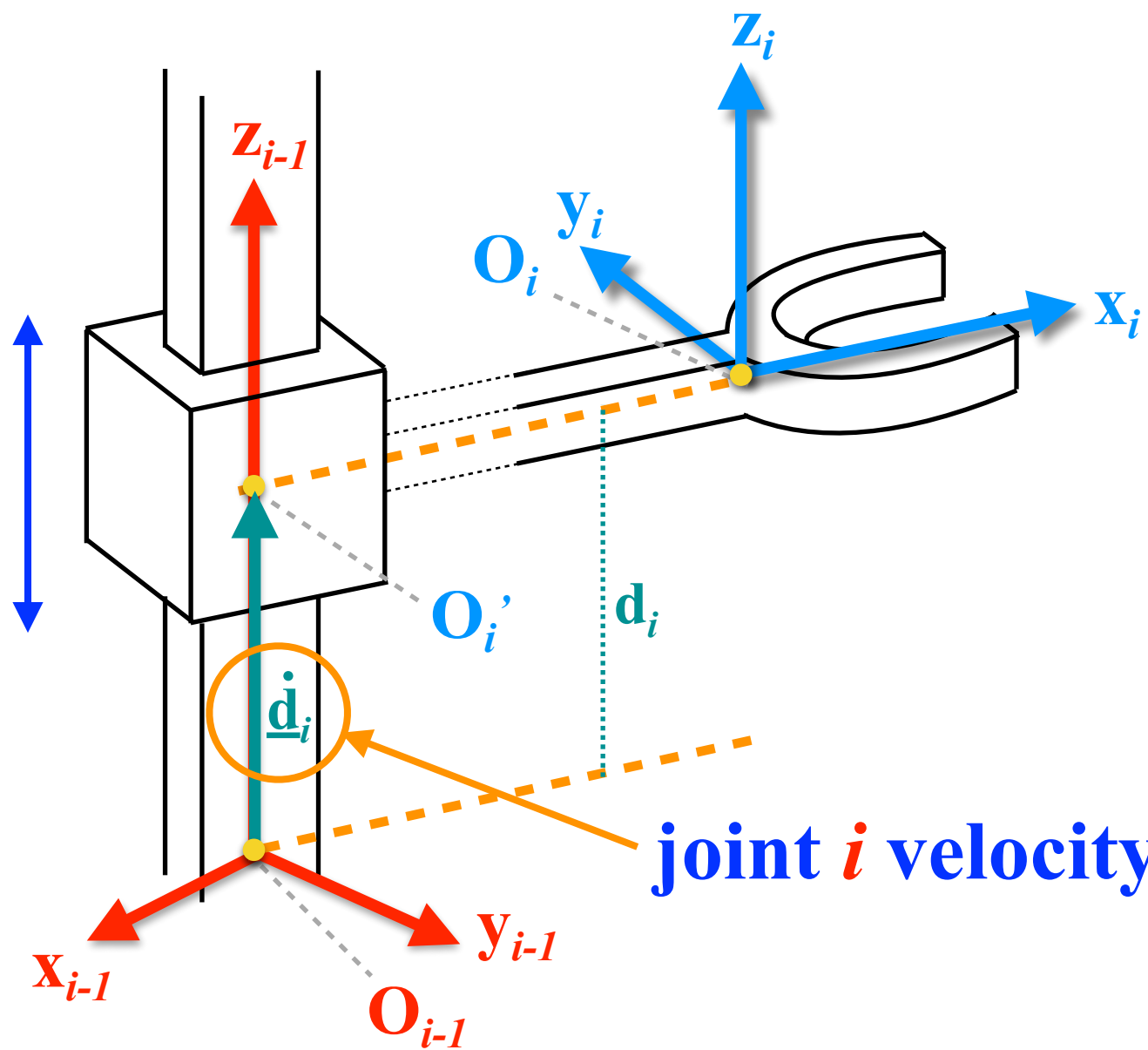
joint i of prismatic type

How to Determine the Geometric Jacobian Matrix Column i with i prismatic joint



joint i of prismatic type

How to Determine the Geometric Jacobian Matrix Column i with i prismatic joint



joint i velocity vector

$\underline{v}(t) \mid_{joint\ i} =$

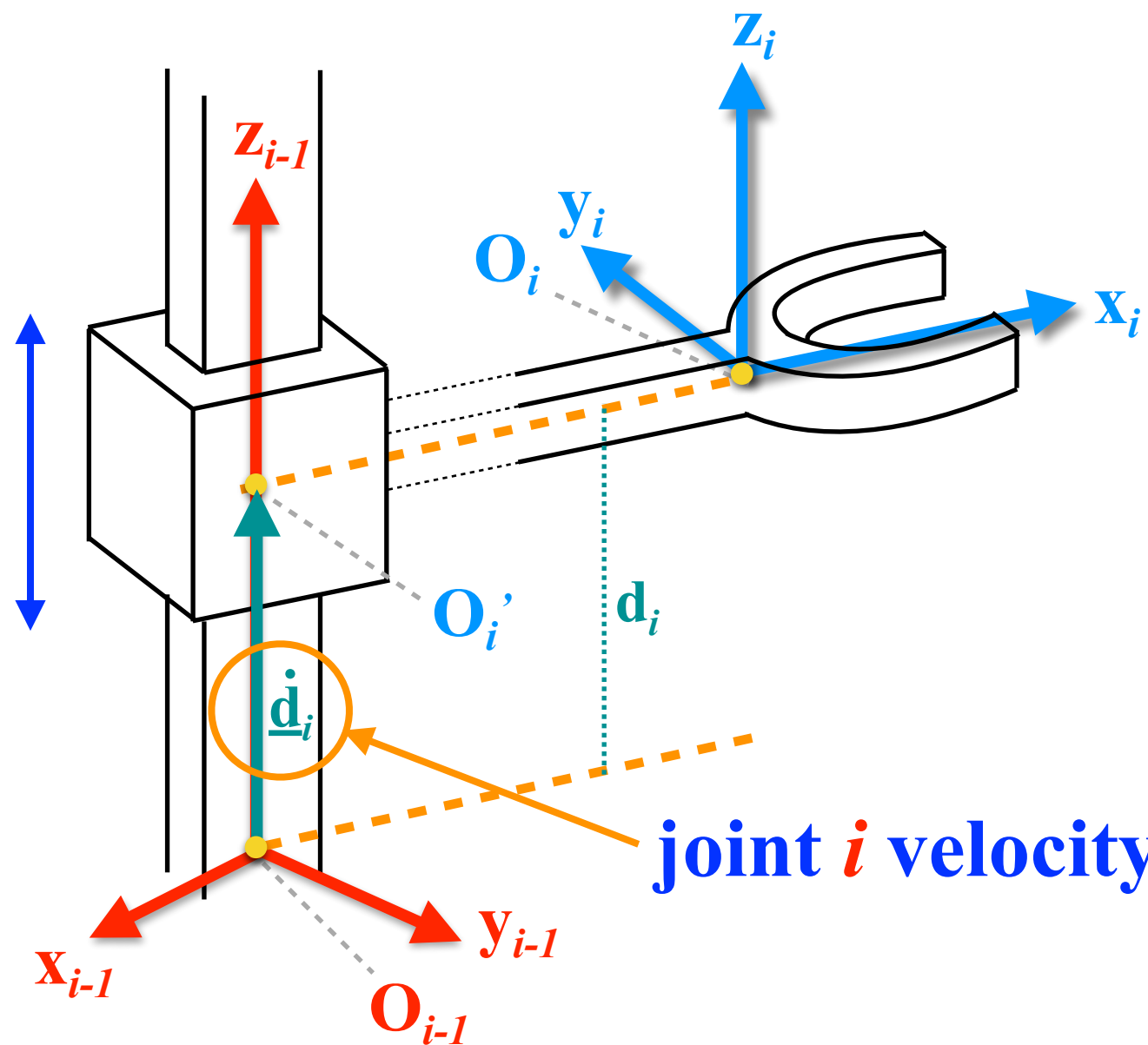
$$\begin{bmatrix} J_{P_i}(\underline{q}) \\ \vdots \\ J_{O_i}(\underline{q}) \end{bmatrix} \dot{q}_i(t)$$

$$\underline{\dot{d}}_i = \underline{z}_{i-1} \cdot \dot{d}_i$$

joint i of prismatic type

$$J_{P_i}(\underline{q}) \dot{q}_i(t) = \underline{z}_{i-1} \cdot \dot{d}_i$$

How to Determine the Geometric Jacobian Matrix Column i with i prismatic joint



joint i of prismatic type

joint i velocity vector

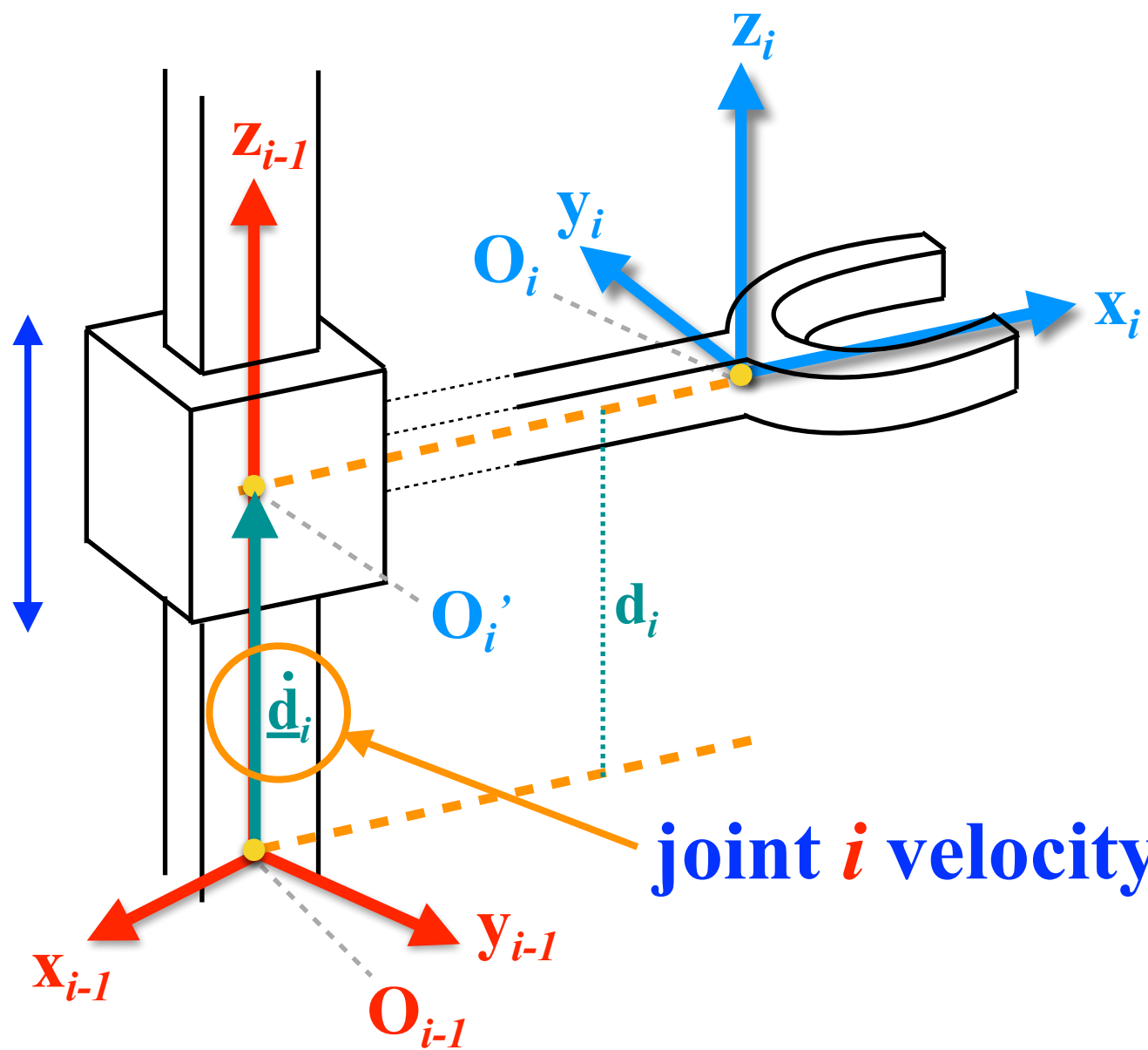
$\underline{v}(t) \mid_{joint\ i} =$

$$\begin{bmatrix} J_{P_i}(\underline{q}) \\ \vdots \\ J_{O_i}(\underline{q}) \end{bmatrix} \dot{q}_i(t)$$

$$\underline{\dot{d}}_i = \underline{z}_{i-1} \cdot \dot{d}_i$$

$$J_{P_i}(\underline{q}) \dot{q}_i(t) = \underline{z}_{i-1} \cdot \dot{d}_i$$

How to Determine the Geometric Jacobian Matrix Column i with i prismatic joint



$$\underline{v}(t) \mid_{joint\ i=} \begin{bmatrix} J_{P_i}(\underline{q}) \\ \vdots \\ J_{O_i}(\underline{q}) \end{bmatrix} \dot{q}_i(t)$$

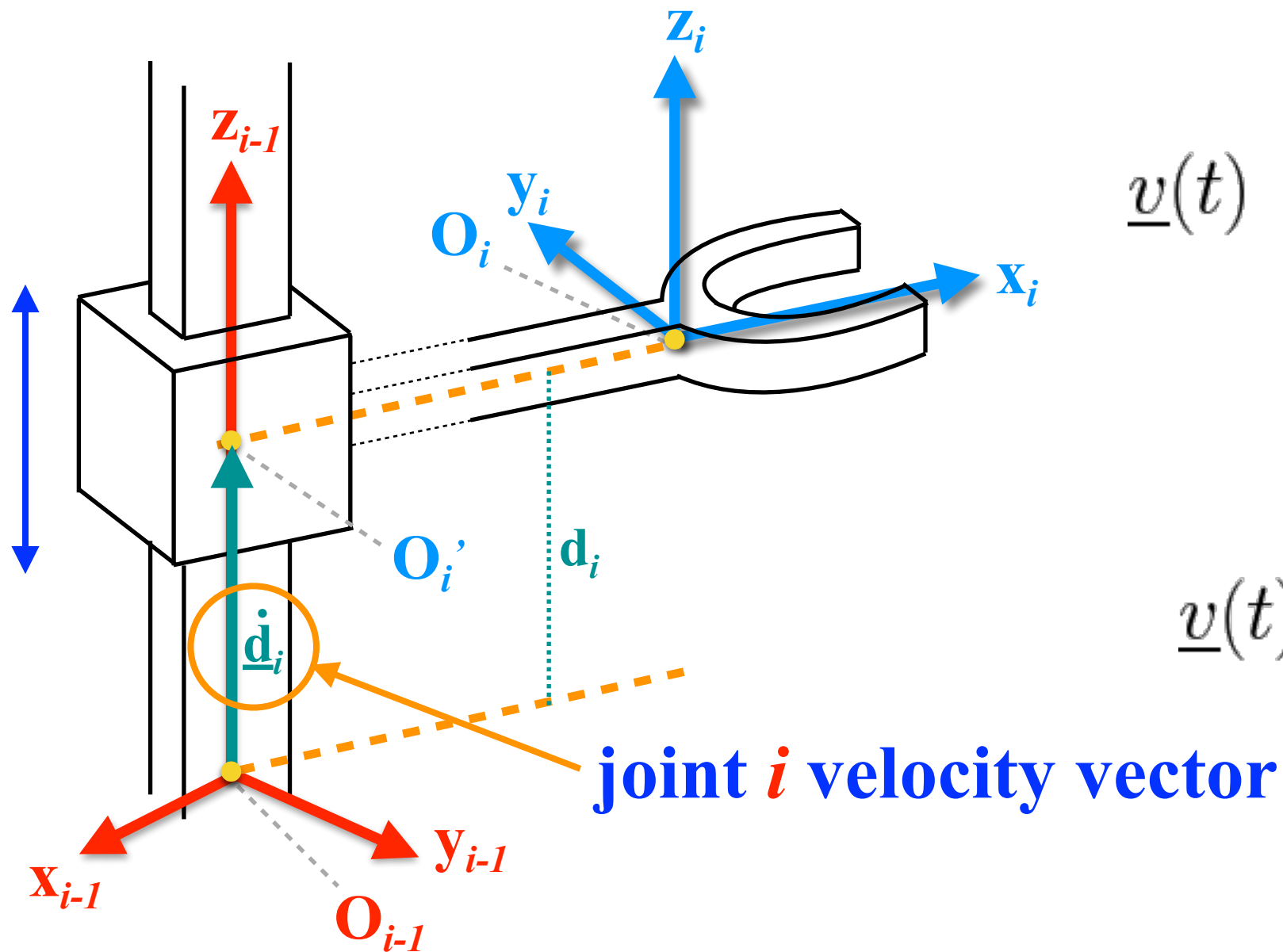
joint i velocity vector $\underline{d}_i = \underline{z}_{i-1} \cdot \dot{d}_i$

joint i of prismatic type

$$J_{P_i}(\underline{q}) \dot{q}_i(t) = \underline{z}_{i-1} \cdot \dot{d}_i$$

How to Determine the Geometric Jacobian

Matrix Column i with i prismatic joint

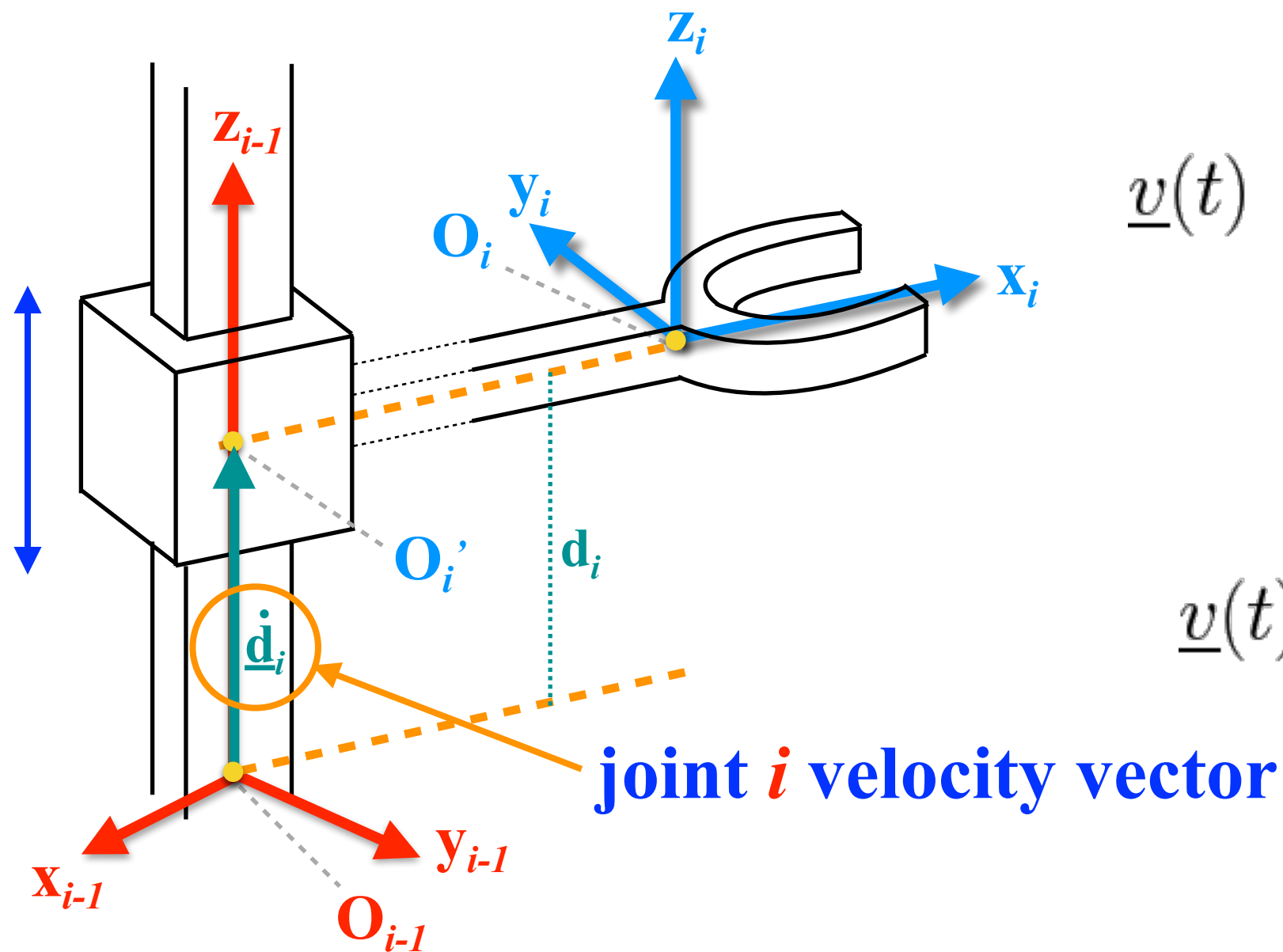


$$\underline{v}(t) \mid_{joint\ i=} \begin{bmatrix} J_{P_i}(\underline{q}) \\ \dots \\ J_{O_i}(\underline{q}) \end{bmatrix} \dot{q}_i(t)$$

$$\underline{v}(t) \mid_{joint\ i=} \begin{bmatrix} \underline{z}_{i-1} \\ \dots \\ 0 \\ 0 \\ 0 \end{bmatrix} \dot{q}_i(t)$$

joint i of prismatic type

How to Determine the Geometric Jacobian Matrix Column i with i prismatic joint



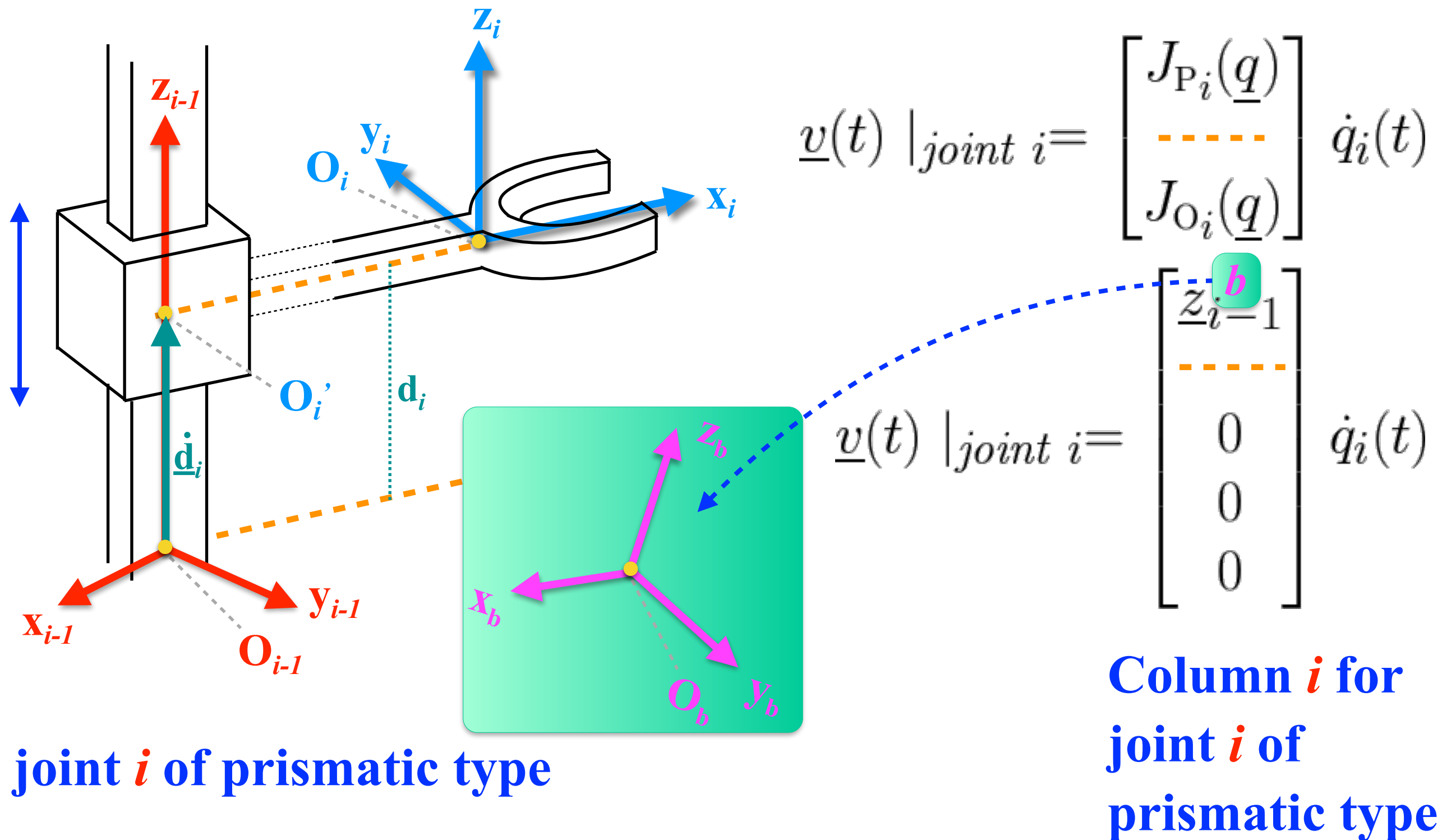
joint i of prismatic type

$$\underline{v}(t) \mid_{joint\ i} = \begin{bmatrix} J_{P_i}(\underline{q}) \\ \vdots \\ J_{O_i}(\underline{q}) \end{bmatrix} \dot{q}_i(t)$$

$$\underline{v}(t) \mid_{joint\ i} = \begin{bmatrix} z_{i-1} \\ \vdots \\ 0 \\ 0 \\ 0 \end{bmatrix} \dot{q}_i(t)$$

Column i for joint i of prismatic type

How to Determine the Geometric Jacobian Matrix **Column i with i prismatic joint**





UNIVERSITÀ DI PAVIA
Dipartimento di
Ingegneria Industriale
e dell'Informazione

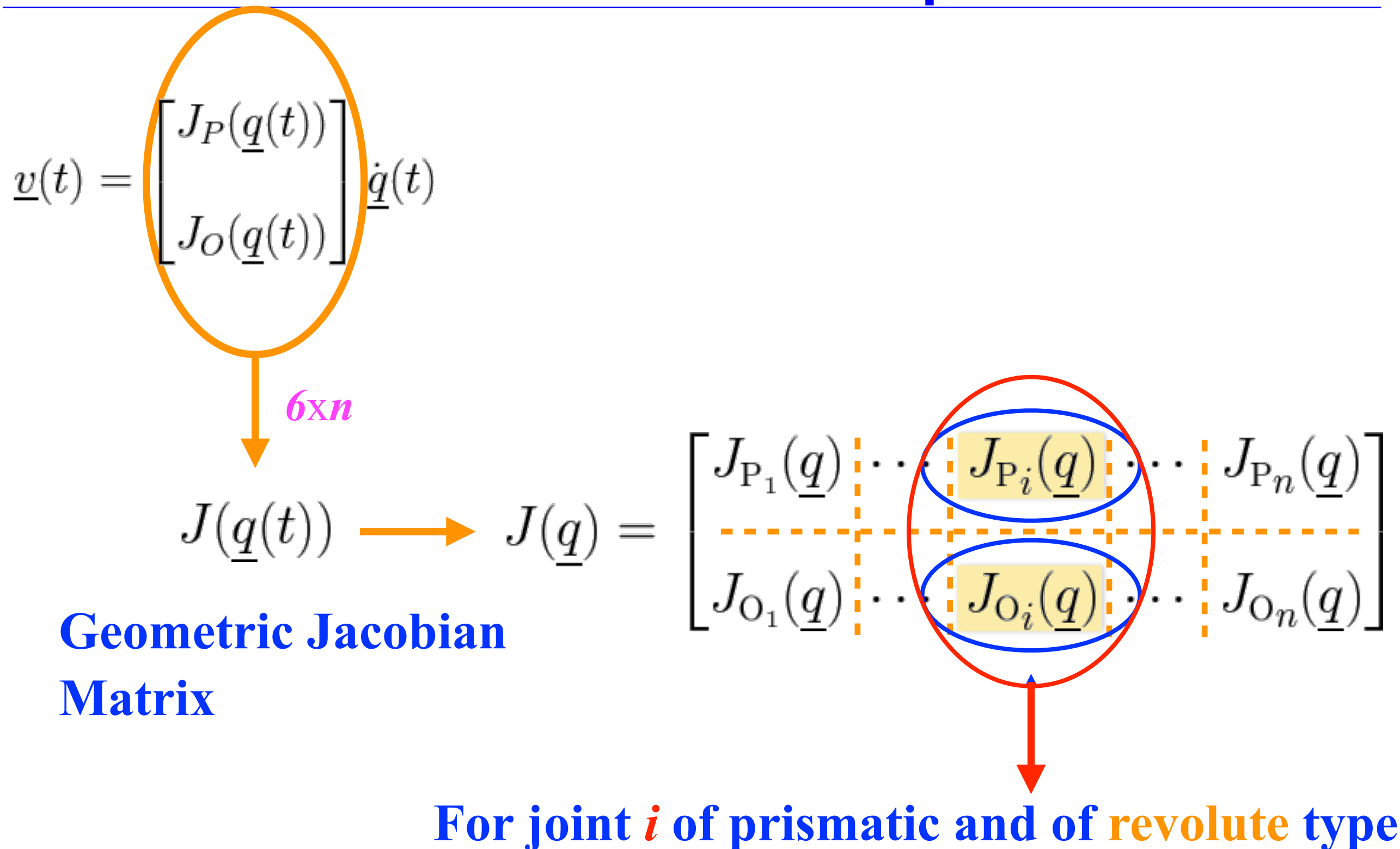
Course of Robot Control

Prof. Antonella Ferrara

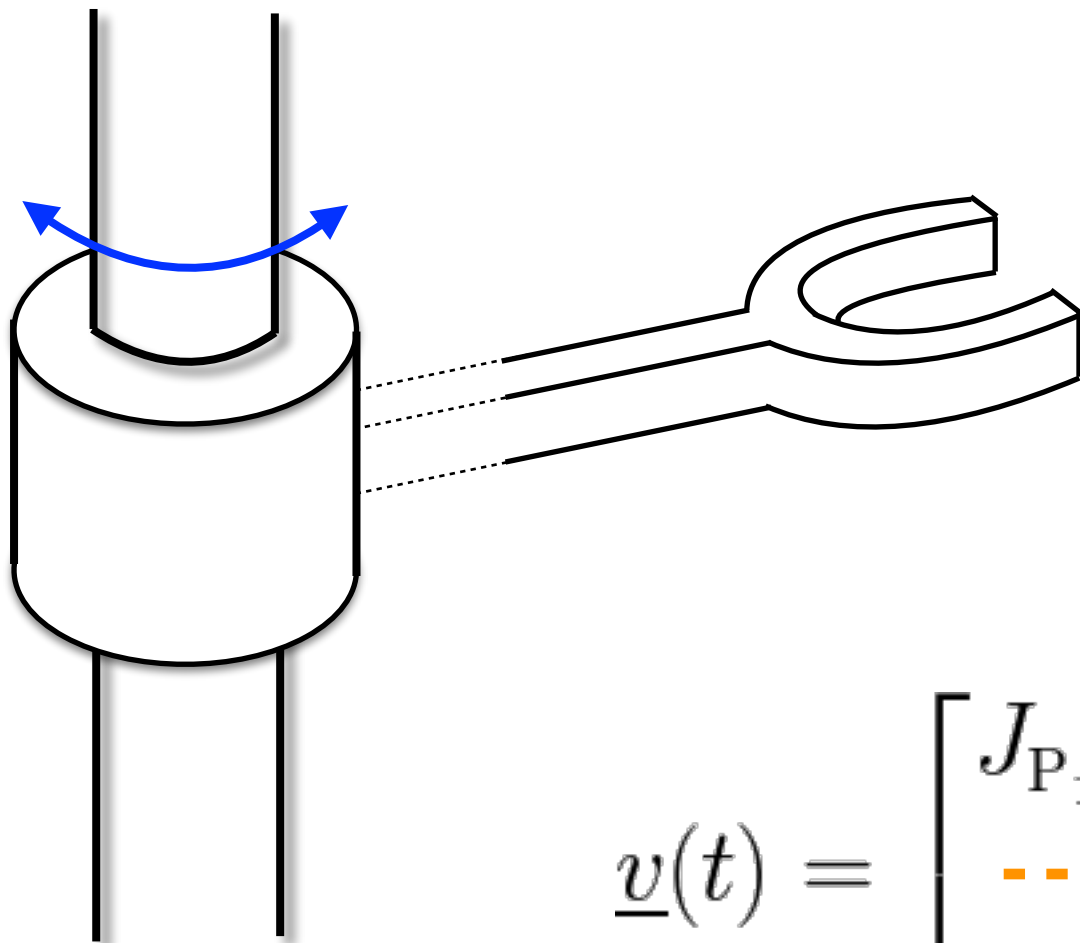


Geometric Jacobian Matrix: Revolute Joint Case

How to Determine the Geometric Jacobian Matrix of a Robot Manipulator



How to Determine the Geometric Jacobian Matrix **Column i with i revolute joint**

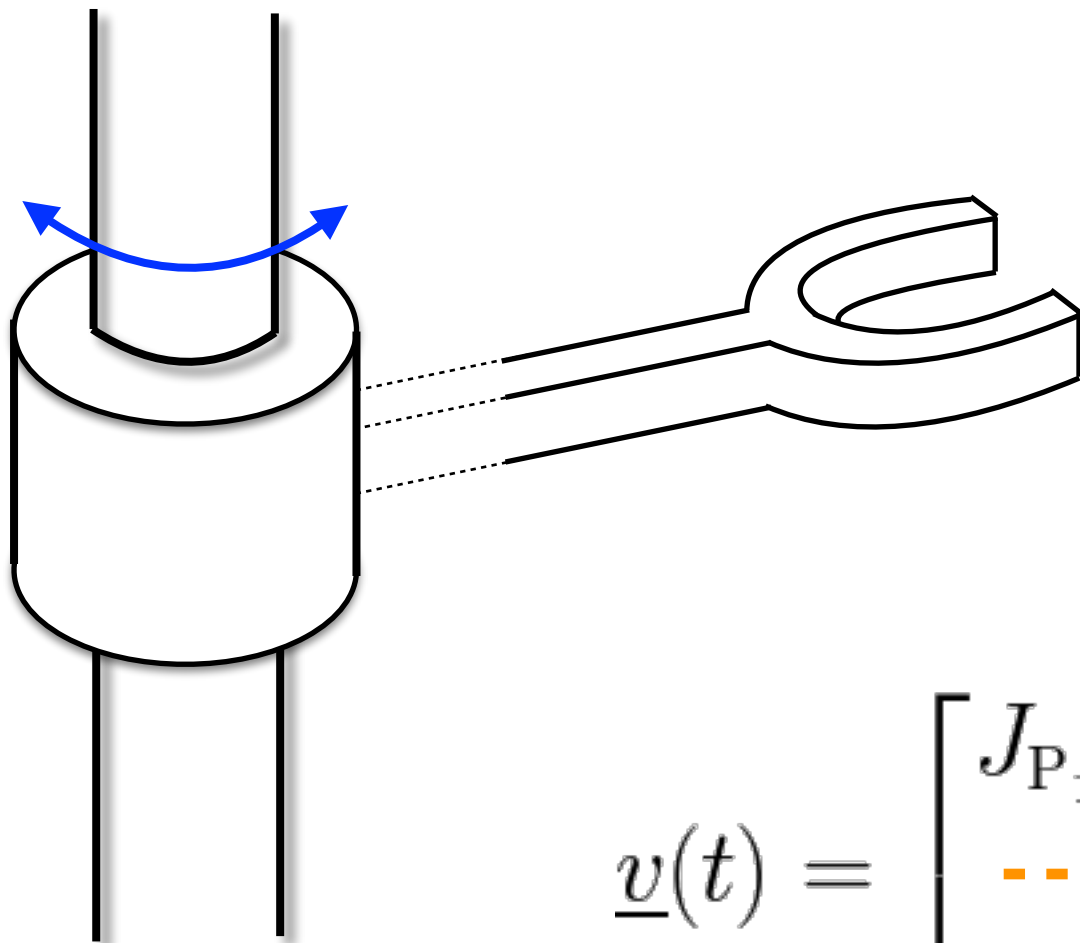


Assume joint i of revolute type

$$\underline{v}(t) = \begin{bmatrix} J_{P_1}(\underline{q}) & \cdots & J_{P_i}(\underline{q}) & \cdots & J_{P_n}(\underline{q}) \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ J_{O_1}(\underline{q}) & \cdots & J_{O_i}(\underline{q}) & \cdots & J_{O_n}(\underline{q}) \end{bmatrix} \underline{\dot{q}}(t)$$

Geometric Jacobian Matrix

How to Determine the Geometric Jacobian Matrix **Column i with i revolute joint**

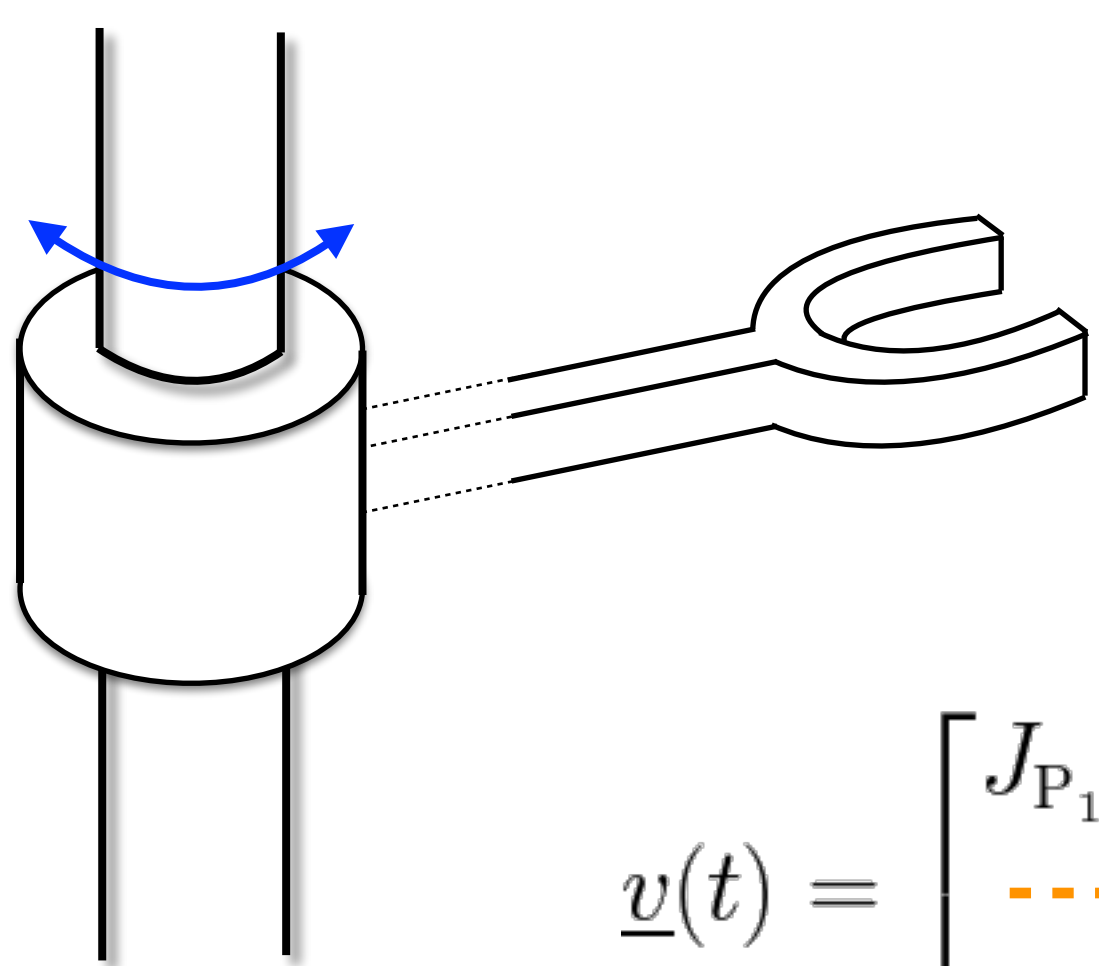


Assume joint i of revolute type

$$\underline{v}(t) = \begin{bmatrix} J_{P_1}(\underline{q}) & \cdots & \boxed{J_{P_i}(\underline{q})} & \cdots & J_{P_n}(\underline{q}) \\ \vdots & & & & \\ J_{O_1}(\underline{q}) & \cdots & \boxed{J_{O_i}(\underline{q})} & \cdots & J_{O_n}(\underline{q}) \end{bmatrix} \cdot \begin{bmatrix} \dot{q}_1(t) \\ \vdots \\ \boxed{\dot{q}_i(t)} \\ \vdots \\ \dot{q}_n(t) \end{bmatrix}$$

Geometric Jacobian Matrix

How to Determine the Geometric Jacobian Matrix **Column i with i revolute joint**



The diagram on the left shows a revolute joint (a cylinder) with a blue curved arrow indicating rotation. A mechanical arm is attached to the joint. Dotted lines connect the joint to the Jacobian matrix equations.

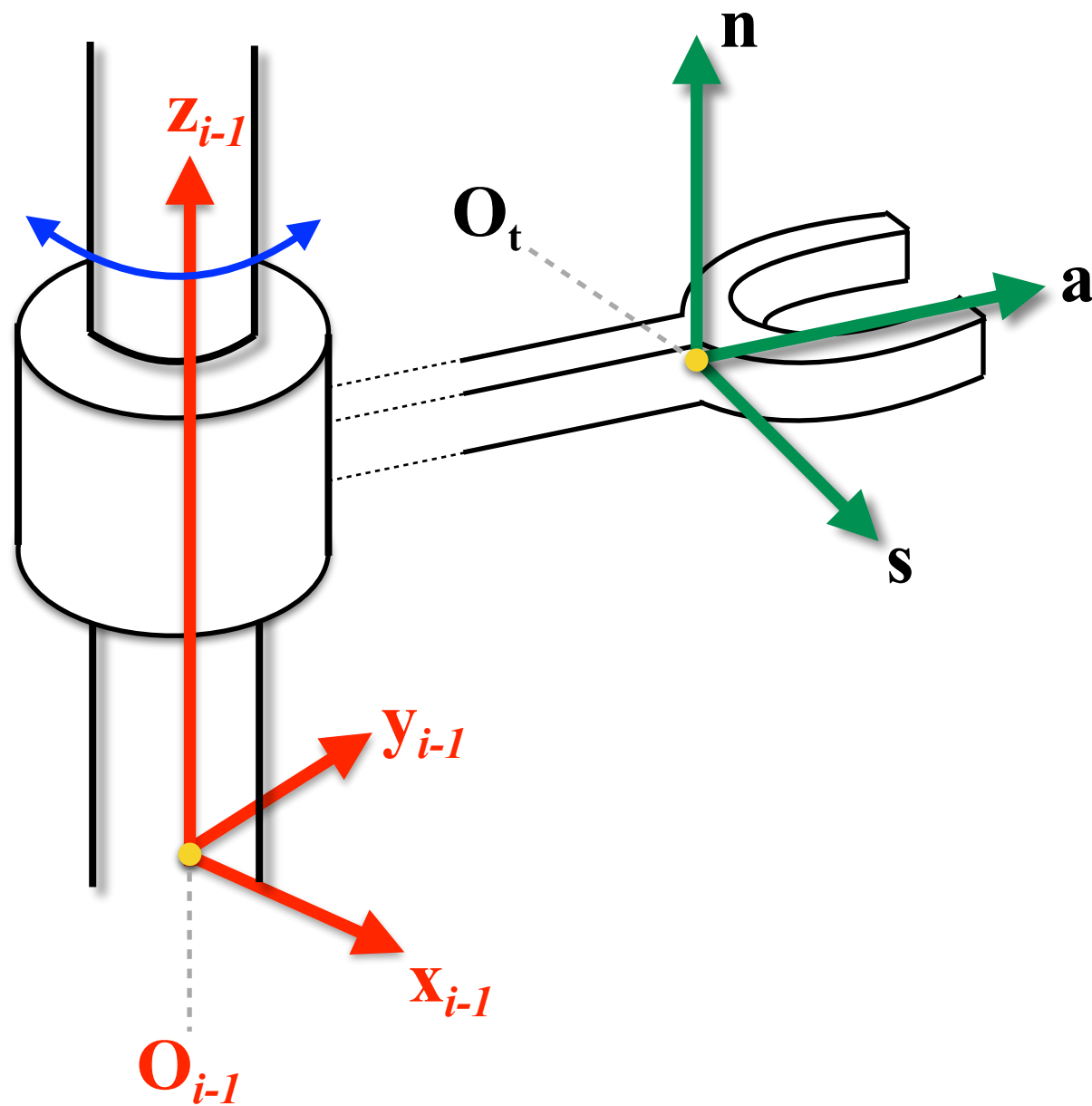
$$\underline{v}(t) \mid_{joint\ i} = \begin{bmatrix} J_{P_i}(\underline{q}) \\ \vdots \\ J_{O_i}(\underline{q}) \end{bmatrix} \dot{q}_i(t)$$

$$\underline{v}(t) = \begin{bmatrix} J_{P_1}(\underline{q}) & \cdots & J_{P_i}(\underline{q}) & \cdots & J_{P_n}(\underline{q}) \\ \vdots & & \vdots & & \vdots \\ J_{O_1}(\underline{q}) & \cdots & J_{O_i}(\underline{q}) & \cdots & J_{O_n}(\underline{q}) \end{bmatrix} \cdot \begin{bmatrix} \dot{q}_1(t) \\ \vdots \\ \dot{q}_i(t) \\ \vdots \\ \dot{q}_n(t) \end{bmatrix}$$

A red circle highlights the i -th column of the Jacobian matrix, which is enclosed in a yellow box. A red arrow points from this column to the equation above. A blue dotted arc is at the bottom.

Geometric Jacobian Matrix

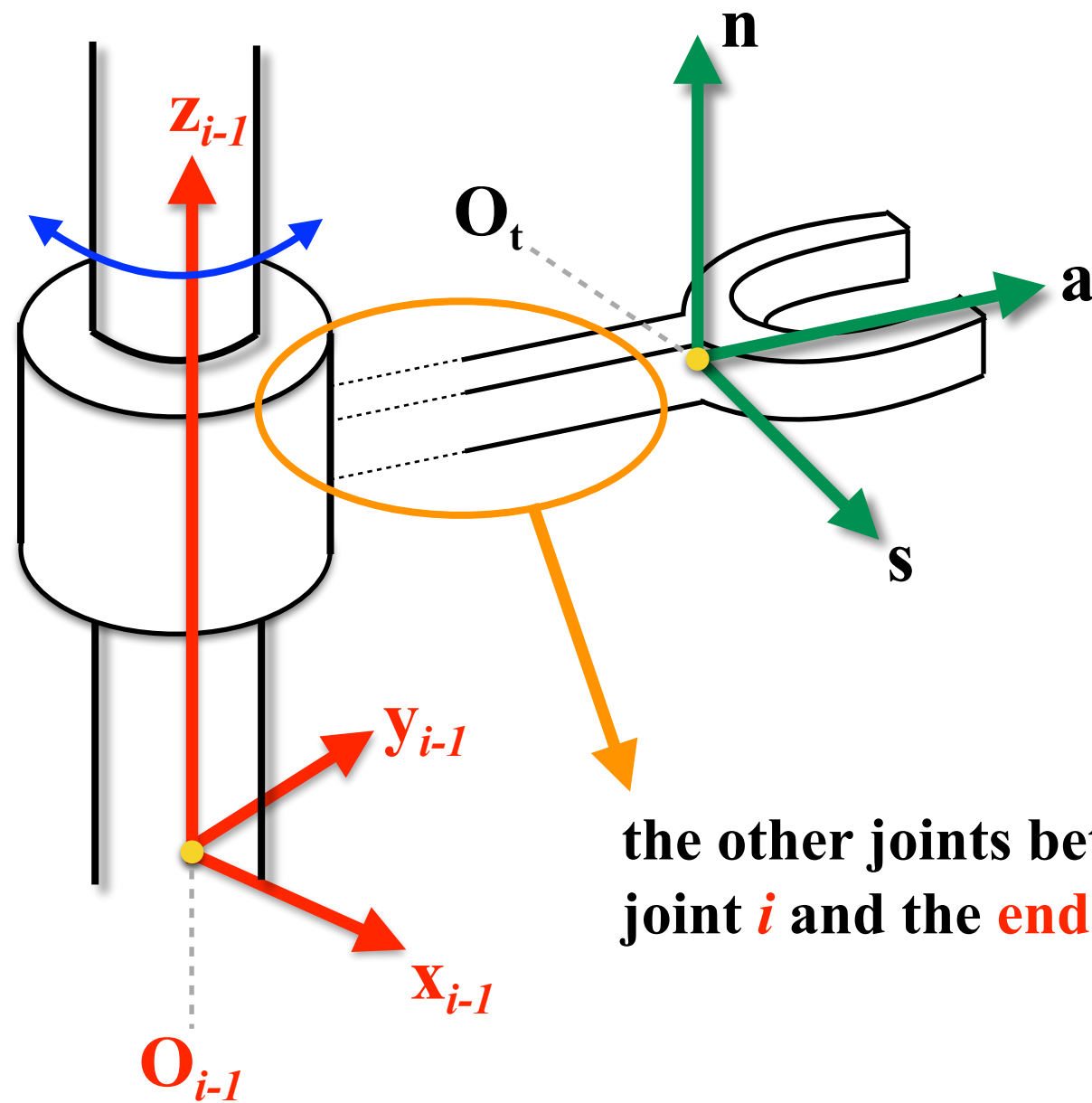
How to Determine the Geometric Jacobian Matrix Column i with i revolute joint



$$\underline{v}(t) \mid_{joint\ i} = \begin{bmatrix} J_{P_i}(\underline{q}) \\ \text{---} \\ J_{O_i}(\underline{q}) \end{bmatrix} \dot{q}_i(t)$$

joint i of revolute type

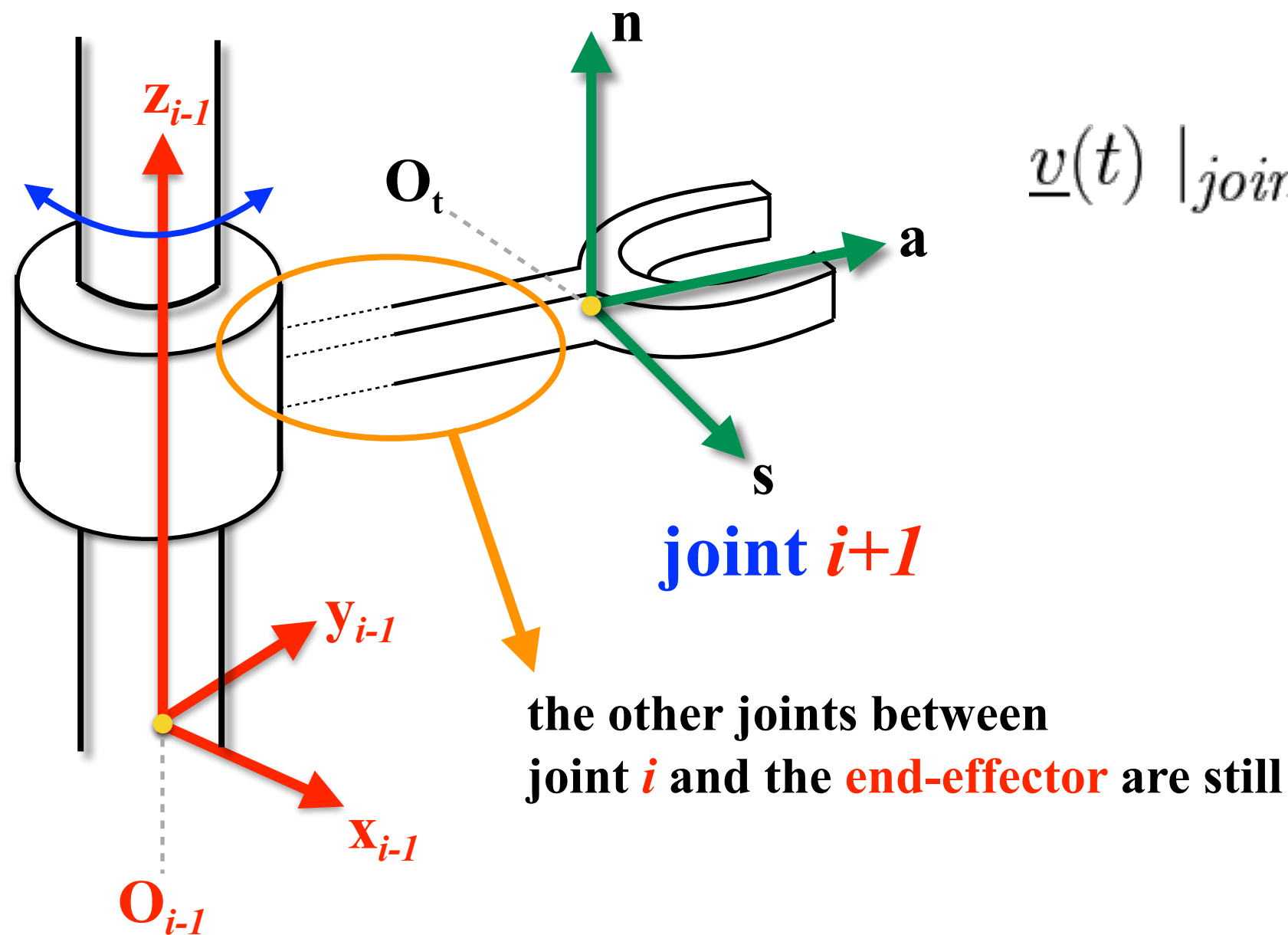
How to Determine the Geometric Jacobian Matrix Column i with i revolute joint



$$\underline{v}(t) \mid_{joint \ i} = \begin{bmatrix} J_{P_i}(\underline{q}) \\ \text{---} \\ J_{O_i}(\underline{q}) \end{bmatrix} \dot{q}_i(t)$$

joint i of revolute type

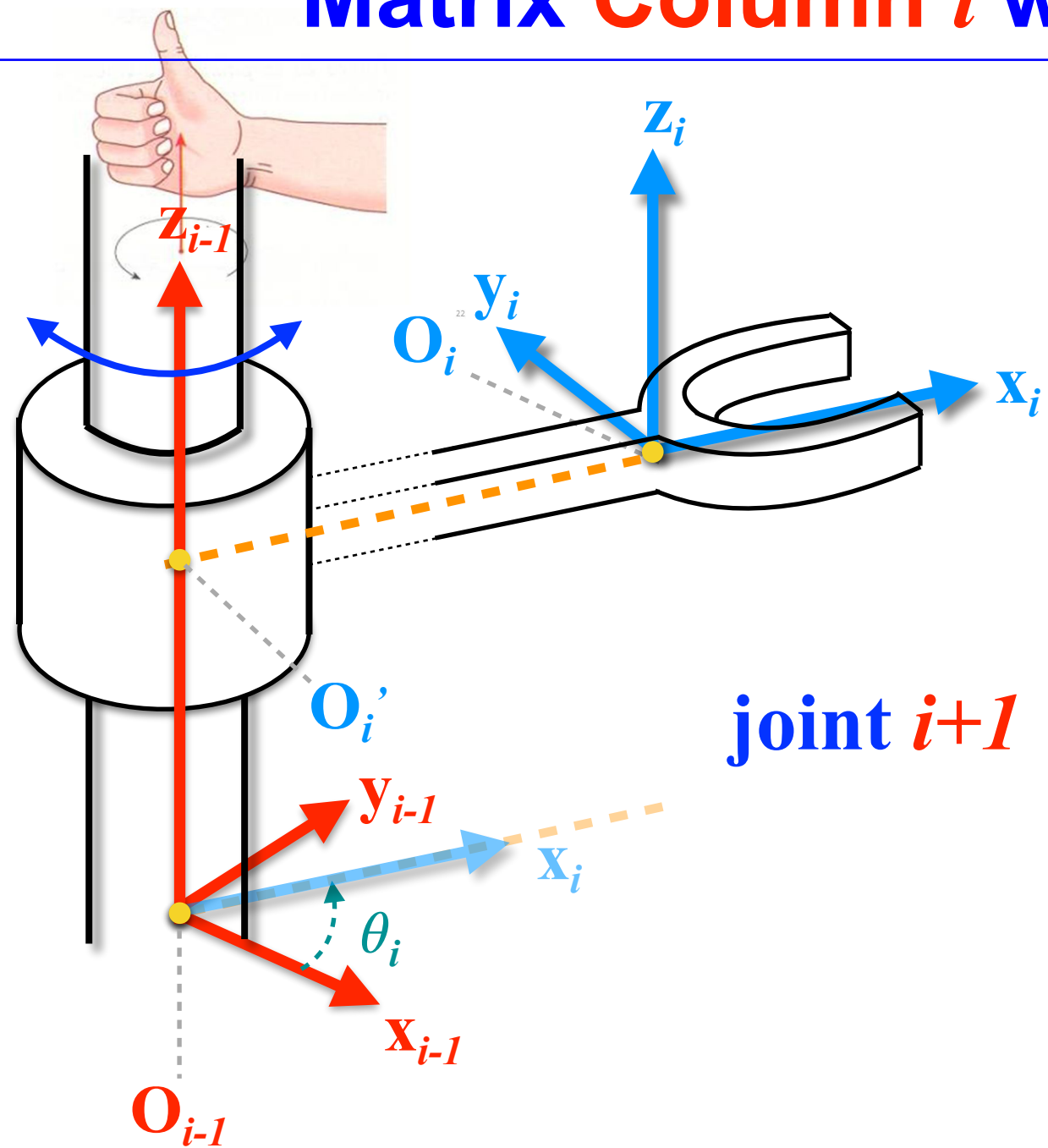
How to Determine the Geometric Jacobian Matrix Column i with i revolute joint



$$\underline{v}(t) \mid_{joint\ i} = \begin{bmatrix} J_{P_i}(\underline{q}) \\ \vdots \\ J_{O_i}(\underline{q}) \end{bmatrix} \dot{q}_i(t)$$

joint i of revolute type

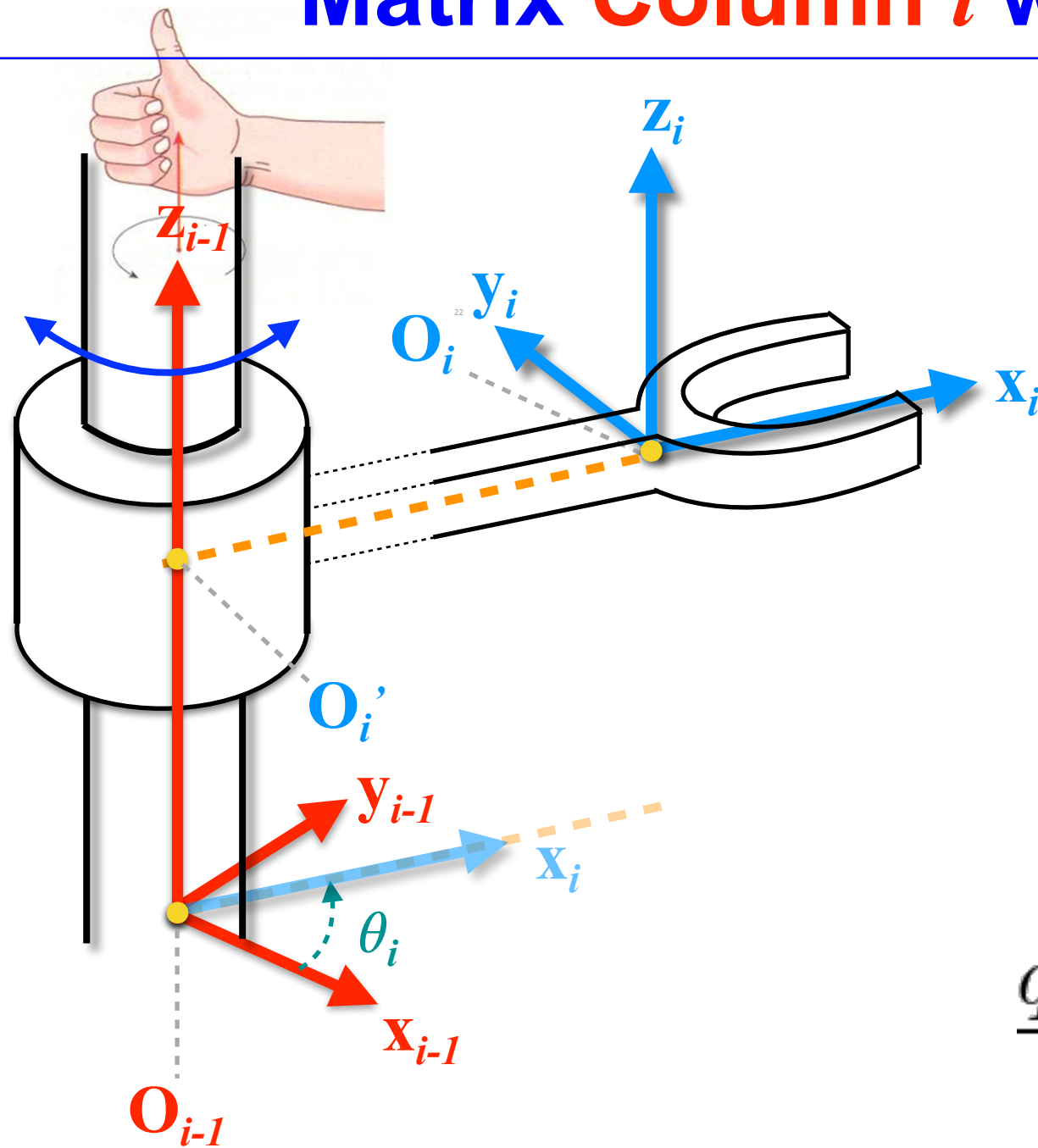
How to Determine the Geometric Jacobian Matrix **Column i with i revolute joint**



$$\underline{v}(t) \mid_{joint\ i} = \begin{bmatrix} J_{P_i}(\underline{q}) \\ \vdots \\ J_{O_i}(\underline{q}) \end{bmatrix} \dot{q}_i(t)$$

joint i of revolute type

How to Determine the Geometric Jacobian Matrix Column i with i revolute joint

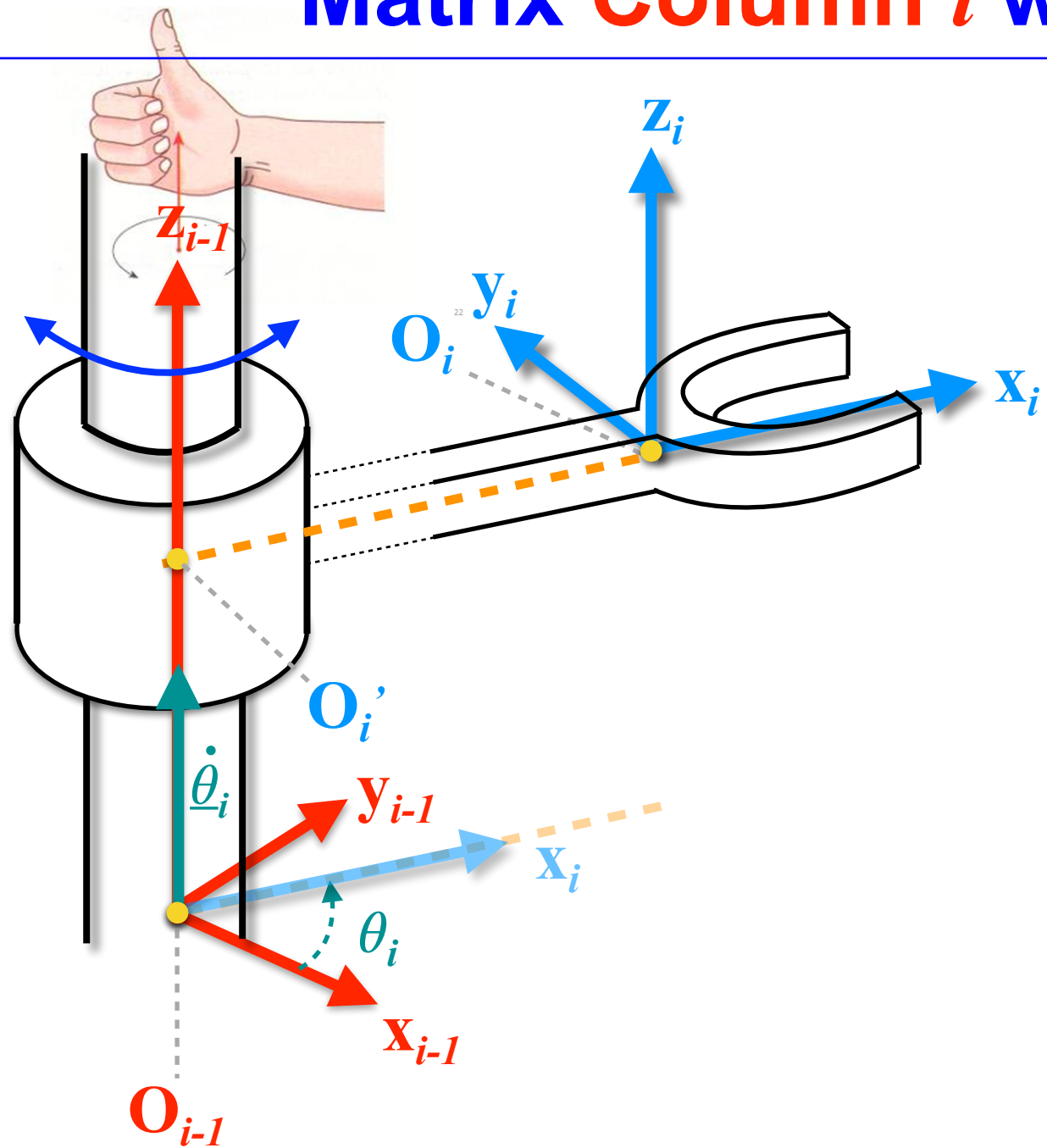


$$\underline{v}(t) \mid_{\text{joint } i} = \begin{bmatrix} J_{P_i}(\underline{q}) \\ \vdots \\ J_{O_i}(\underline{q}) \end{bmatrix} \dot{q}_i(t)$$

$$\underline{q}(t) = \begin{bmatrix} q_1(t) \\ \vdots \\ q_i(t) \\ \vdots \\ q_n(t) \end{bmatrix} = \begin{bmatrix} q_1(t) \\ \vdots \\ \theta_i(t) \\ \vdots \\ q_n(t) \end{bmatrix}$$

joint i of revolute type

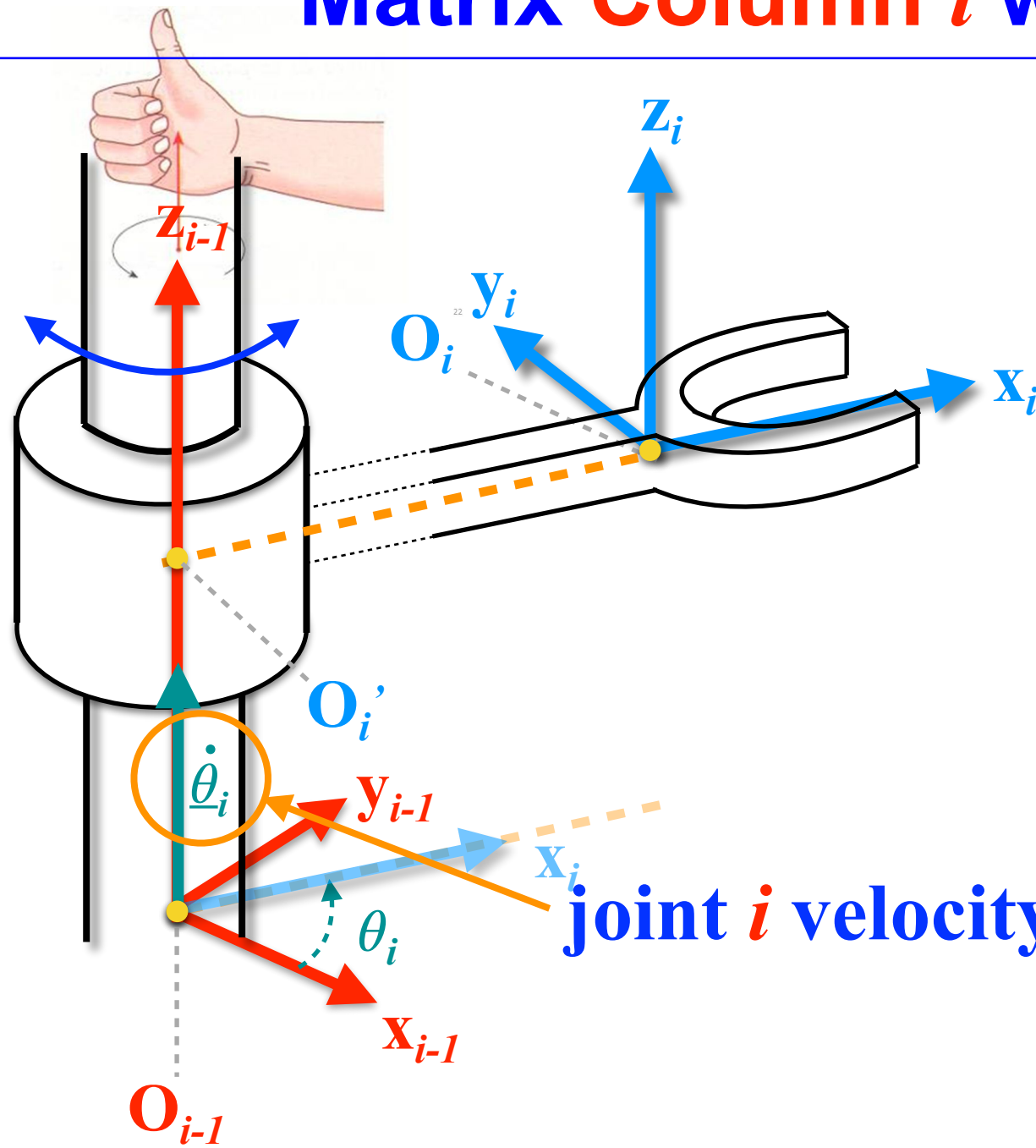
How to Determine the Geometric Jacobian Matrix Column i with i revolute joint



$$\underline{v}(t) \mid_{joint\ i} = \begin{bmatrix} J_{P_i}(\underline{q}) \\ \vdots \\ J_{O_i}(\underline{q}) \end{bmatrix} \dot{q}_i(t)$$

joint i of revolute type

How to Determine the Geometric Jacobian Matrix Column i with i revolute joint

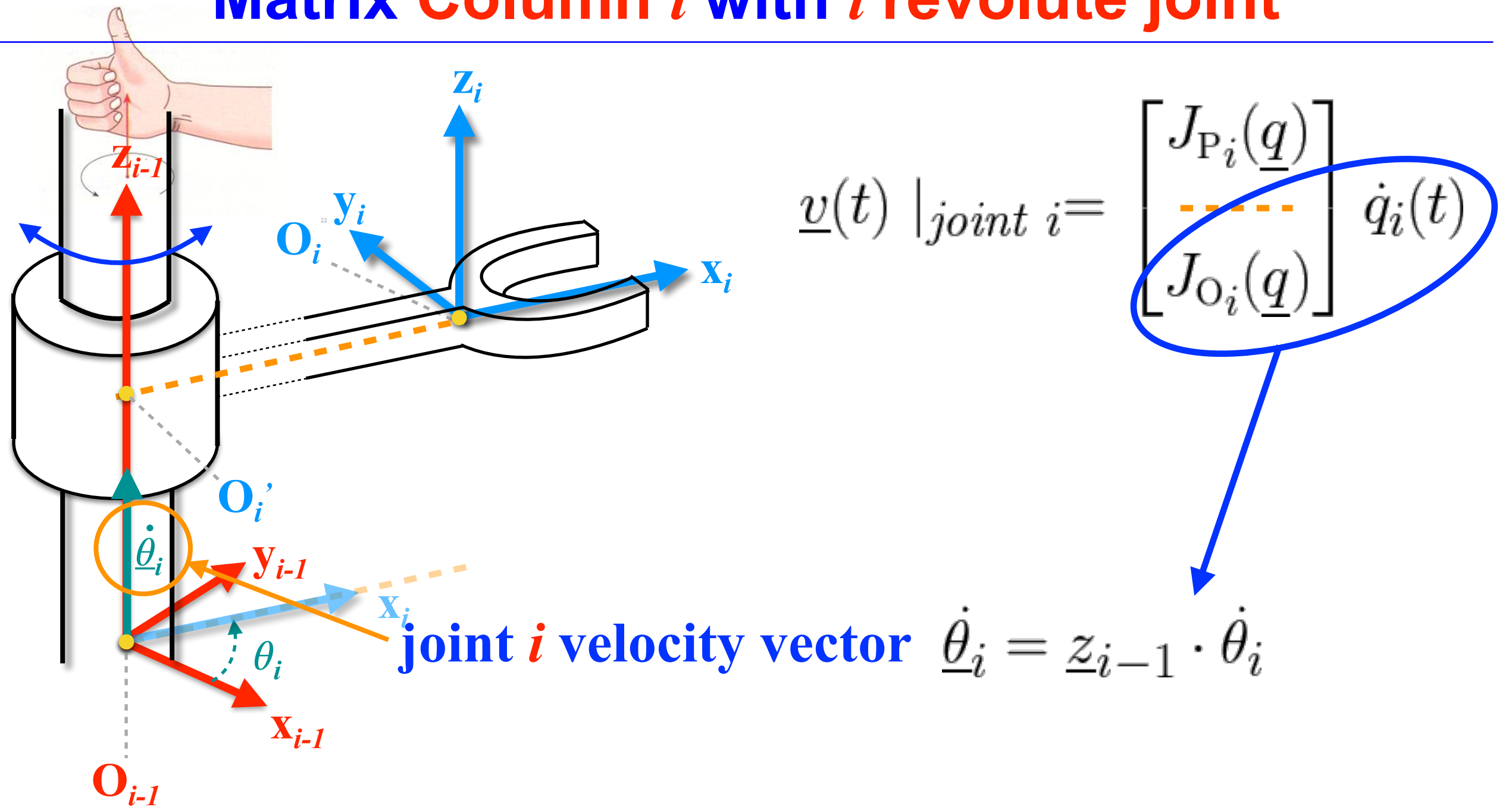


$$\underline{v}(t) \mid_{joint \ i} = \begin{bmatrix} J_{P_i}(\underline{q}) \\ \vdots \\ J_{O_i}(\underline{q}) \end{bmatrix} \dot{q}_i(t)$$

joint i velocity vector $\dot{\underline{\theta}}_i = \underline{z}_{i-1} \cdot \dot{\theta}_i$

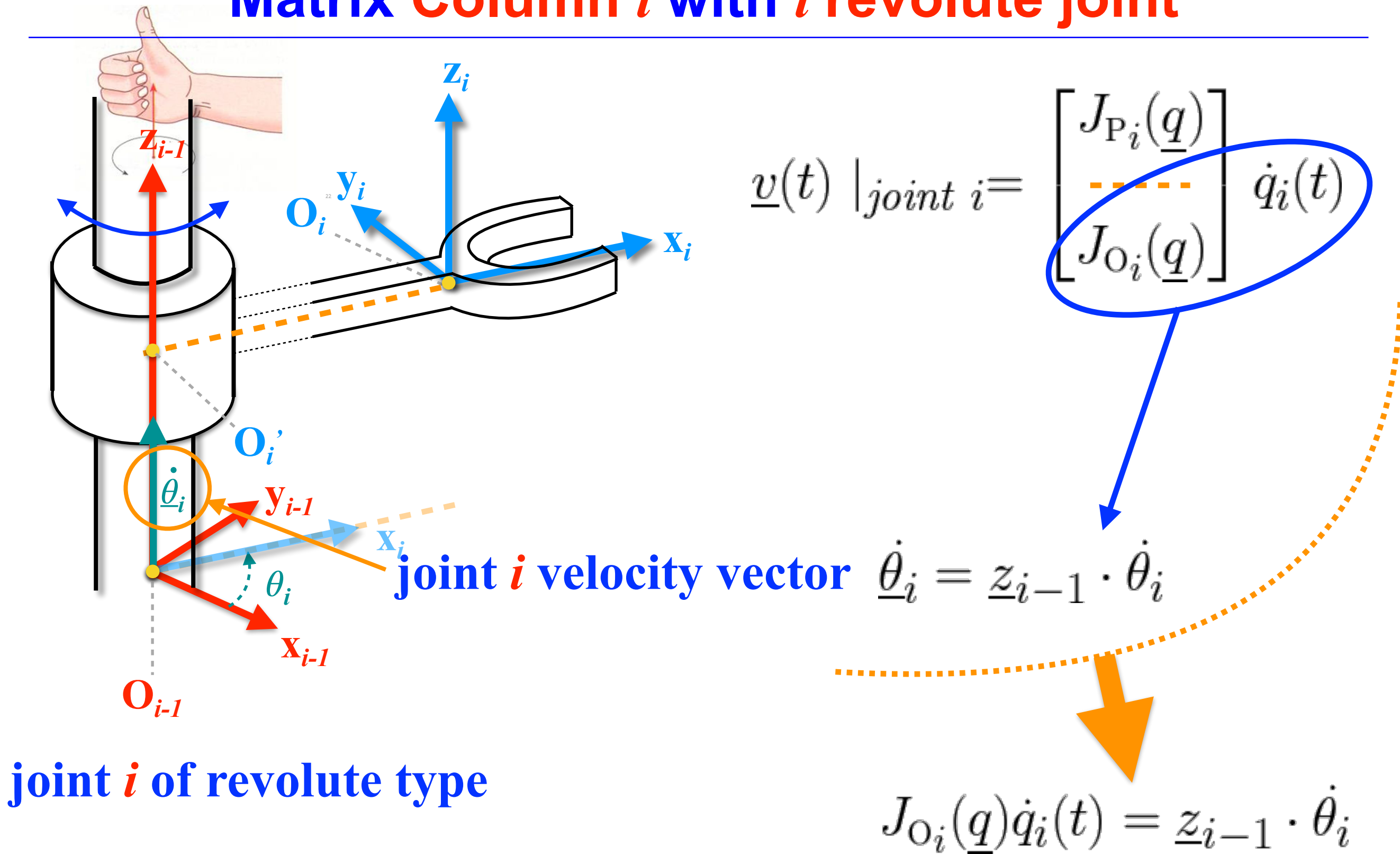
joint i of revolute type

How to Determine the Geometric Jacobian Matrix Column i with i revolute joint

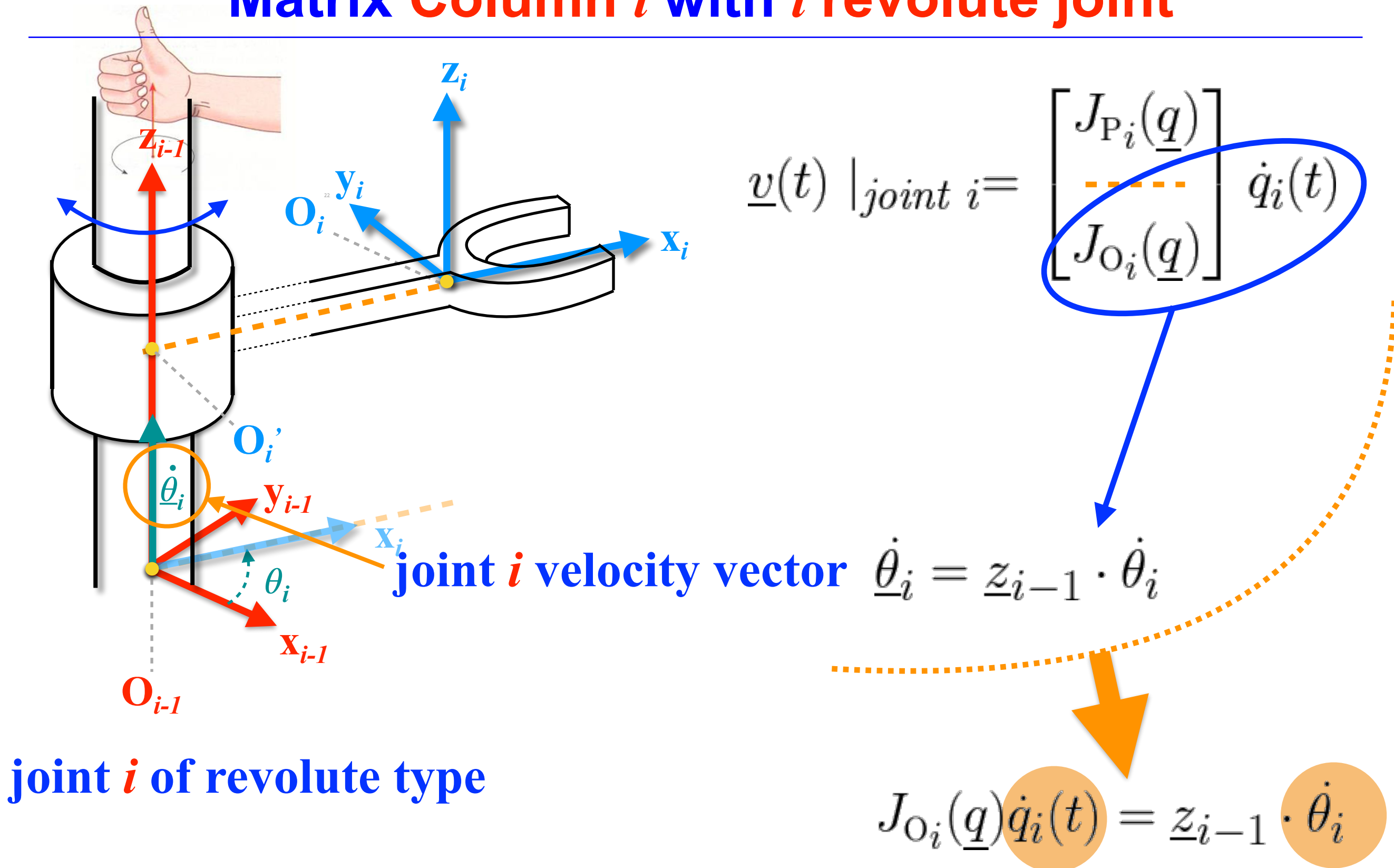


joint i of revolute type

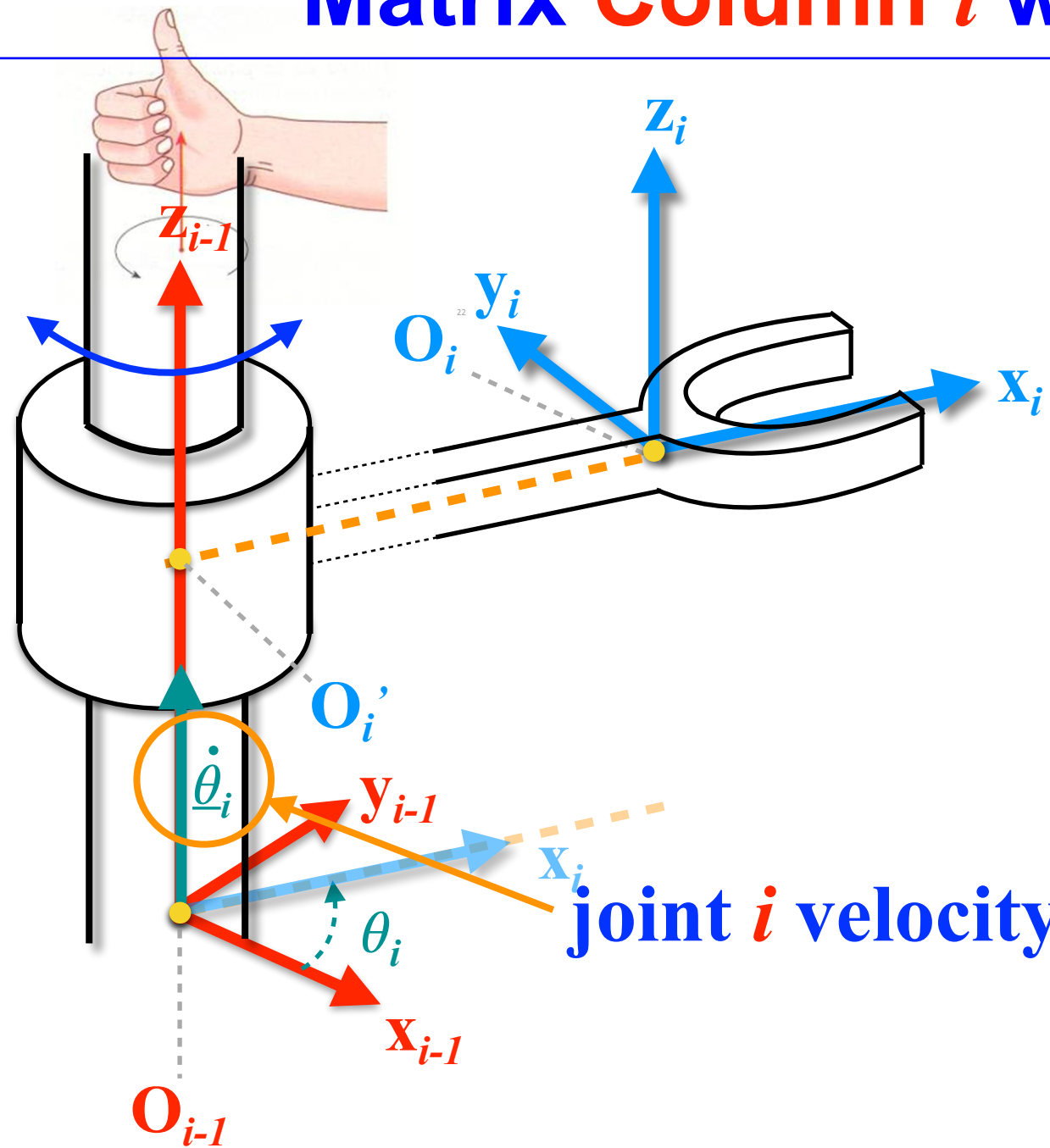
How to Determine the Geometric Jacobian Matrix Column i with i revolute joint



How to Determine the Geometric Jacobian Matrix Column i with i revolute joint



How to Determine the Geometric Jacobian Matrix Column i with i revolute joint



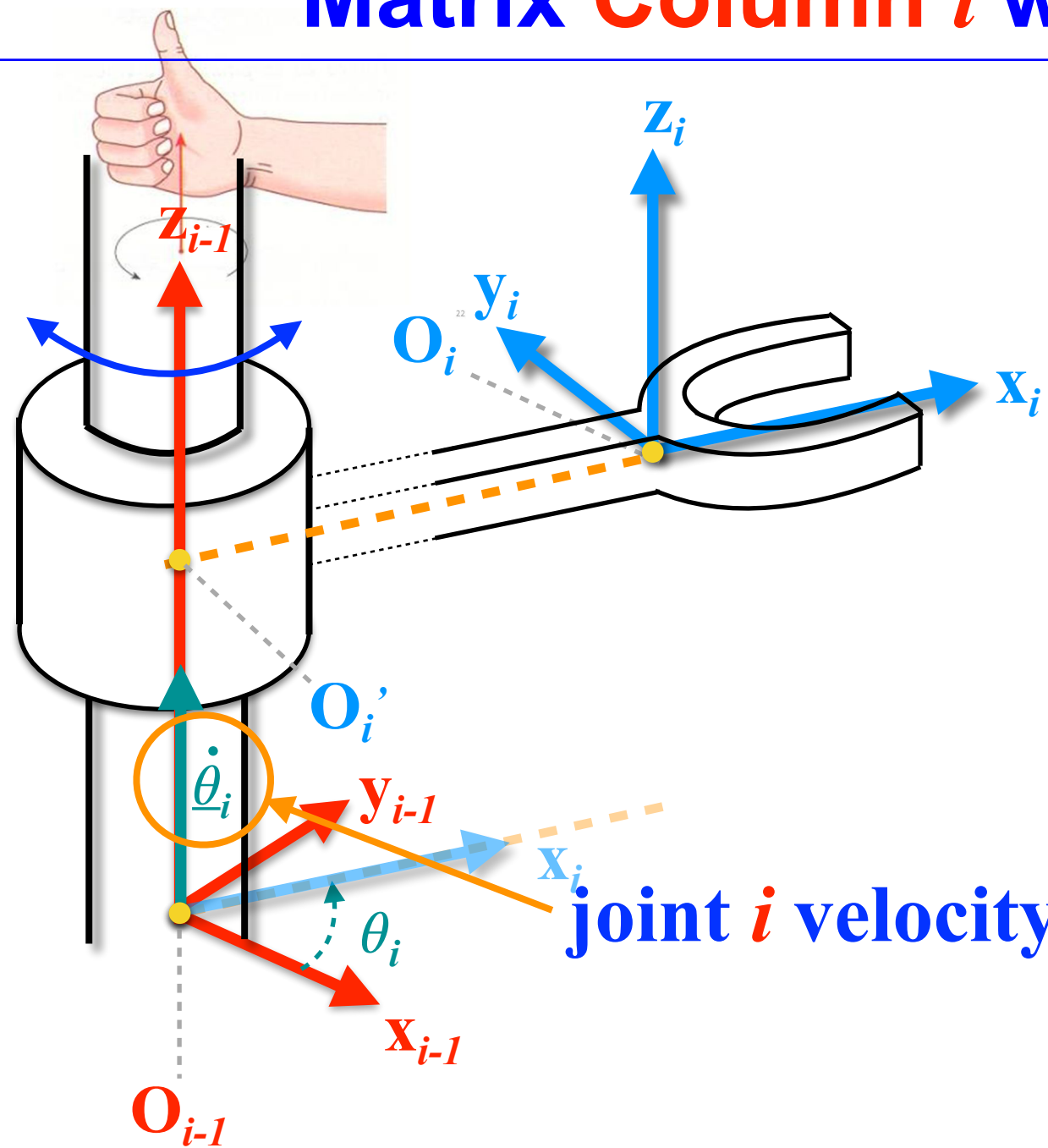
$$\underline{v}(t) \mid_{joint\ i} = \begin{bmatrix} J_{P_i}(\underline{q}) \\ \vdots \\ J_{O_i}(\underline{q}) \end{bmatrix} \dot{q}_i(t)$$

joint i velocity vector $\underline{\dot{\theta}}_i = \underline{z}_{i-1} \cdot \dot{\theta}_i$

joint i of revolute type

$$J_{O_i}(\underline{q}) \dot{q}_i(t) = \underline{z}_{i-1} \dot{\theta}_i$$

How to Determine the Geometric Jacobian Matrix Column i with i revolute joint

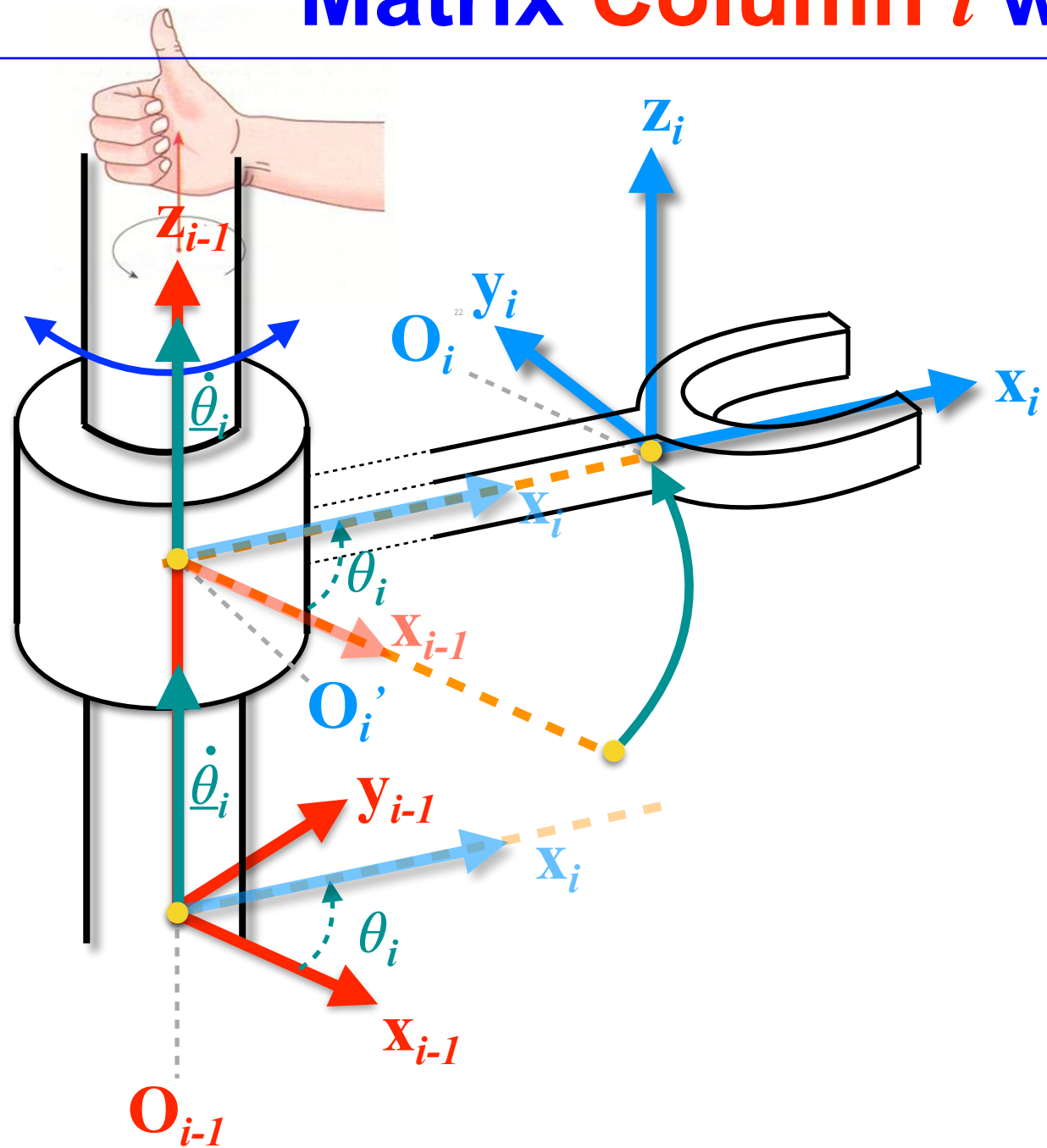


$$\underline{v}(t) \mid_{joint\ i} = \begin{bmatrix} J_{P_i}(\underline{q}) \\ \vdots \\ J_{O_i}(\underline{q}) \end{bmatrix} \dot{q}_i(t) \quad ?$$

joint i velocity vector $\dot{\underline{\theta}}_i = \underline{z}_{i-1} \cdot \dot{\theta}_i$

joint i of revolute type

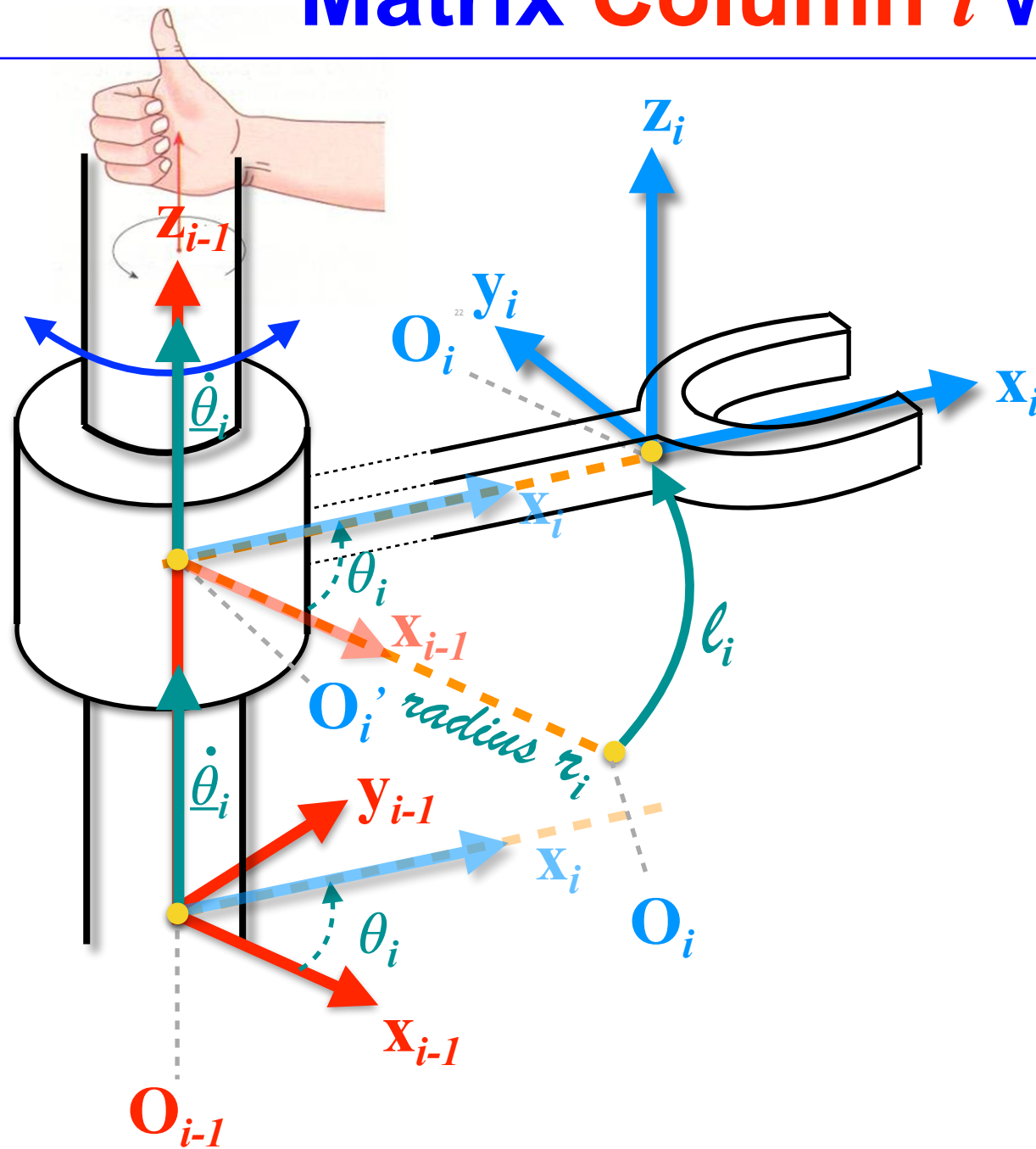
How to Determine the Geometric Jacobian Matrix **Column i with i revolute joint**



$$\underline{v}(t) \mid_{joint\ i} = \begin{bmatrix} J_{P_i}(\underline{q}) \\ \text{---} \\ J_{O_i}(\underline{q}) \end{bmatrix} \dot{q}_i(t)$$

joint i of revolute type

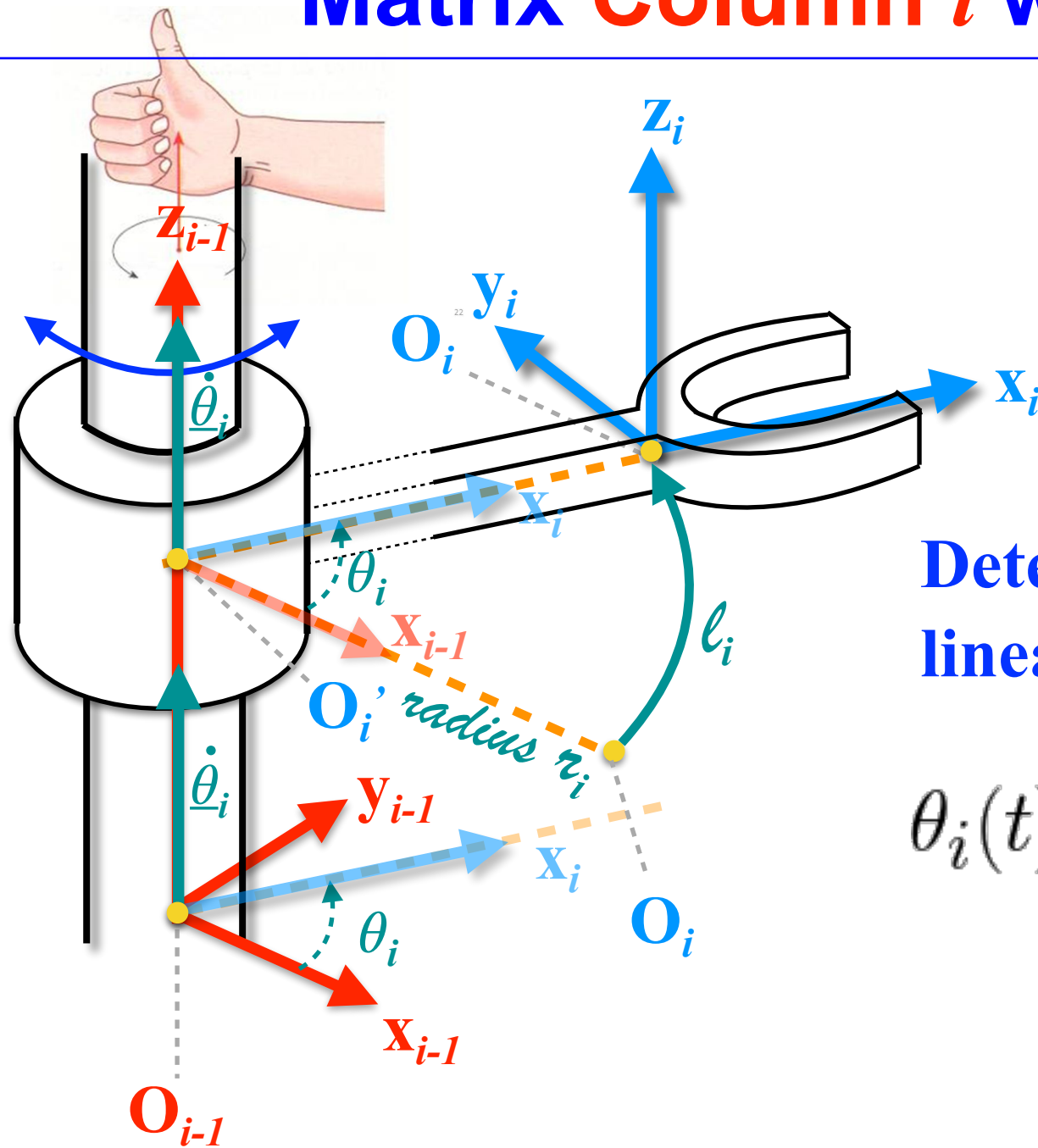
How to Determine the Geometric Jacobian Matrix Column i with i revolute joint



$$\underline{v}(t) \mid_{joint\ i} = \begin{bmatrix} J_{P_i}(\underline{q}) \\ \vdots \\ J_{O_i}(\underline{q}) \end{bmatrix} \dot{q}_i(t)$$

joint i of revolute type

How to Determine the Geometric Jacobian Matrix Column i with i revolute joint



$$\underline{v}(t) \mid_{joint\ i} = \begin{bmatrix} J_{P_i}(\underline{q}) \\ \vdots \\ J_{O_i}(\underline{q}) \end{bmatrix} \dot{q}_i(t)$$

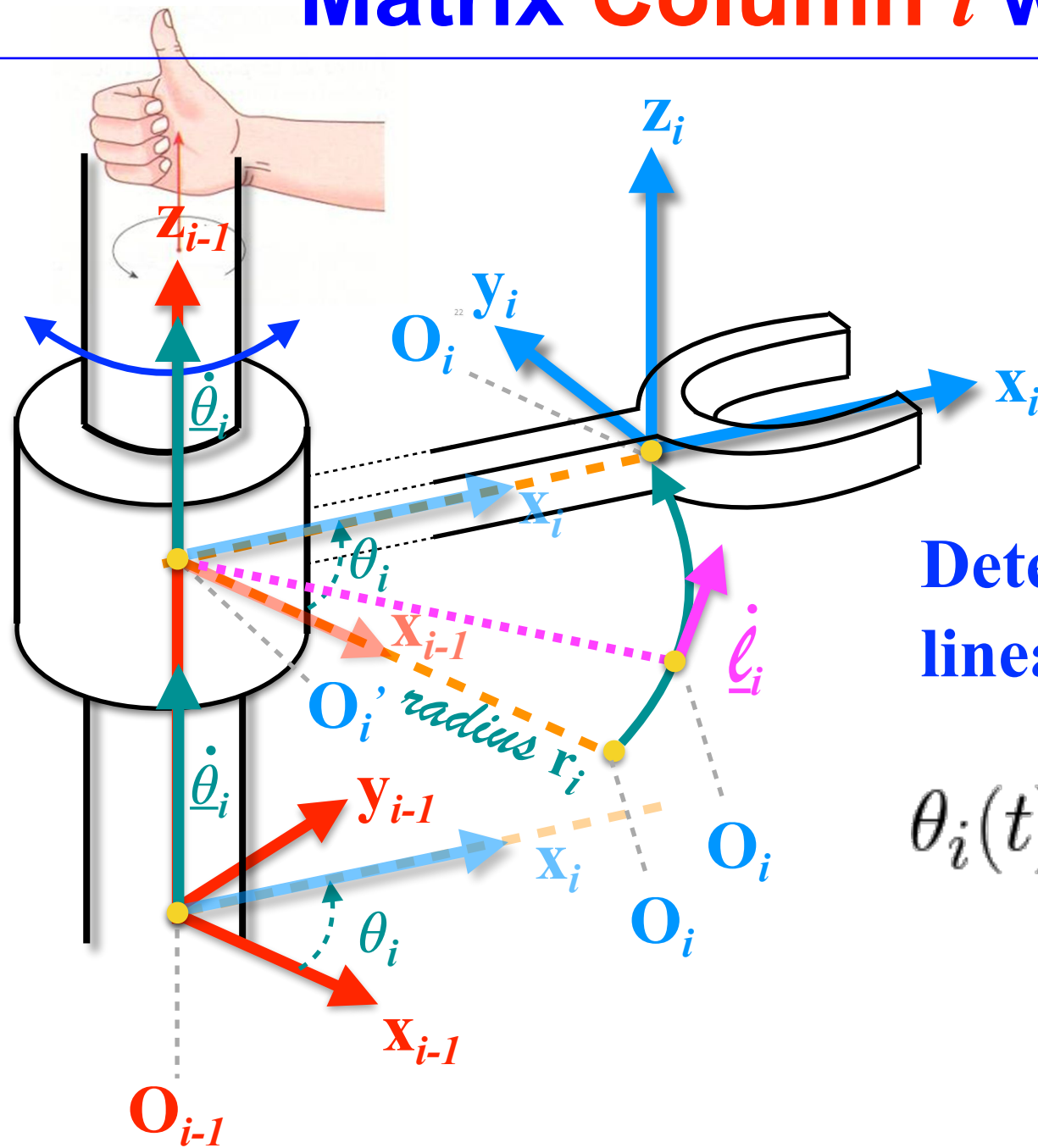
Determination of the magnitude of the linear velocity of the end-effector:

$$\theta_i(t) = \frac{l_i(t)}{r_i} \rightarrow l_i(t) = \theta_i(t) r_i$$

$$\dot{l}_i(t) = \dot{\theta}_i(t) r_i$$

joint i of revolute type

How to Determine the Geometric Jacobian Matrix Column i with i revolute joint



$$\underline{v}(t) \mid_{joint\ i} = \begin{bmatrix} J_{P_i}(\underline{q}) \\ \vdots \\ J_{O_i}(\underline{q}) \end{bmatrix} \dot{q}_i(t)$$

Determination of the magnitude of the linear velocity of the end-effector:

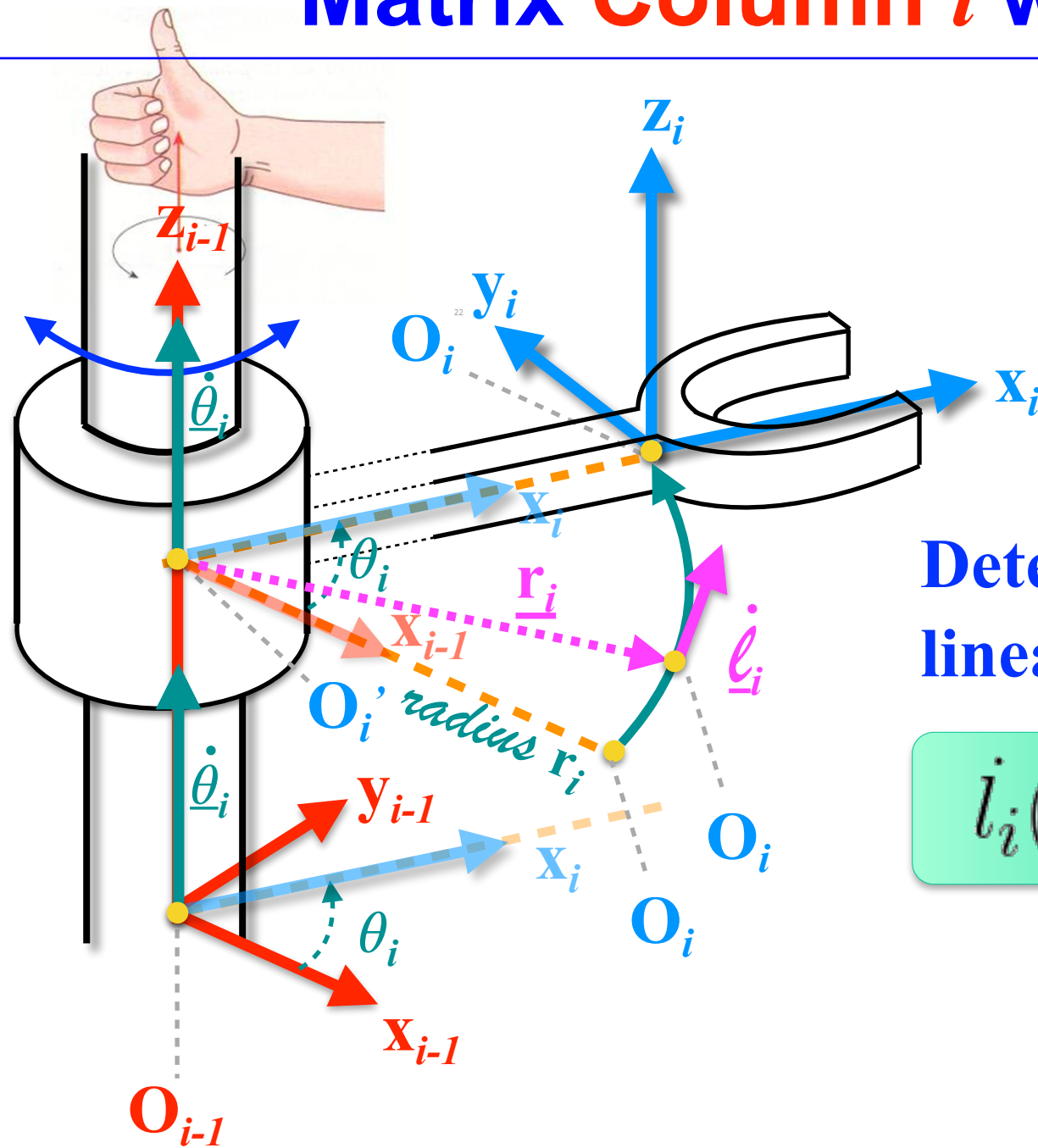
$$\theta_i(t) = \frac{l_i(t)}{r_i} \rightarrow l_i(t) = \theta_i(t) r_i$$

$$\dot{l}_i(t) = \dot{\theta}_i(t) r_i$$

joint i of revolute type

Direction of the linear velocity of the end-effector: *tangent to the arc traveled*

How to Determine the Geometric Jacobian Matrix Column i with i revolute joint



joint i of revolute type

$$\underline{v}(t) \mid_{\text{joint } i} = \begin{bmatrix} J_{P_i}(\underline{q}) \\ \vdots \\ J_{O_i}(\underline{q}) \end{bmatrix} \dot{q}_i(t)$$

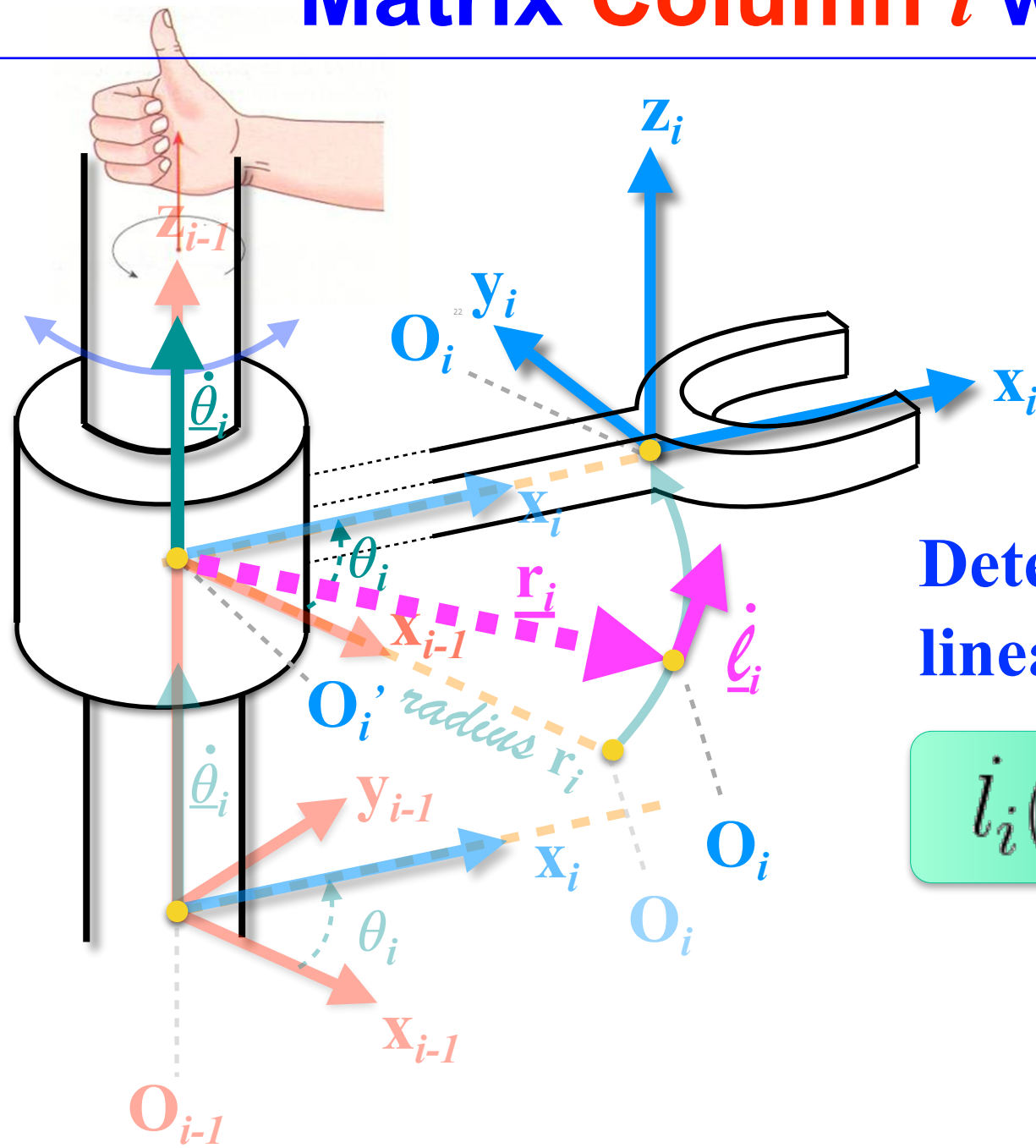
Determination of the vector of the linear velocity of the end-effector:

$$\dot{l}_i(t) = \dot{\theta}_i(t) r_i$$

$$\underline{\dot{\theta}}_i = \underline{z}_{i-1} \cdot \dot{\theta}_i$$

$$\dot{l}_i(t) = [\underline{z}_{i-1} \times \underline{r}_i] \cdot \dot{\theta}_i$$

How to Determine the Geometric Jacobian Matrix Column i with i revolute joint



joint i of revolute type

$\underline{v}(t) |_{joint\ i} =$

$$\begin{bmatrix} J_{P_i}(\underline{q}) \\ \vdots \\ J_{O_i}(\underline{q}) \end{bmatrix} \dot{q}_i(t)$$

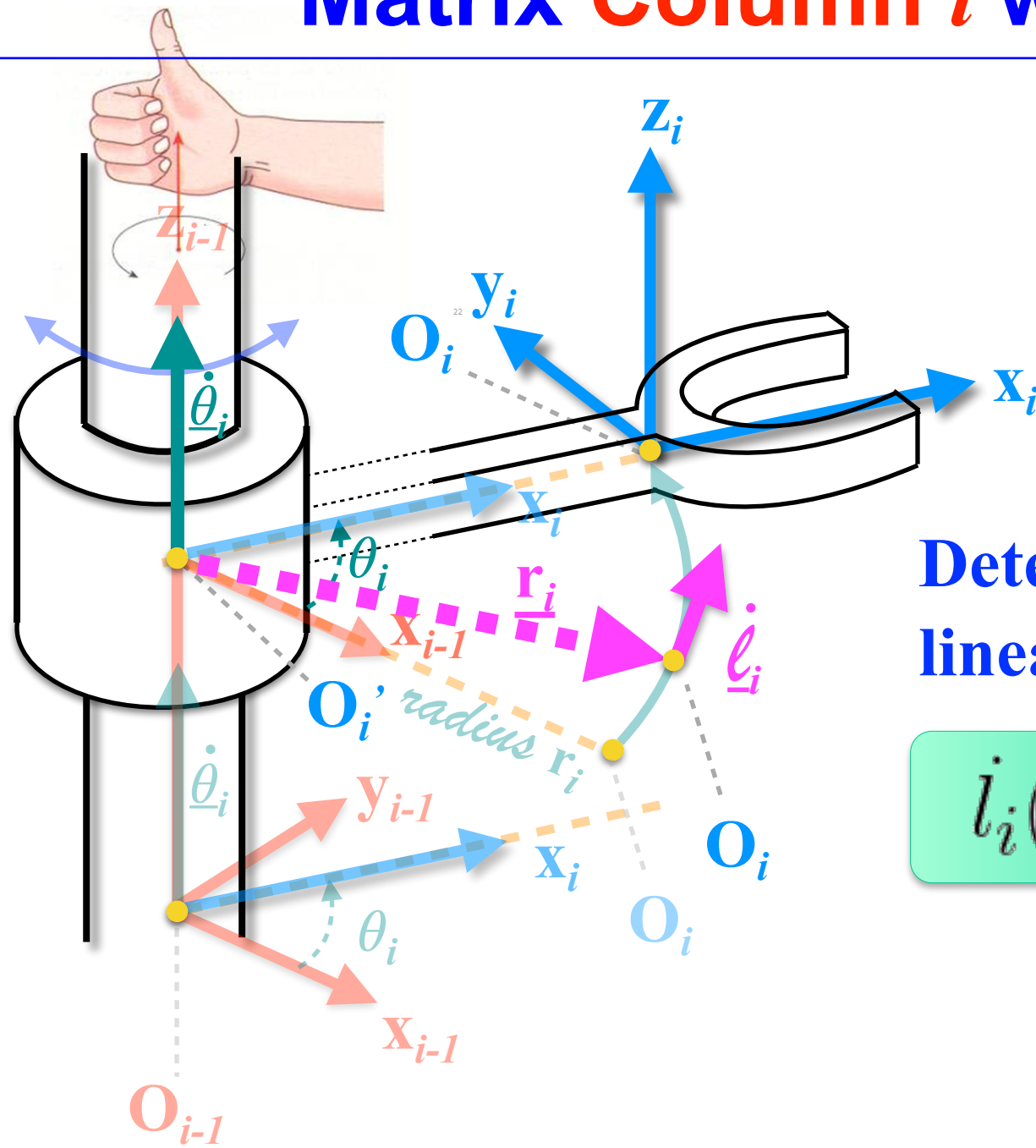
Determination of the vector of the linear velocity of the end-effector:

$$\dot{\underline{l}}_i(t) = \dot{\theta}_i(t) r_i$$

$$\underline{\dot{\theta}}_i = \underline{z}_{i-1} \cdot \dot{\theta}_i$$

$$\dot{\underline{l}}_i(t) = [\underline{z}_{i-1} \times \underline{r}_i] \cdot \dot{\theta}_i$$

How to Determine the Geometric Jacobian Matrix Column i with i revolute joint



joint i of revolute type

$$\underline{v}(t) \mid_{joint\ i} =$$

$$\begin{bmatrix} J_{P_i}(\underline{q}) \\ \vdots \\ J_{O_i}(\underline{q}) \end{bmatrix} \dot{q}_i(t)$$

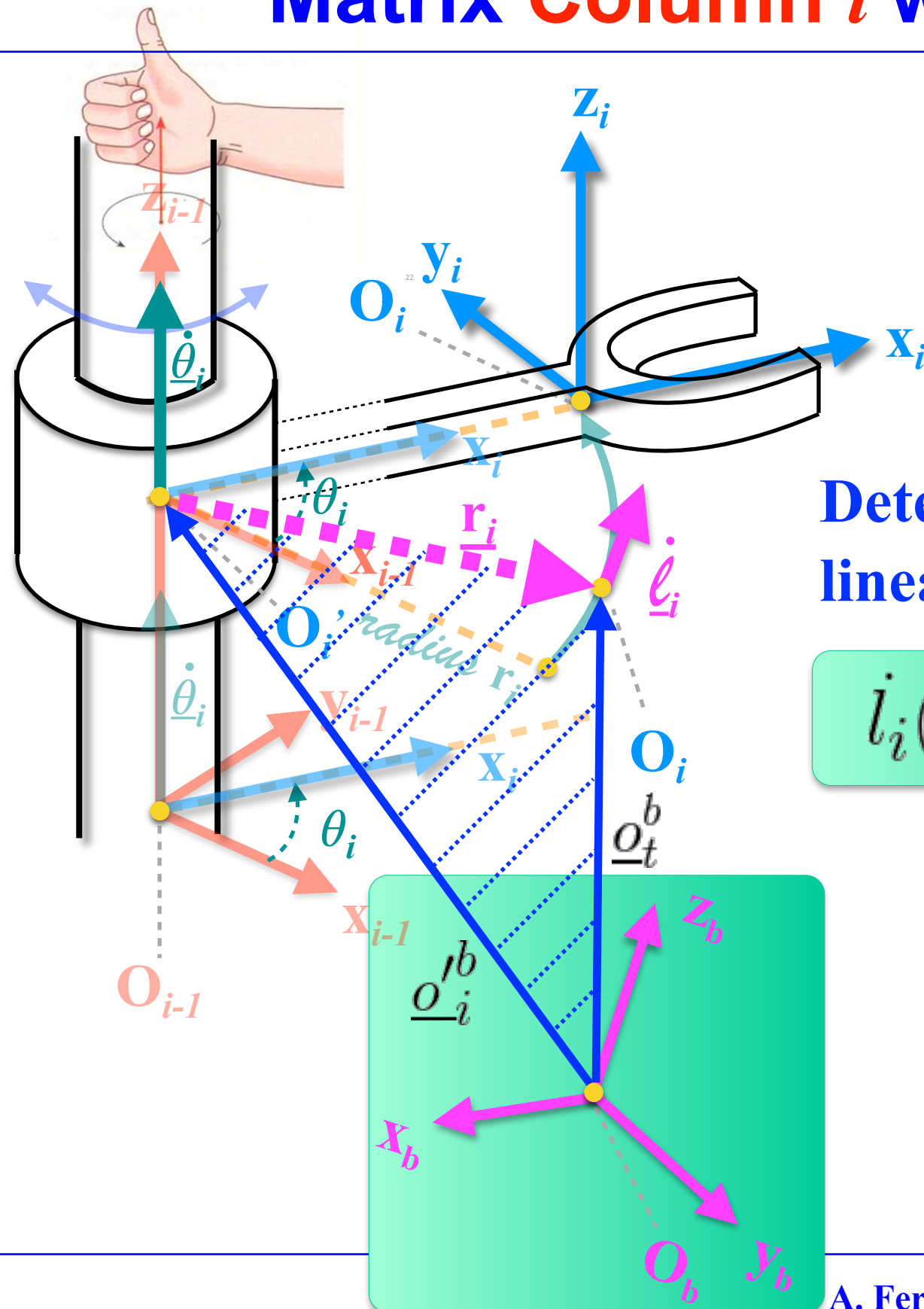
Determination of the vector of the linear velocity of the end-effector:

$$\dot{l}_i(t) = \dot{\theta}_i(t) r_i$$

$$\underline{\dot{\theta}}_i = \underline{z}_{i-1} \cdot \dot{\theta}_i$$

$$\dot{l}_i(t) = [\underline{z}_{i-1} \times \underline{r}_i] \cdot \dot{\theta}_i$$

How to Determine the Geometric Jacobian Matrix **Column i with i revolute joint**



$$\underline{v}(t) \mid_{\text{joint } i} = \begin{bmatrix} J_{P_i}(\underline{q}) \\ \vdots \\ J_{O_i}(\underline{q}) \end{bmatrix} \dot{q}_i(t)$$

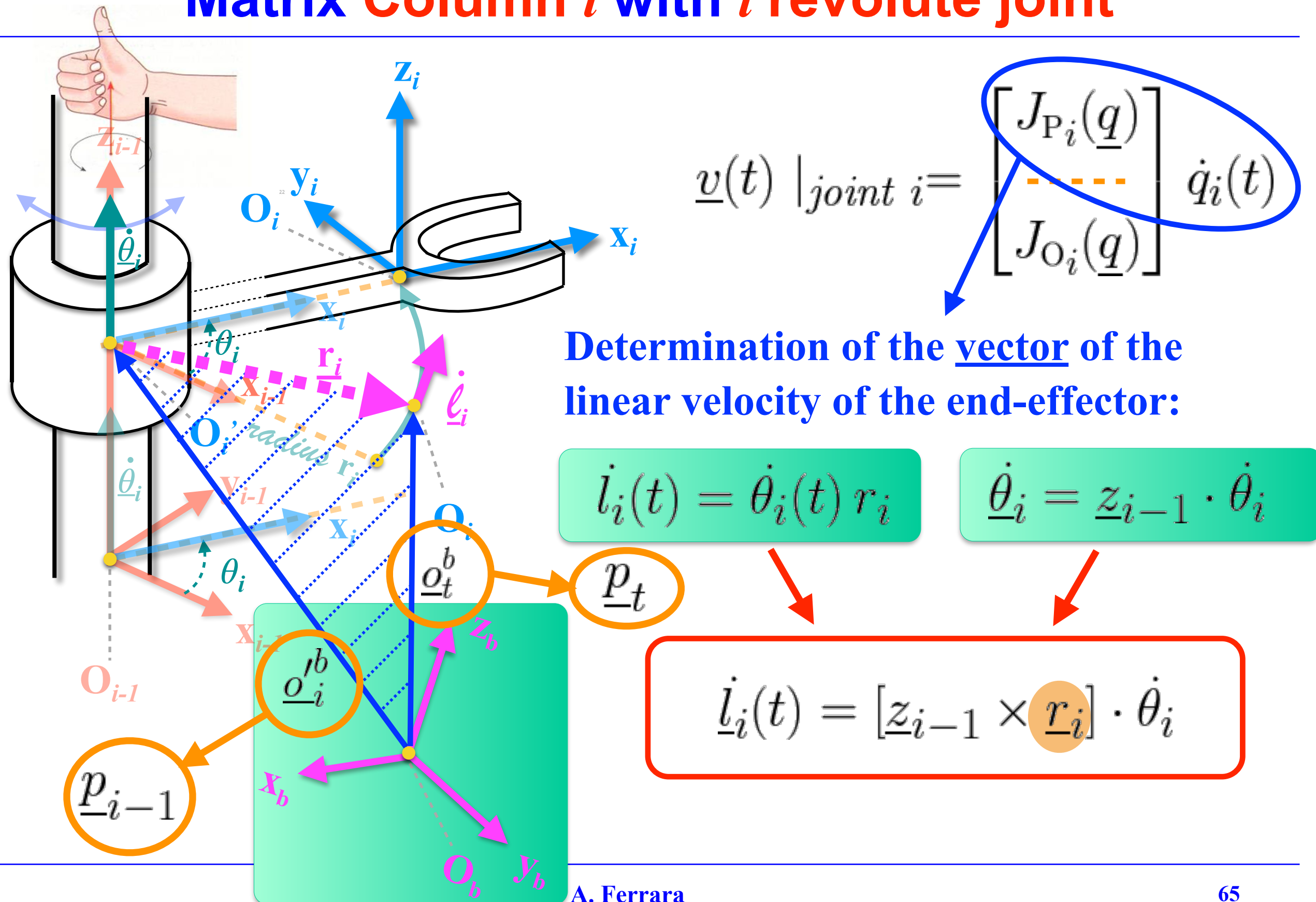
Determination of the vector of the linear velocity of the end-effector:

$$\dot{l}_i(t) = \dot{\theta}_i(t) r_i$$

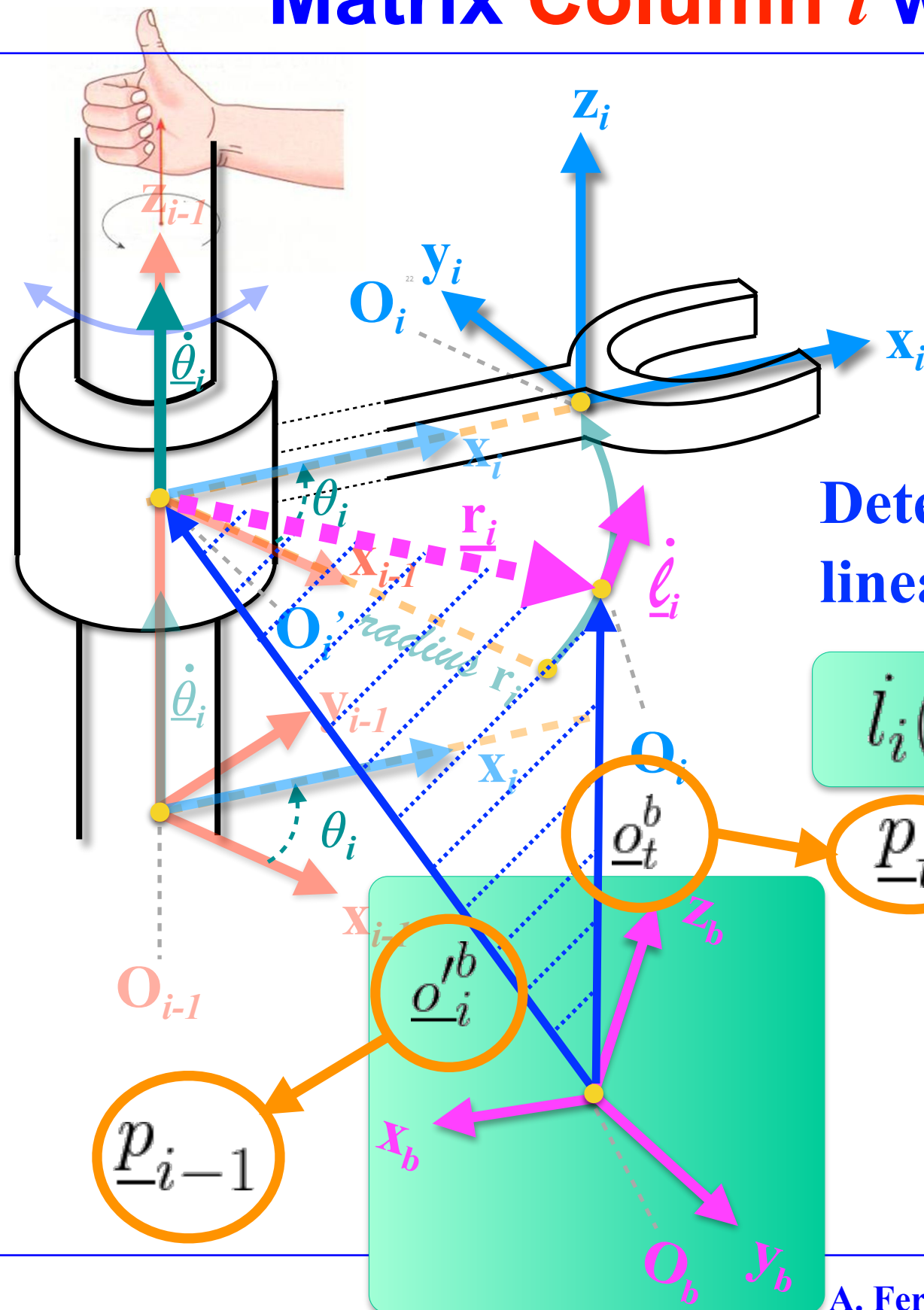
$$\underline{\dot{\theta}}_i = \underline{z}_{i-1} \cdot \dot{\theta}_i$$

$$\underline{\dot{l}}_i(t) = [\underline{z}_{i-1} \times \underline{r}_i] \cdot \dot{\theta}_i$$

How to Determine the Geometric Jacobian Matrix Column i with i revolute joint



How to Determine the Geometric Jacobian Matrix Column i with i revolute joint



$$\underline{v}(t) |_{joint\ i} = \begin{bmatrix} J_{P_i}(\underline{q}) \\ \vdots \\ J_{O_i}(\underline{q}) \end{bmatrix} \dot{q}_i(t)$$

Determination of the vector of the linear velocity of the end-effector:

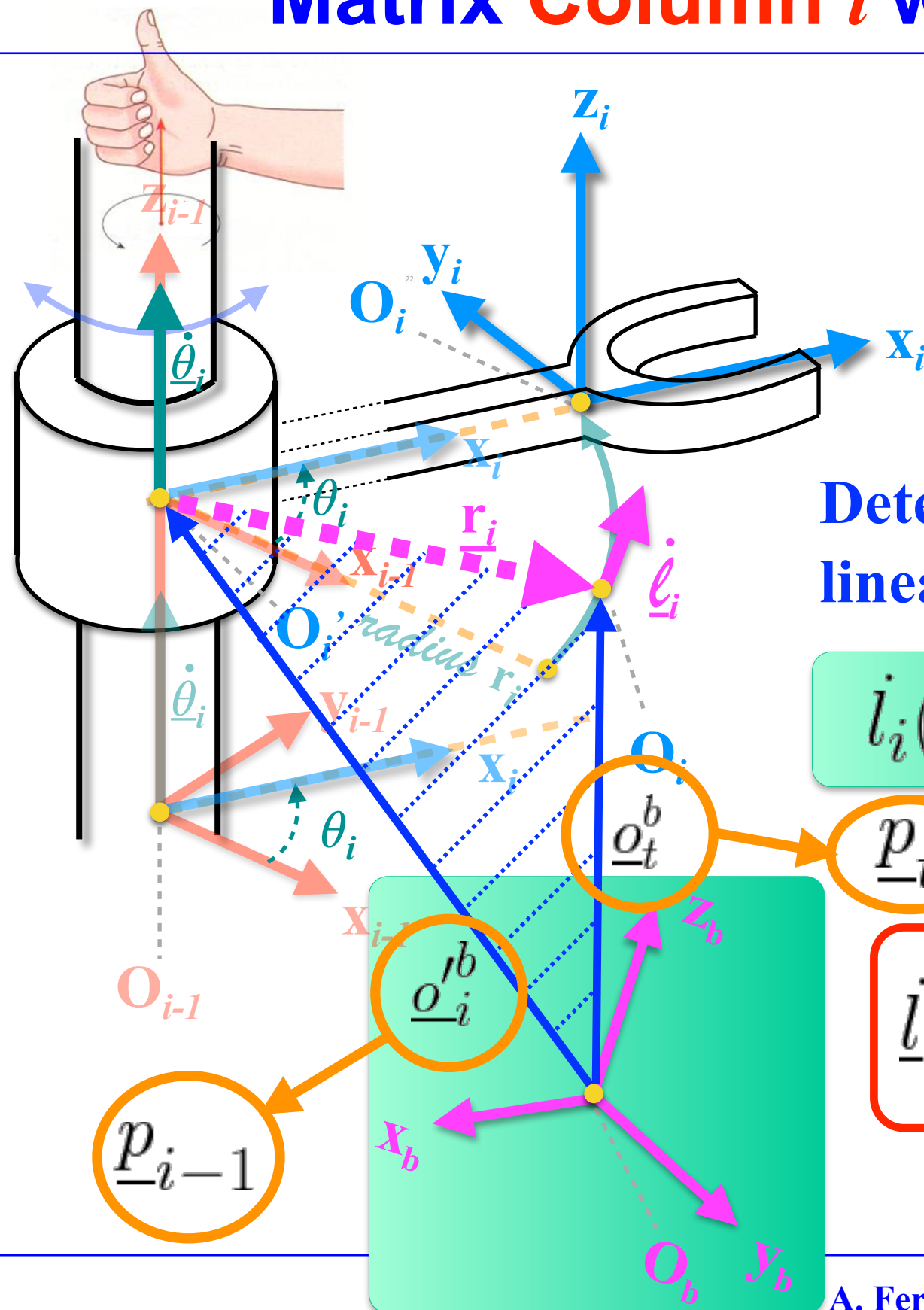
$$\dot{l}_i(t) = \dot{\theta}_i(t) r_i$$

$$\underline{\dot{\theta}}_i = \underline{z}_{i-1} \cdot \dot{\theta}_i$$

$$\dot{l}_i(t) = [\underline{z}_{i-1} \times \underline{r}_i] \cdot \dot{\theta}_i$$

$$(\underline{p}_t - \underline{p}_{i-1})$$

How to Determine the Geometric Jacobian Matrix Column i with i revolute joint



$$\underline{v}(t) |_{joint\ i} = \begin{bmatrix} J_{P_i}(\underline{q}) \\ \vdots \\ J_{O_i}(\underline{q}) \end{bmatrix} \dot{q}_i(t)$$

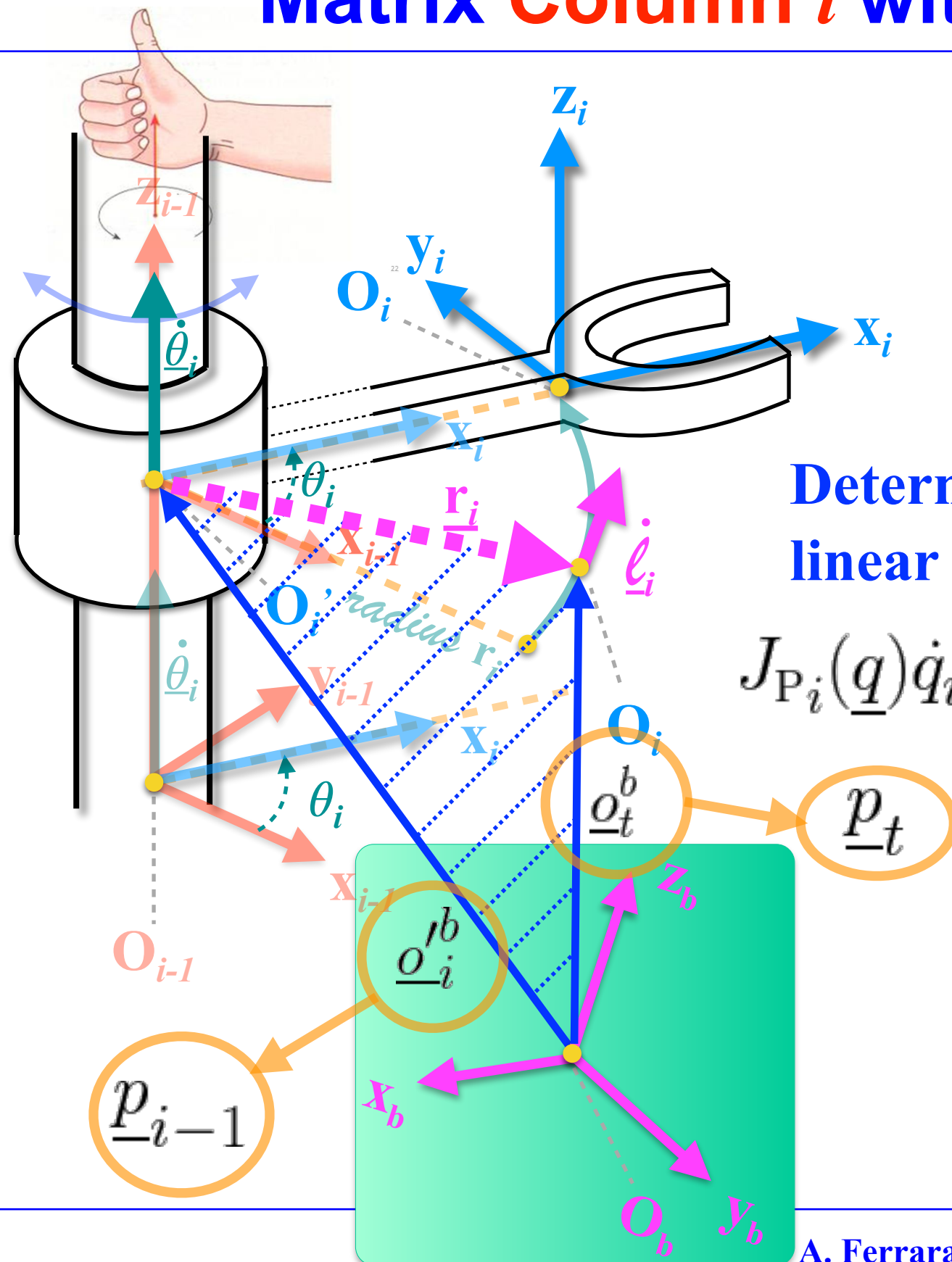
Determination of the vector of the linear velocity of the end-effector:

$$\dot{l}_i(t) = \dot{\theta}_i(t) r_i$$

$$\underline{\dot{\theta}}_i = \underline{z}_{i-1} \cdot \dot{\theta}_i$$

$$\dot{l}_i(t) = [\underline{z}_{i-1} \times (\underline{p}_t - \underline{p}_{i-1})] \cdot \dot{\theta}_i$$

How to Determine the Geometric Jacobian Matrix Column i with i revolute joint

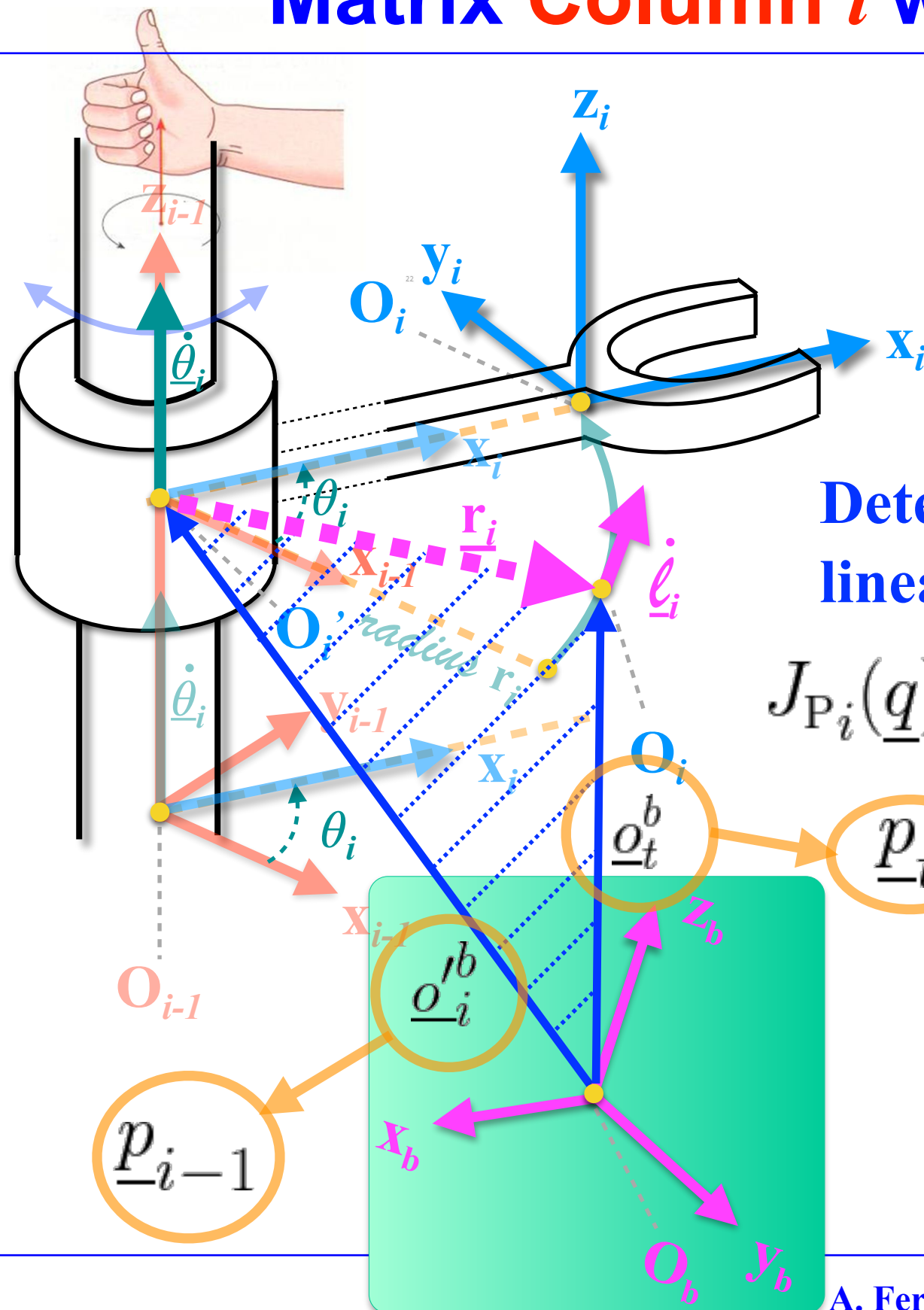


$$\underline{v}(t) \mid_{joint\ i} = \begin{bmatrix} J_{P_i}(\underline{q}) \\ \vdots \\ J_{O_i}(\underline{q}) \end{bmatrix} \dot{q}_i(t)$$

Determination of the vector of the linear velocity of the end-effector:

$$J_{P_i}(\underline{q}) \dot{q}_i(t) = [\underline{z}_{i-1} \times (\underline{p}_t - \underline{p}_{i-1})] \cdot \dot{\theta}_i$$

How to Determine the Geometric Jacobian Matrix **Column i with i revolute joint**

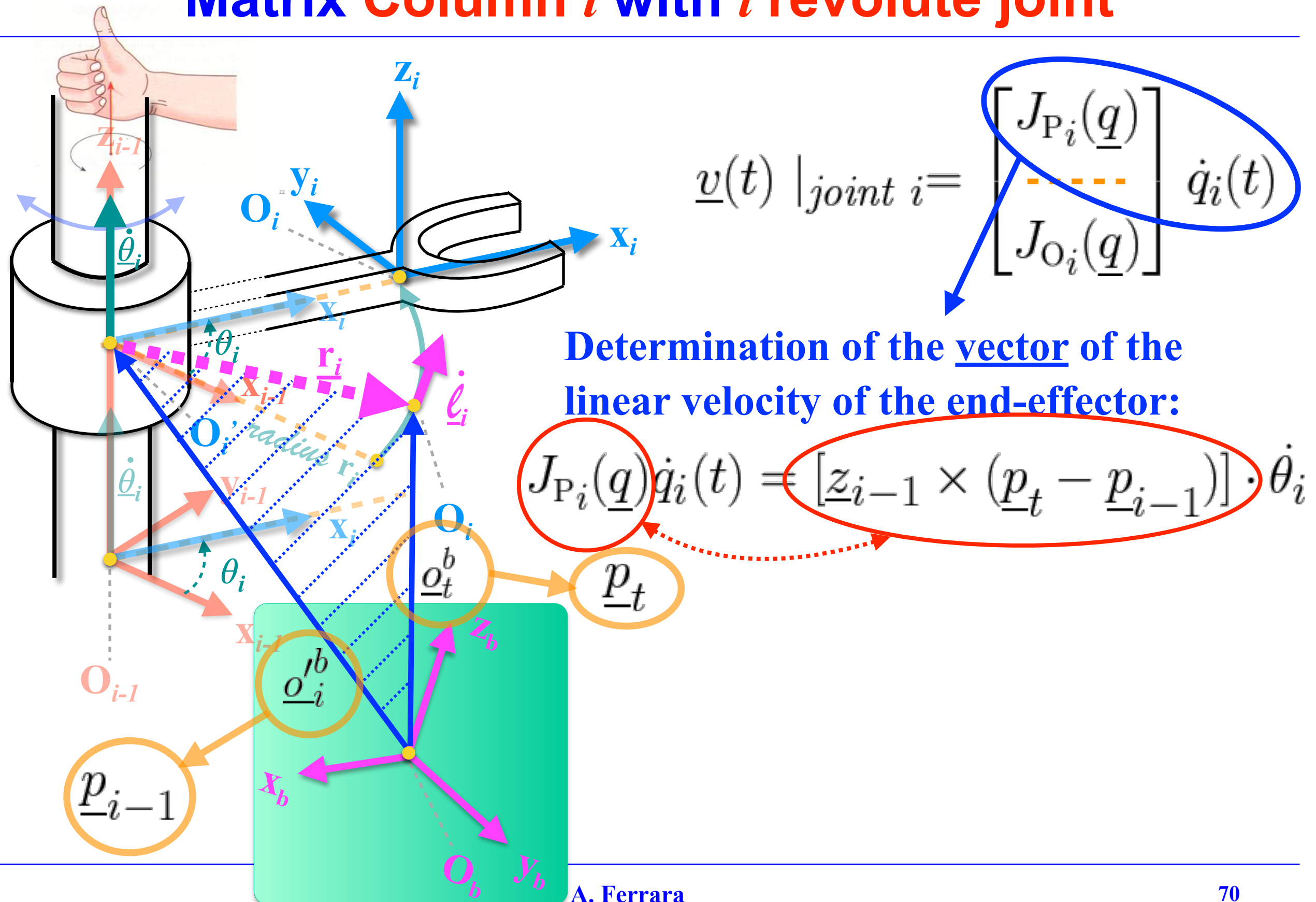


$$\underline{v}(t) \mid_{joint \ i} = \begin{bmatrix} J_{P_i}(\underline{q}) \\ \vdots \\ J_{O_i}(\underline{q}) \end{bmatrix} \dot{q}_i(t)$$

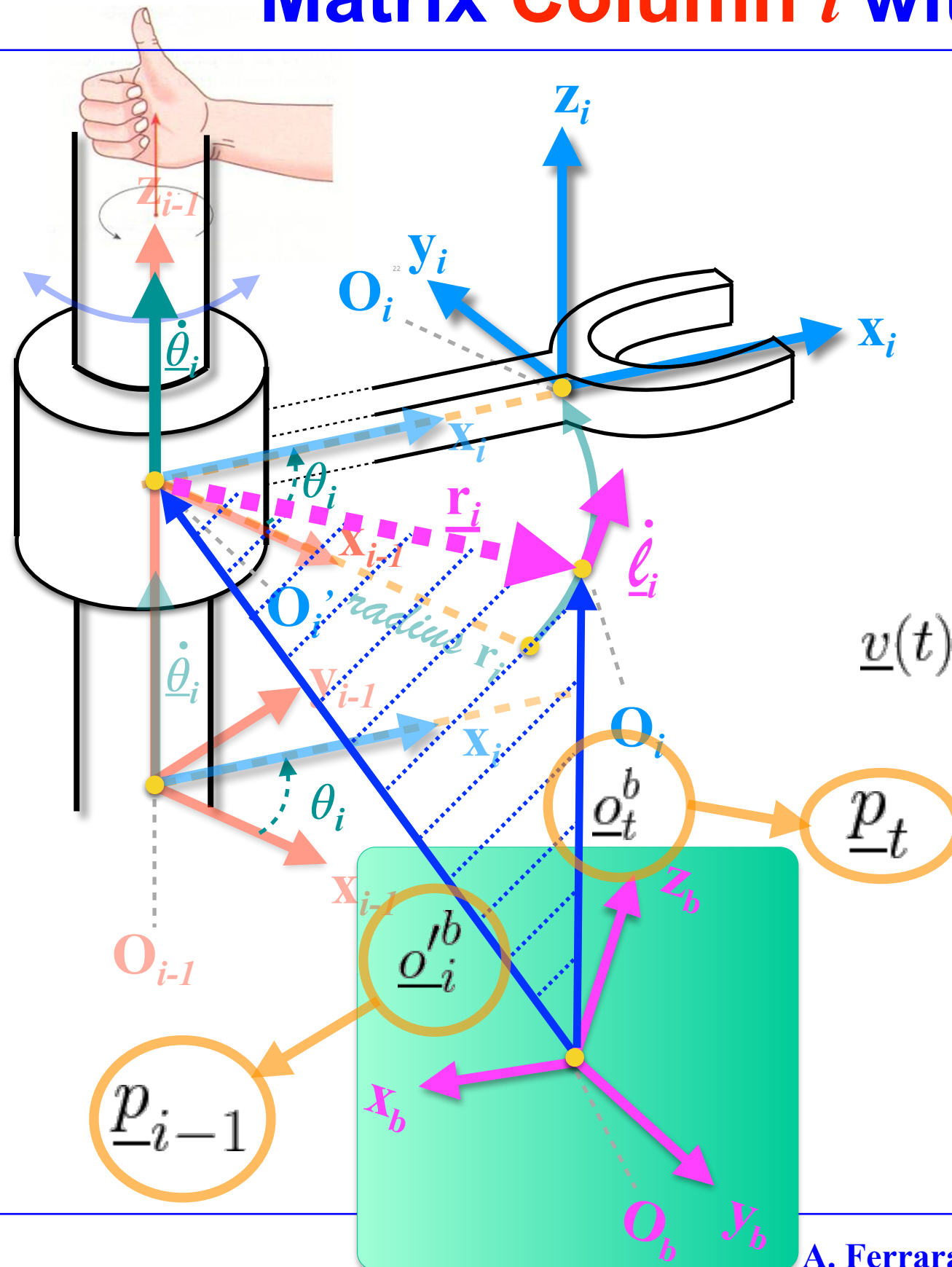
Determination of the vector of the linear velocity of the end-effector:

$$J_{P_i}(\underline{q}) \dot{q}_i(t) = [\underline{z}_{i-1} \times (\underline{p}_t - \underline{p}_{i-1})] \cdot \dot{\theta}_i$$

How to Determine the Geometric Jacobian Matrix Column i with i revolute joint



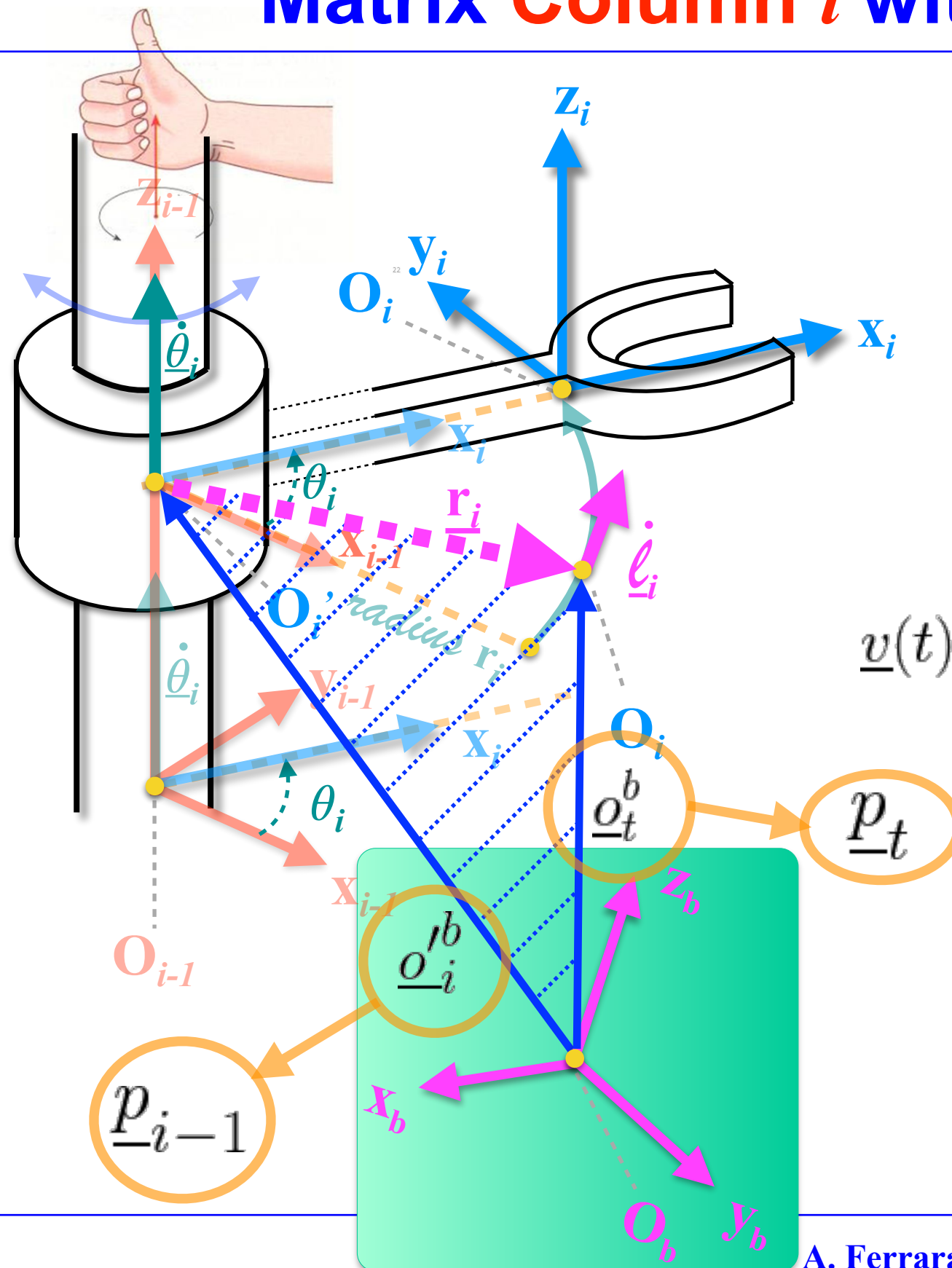
How to Determine the Geometric Jacobian Matrix Column i with i revolute joint



$$\underline{v}(t) \mid_{\text{joint } i} = \begin{bmatrix} J_{P_i}(\underline{q}) \\ \dots \\ J_{O_i}(\underline{q}) \end{bmatrix} \dot{q}_i(t)$$

$$\underline{v}(t) \mid_{\text{joint } i} = \begin{bmatrix} [\underline{z}_{i-1} \times (\underline{p}_t - \underline{p}_{i-1})] \\ \dots \\ \underline{z}_{i-1} \end{bmatrix} \dot{q}_i(t)$$

How to Determine the Geometric Jacobian Matrix Column i with i revolute joint



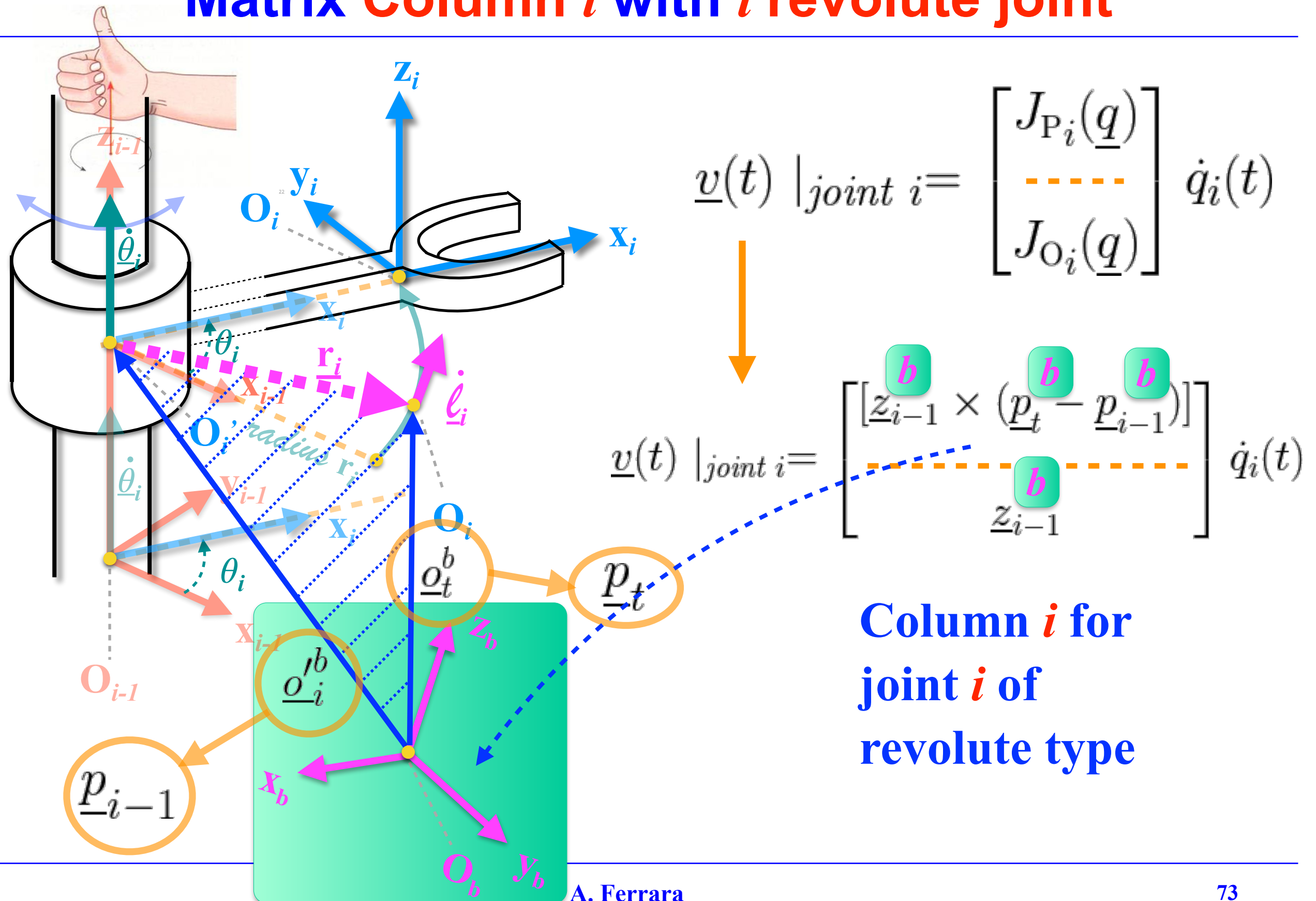
$$\underline{v}(t) \mid_{\text{joint } i} = \begin{bmatrix} J_{P_i}(\underline{q}) \\ \vdots \\ J_{O_i}(\underline{q}) \end{bmatrix} \dot{q}_i(t)$$

$$\underline{v}(t) \mid_{\text{joint } i} =$$

$$\begin{bmatrix} [\underline{z}_{i-1} \times (\underline{p}_t - \underline{p}_{i-1})] \\ \vdots \\ \underline{z}_{i-1} \end{bmatrix} \dot{q}_i(t)$$

Column i for joint i of revolute type

How to Determine the Geometric Jacobian Matrix **Column i with i revolute joint**





UNIVERSITÀ DI PAVIA
Dipartimento di
Ingegneria Industriale
e dell'Informazione

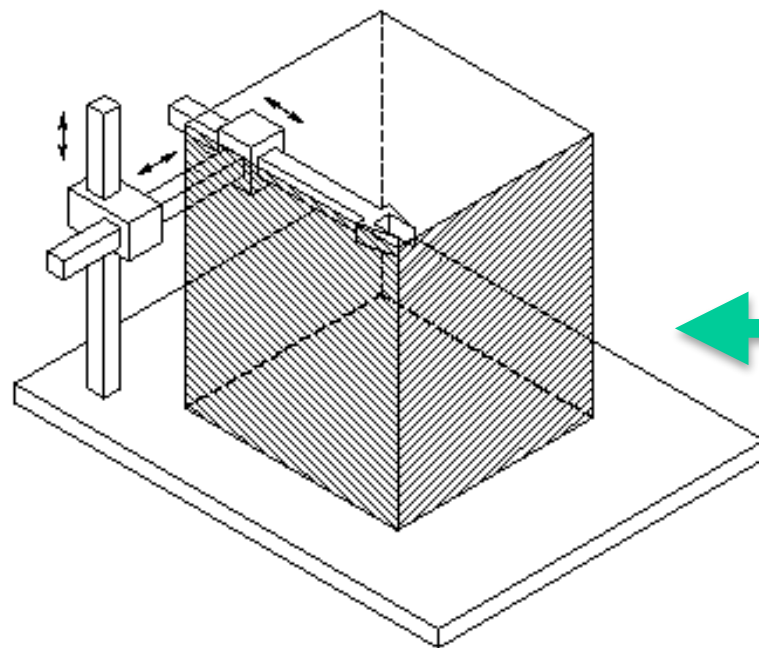
Course of Robot Control

Prof. Antonella Ferrara

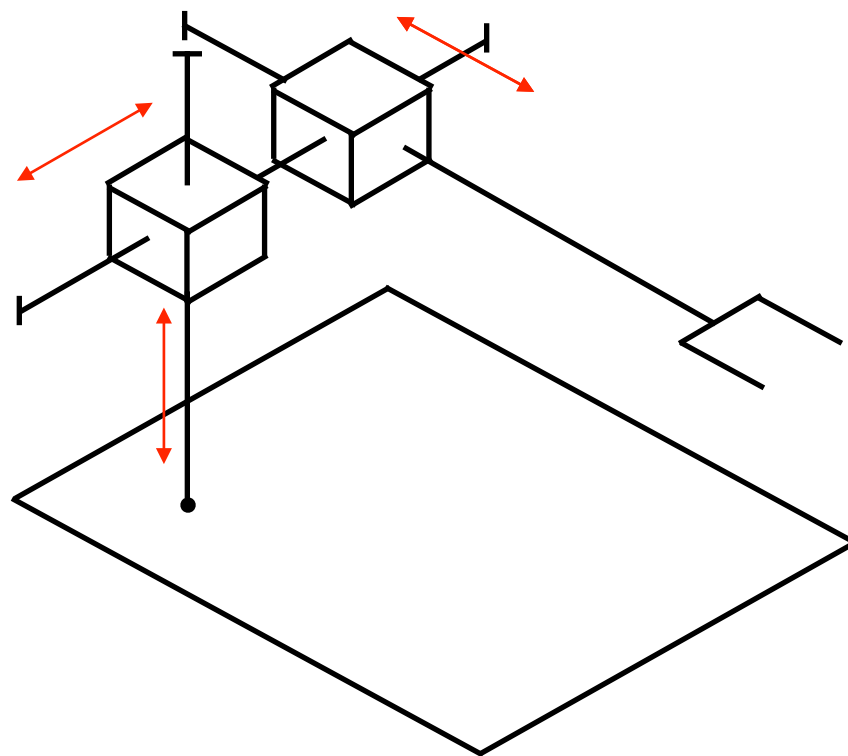


Robot Differential Kinematics: Cartesian Robots

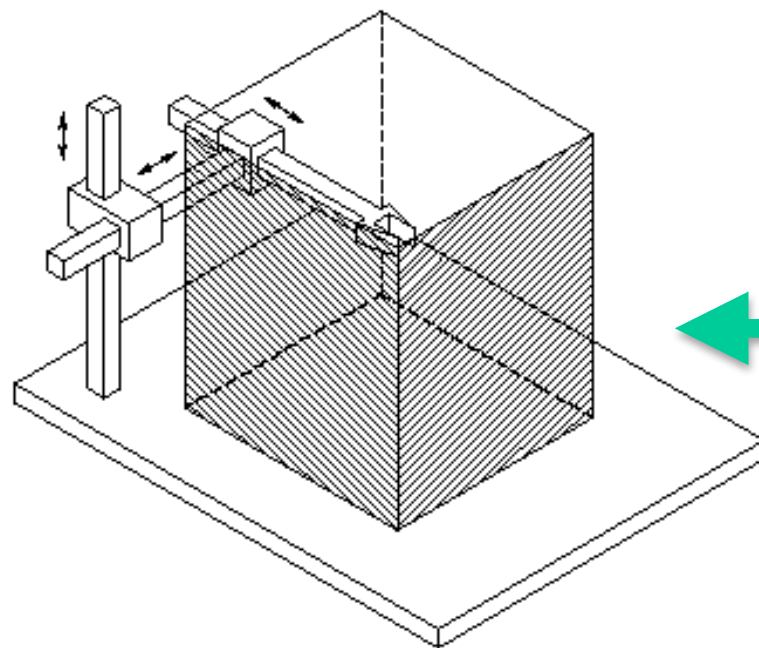
The Geometric Jacobian in Cartesian Robots



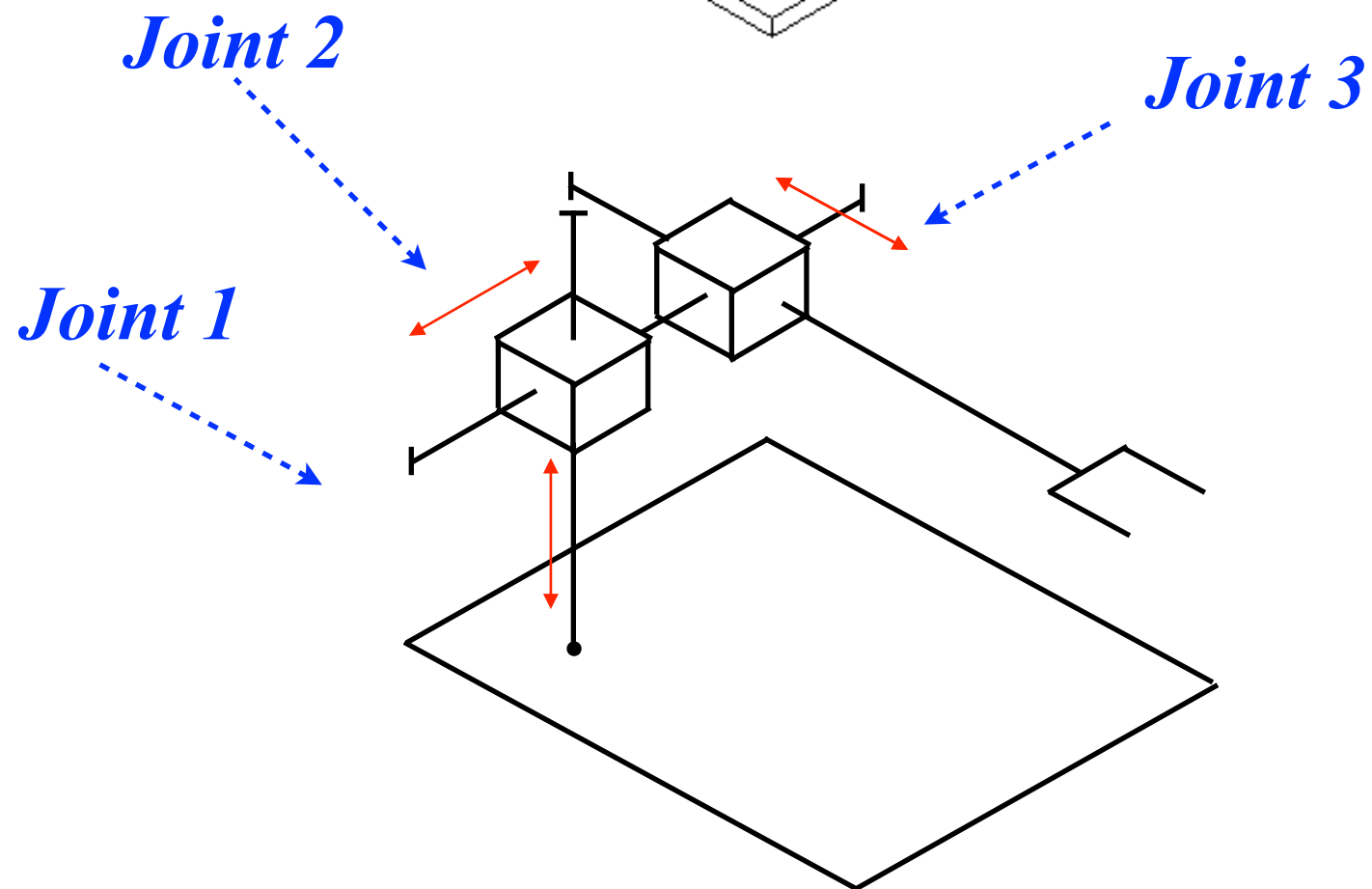
**Cartesian Robot
(3 DoM)**



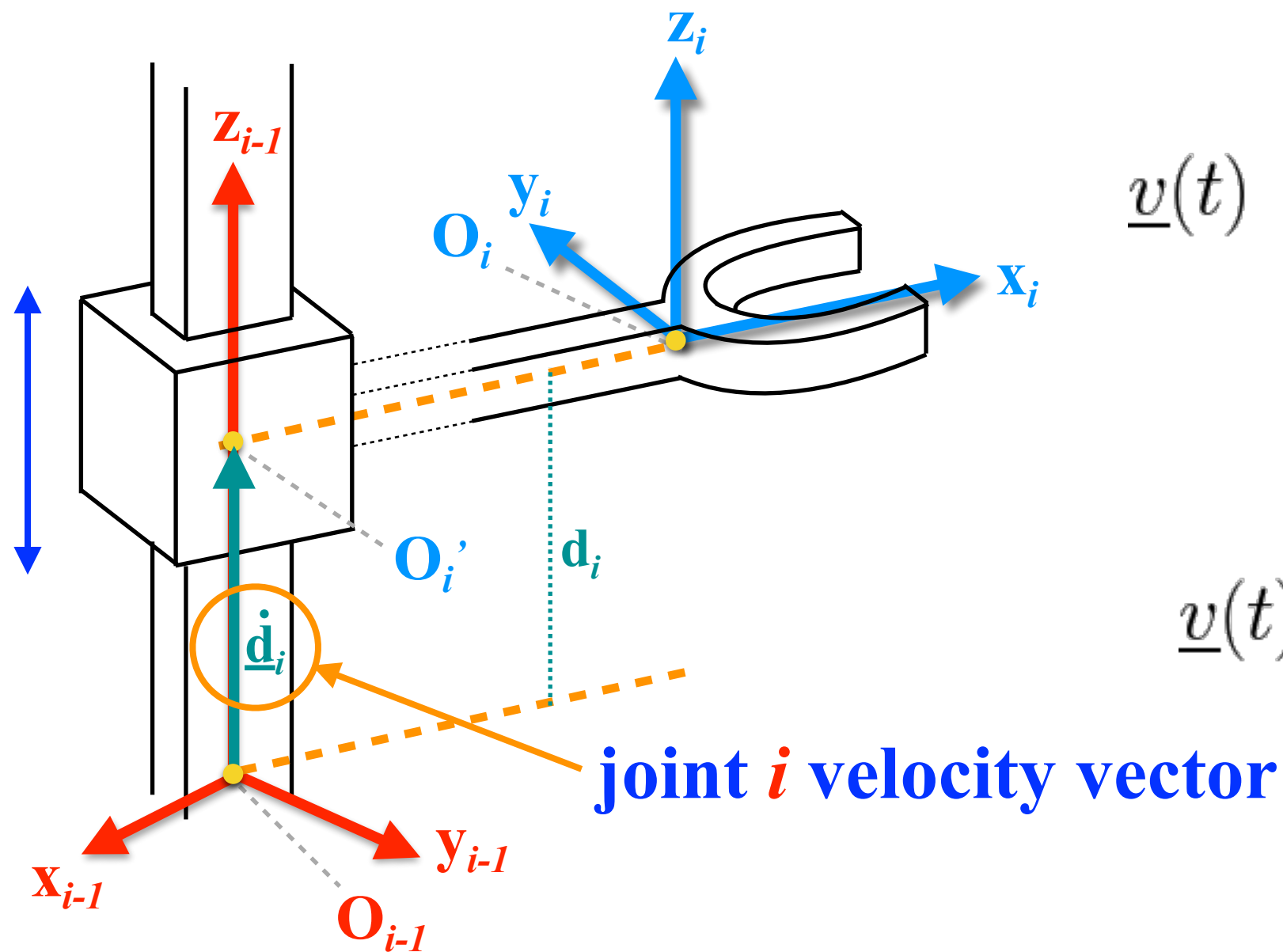
The Geometric Jacobian in Cartesian Robots



**Cartesian Robot
(3 DoM)**



How to Determine the Geometric Jacobian Matrix Column i with i prismatic joint

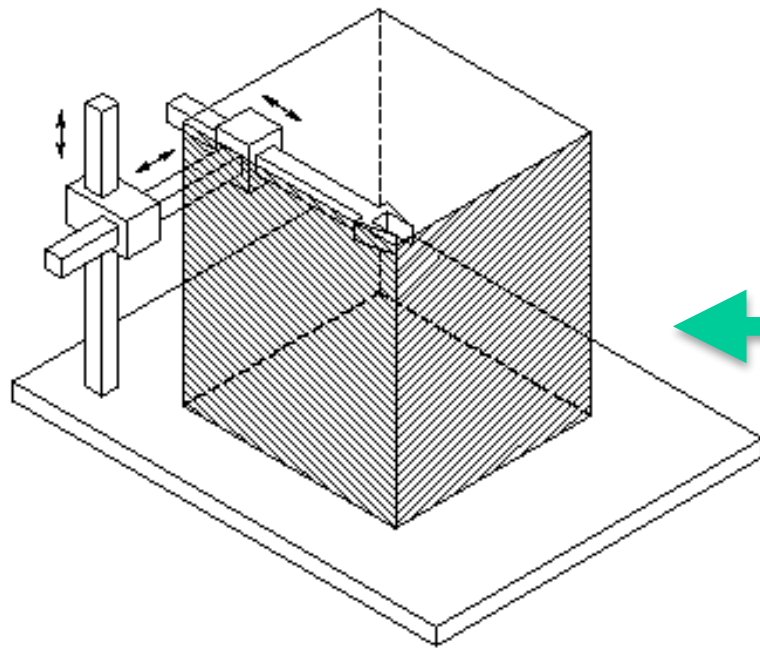


$$\underline{v}(t) \mid_{joint\ i} = \begin{bmatrix} J_{P_i}(\underline{q}) \\ \vdots \\ J_{O_i}(\underline{q}) \end{bmatrix} \dot{q}_i(t)$$

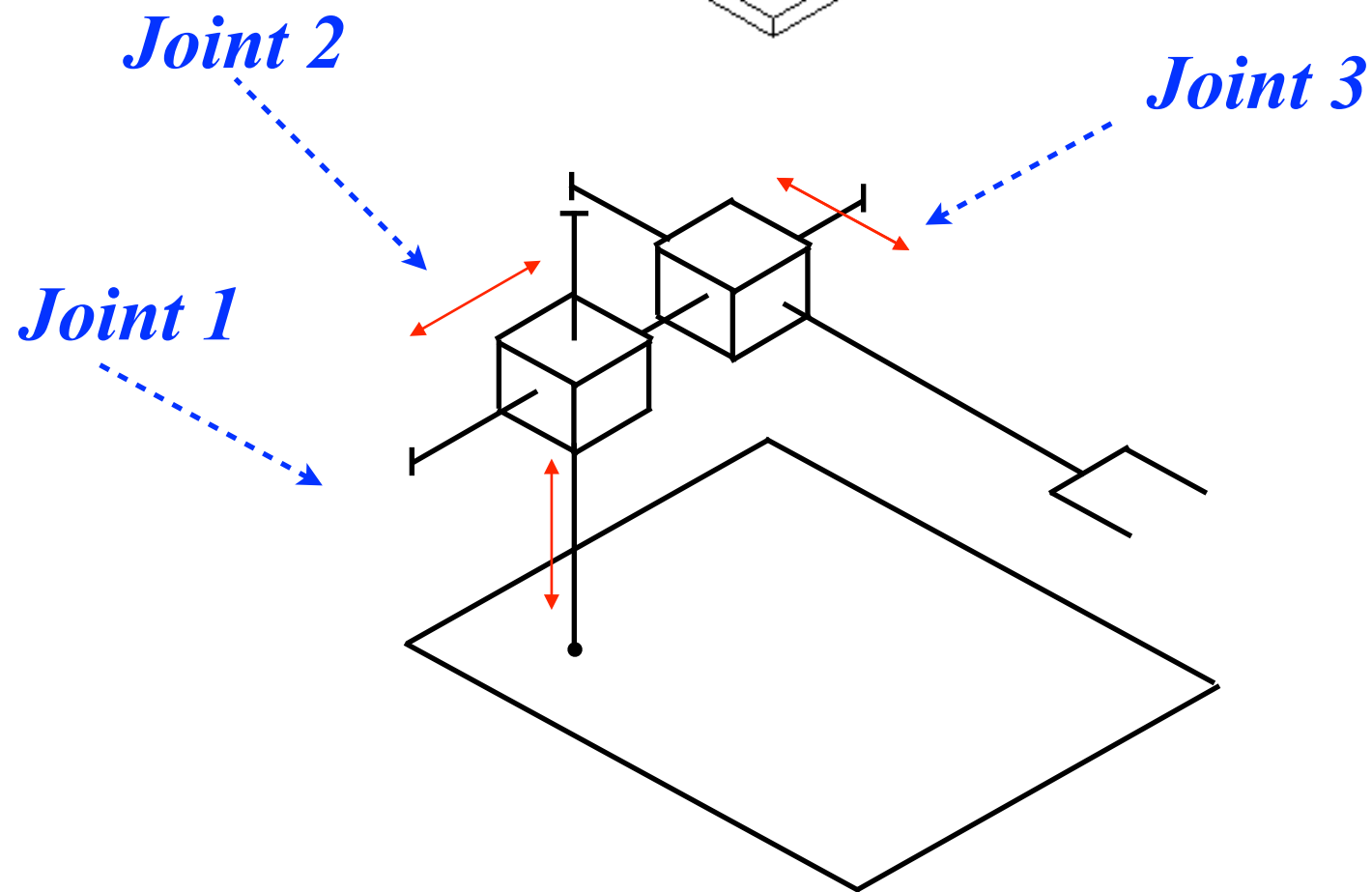
$$\underline{v}(t) \mid_{joint\ i} = \begin{bmatrix} z_{i-1} \\ \vdots \\ 0 \\ 0 \\ 0 \end{bmatrix} \dot{q}_i(t)$$

Column i for joint i of prismatic type

The Geometric Jacobian in Cartesian Robots

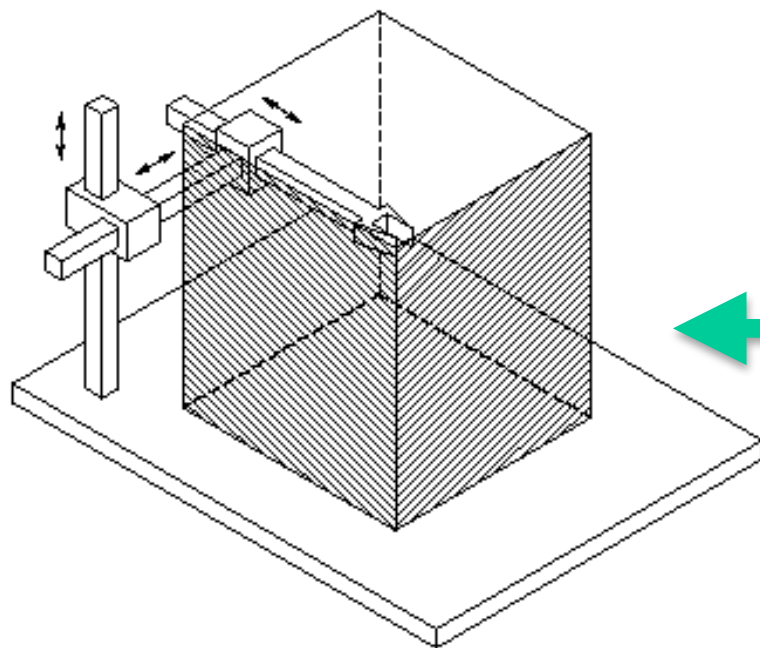


**Cartesian Robot
(3 DoM)**

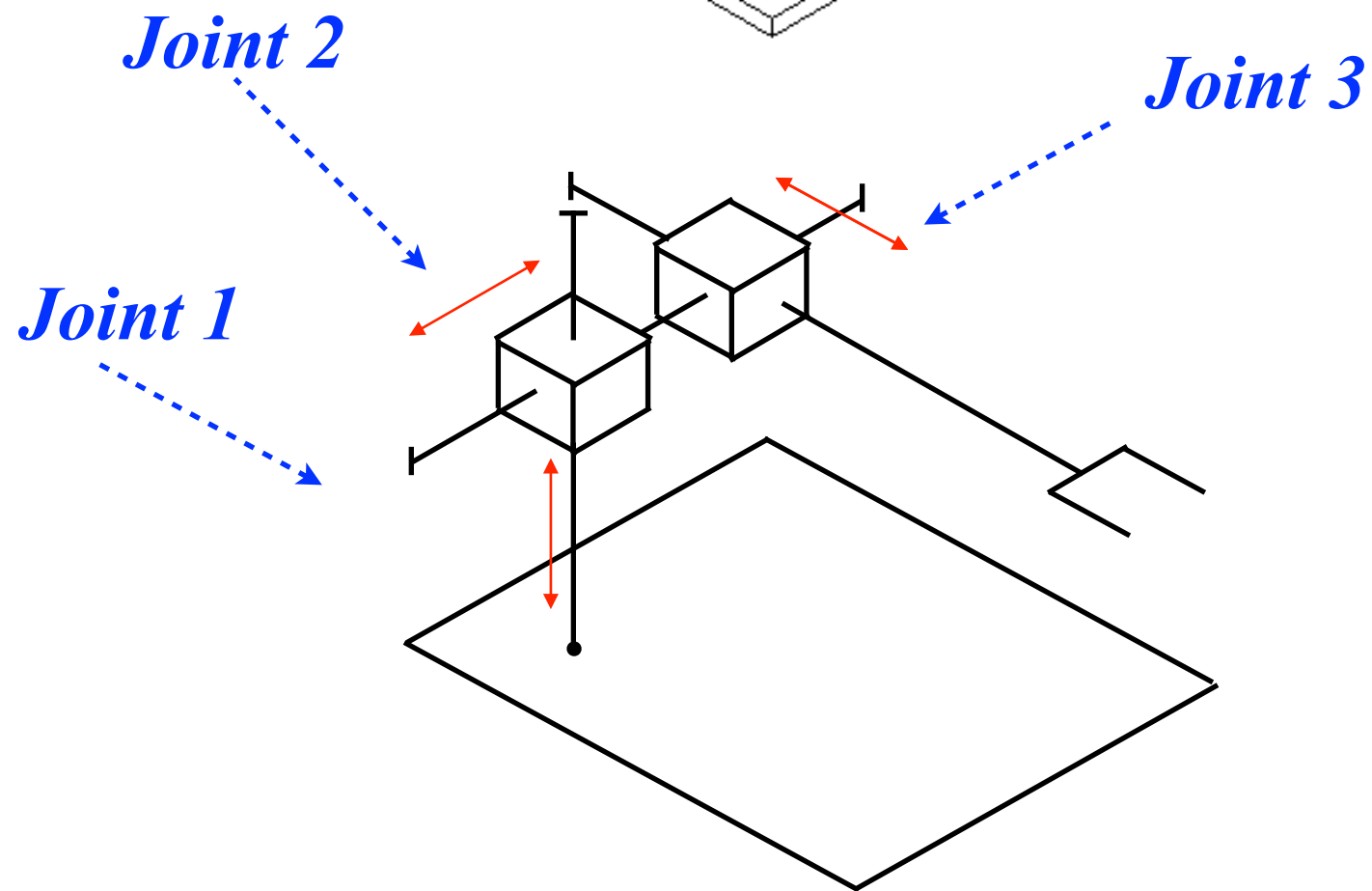


$$\underline{v}(t) = \begin{bmatrix} \underline{z}_0 & \underline{z}_1 & \underline{z}_2 \\ \hline 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \underline{\dot{q}}(t)$$

The Geometric Jacobian in Cartesian Robots

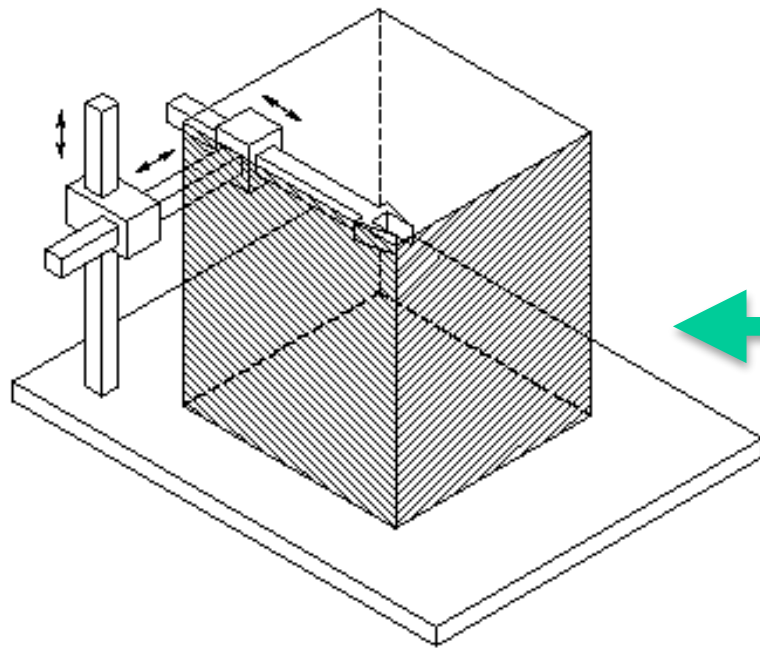


**Cartesian Robot
(3 DoM)**

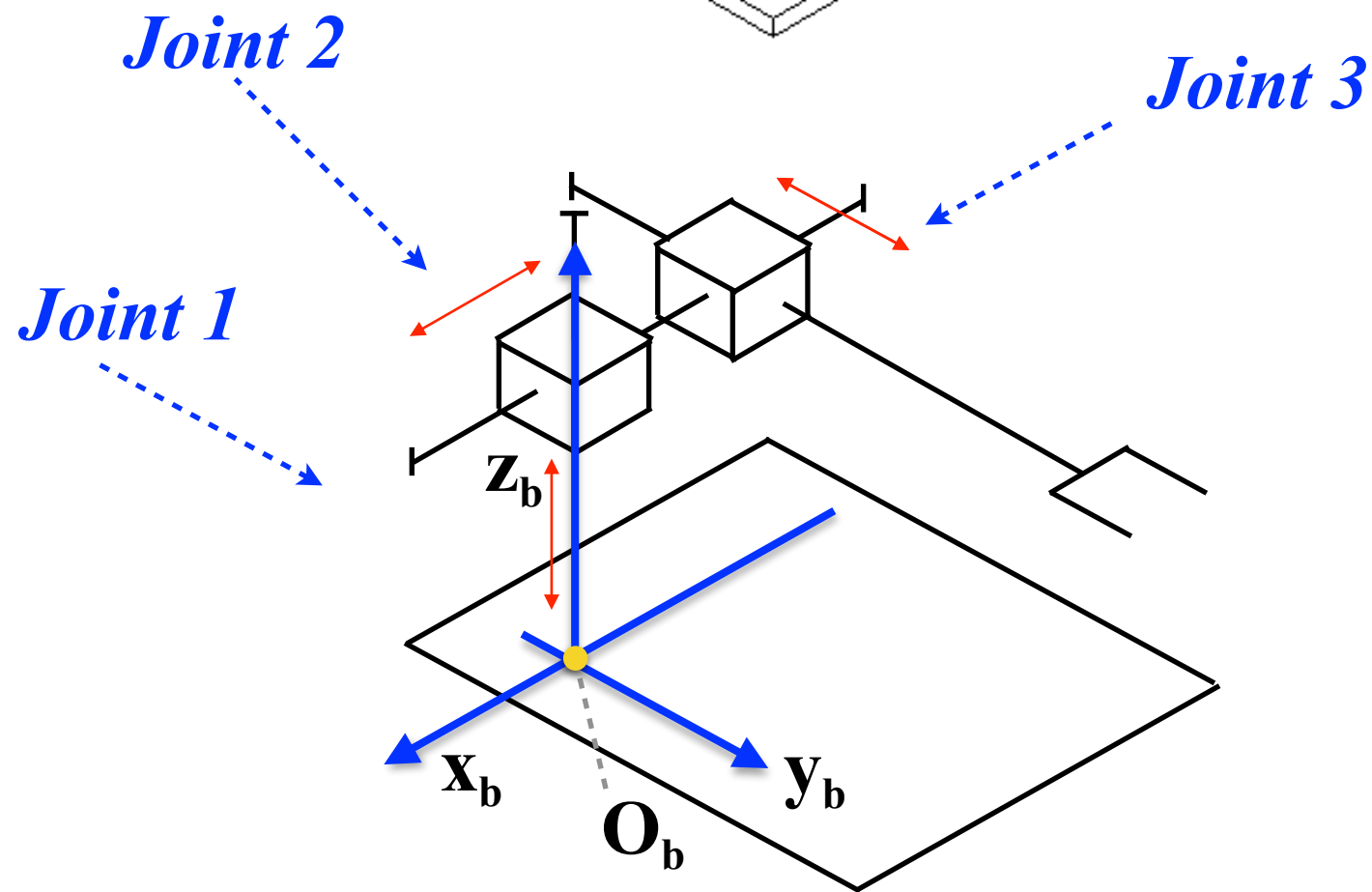


$$\underline{v}(t) = \begin{bmatrix} \underline{z}_0^b & \underline{z}_1^b & \underline{z}_2^b \\ \hline 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \underline{\dot{q}}(t)$$

The Geometric Jacobian in Cartesian Robots

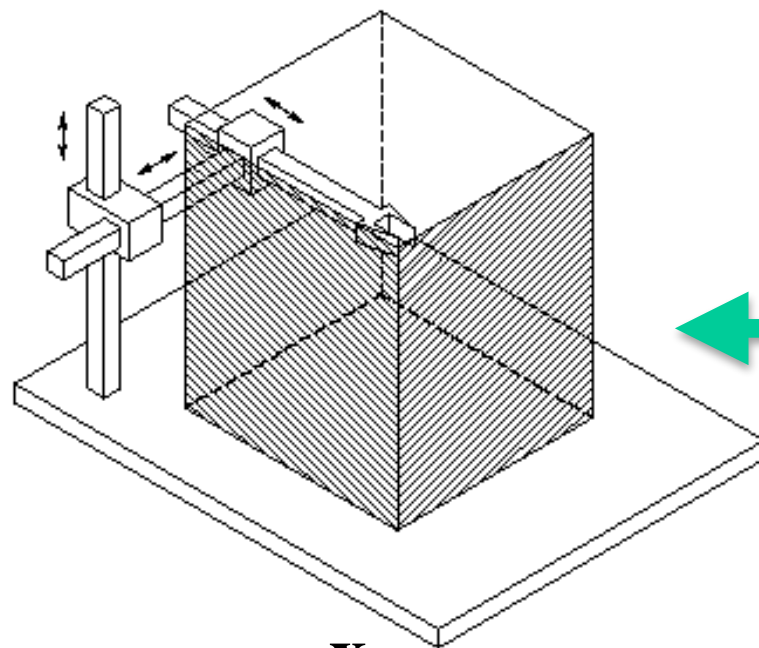


**Cartesian Robot
(3 DoM)**

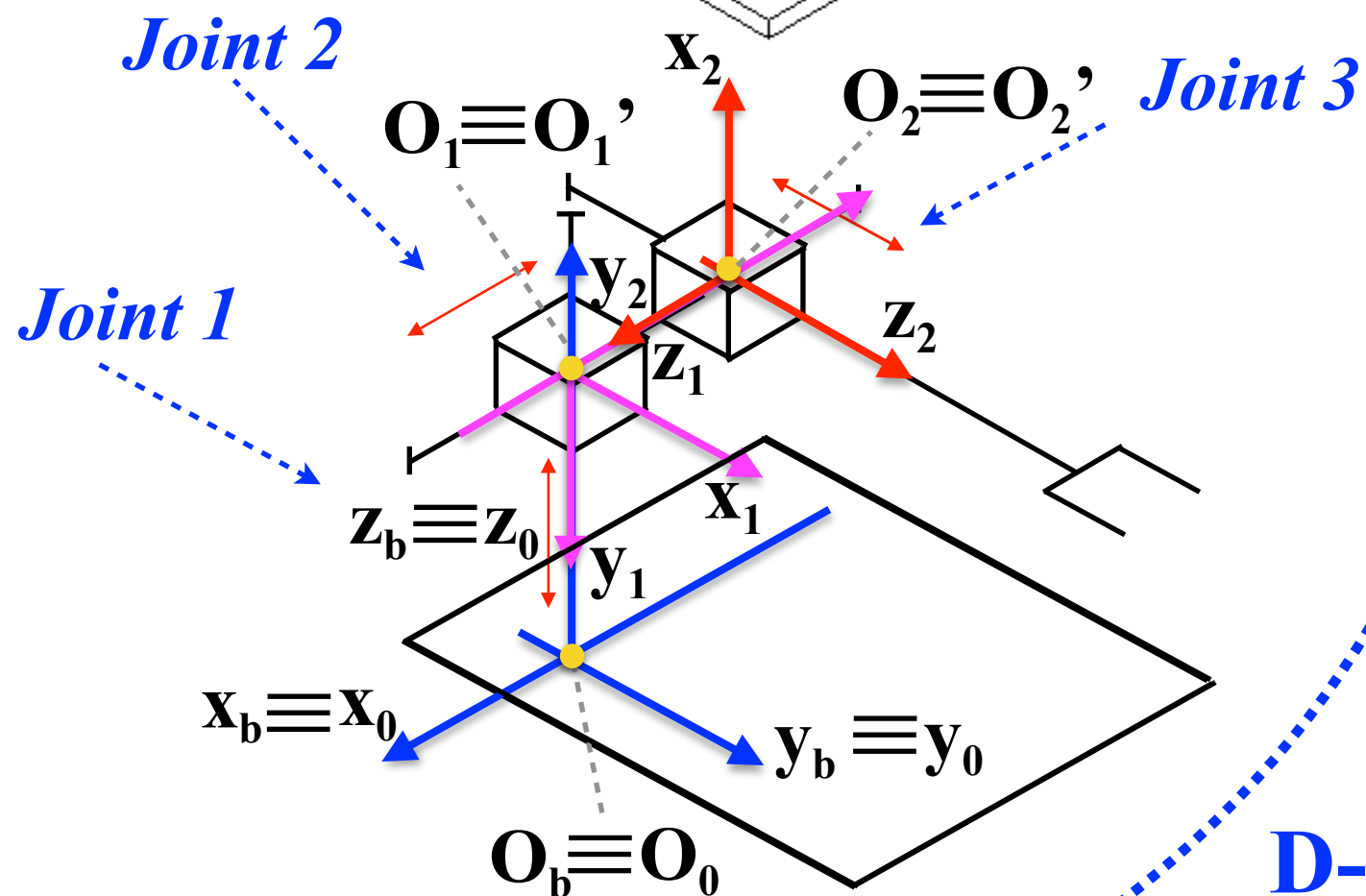


$$\underline{v}(t) = \begin{bmatrix} \underline{z}_0^b & \underline{z}_1^b & \underline{z}_2^b \\ \hline 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \underline{\dot{q}}(t)$$

The Geometric Jacobian in Cartesian Robots



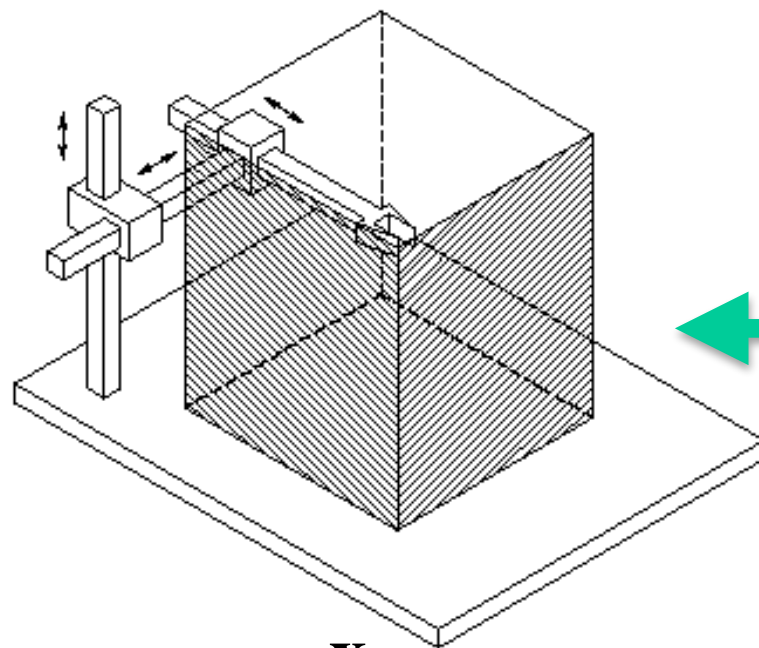
**Cartesian Robot
(3 DoM)**



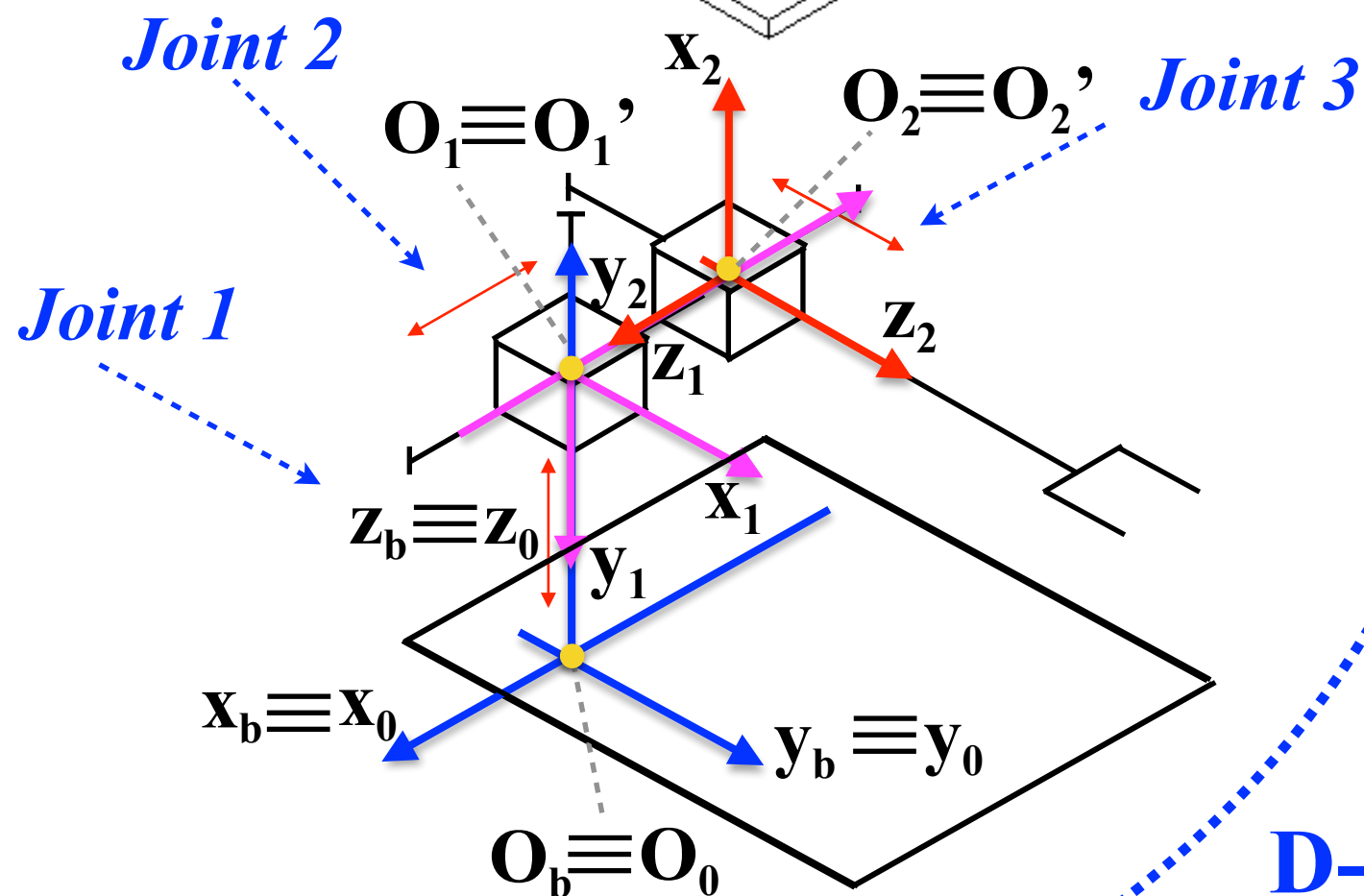
$$\underline{v}(t) = \begin{bmatrix} \underline{z}_0^b & \underline{z}_1^b & \underline{z}_2^b \\ \hline 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \dot{\underline{q}}(t)$$

D-H reference frames

The Geometric Jacobian in Cartesian Robots



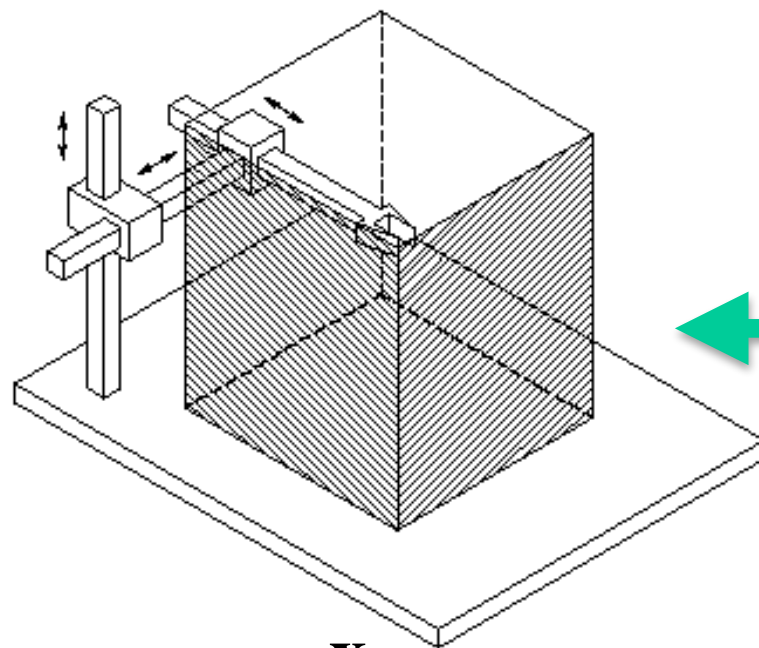
**Cartesian Robot
(3 DoM)**



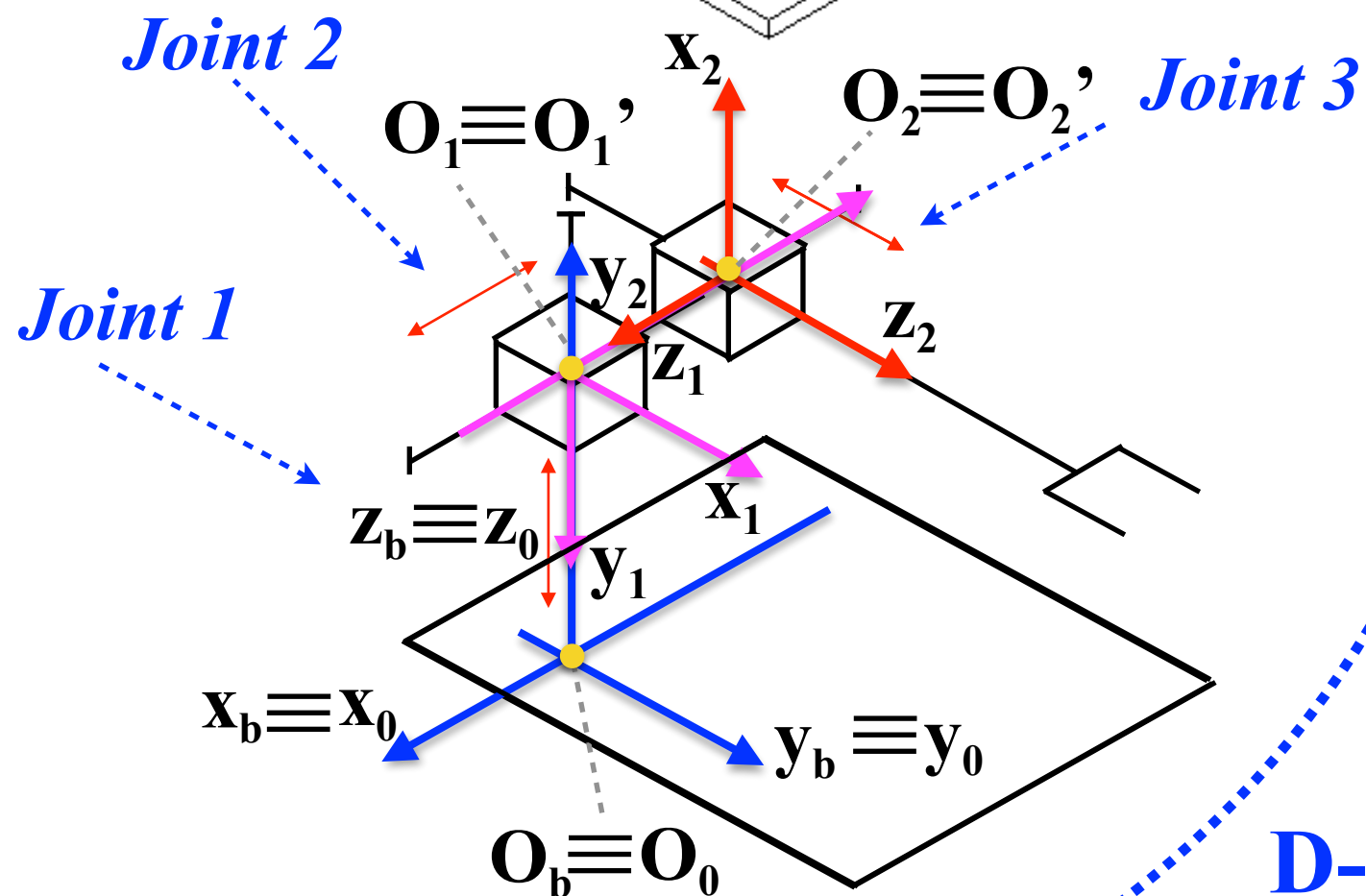
$$\underline{v}(t) = \begin{bmatrix} \underline{z}_0^b & \underline{z}_1^b & \underline{z}_2^b \\ \hline 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \underline{\dot{q}}(t)$$

D-H reference frames

The Geometric Jacobian in Cartesian Robots



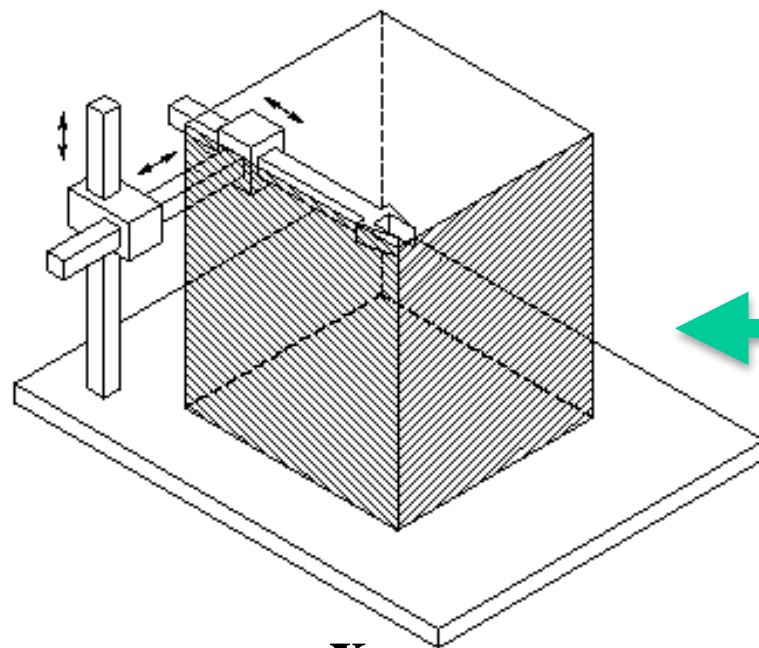
**Cartesian Robot
(3 DoM)**



$$\underline{v}(t) = \begin{bmatrix} 0 & -1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \\ \hline 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \underline{\dot{q}}(t)$$

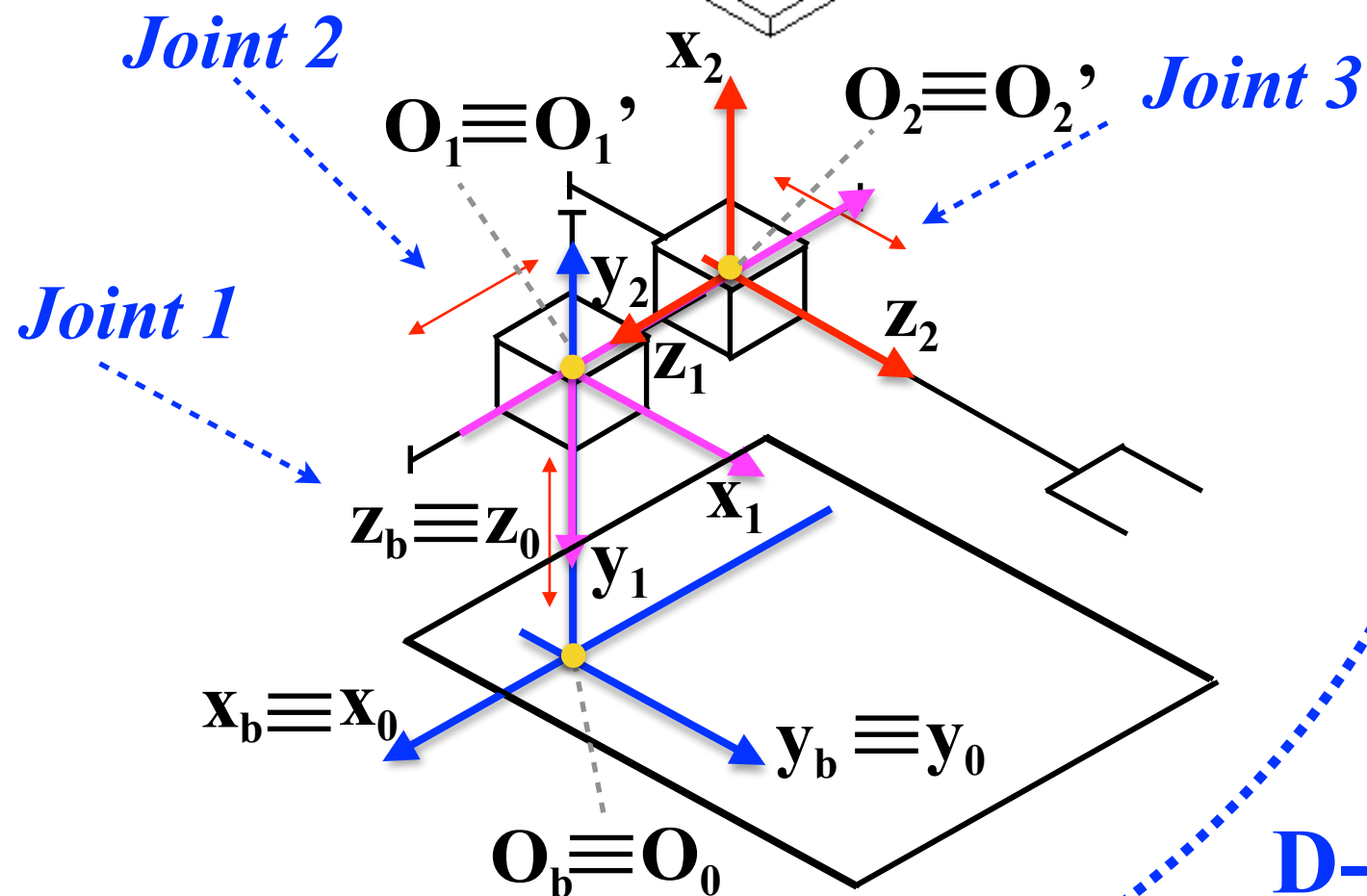
D-H reference frames

The Geometric Jacobian in Cartesian Robots



**Cartesian Robot
(3 DoM)**

$$J(\underline{q})$$



$$\underline{v}(t) =$$

$$\begin{bmatrix} 0 & -1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \\ \hline 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \underline{\dot{q}}(t)$$

D-H reference frames



UNIVERSITÀ DI PAVIA
Dipartimento di
Ingegneria Industriale
e dell'Informazione

Course of Robot Control

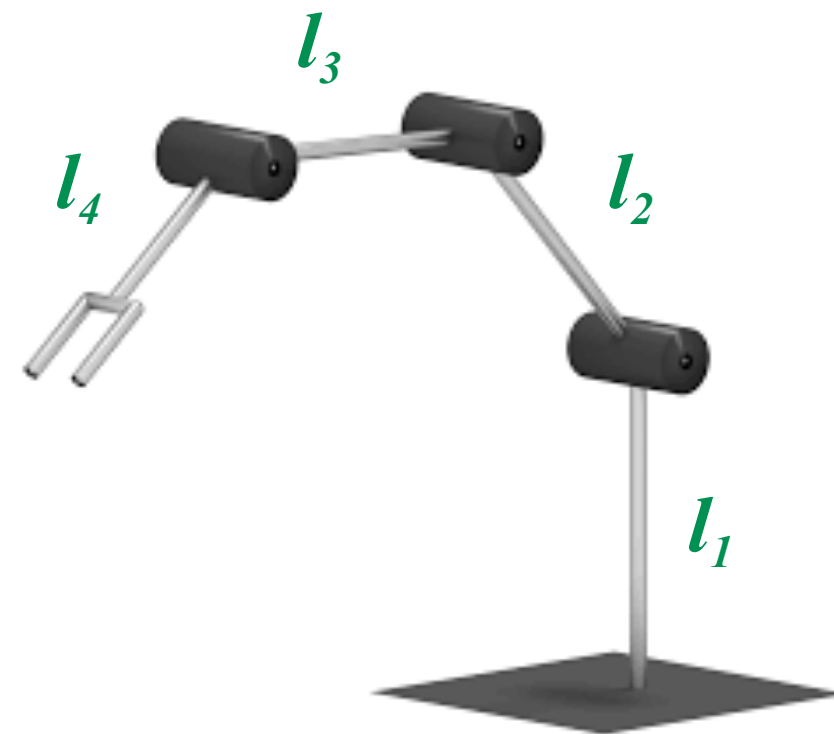
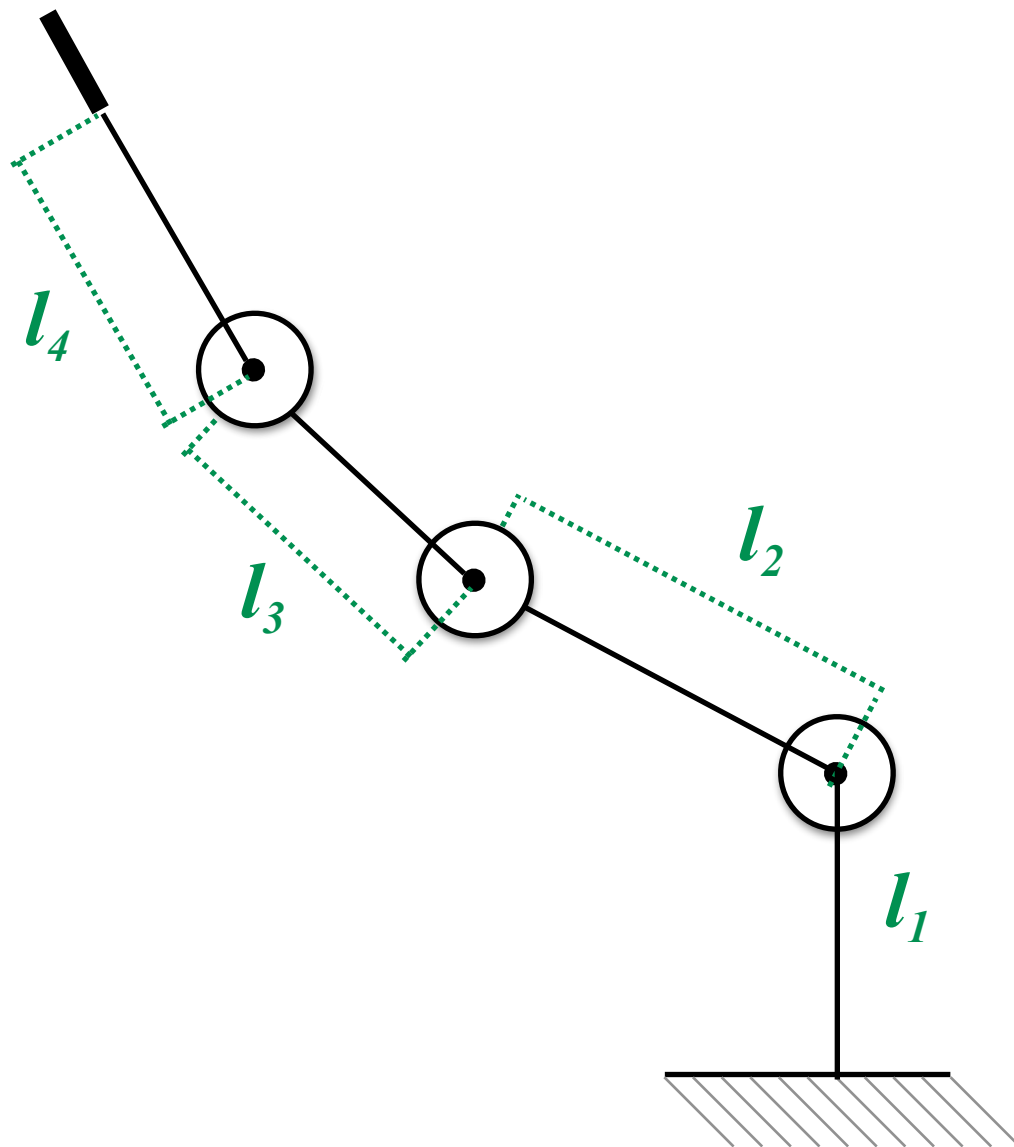
Prof. Antonella Ferrara



Robot Differential Kinematics: Anthropomorphic Robots

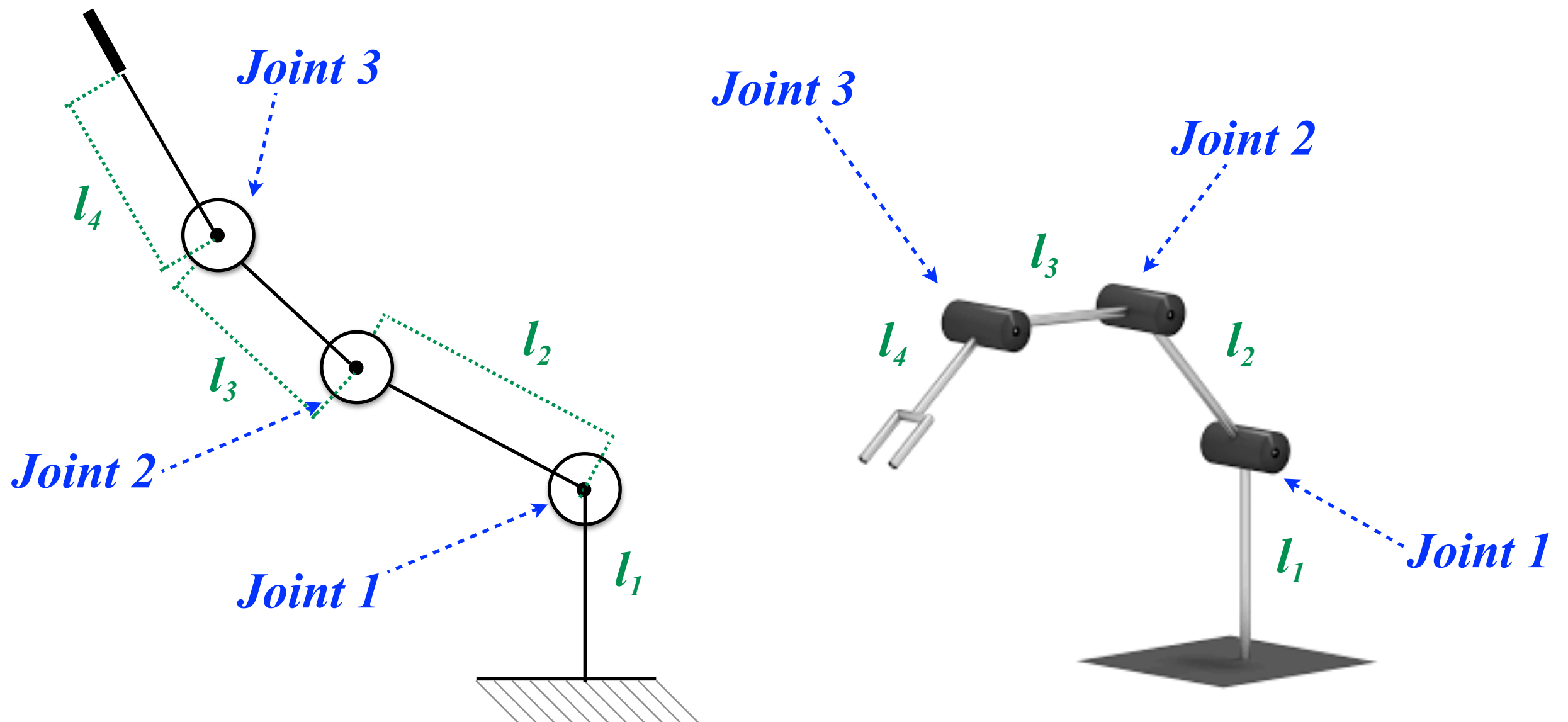
The Geometric Jacobian in Anthropomorphic Robots

Planar Anthropomorphic Robot (3 DoM)

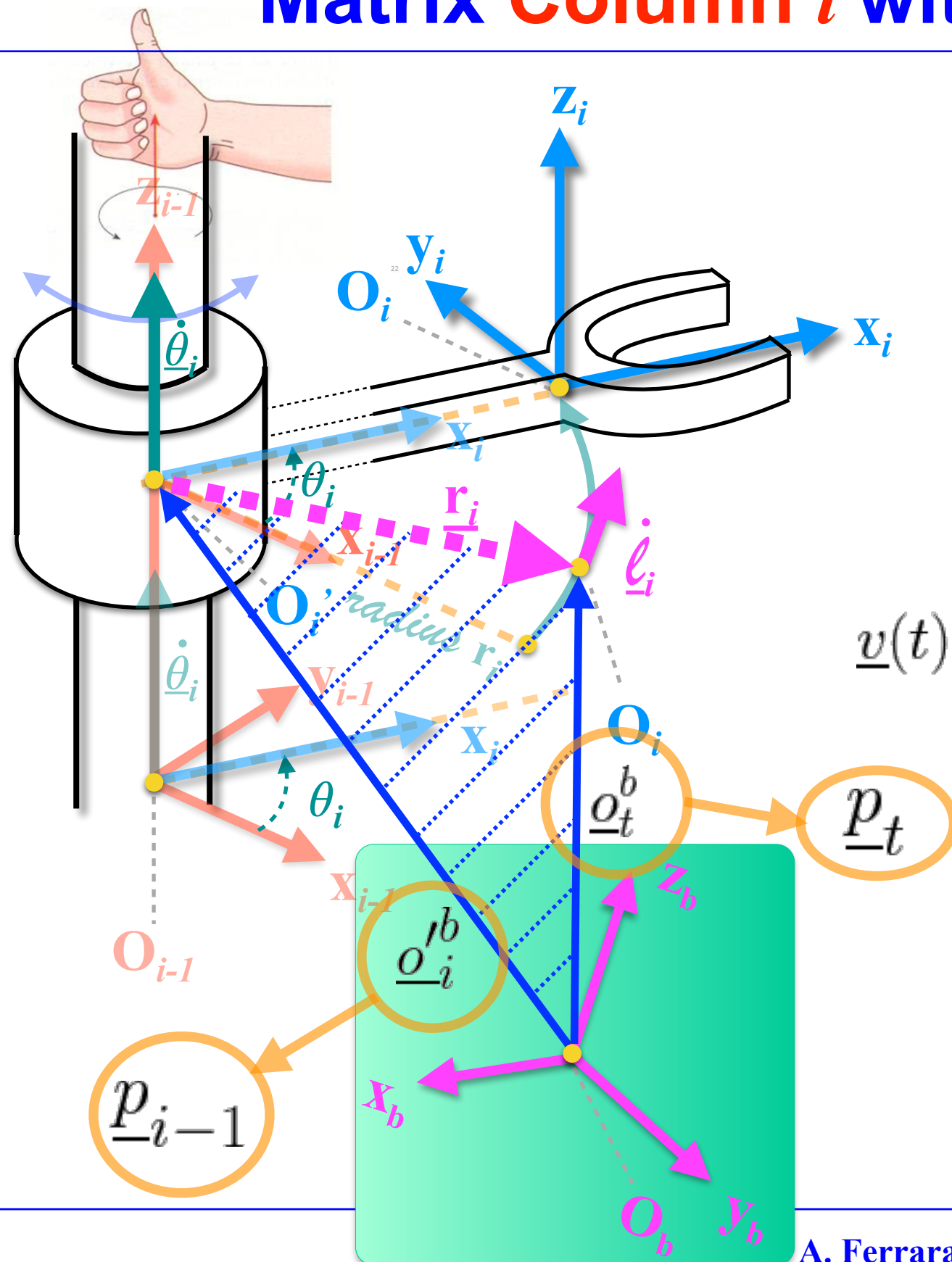


The Geometric Jacobian in Anthropomorphic Robots

Planar Anthropomorphic Robot (3 DoM)



How to Determine the Geometric Jacobian Matrix Column i with i revolute joint



$$\underline{v}(t) \mid_{\text{joint } i} = \begin{bmatrix} J_{P_i}(\underline{q}) \\ \vdots \\ J_{O_i}(\underline{q}) \end{bmatrix} \dot{q}_i(t)$$

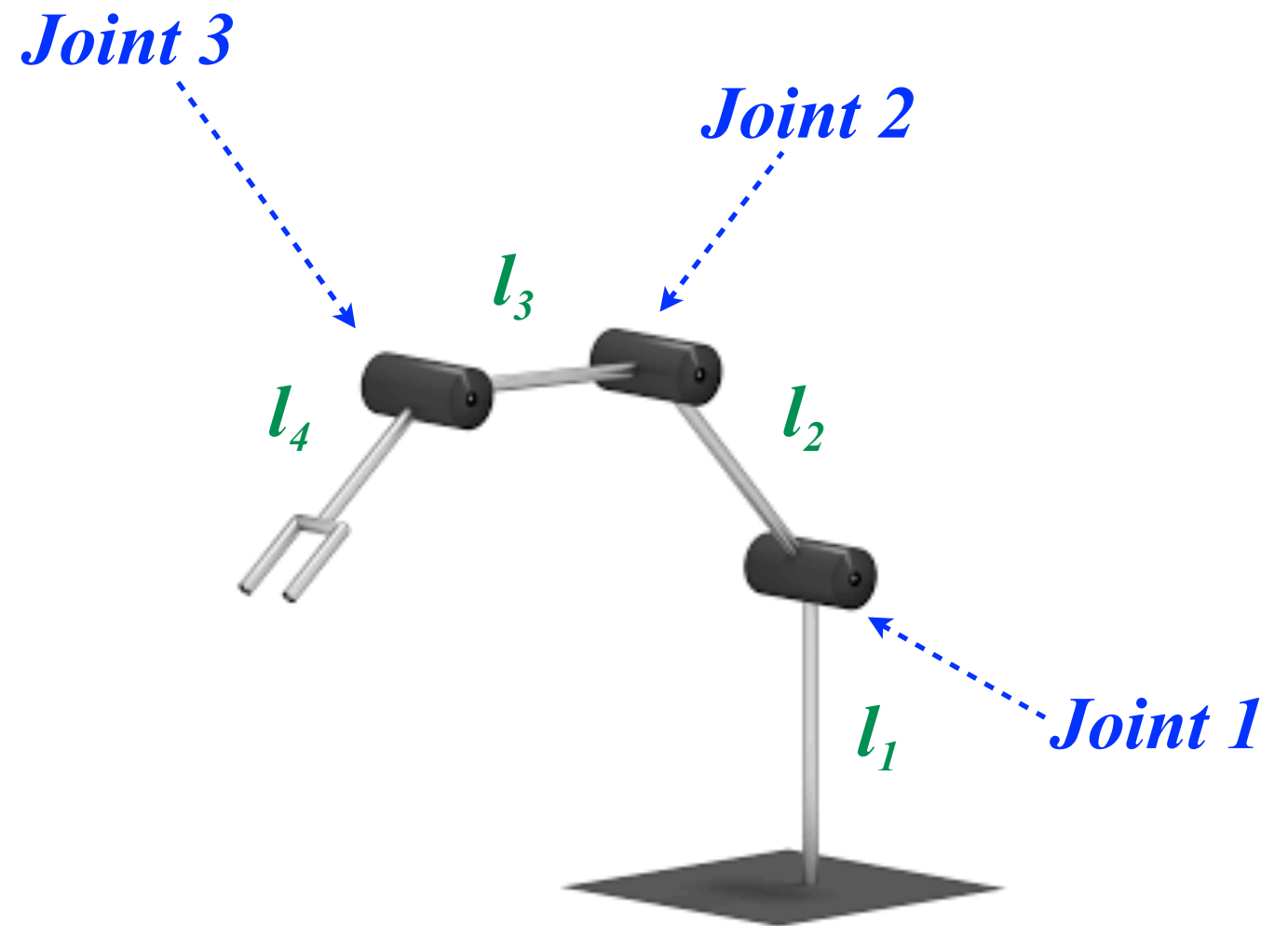
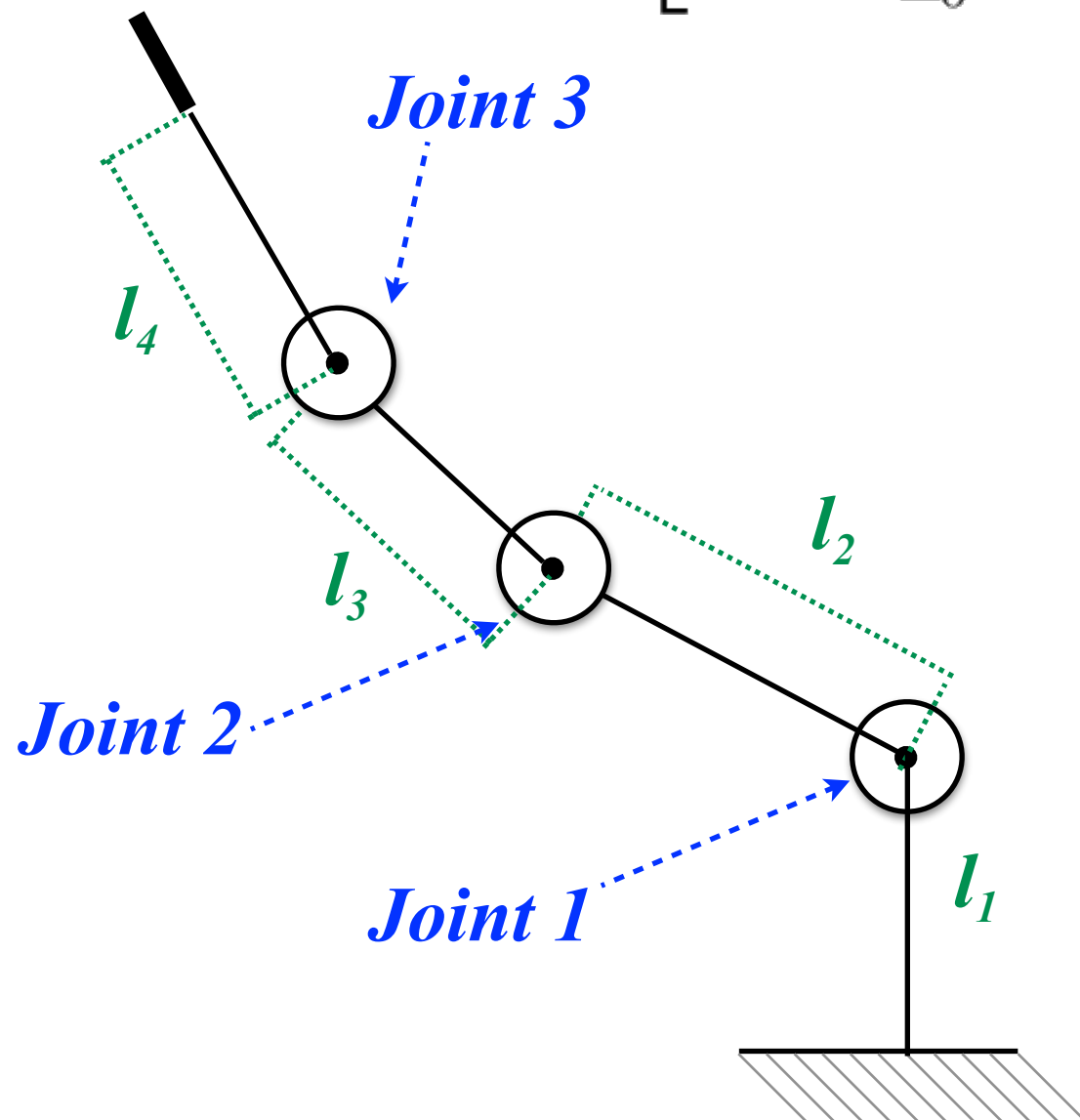
$$\underline{v}(t) \mid_{\text{joint } i} =$$

$$\begin{bmatrix} [\underline{z}_{i-1} \times (\underline{p}_t - \underline{p}_{i-1})] \\ \vdots \\ \underline{z}_{i-1} \end{bmatrix} \dot{q}_i(t)$$

Column i for joint i of revolute type

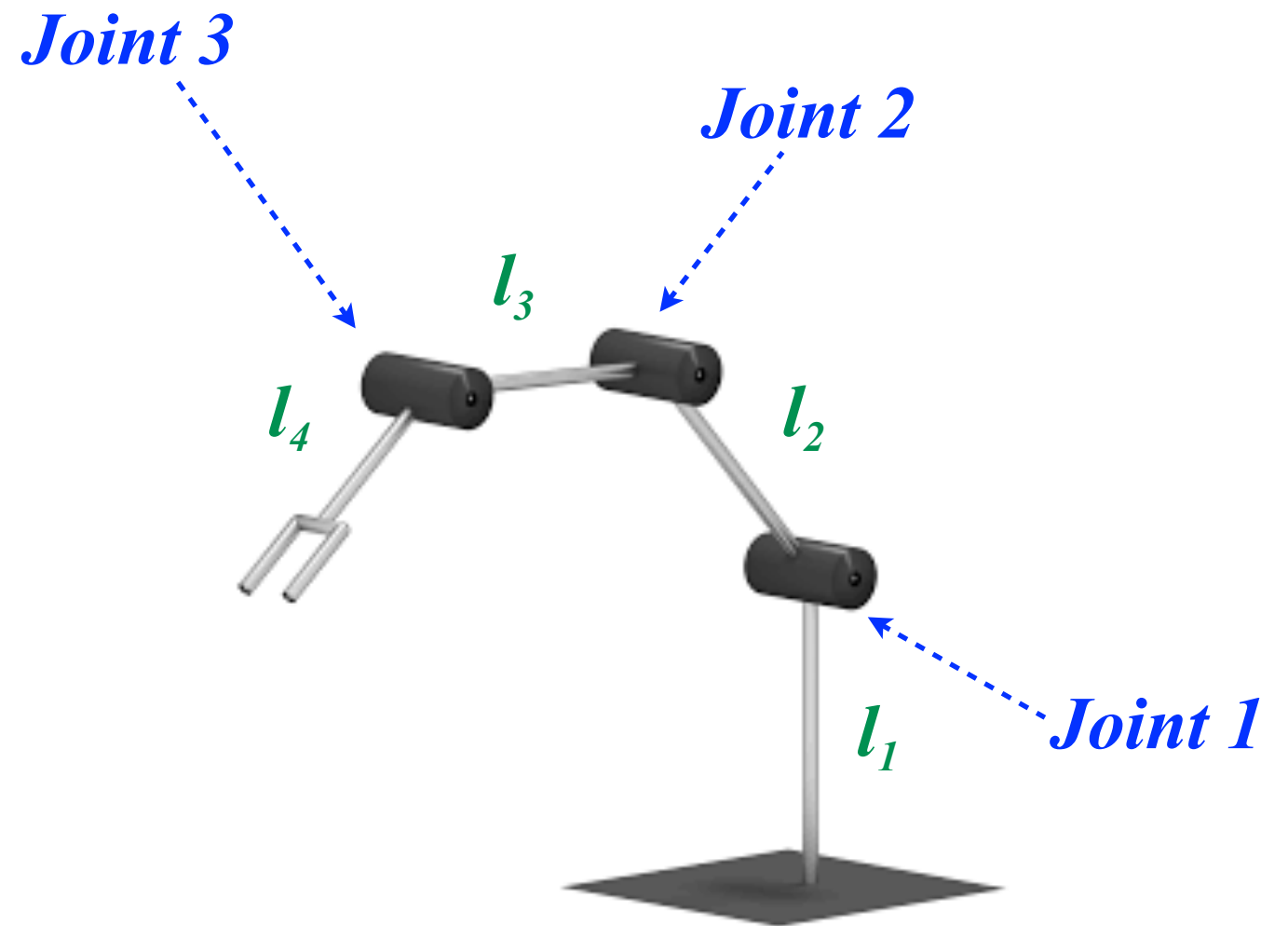
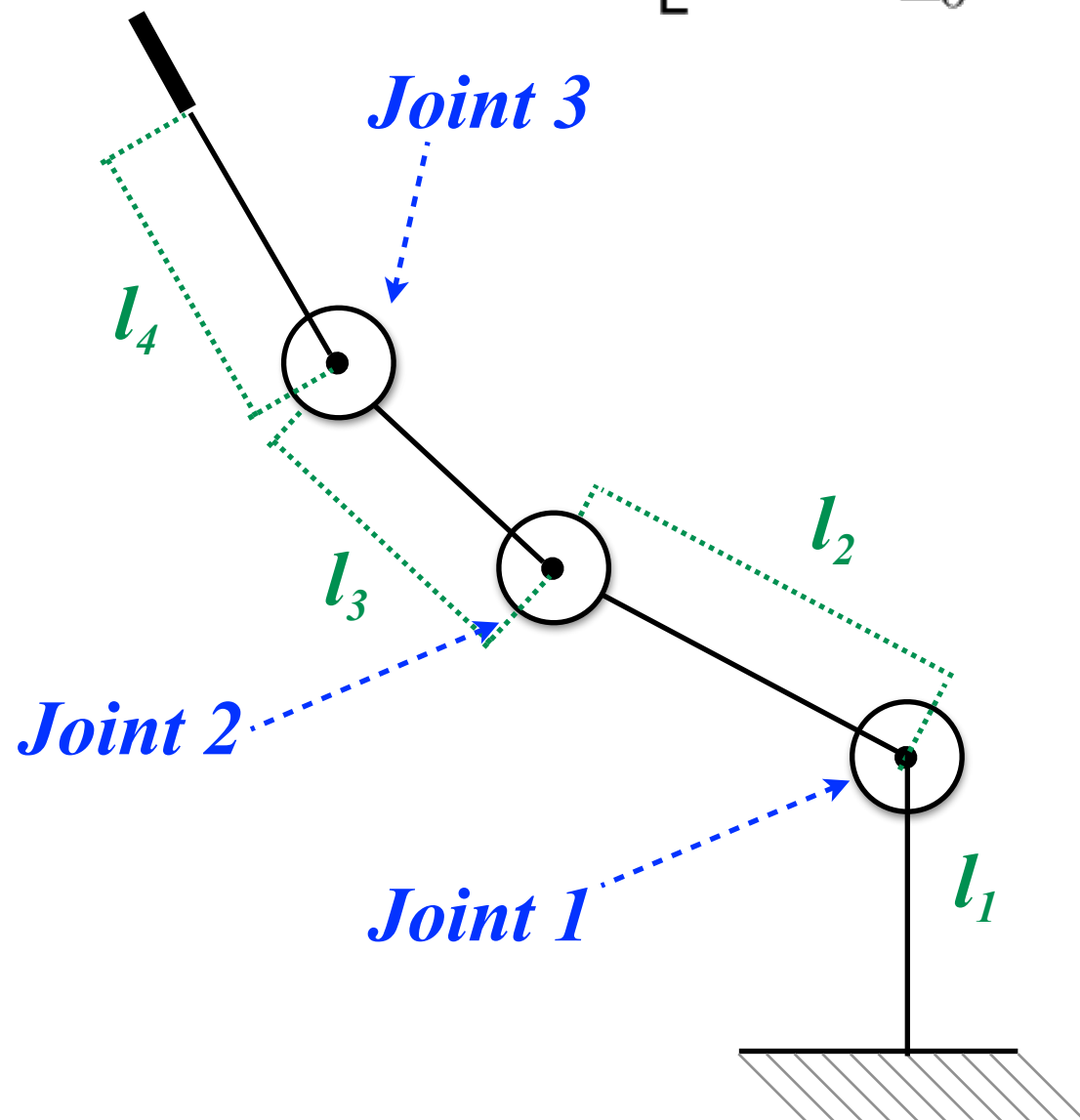
The Geometric Jacobian in Anthropomorphic Robots

$$\underline{v}(t) = \begin{bmatrix} [\underline{z}_0 \times (\underline{p}_t - \underline{p}_0)] & [\underline{z}_1 \times (\underline{p}_t - \underline{p}_1)] & [\underline{z}_2 \times (\underline{p}_t - \underline{p}_2)] \\ \hline \underline{z}_0 & \underline{z}_1 & \underline{z}_2 \end{bmatrix} \dot{\underline{q}}(t)$$



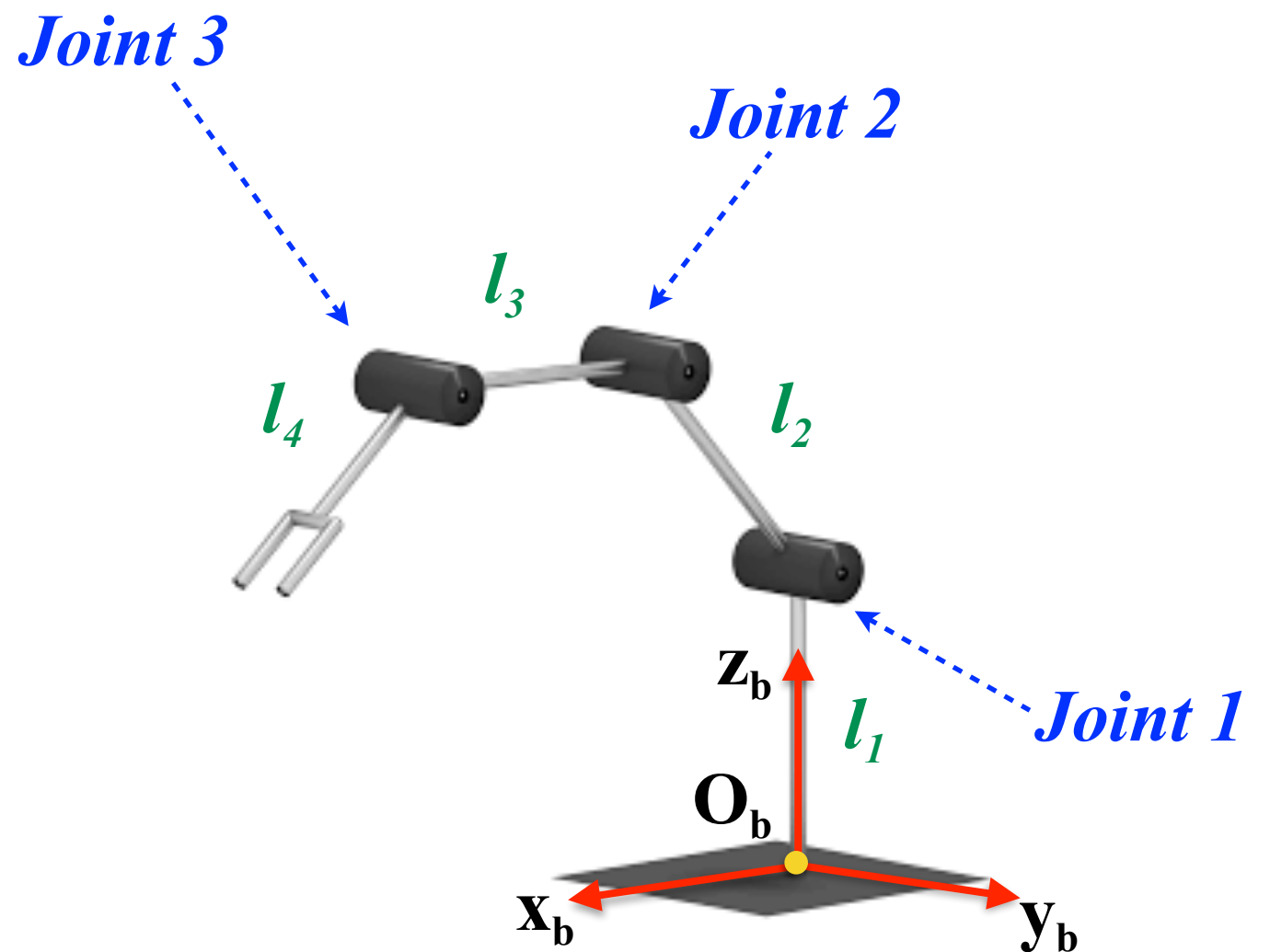
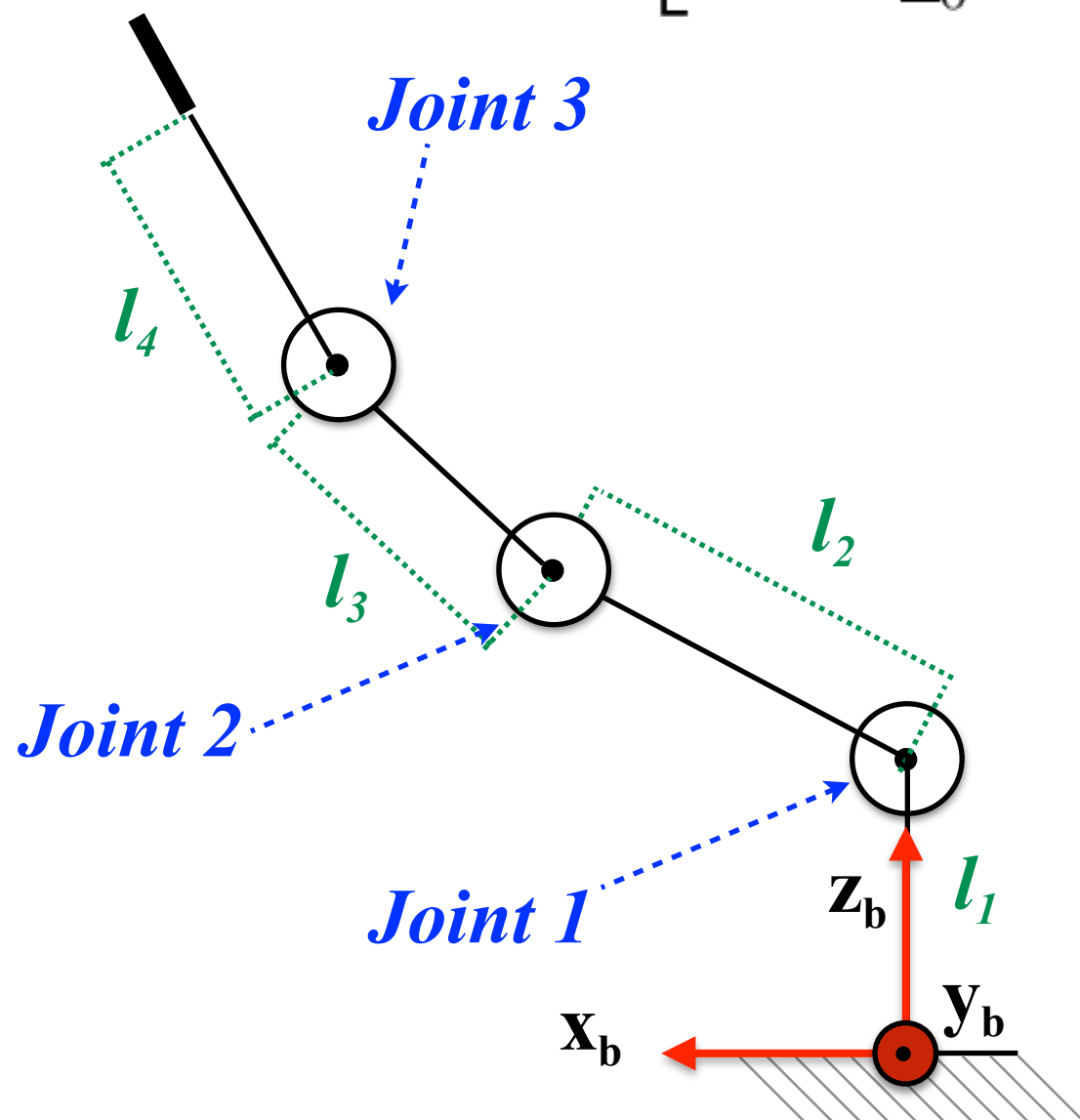
The Geometric Jacobian in Anthropomorphic Robots

$$\underline{v}(t) = \begin{bmatrix} [\underline{z}_0^b \times (\underline{p}_t^b - \underline{p}_0^b)] & [\underline{z}_1^b \times (\underline{p}_t^b - \underline{p}_1^b)] & [\underline{z}_2^b \times (\underline{p}_t^b - \underline{p}_2^b)] \\ \hline \underline{z}_0^b & \underline{z}_1^b & \underline{z}_2^b \end{bmatrix} \dot{\underline{q}}(t)$$



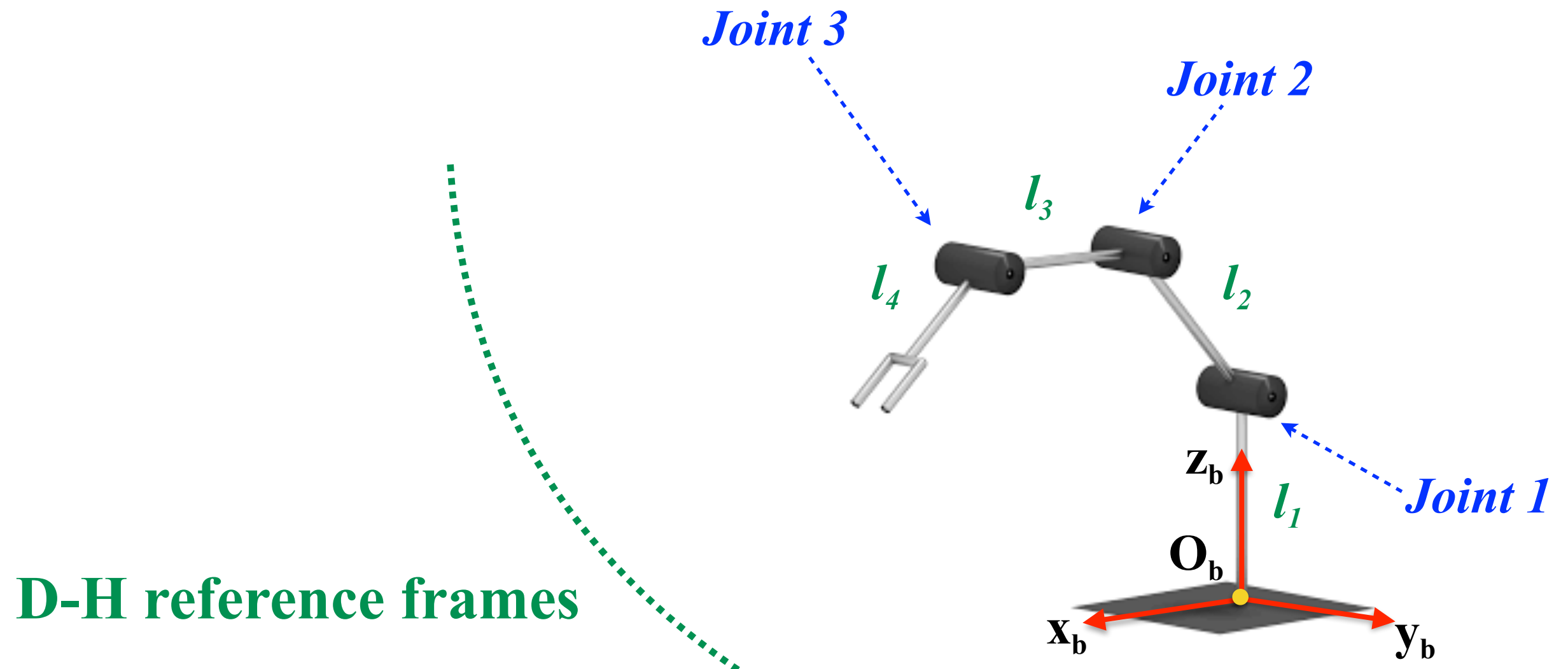
The Geometric Jacobian in Anthropomorphic Robots

$$\underline{v}(t) = \begin{bmatrix} [\underline{z}_0^b \times (\underline{p}_t^b - \underline{p}_0^b)] & [\underline{z}_1^b \times (\underline{p}_t^b - \underline{p}_1^b)] & [\underline{z}_2^b \times (\underline{p}_t^b - \underline{p}_2^b)] \\ \hline \underline{z}_0^b & \underline{z}_1^b & \underline{z}_2^b \end{bmatrix} \dot{\underline{q}}(t)$$



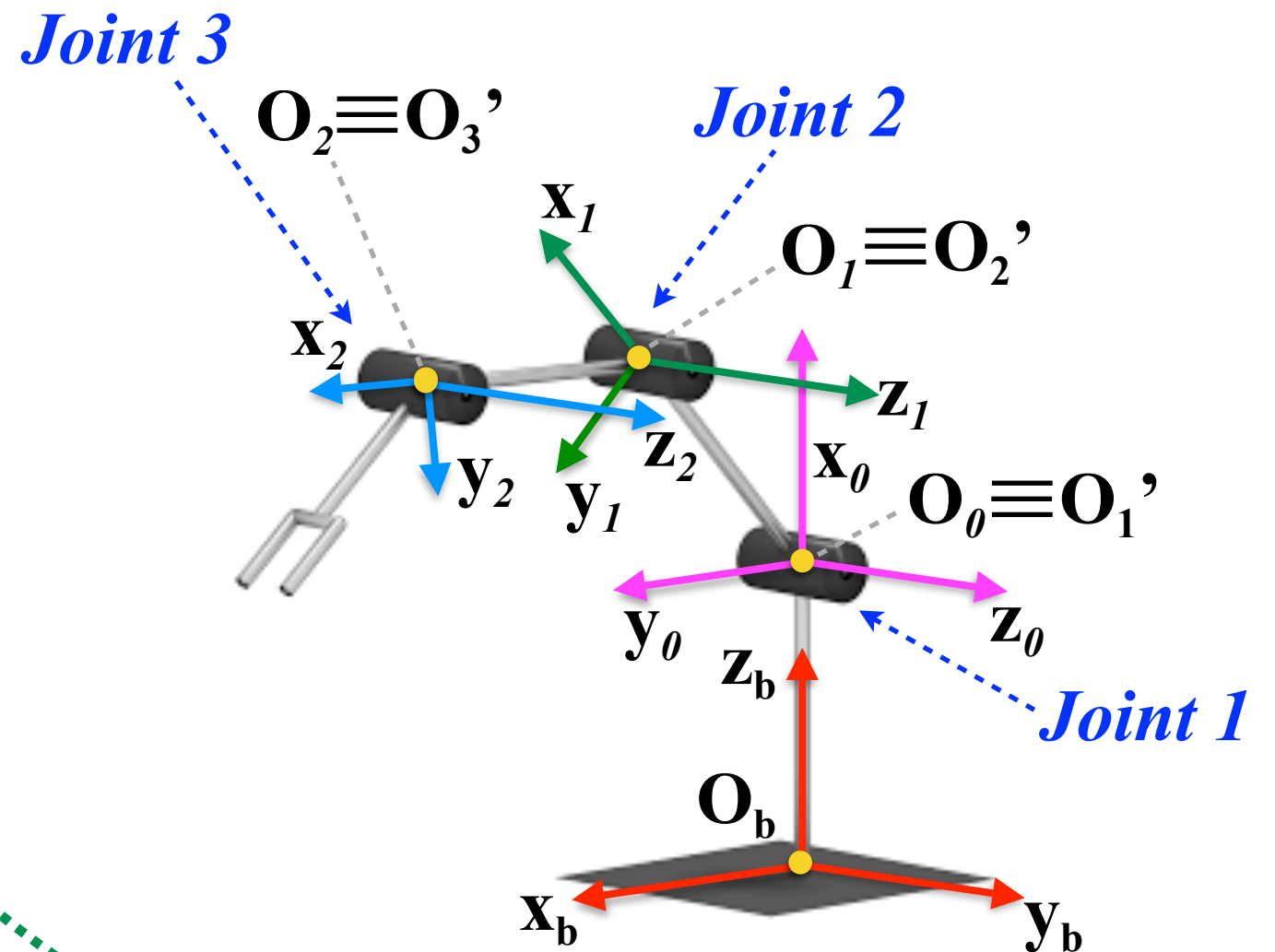
The Geometric Jacobian in Anthropomorphic Robots

$$\underline{v}(t) = \begin{bmatrix} [\underline{z}_0^b \times (\underline{p}_t^b - \underline{p}_0^b)] & [\underline{z}_1^b \times (\underline{p}_t^b - \underline{p}_1^b)] & [\underline{z}_2^b \times (\underline{p}_t^b - \underline{p}_2^b)] \\ \hline \underline{z}_0^b & \underline{z}_1^b & \underline{z}_2^b \end{bmatrix} \underline{\dot{q}}(t)$$



The Geometric Jacobian in Anthropomorphic Robots

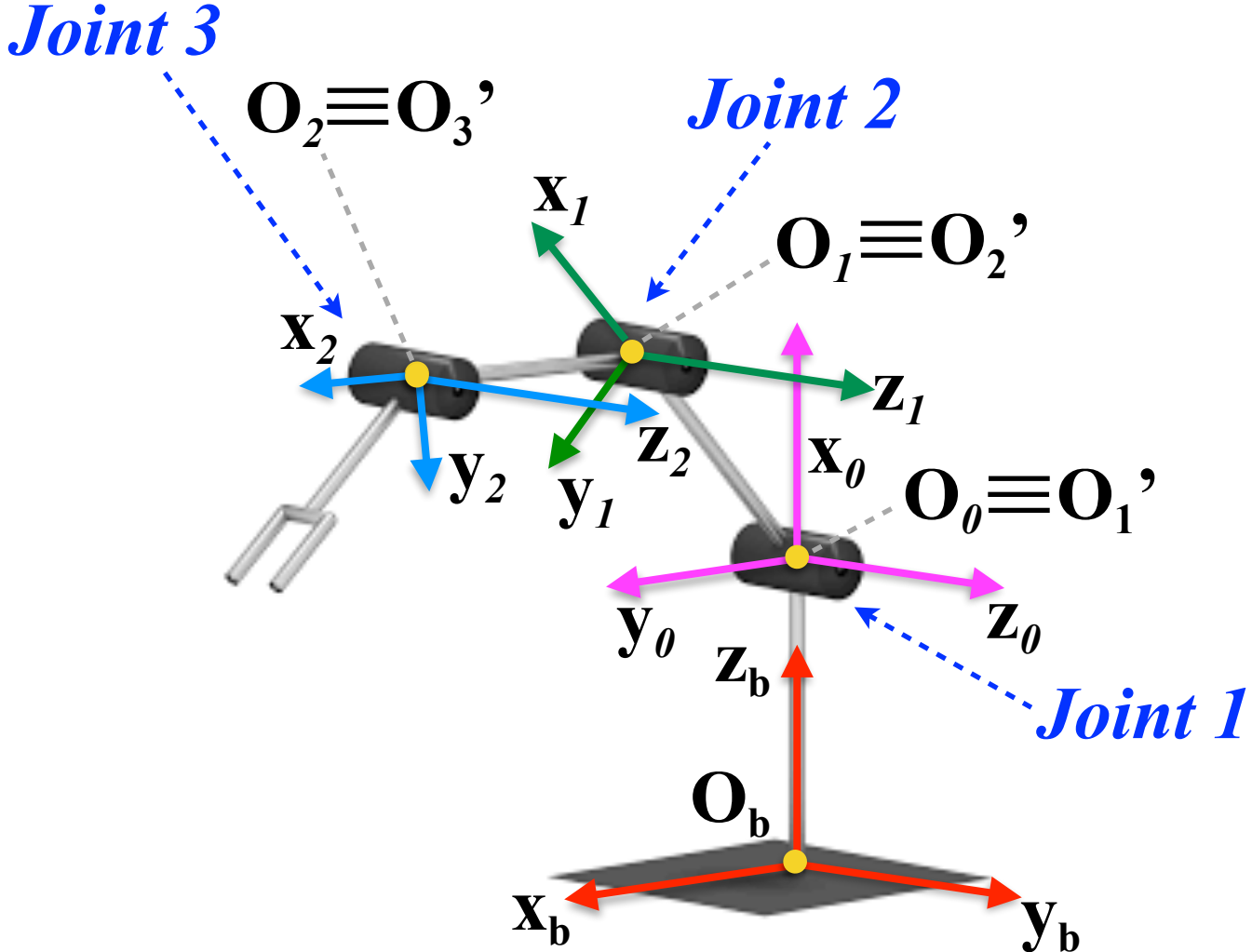
$$\underline{v}(t) = \begin{bmatrix} [\underline{z}_0^b \times (\underline{p}_t^b - \underline{p}_0^b)] & [\underline{z}_1^b \times (\underline{p}_t^b - \underline{p}_1^b)] & [\underline{z}_2^b \times (\underline{p}_t^b - \underline{p}_2^b)] \\ \hline \underline{z}_0^b & \underline{z}_1^b & \underline{z}_2^b \end{bmatrix} \dot{\underline{q}}(t)$$



D-H reference frames

The Geometric Jacobian in Anthropomorphic Robots

$$\underline{v}(t) = \begin{bmatrix} [\underline{z}_0^b \times (\underline{p}_t^b - \underline{p}_0^b)] & [\underline{z}_1^b \times (\underline{p}_t^b - \underline{p}_1^b)] & [\underline{z}_2^b \times (\underline{p}_t^b - \underline{p}_2^b)] \\ \underline{z}_0^b & \underline{z}_1^b & \underline{z}_2^b \end{bmatrix} \dot{\underline{q}}(t)$$

$$\underline{z}_0^b = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$


The diagram illustrates a 3-DOF anthropomorphic robot arm with three joints labeled Joint 1, Joint 2, and Joint 3. The base is at origin O_b with a red coordinate frame $\{x_b, y_b, z_b\}$. Joint 1 is at origin $O_0 \equiv O_1$ with a magenta frame $\{x_0, y_0, z_0\}$. Joint 2 is at origin $O_1 \equiv O_2$ with a green frame $\{x_1, y_1, z_1\}$. Joint 3 is at origin $O_2 \equiv O_3$ with a blue frame $\{x_2, y_2, z_2\}$. Dashed blue lines indicate the joint axes. The end effector is a gripper.

The Geometric Jacobian in Anthropomorphic Robots

$$\underline{v}(t) = \begin{bmatrix} [\underline{z}_0^b \times (\underline{p}_t^b - \underline{p}_0^b)] & [\underline{z}_1^b \times (\underline{p}_t^b - \underline{p}_1^b)] & [\underline{z}_2^b \times (\underline{p}_t^b - \underline{p}_2^b)] \\ \underline{z}_0^b & \underline{z}_1^b & \underline{z}_2^b \end{bmatrix} \dot{\underline{q}}(t)$$

$$\underline{z}_0^b = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

$$\underline{z}_1^b = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

Joint 3

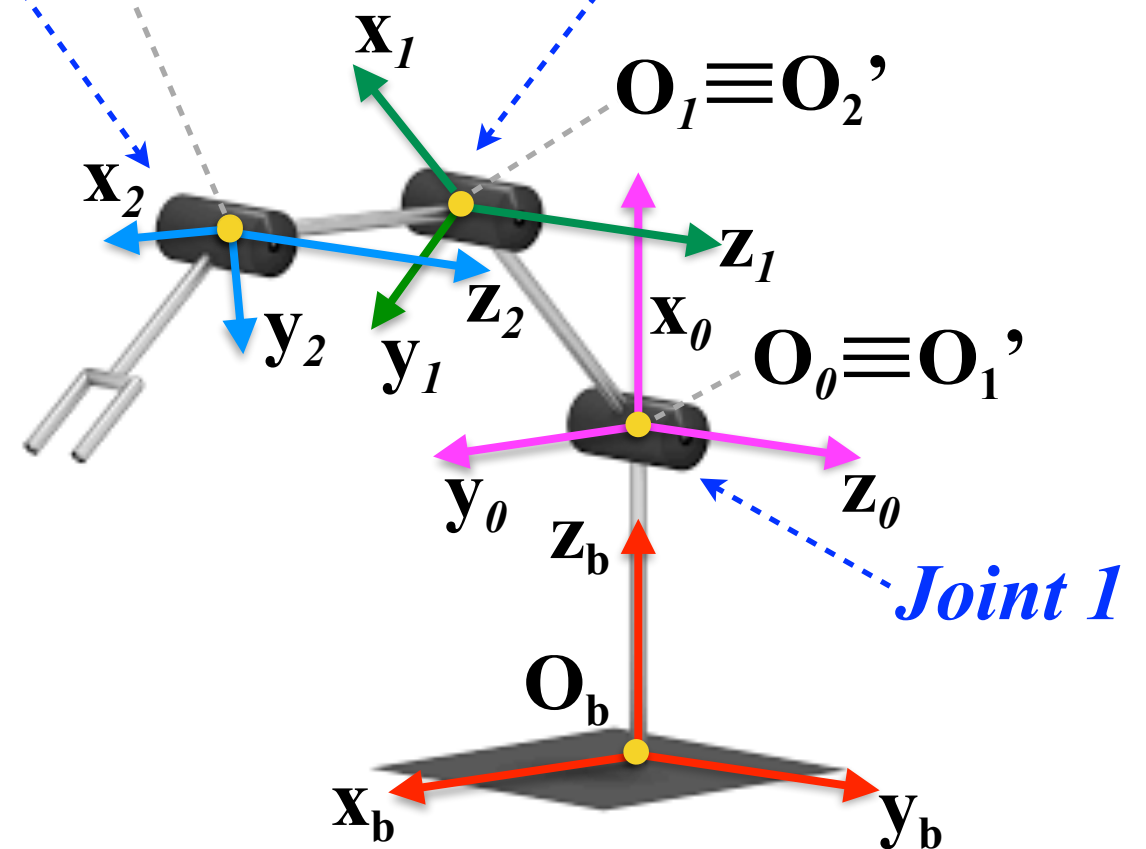
$\mathbf{O}_2 \equiv \mathbf{O}_3'$

Joint 2

$\mathbf{O}_1 \equiv \mathbf{O}_2'$

$\mathbf{O}_0 \equiv \mathbf{O}_1'$

Joint 1



The Geometric Jacobian in Anthropomorphic Robots

$$\underline{v}(t) = \begin{bmatrix} [\underline{z}_0^b \times (\underline{p}_t^b - \underline{p}_0^b)] & [\underline{z}_1^b \times (\underline{p}_t^b - \underline{p}_1^b)] & [\underline{z}_2^b \times (\underline{p}_t^b - \underline{p}_2^b)] \\ \underline{z}_0^b & \underline{z}_1^b & \underline{z}_2^b \end{bmatrix} \dot{\underline{q}}(t)$$

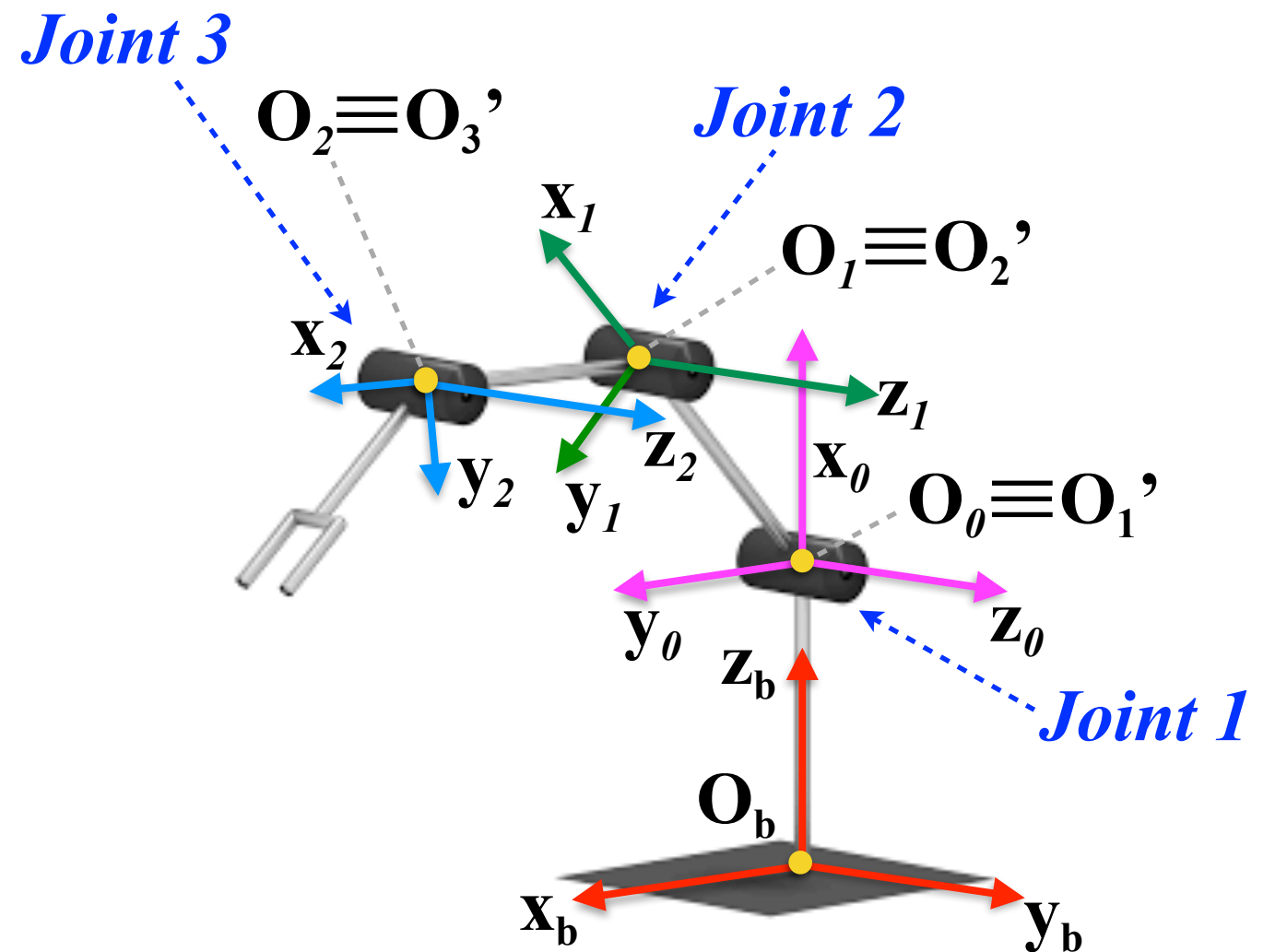
$\underline{z}_0^b = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$

$\underline{z}_1^b = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$

$\underline{z}_2^b = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$

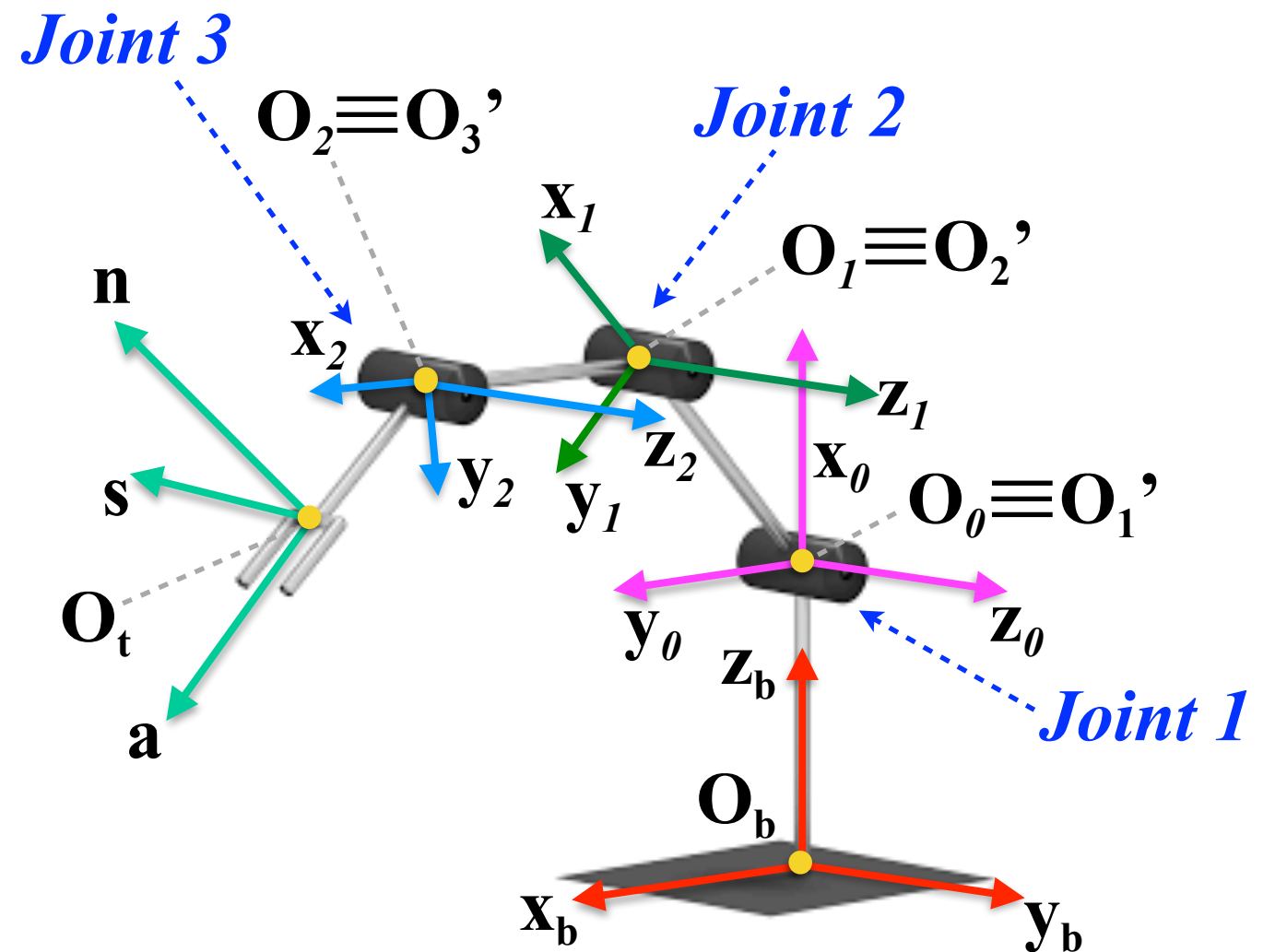
The Geometric Jacobian in Anthropomorphic Robots

$$\underline{v}(t) = \begin{bmatrix} [\underline{z}_0^b \times (\underline{p}_t^b - \underline{p}_0^b)] & [\underline{z}_1^b \times (\underline{p}_t^b - \underline{p}_1^b)] & [\underline{z}_2^b \times (\underline{p}_t^b - \underline{p}_2^b)] \\ \hline \underline{z}_0^b & \underline{z}_1^b & \underline{z}_2^b \end{bmatrix} \dot{\underline{q}}(t)$$



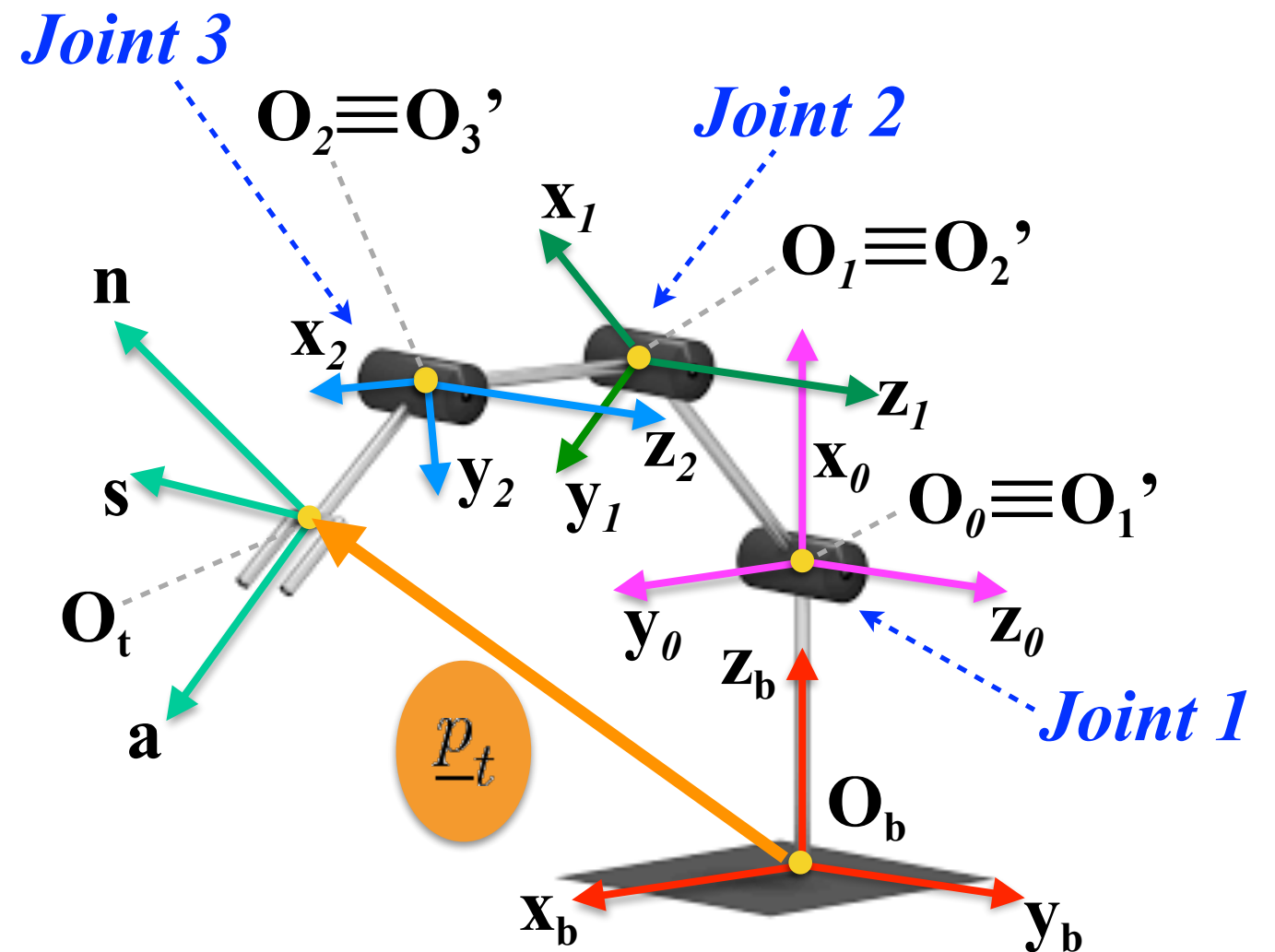
The Geometric Jacobian in Anthropomorphic Robots

$$\underline{v}(t) = \begin{bmatrix} [\underline{z}_0^b \times (\underline{p}_t^b - \underline{p}_0^b)] & [\underline{z}_1^b \times (\underline{p}_t^b - \underline{p}_1^b)] & [\underline{z}_2^b \times (\underline{p}_t^b - \underline{p}_2^b)] \\ \hline \underline{z}_0^b & \underline{z}_1^b & \underline{z}_2^b \end{bmatrix} \dot{\underline{q}}(t)$$

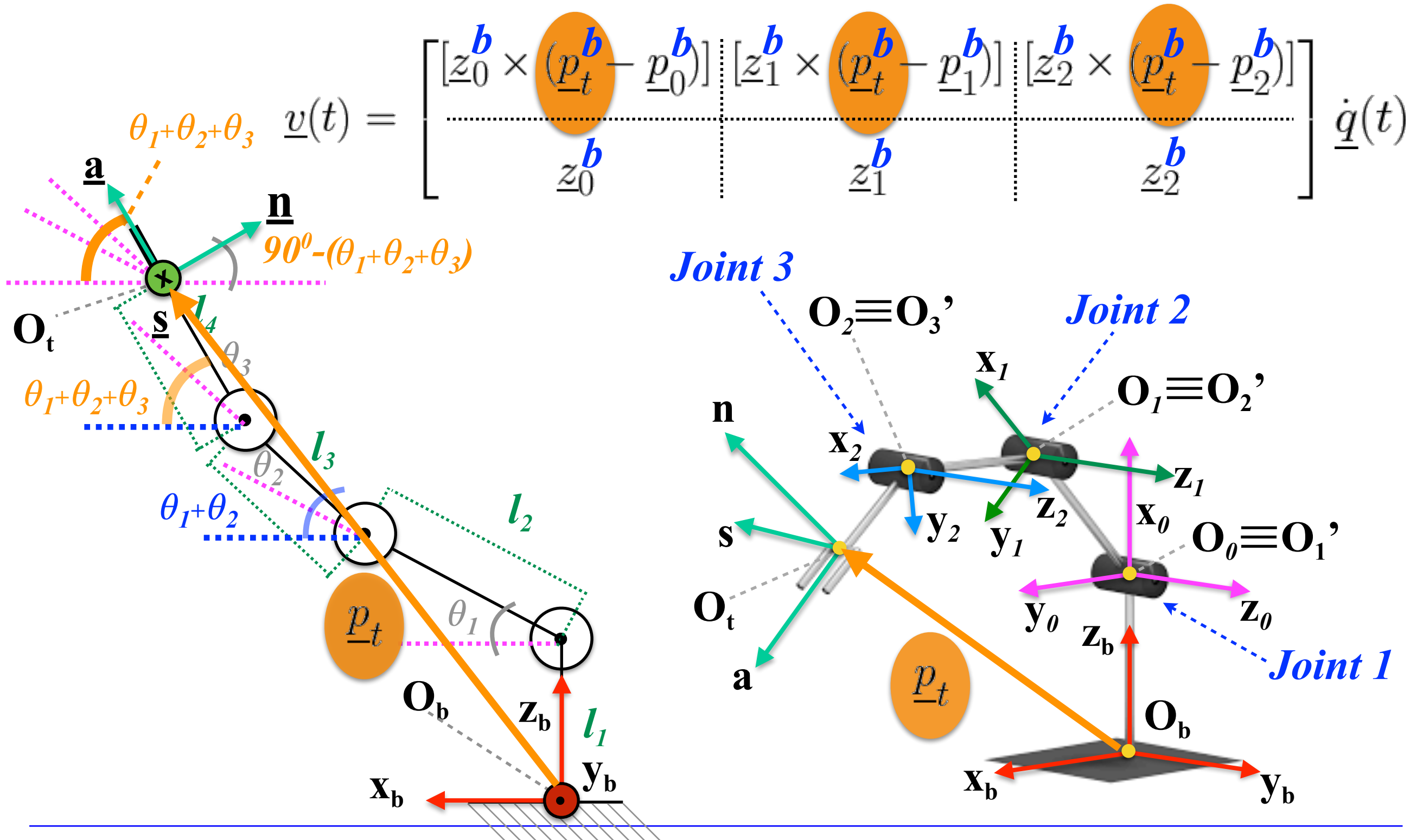


The Geometric Jacobian in Anthropomorphic Robots

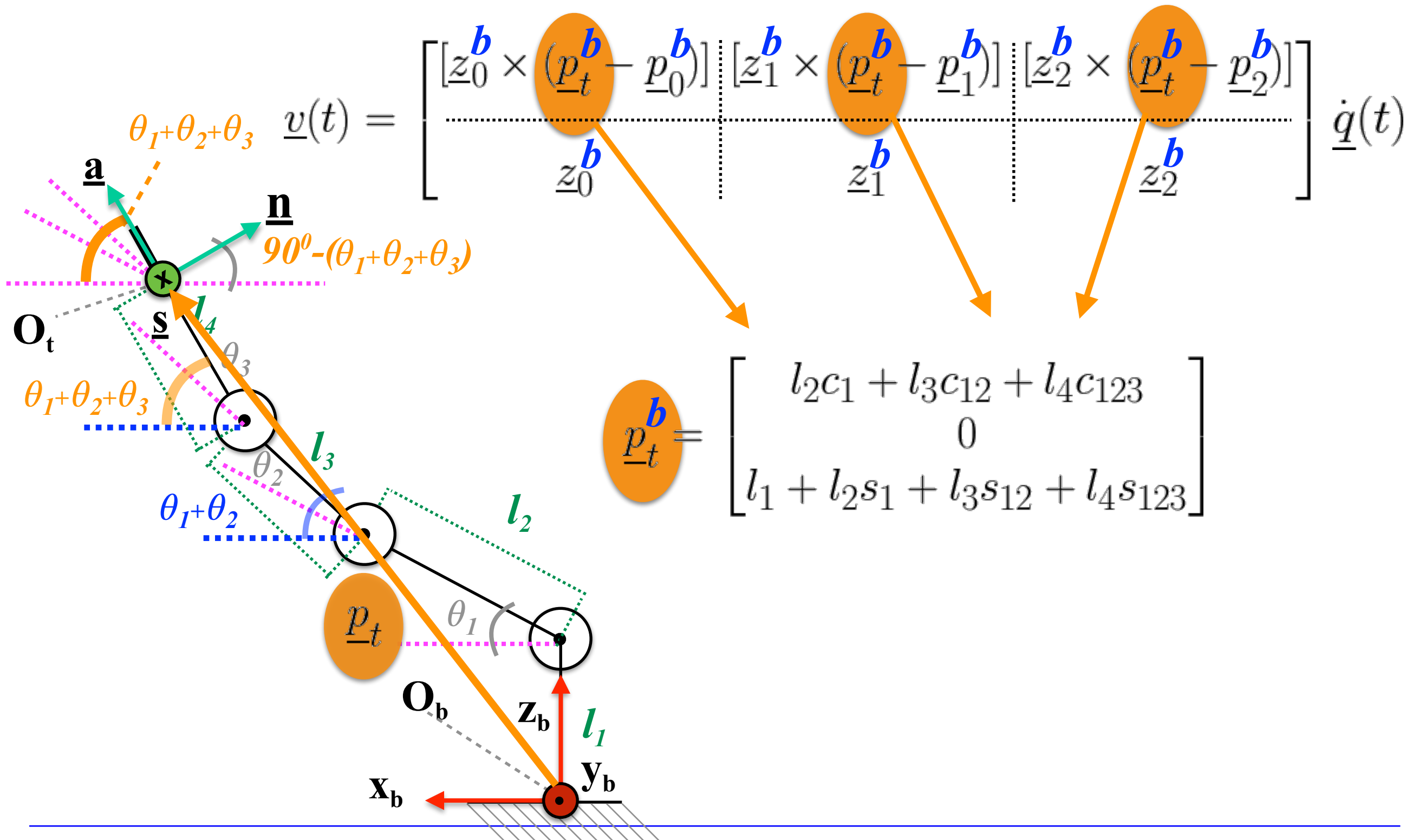
$$\underline{v}(t) = \begin{bmatrix} [\underline{z}_0^b \times (\underline{p}_t^b - \underline{p}_0^b)] & [\underline{z}_1^b \times (\underline{p}_t^b - \underline{p}_1^b)] & [\underline{z}_2^b \times (\underline{p}_t^b - \underline{p}_2^b)] \\ \hline \underline{z}_0^b & \underline{z}_1^b & \underline{z}_2^b \end{bmatrix} \dot{\underline{q}}(t)$$



The Geometric Jacobian in Anthropomorphic Robots



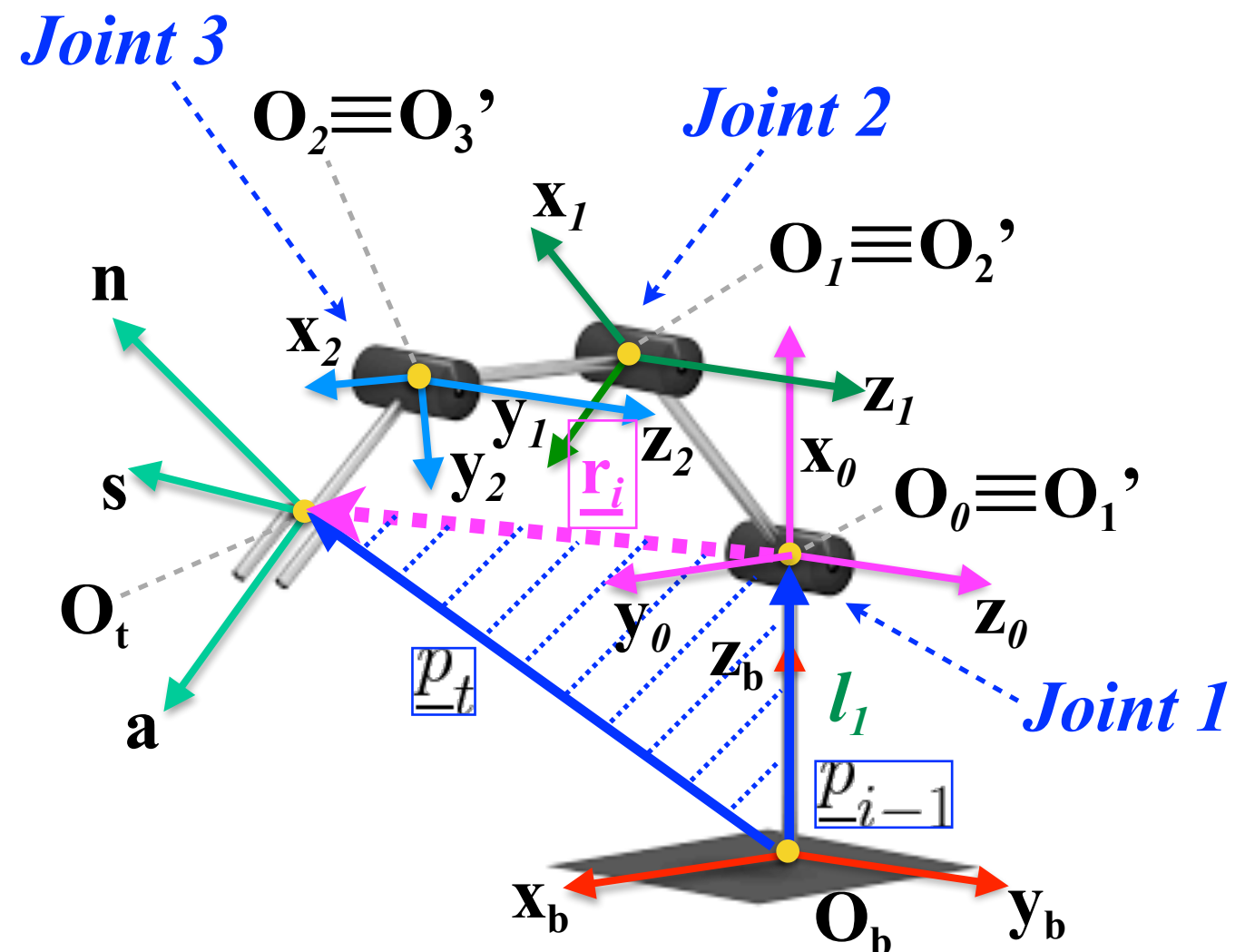
The Geometric Jacobian in Anthropomorphic Robots



The Geometric Jacobian in Anthropomorphic Robots

$$\underline{v}(t) = \begin{bmatrix} [\underline{z}_0^b \times (\underline{p}_t^b - \underline{p}_0^b)] & [\underline{z}_1^b \times (\underline{p}_t^b - \underline{p}_1^b)] & [\underline{z}_2^b \times (\underline{p}_t^b - \underline{p}_2^b)] \\ \hline \underline{z}_0^b & \underline{z}_1^b & \underline{z}_2^b \end{bmatrix} \dot{\underline{q}}(t)$$

Joint 1:

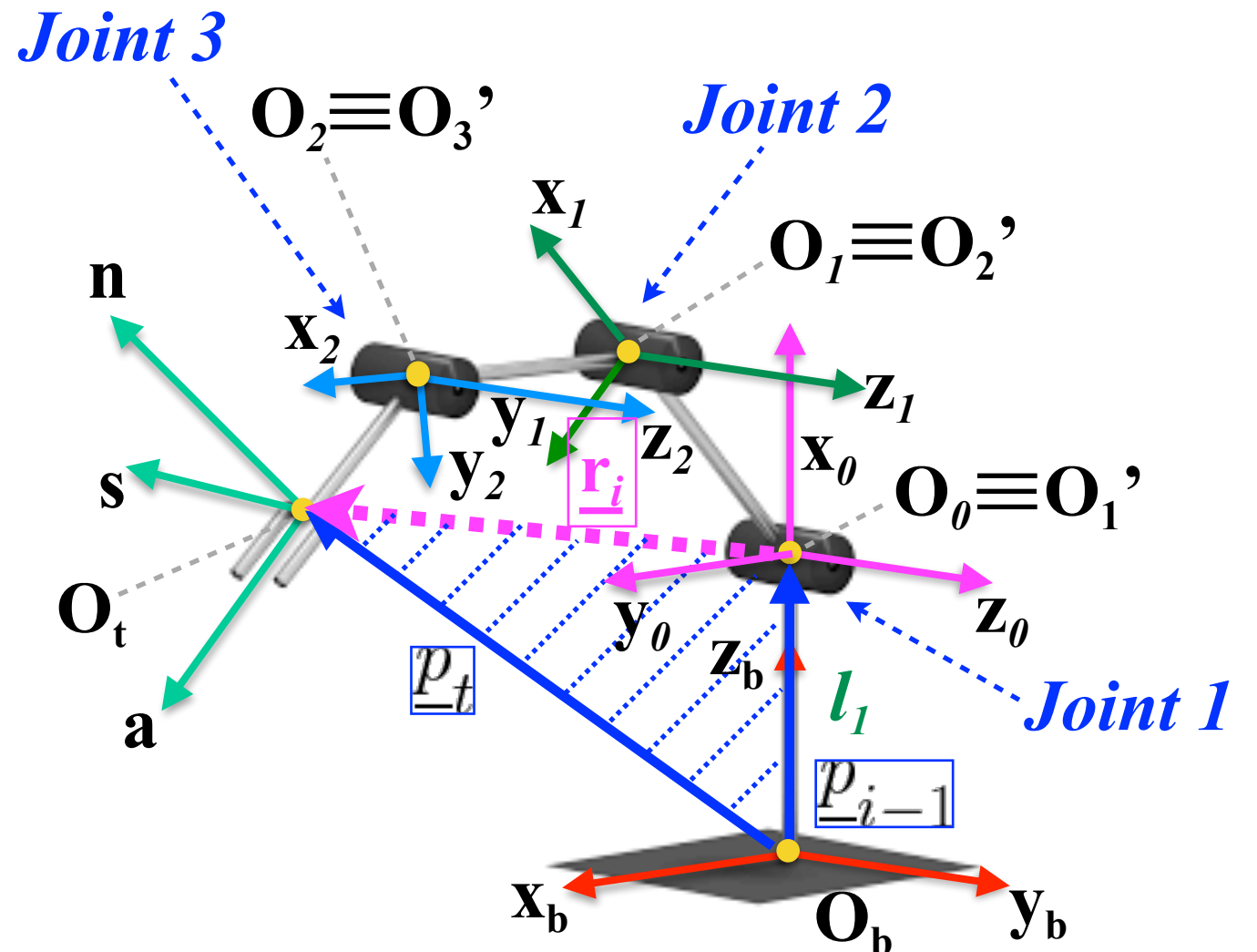


The Geometric Jacobian in Anthropomorphic Robots

$$\underline{v}(t) = \begin{bmatrix} [\underline{z}_0^b \times (\underline{p}_t^b - \underline{p}_0^b)] & [\underline{z}_1^b \times (\underline{p}_t^b - \underline{p}_1^b)] & [\underline{z}_2^b \times (\underline{p}_t^b - \underline{p}_2^b)] \\ \hline \underline{z}_0^b & \underline{z}_1^b & \underline{z}_2^b \end{bmatrix} \dot{\underline{q}}(t)$$

Joint 1:

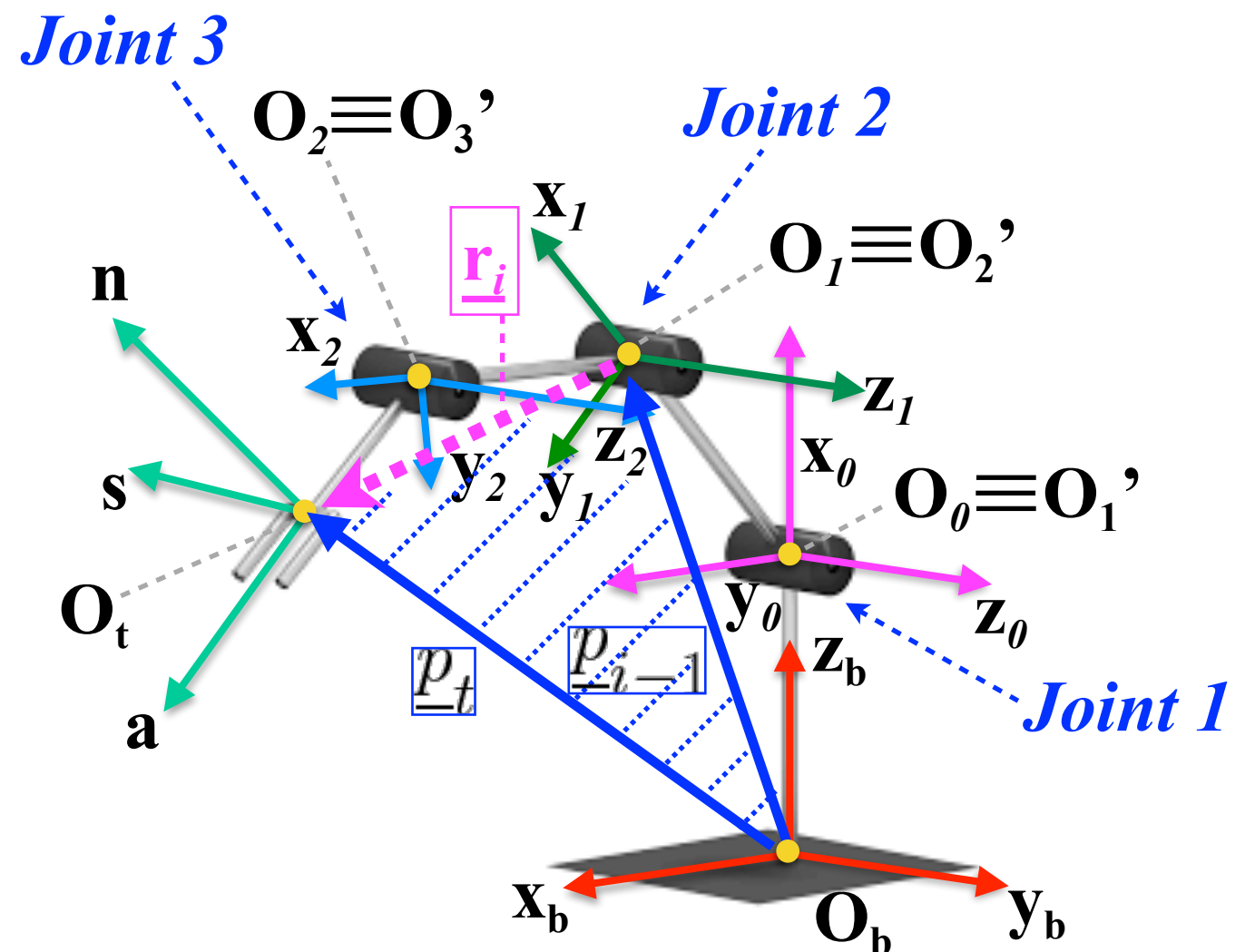
$$\underline{p}_{i-1} = \underline{p}_0^b = \underline{o}_0^b = \begin{bmatrix} 0 \\ 0 \\ l_1 \end{bmatrix}$$



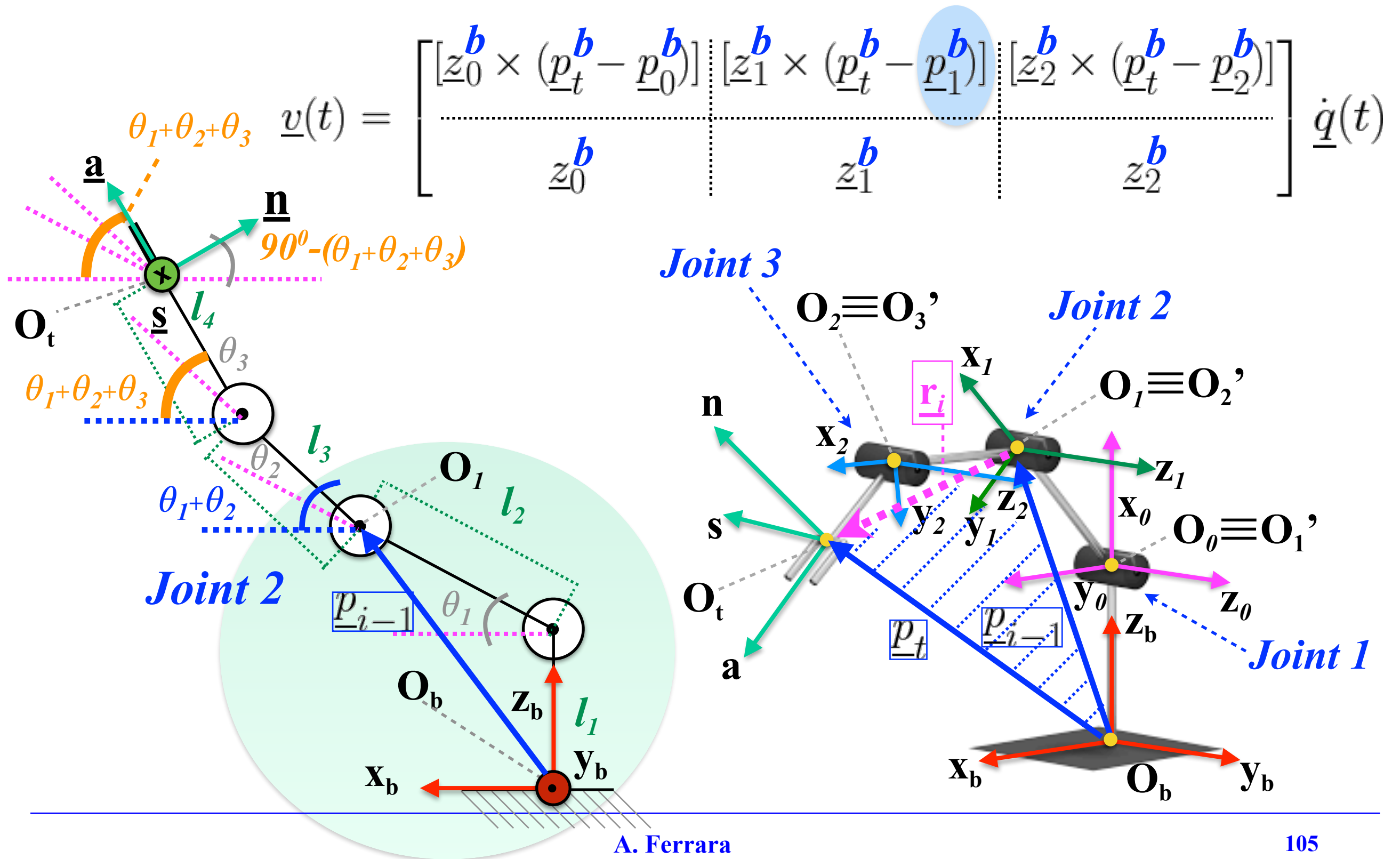
The Geometric Jacobian in Anthropomorphic Robots

$$\underline{v}(t) = \begin{bmatrix} [\underline{z}_0^b \times (\underline{p}_t^b - \underline{p}_0^b)] & [\underline{z}_1^b \times (\underline{p}_t^b - \underline{p}_1^b)] & [\underline{z}_2^b \times (\underline{p}_t^b - \underline{p}_2^b)] \\ \hline \underline{z}_0^b & \underline{z}_1^b & \underline{z}_2^b \end{bmatrix} \dot{\underline{q}}(t)$$

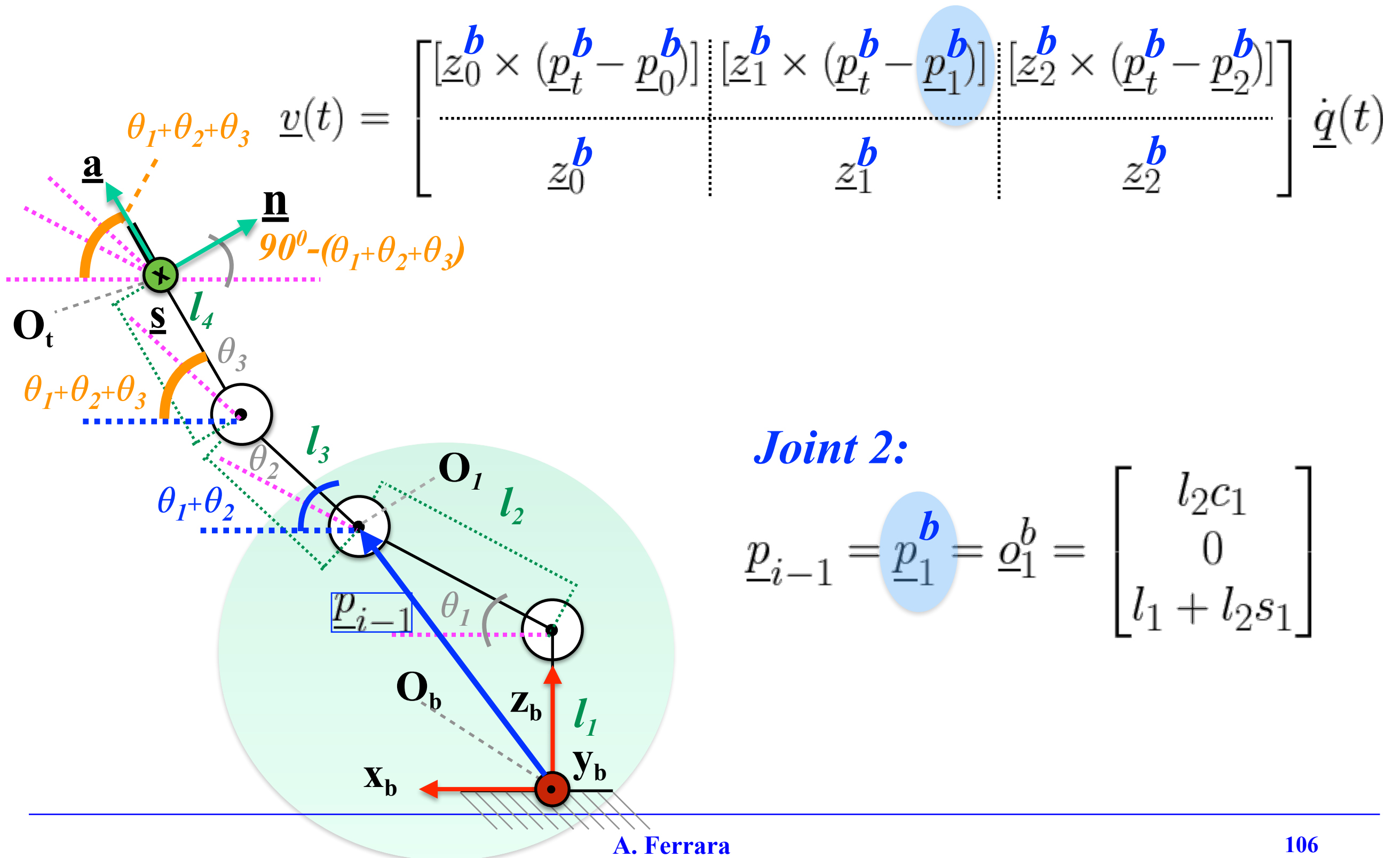
Joint 2:



The Geometric Jacobian in Anthropomorphic Robots



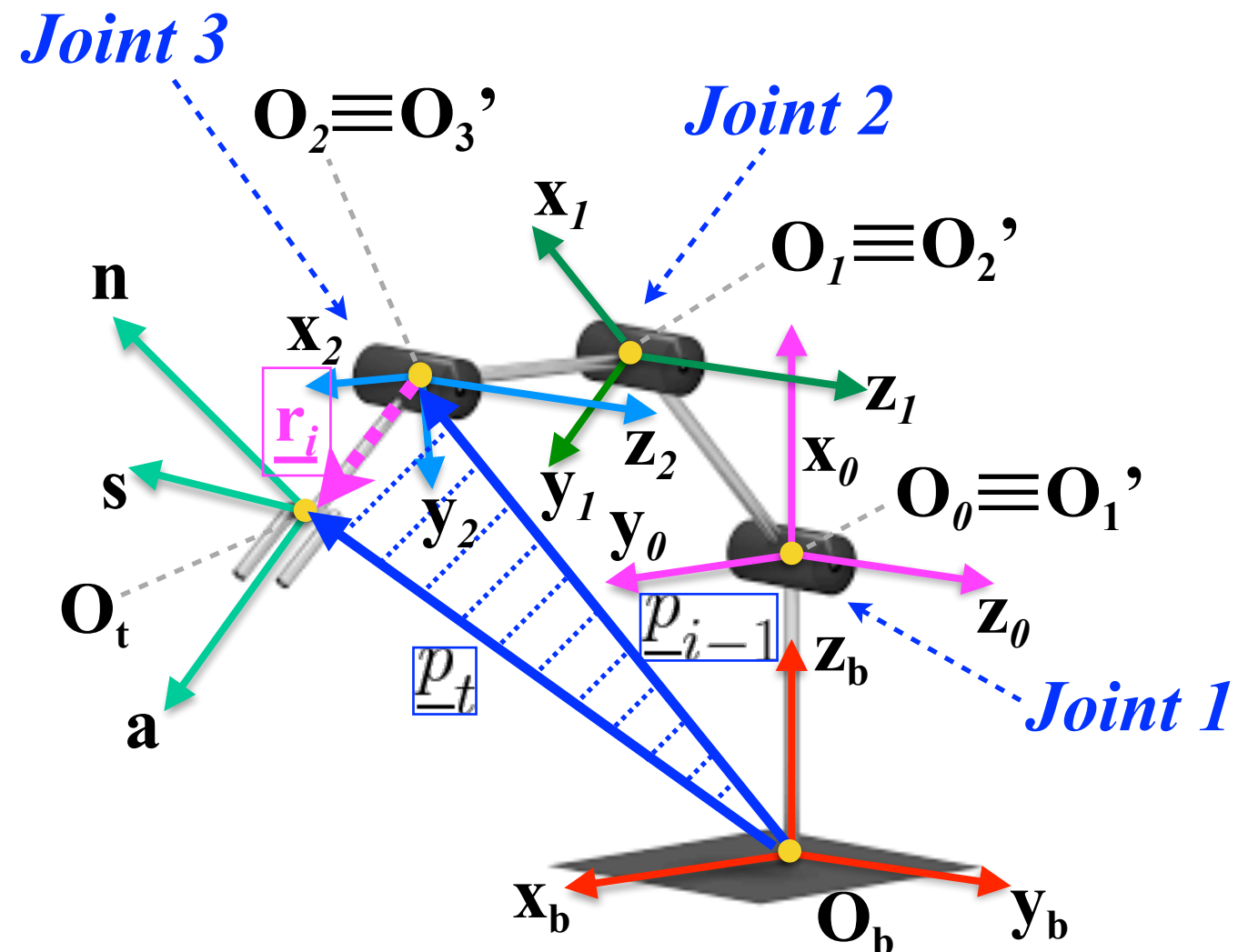
The Geometric Jacobian in Anthropomorphic Robots



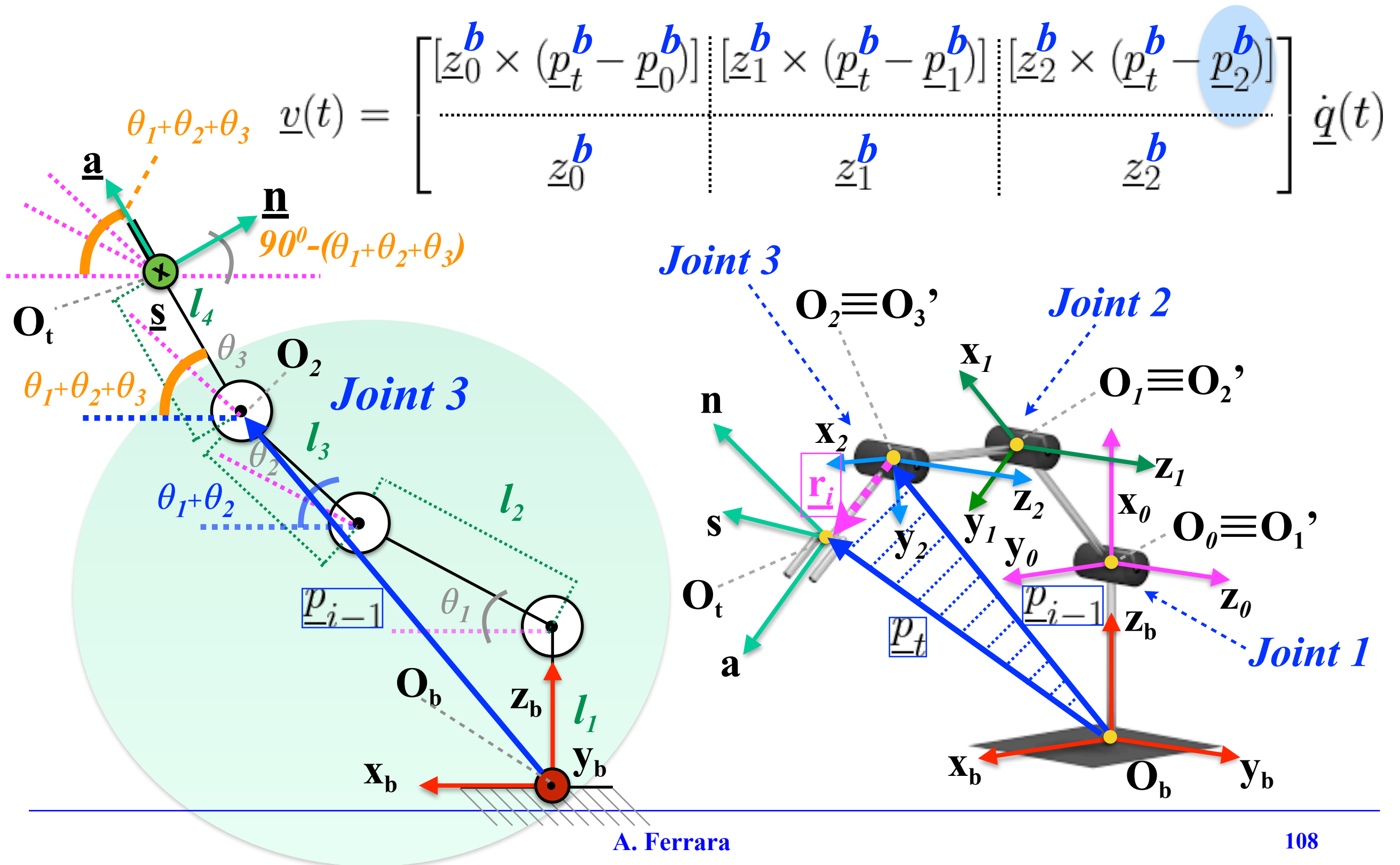
The Geometric Jacobian in Anthropomorphic Robots

$$\underline{v}(t) = \begin{bmatrix} [\underline{z}_0^b \times (\underline{p}_t^b - \underline{p}_0^b)] & [\underline{z}_1^b \times (\underline{p}_t^b - \underline{p}_1^b)] & [\underline{z}_2^b \times (\underline{p}_t^b - \underline{p}_2^b)] \\ \hline \underline{z}_0^b & \underline{z}_1^b & \underline{z}_2^b \end{bmatrix} \dot{\underline{q}}(t)$$

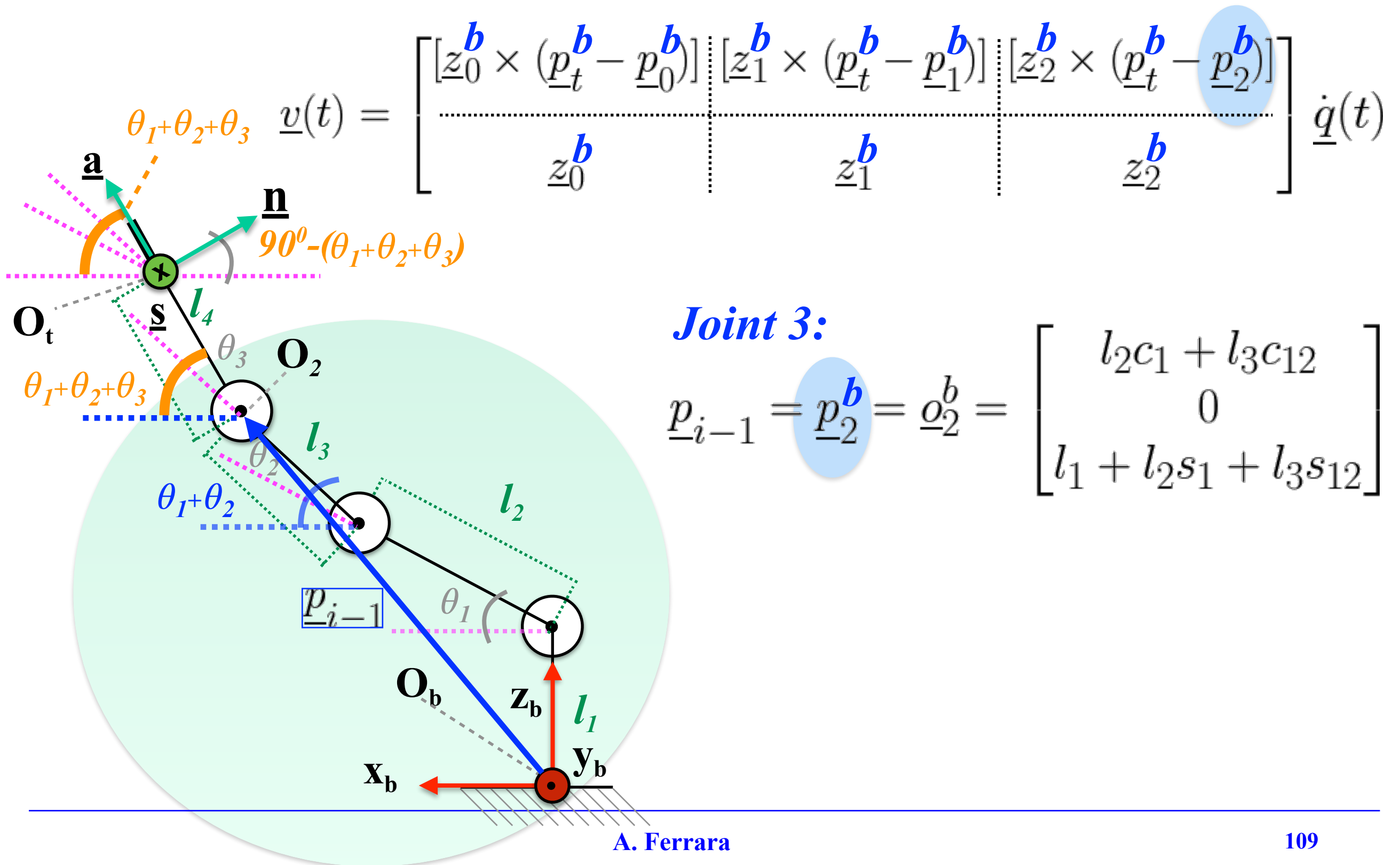
Joint 3:



The Geometric Jacobian in Anthropomorphic Robots



The Geometric Jacobian in Anthropomorphic Robots



The Geometric Jacobian in Anthropomorphic Robots

$$\underline{v}(t) = \begin{bmatrix} [\underline{z}_0^b \times (\underline{p}_t^b - \underline{p}_0^b)] & [\underline{z}_1^b \times (\underline{p}_t^b - \underline{p}_1^b)] & [\underline{z}_2^b \times (\underline{p}_t^b - \underline{p}_2^b)] \\ \underline{z}_0^b & \underline{z}_1^b & \underline{z}_2^b \end{bmatrix} \dot{\underline{q}}(t)$$

$$\underline{z}_0^b = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \quad \underline{p}_t^b = \begin{bmatrix} l_2 c_1 + l_3 c_{12} + l_4 c_{123} \\ 0 \\ l_1 + l_2 s_1 + l_3 s_{12} + l_4 s_{123} \end{bmatrix} \quad \underline{p}_{i-1} = \underline{p}_0^b = \underline{o}_0^b = \begin{bmatrix} 0 \\ 0 \\ l_1 \end{bmatrix}$$

$$\underline{z}_1^b = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \quad \underline{p}_{i-1} = \underline{p}_1^b = \underline{o}_1^b = \begin{bmatrix} l_2 c_1 \\ 0 \\ l_1 + l_2 s_1 \end{bmatrix}$$

$$\underline{z}_2^b = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \quad \underline{p}_{i-1} = \underline{p}_2^b = \underline{o}_2^b = \begin{bmatrix} l_2 c_1 + l_3 c_{12} \\ 0 \\ l_1 + l_2 s_1 + l_3 s_{12} \end{bmatrix}$$



UNIVERSITÀ DI PAVIA
Dipartimento di
Ingegneria Industriale
e dell'Informazione

Course of Robot Control

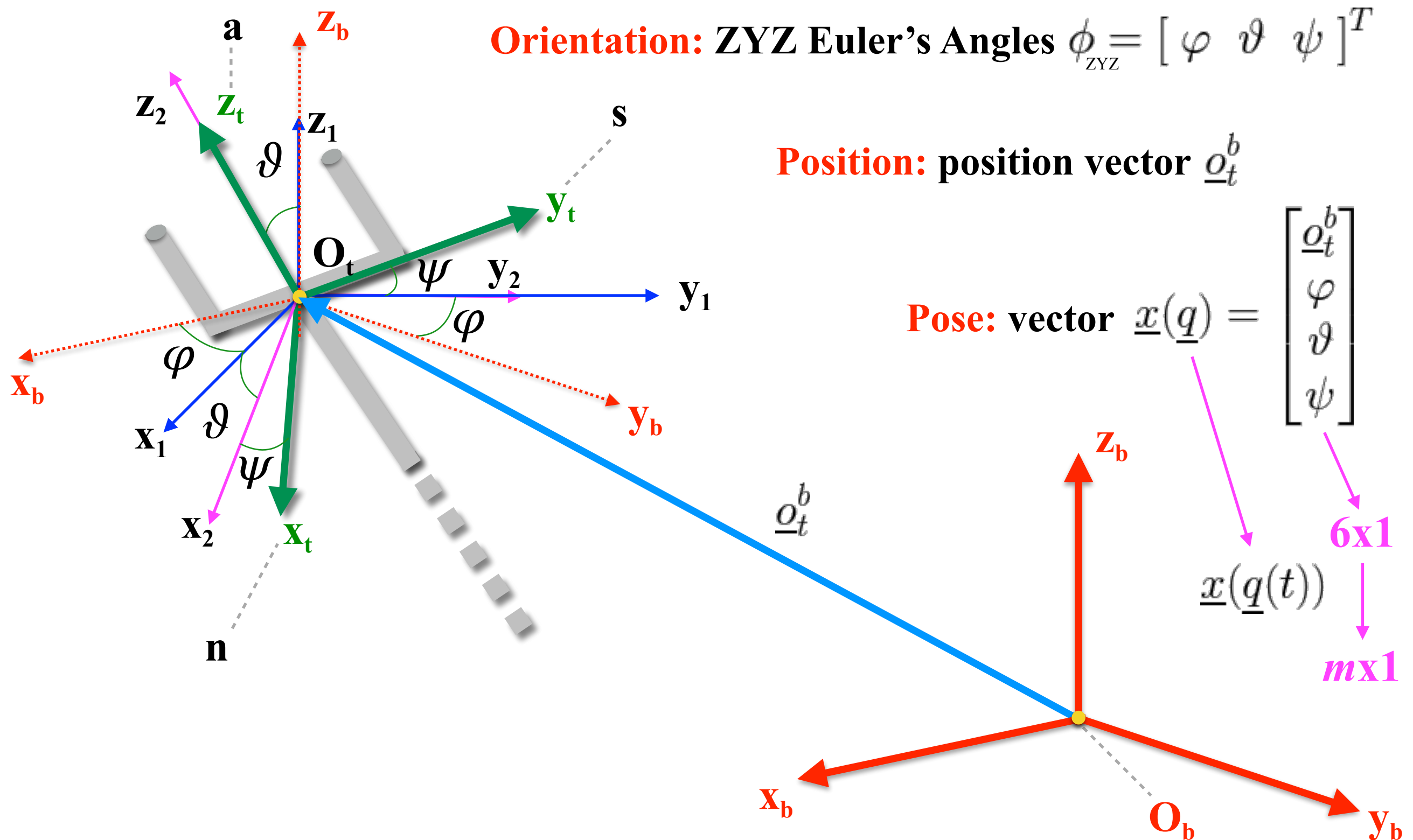
Prof. Antonella Ferrara



Robot Differential Kinematics: Analytic Approach

In a previous lecture...

Euler Angles and a New Pose Description



A Useful Definition: Analytic Jacobian Matrix

Pose vector $\underline{x}(\underline{q}) = \underline{x}(\underline{q}(t))$  **Direct Kinematics Function**



by differentiating w.r.t. time

$$\underline{x}(\underline{q}(t)) = \begin{bmatrix} \underline{o}_t^b(\underline{q}(t)) \\ \phi(\underline{q}(t)) \end{bmatrix} \quad \xrightarrow{\text{green arrow}} \quad \dot{\underline{x}}(\underline{q}(t)) = \begin{bmatrix} \frac{\partial \underline{o}_t^b(\underline{q}(t))}{\partial \underline{q}(t)} \\ \frac{\partial \phi(\underline{q}(t))}{\partial \underline{q}(t)} \end{bmatrix} \dot{\underline{q}}(t)$$

A Useful Definition: Analytic Jacobian Matrix

Pose vector $\underline{x}(\underline{q}) = \underline{x}(\underline{q}(t))$ $\longrightarrow$ **Direct Kinematics Function**

by differentiating w.r.t. time

$$\underline{x}(\underline{q}(t)) = \begin{bmatrix} \underline{o}_t^b(\underline{q}(t)) \\ \phi(\underline{q}(t)) \end{bmatrix} \xrightarrow{\quad} \dot{\underline{x}}(\underline{q}(t)) = \begin{bmatrix} \frac{\partial \underline{o}_t^b(\underline{q}(t))}{\partial \underline{q}(t)} \\ \frac{\partial \phi(\underline{q}(t))}{\partial \underline{q}(t)} \end{bmatrix} \underline{\dot{q}}(t)$$

$\dot{\underline{o}}_t^b(\underline{q})$

A Useful Definition: Analytic Jacobian Matrix

Pose vector $\underline{x}(\underline{q}) = \underline{x}(\underline{q}(t))$ $\longrightarrow$ **Direct Kinematics Function**



by differentiating w.r.t. time

$$\underline{x}(\underline{q}(t)) = \begin{bmatrix} \underline{o}_t^b(\underline{q}(t)) \\ \phi(\underline{q}(t)) \end{bmatrix} \longrightarrow \dot{\underline{x}}(\underline{q}(t)) = \begin{bmatrix} \frac{\partial \underline{o}_t^b(\underline{q}(t))}{\partial \underline{q}(t)} \\ \frac{\partial \phi(\underline{q}(t))}{\partial \underline{q}(t)} \end{bmatrix} \dot{\underline{q}}(t)$$

A blue oval highlights the Jacobian matrix $\begin{bmatrix} \frac{\partial \underline{o}_t^b(\underline{q}(t))}{\partial \underline{q}(t)} \\ \frac{\partial \phi(\underline{q}(t))}{\partial \underline{q}(t)} \end{bmatrix}$, with an arrow pointing to $\dot{\phi}(\underline{q})$.

A Useful Definition: Analytic Jacobian Matrix

Pose vector $\underline{x}(\underline{q}) = \underline{x}(\underline{q}(t))$ $\longrightarrow$ **Direct Kinematics Function**

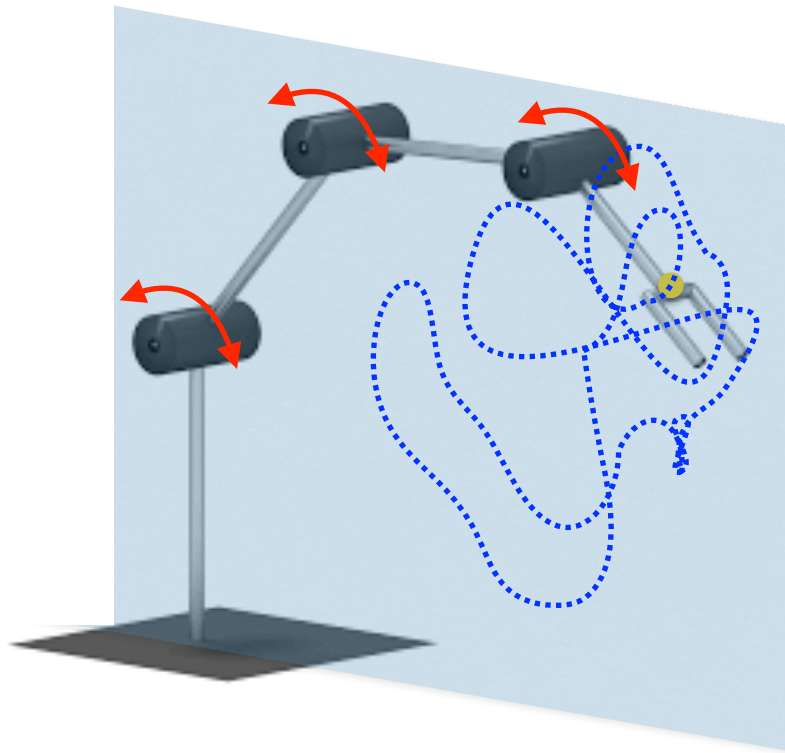
by differentiating w.r.t. time

**Analytic
Jacobian
Matrix**

$$J_A(\underline{q}(t))$$

$$\dot{\underline{x}}(\underline{q}(t)) = \begin{bmatrix} \frac{\partial o_t^b(\underline{q}(t))}{\partial \underline{q}(t)} \\ \frac{\partial \phi(\underline{q}(t))}{\partial \underline{q}(t)} \end{bmatrix} \dot{\underline{q}}(t)$$

Differential Kinematics: Geometric Approach

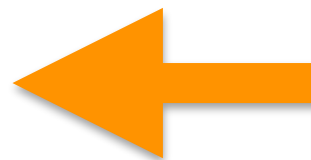


$$\underline{\dot{q}}(t)$$

“Velocity” of the joints



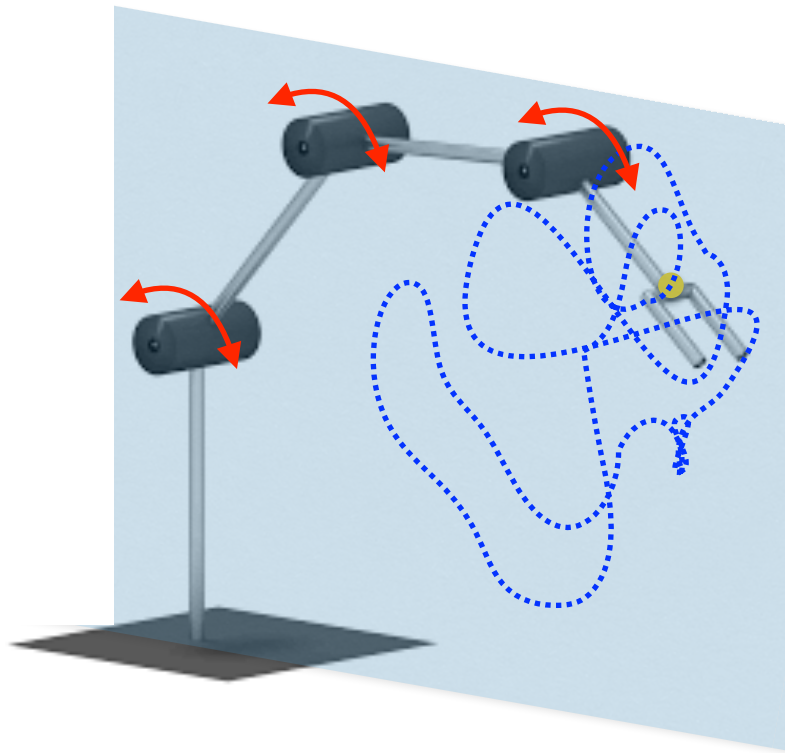
$$\underline{v}(t) = \begin{bmatrix} J_P(\underline{q}(t)) \\ J_O(\underline{q}(t)) \end{bmatrix} \underline{\dot{q}}(t)$$



“Velocity” of the end-effector

?

Differential Kinematics: Analytic Approach



$$\underline{\dot{q}}(t)$$

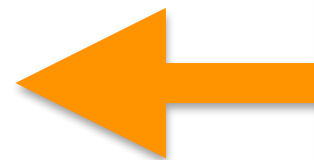
“Velocity” of the joints



“Velocity” of the end-effector

?

$$\underline{\dot{x}}(\underline{q}(t)) = \begin{bmatrix} \frac{\partial o_t^b(\underline{q}(t))}{\partial \underline{q}(t)} \\ \frac{\partial \phi(\underline{q}(t))}{\partial \underline{q}(t)} \end{bmatrix} \underline{\dot{q}}(t)$$



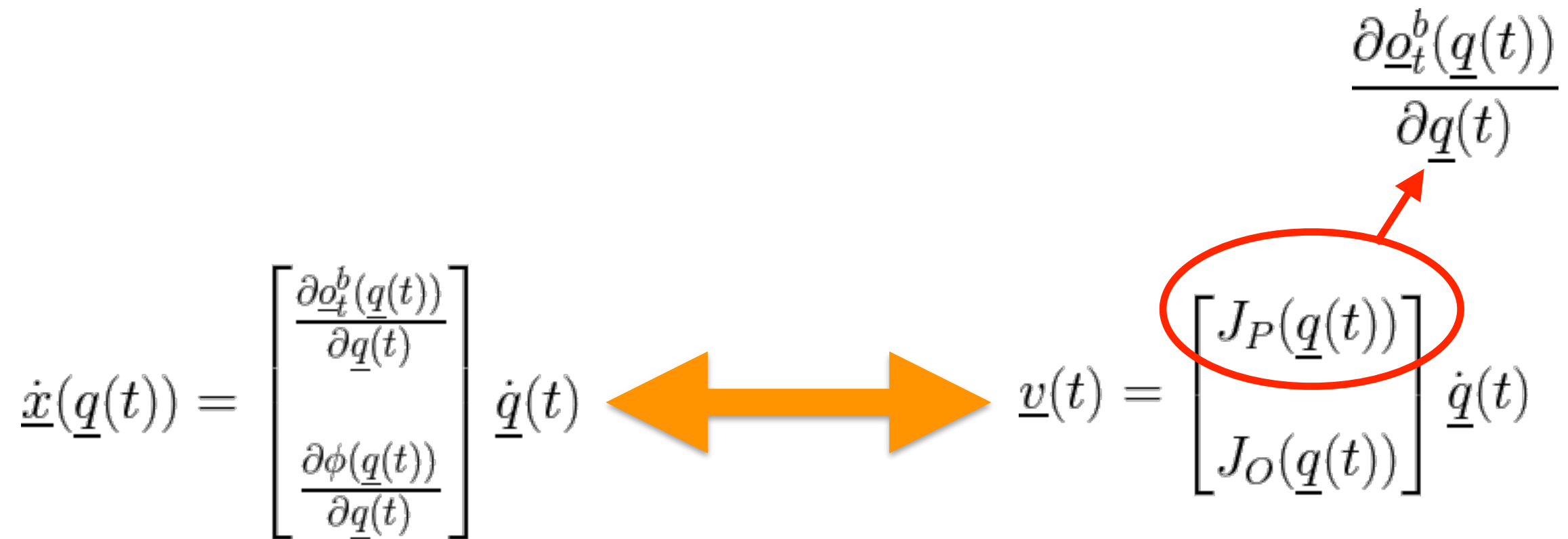
Differential Kinematics: Analytic Approach vs Geometric Approach

$$\underline{\dot{x}}(\underline{q}(t)) = \begin{bmatrix} \frac{\partial o_t^b(\underline{q}(t))}{\partial \underline{q}(t)} \\ \frac{\partial \phi(\underline{q}(t))}{\partial \underline{q}(t)} \end{bmatrix} \underline{\dot{q}}(t) \quad \longleftrightarrow \quad \underline{v}(t) = \begin{bmatrix} J_P(\underline{q}(t)) \\ J_O(\underline{q}(t)) \end{bmatrix} \underline{\dot{q}}(t)$$

Differential Kinematics: Analytic Approach vs Geometric Approach

$$\underline{\dot{x}}(\underline{q}(t)) = \begin{bmatrix} \frac{\partial o_t^b(\underline{q}(t))}{\partial \underline{q}(t)} \\ \frac{\partial \phi(\underline{q}(t))}{\partial \underline{q}(t)} \end{bmatrix} \underline{\dot{q}}(t) \quad \longleftrightarrow \quad \underline{v}(t) = \begin{bmatrix} J_P(\underline{q}(t)) \\ J_O(\underline{q}(t)) \end{bmatrix} \underline{\dot{q}}(t)$$

$\frac{\partial o_t^b(\underline{q}(t))}{\partial \underline{q}(t)}$



Differential Kinematics: Analytic Approach vs Geometric Approach

The diagram illustrates the relationship between the Analytic and Geometric approaches to differential kinematics. It features two main equations connected by a large orange double-headed arrow, with additional annotations showing the internal structure of the Jacobian matrices.

Analytic Approach (Left):

$$\dot{\underline{x}}(\underline{q}(t)) = \begin{bmatrix} \frac{\partial o_t^b(\underline{q}(t))}{\partial \underline{q}(t)} \\ \frac{\partial \phi(\underline{q}(t))}{\partial \underline{q}(t)} \end{bmatrix} \dot{\underline{q}}(t)$$

The top element of the Jacobian matrix, $\frac{\partial o_t^b(\underline{q}(t))}{\partial \underline{q}(t)}$, is enclosed in a red dotted oval. A red dotted arrow points from this oval to the top element of the Jacobian matrix in the Geometric approach.

Geometric Approach (Right):

$$\underline{v}(t) = \begin{bmatrix} J_P(\underline{q}(t)) \\ J_O(\underline{q}(t)) \end{bmatrix} \dot{\underline{q}}(t)$$

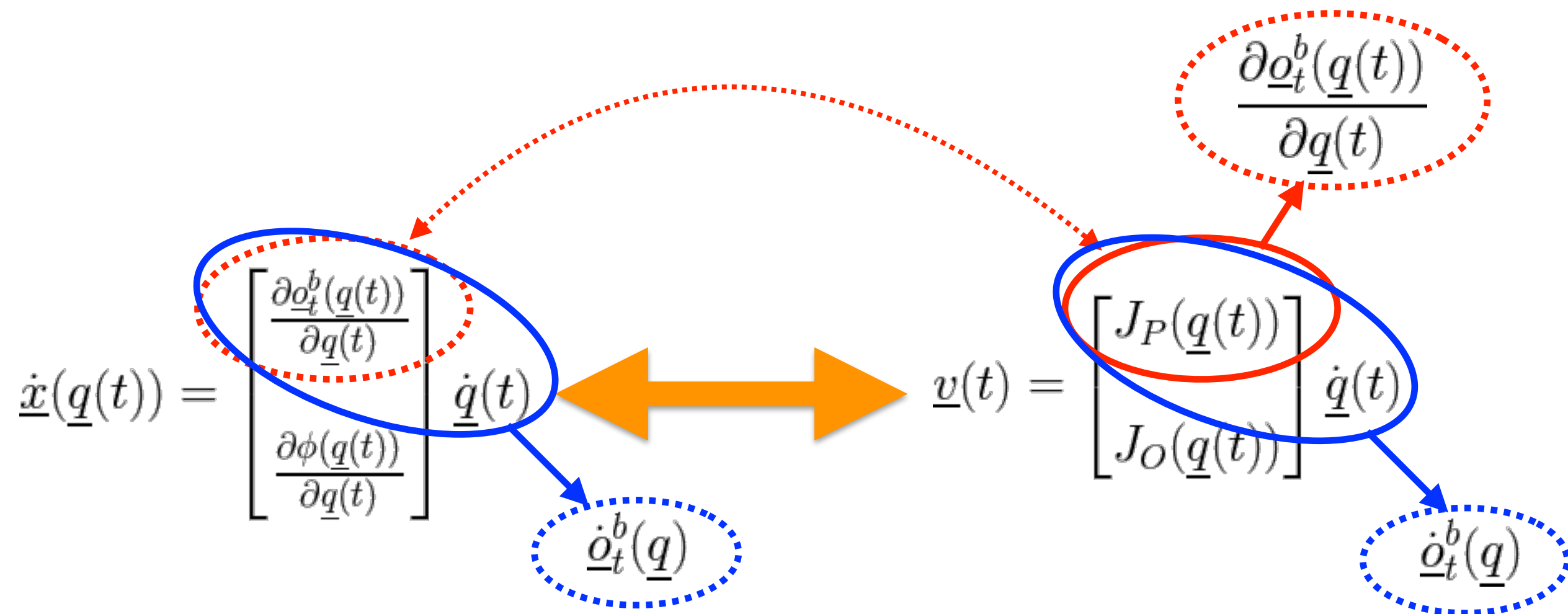
The entire Jacobian matrix $\begin{bmatrix} J_P(\underline{q}(t)) \\ J_O(\underline{q}(t)) \end{bmatrix}$ is enclosed in a solid red oval. A red solid arrow points from this oval to the top element of the Jacobian matrix in the Analytic approach.

Geometric Jacobian Detail (Top Right):

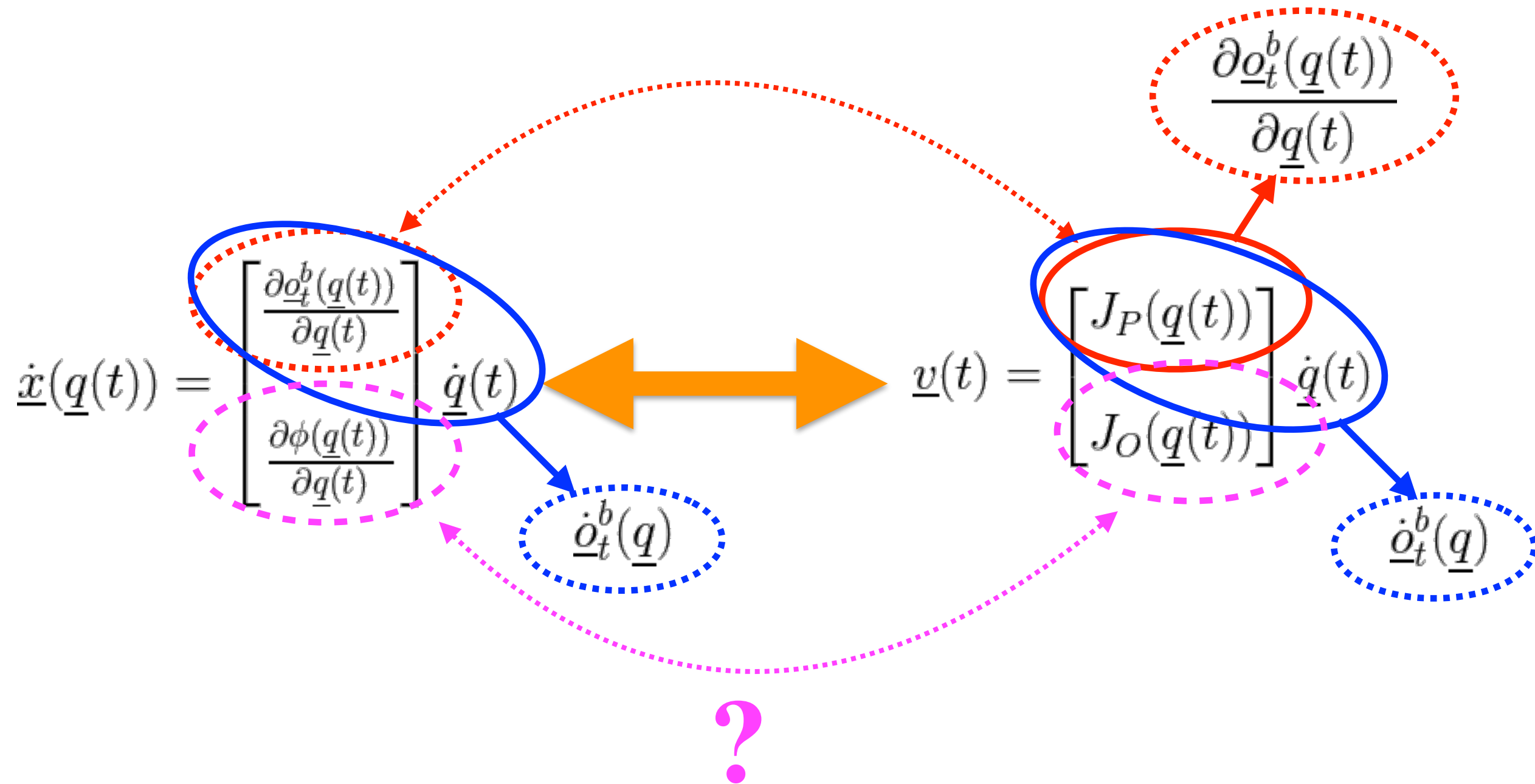
$$\frac{\partial o_t^b(\underline{q}(t))}{\partial \underline{q}(t)}$$

This expression is enclosed in a red dotted oval, representing the top element of the Jacobian matrix in the Geometric approach.

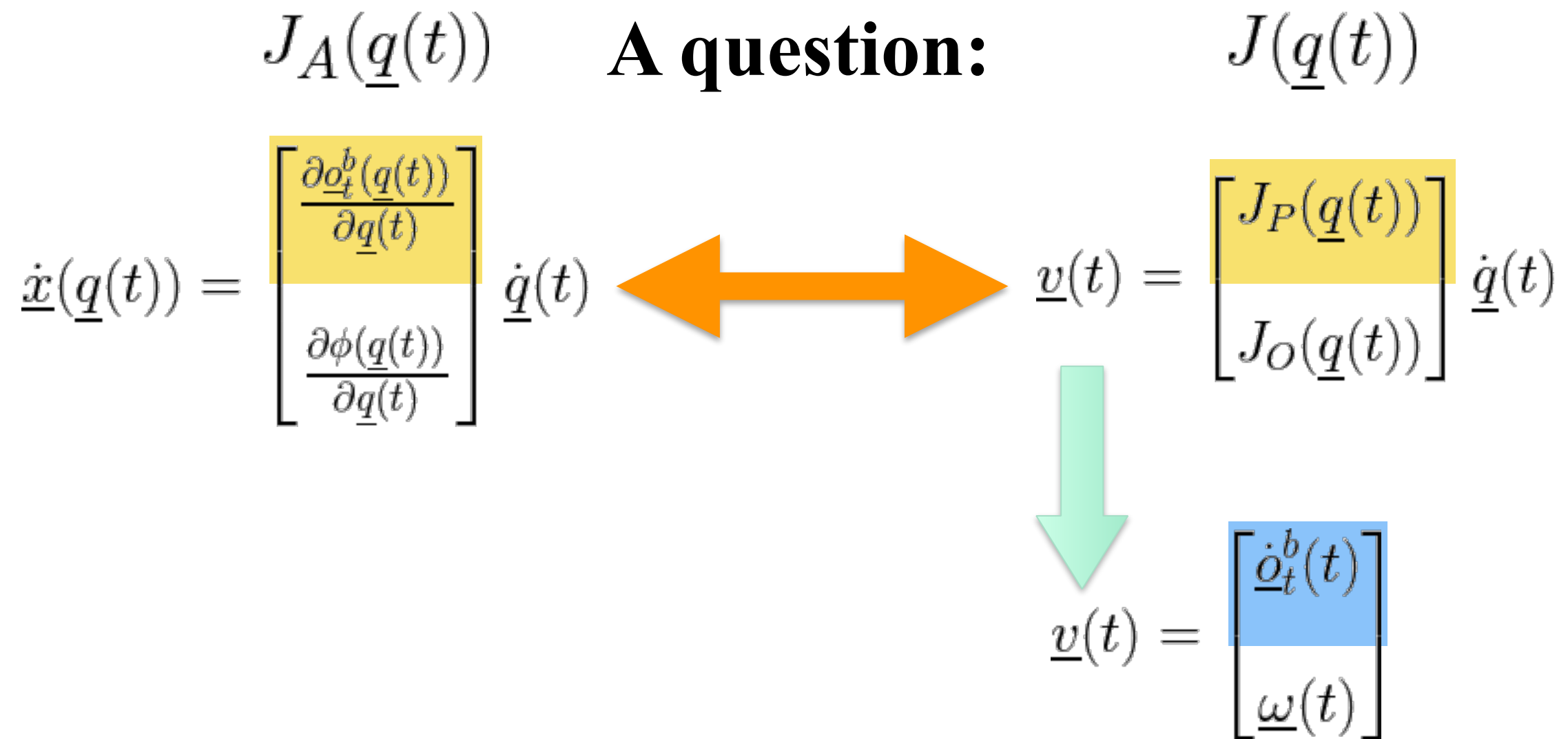
Differential Kinematics: Analytic Approach vs Geometric Approach



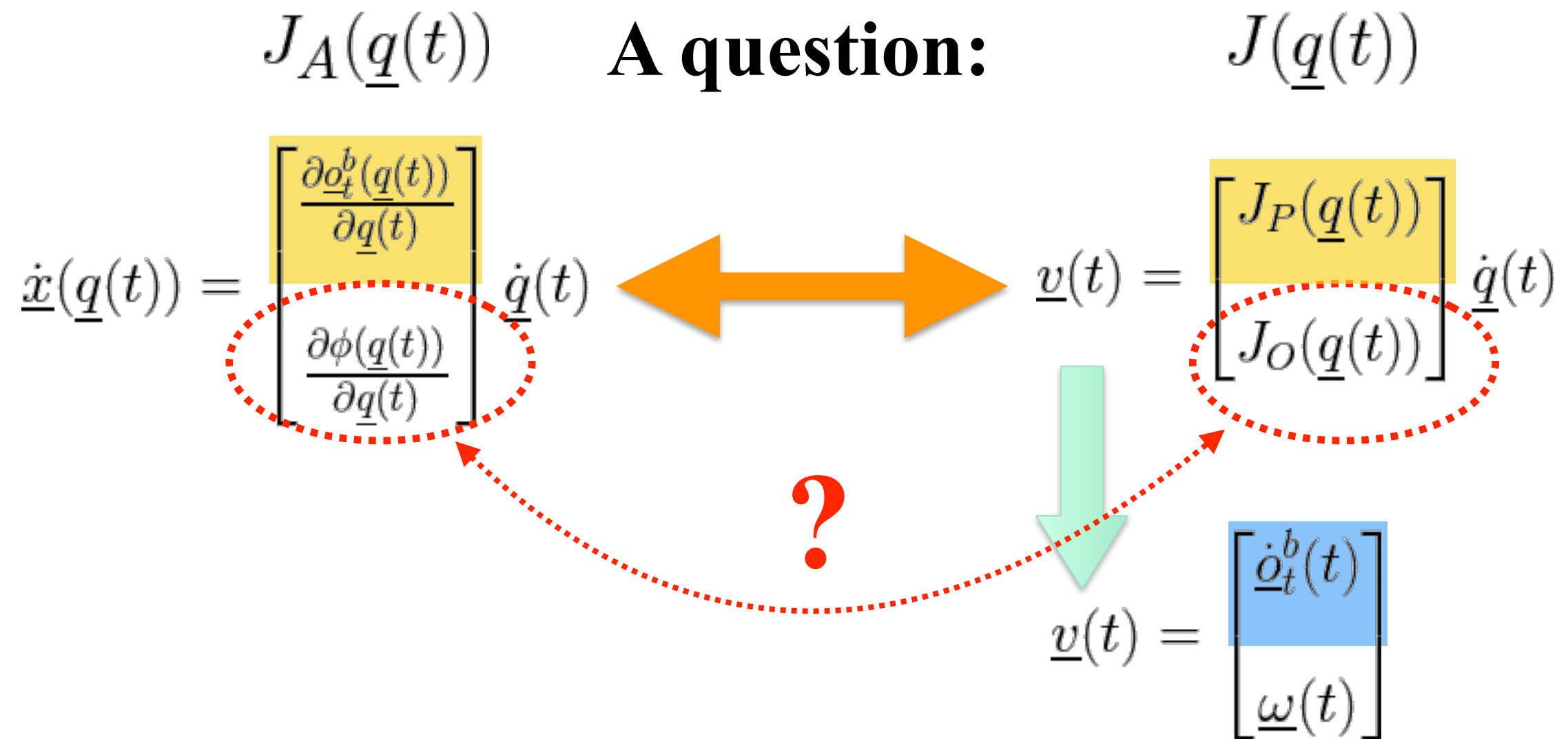
Differential Kinematics: Analytic Approach vs Geometric Approach



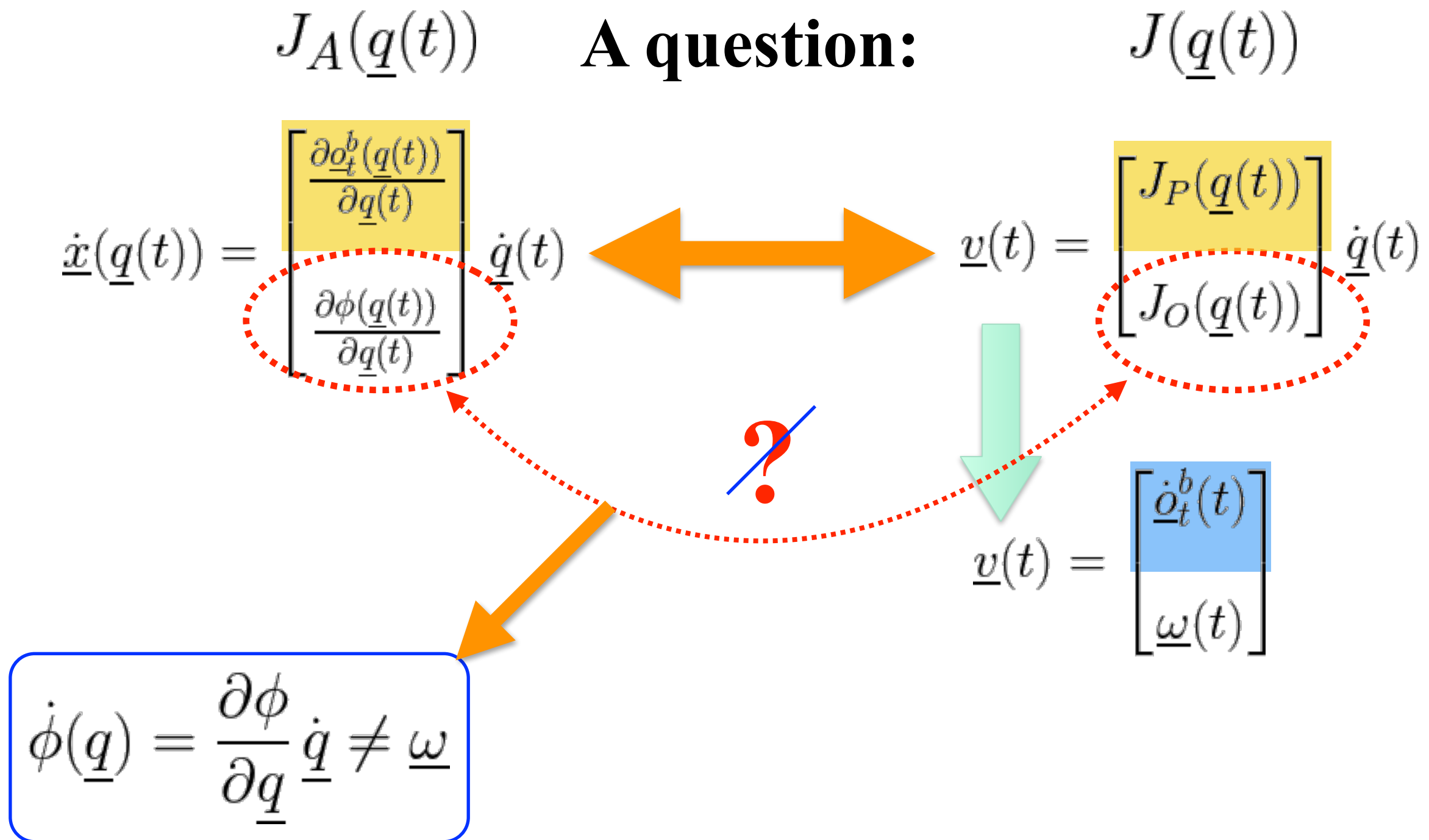
Differential Kinematics: Analytic Approach vs Geometric Approach



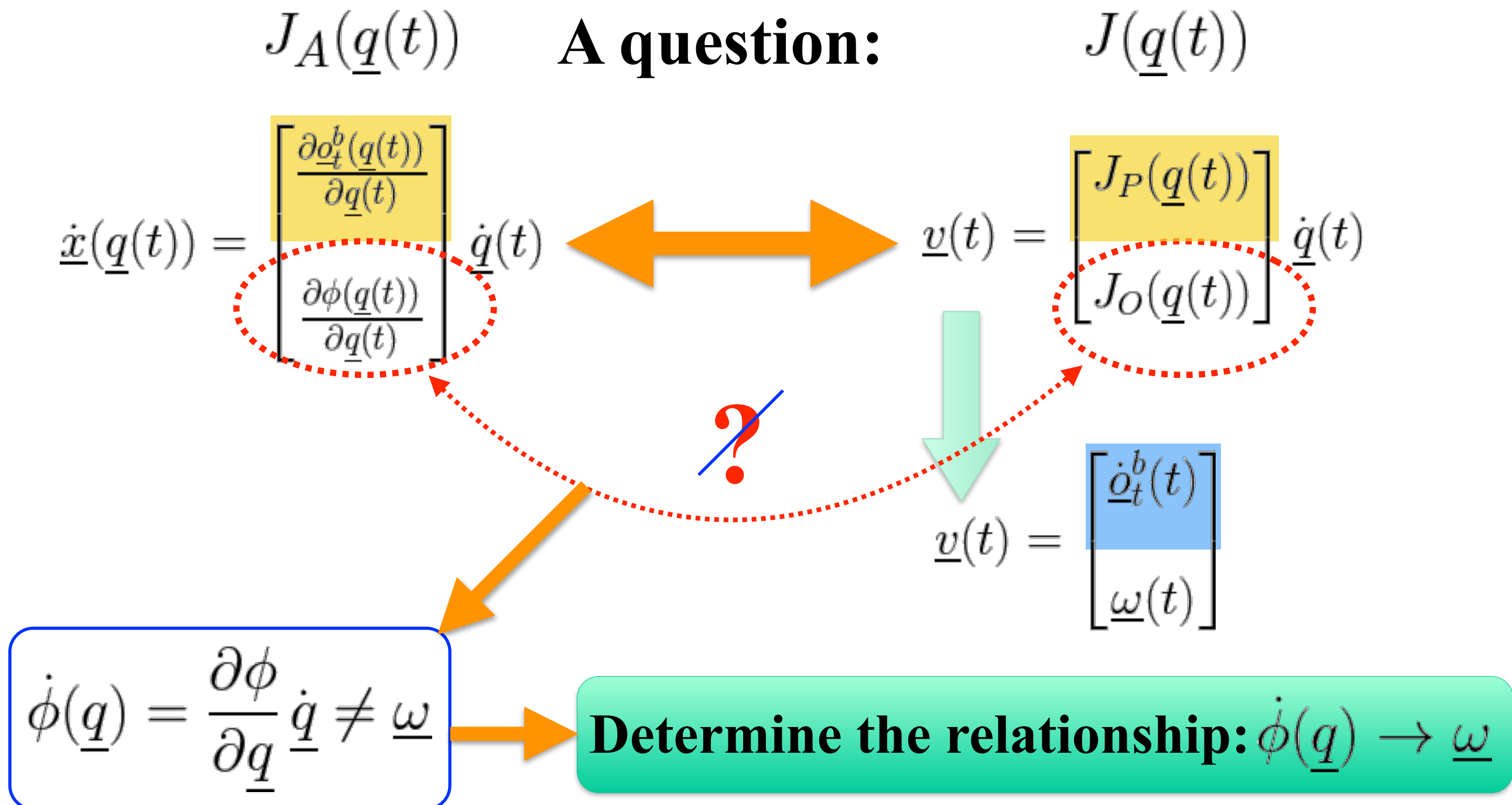
Differential Kinematics: Analytic Approach vs Geometric Approach



Differential Kinematics: Analytic Approach vs Geometric Approach



Differential Kinematics: Analytic Approach vs Geometric Approach



Differential Kinematics: Analytic Approach vs Geometric Approach

$$\dot{\phi}(\underline{q}) = \frac{\partial \phi}{\partial \underline{q}} \dot{\underline{q}} \neq \underline{\omega}$$

Determine the relationship: $\dot{\phi}(\underline{q}) \rightarrow \underline{\omega}$

Differential Kinematics: Analytic Approach vs Geometric Approach

$$\dot{\phi}(\underline{q}) = \frac{\partial \phi}{\partial \underline{q}} \dot{\underline{q}} \neq \underline{\omega}$$

Determine the relationship: $\dot{\phi}(\underline{q}) \rightarrow \underline{\omega}$

In general:

$$\underline{\omega} = T(\phi) \cdot \dot{\phi}$$

Differential Kinematics: Analytic Approach vs Geometric Approach

$$\dot{\phi}(\underline{q}) = \frac{\partial \phi}{\partial \underline{q}} \dot{\underline{q}} \neq \underline{\omega}$$

Determine the relationship: $\dot{\phi}(\underline{q}) \rightarrow \underline{\omega}$

In general:

$$\underline{\omega} = T(\phi) \dot{\phi}$$

?

Differential Kinematics: Analytic Approach vs Geometric Approach

$$\dot{\phi}(\underline{q}) = \frac{\partial \phi}{\partial \underline{q}} \dot{\underline{q}} \neq \underline{\omega}$$

Determine the relationship: $\dot{\phi}(\underline{q}) \rightarrow \underline{\omega}$

In general:

$$\underline{\omega} = T(\phi) \cdot \dot{\phi}$$

?

$$\underline{\omega} = T_{\text{EUL}}(\phi) \cdot \dot{\phi}$$

$$\underline{\omega} = T_{\text{RPY}}(\phi) \cdot \dot{\phi}$$

$$\underline{\omega} = T_{\text{EULZYZ}}(\phi) \cdot \dot{\phi}$$

Differential Kinematics: Analytic Approach vs Geometric Approach

$$\dot{\phi}(\underline{q}) = \frac{\partial \phi}{\partial \underline{q}} \dot{\underline{q}} \neq \underline{\omega}$$

Determine the relationship: $\dot{\phi}(\underline{q}) \rightarrow \underline{\omega}$

In general:

$$\underline{\omega}(t) = T(\phi) \cdot \dot{\phi}(t) \quad \cdots \longrightarrow \quad \underline{v}(t) = \begin{bmatrix} \dot{\phi}^b(t) \\ \underline{\omega}(t) \end{bmatrix}$$

Differential Kinematics: Analytic Approach vs Geometric Approach

$$\dot{\phi}(\underline{q}) = \frac{\partial \phi}{\partial \underline{q}} \dot{\underline{q}} \neq \underline{\omega}$$

Determine the relationship: $\dot{\phi}(\underline{q}) \rightarrow \underline{\omega}$

In general:

$$\underline{\omega}(t) = T(\phi) \cdot \dot{\phi}(t) \quad \cdots \cdots \cdots \rightarrow \underline{v}(t) = \begin{bmatrix} \dot{\underline{q}}^b(t) \\ \underline{\omega}(t) \end{bmatrix}$$

$$\underline{v}(t) = \begin{bmatrix} I & \underline{0} \\ \underline{0} & T(\phi) \end{bmatrix} \dot{\underline{x}}(t)$$

Differential Kinematics: Analytic Approach vs Geometric Approach

$$\dot{\phi}(\underline{q}) = \frac{\partial \phi}{\partial \underline{q}} \dot{\underline{q}} \neq \underline{\omega}$$

Determine the relationship: $\dot{\phi}(\underline{q}) \rightarrow \underline{\omega}$

In general:

$$\underline{\omega}(t) = T(\phi) \cdot \dot{\phi}(t) \quad \text{.....} \rightarrow \quad \underline{v}(t) = \begin{bmatrix} \dot{\underline{q}}^b(t) \\ \underline{\omega}(t) \end{bmatrix}$$

$$\underline{v}(t) = \begin{bmatrix} I & \underline{0} \\ \underline{0} & T(\phi) \end{bmatrix} \dot{\underline{x}}(t) = T_A(\phi) \dot{\underline{x}}(t)$$

Differential Kinematics: Analytic Approach vs Geometric Approach

$$\dot{\phi}(\underline{q}) = \frac{\partial \phi}{\partial \underline{q}} \dot{\underline{q}} \neq \underline{\omega}$$

Determine the relationship: $\dot{\phi}(\underline{q}) \rightarrow \underline{\omega}$

In general:

$$\underline{\omega}(t) = T(\phi) \cdot \dot{\phi}(t) \quad \longrightarrow \quad \underline{v}(t) = \begin{bmatrix} \dot{\underline{q}}^b(t) \\ \underline{\omega}(t) \end{bmatrix}$$

$$\underline{v}(t) = \begin{bmatrix} I & \underline{0} \\ \underline{0} & T(\phi) \end{bmatrix}$$

$$\dot{\underline{x}}(t) = T_A(\phi) \dot{\underline{x}}(t)$$

$$\dot{\underline{x}}(\underline{q}(t)) = J_A(\underline{q}(t)) \dot{\underline{q}}(t)$$

Differential Kinematics: Analytic Approach vs Geometric Approach

$$\dot{\phi}(\underline{q}) = \frac{\partial \phi}{\partial \underline{q}} \dot{\underline{q}} \neq \underline{\omega}$$

Determine the relationship: $\dot{\phi}(\underline{q}) \rightarrow \underline{\omega}$

In general:

$$\underline{\omega}(t) = T(\phi) \cdot \dot{\phi}(t) \quad \dots \rightarrow \quad \underline{v}(t) = \begin{bmatrix} J_P(\underline{q}(t)) \\ J_O(\underline{q}(t)) \end{bmatrix} \dot{\underline{q}}(t)$$

$J(\underline{q}(t))$

$$\underline{v}(t) = \begin{bmatrix} I & \underline{0} \\ \underline{0} & T(\phi) \end{bmatrix} \dot{\underline{x}}(t) = T_A(\phi) J_A(\underline{q}(t)) \dot{\underline{q}}(t)$$

Differential Kinematics: Analytic Approach vs Geometric Approach

$$\dot{\phi}(\underline{q}) = \frac{\partial \phi}{\partial \underline{q}} \dot{\underline{q}} \neq \underline{\omega}$$

Determine the relationship: $\dot{\phi}(\underline{q}) \rightarrow \underline{\omega}$

In general:

$$\underline{\omega}(t) = T(\phi) \cdot \dot{\phi}(t) \quad \cdots \cdots \cdots \rightarrow \quad \underline{v}(t) = \begin{bmatrix} J_P(\underline{q}(t)) \\ J_O(\underline{q}(t)) \end{bmatrix} \dot{\underline{q}}(t)$$

$J(\underline{q}(t))$

$$J(\underline{q}(t)) = T_A(\phi(t)) J_A(\underline{q}(t))$$

Differential Kinematics: Analytic Approach vs Geometric Approach

$$\dot{\phi}(\underline{q}) = \frac{\partial \phi}{\partial \underline{q}} \dot{\underline{q}} \neq \underline{\omega}$$

Determine the relationship: $\dot{\phi}(\underline{q}) \rightarrow \underline{\omega}$

In general:

$$\underline{\omega}(t) = T(\phi) \cdot \dot{\phi}(t) \quad \longrightarrow \quad \underline{v}(t) = \begin{bmatrix} J_P(\underline{q}(t)) \\ J_O(\underline{q}(t)) \end{bmatrix} \dot{\underline{q}}(t)$$

$J(\underline{q}(t))$

$$J(\underline{q}(t)) = T_A(\phi(t)) J_A(\underline{q}(t))$$



UNIVERSITÀ DI PAVIA
Dipartimento di
Ingegneria Industriale
e dell'Informazione

Course of Robot Control

Prof. Antonella Ferrara



Determination of the Transformation Matrix

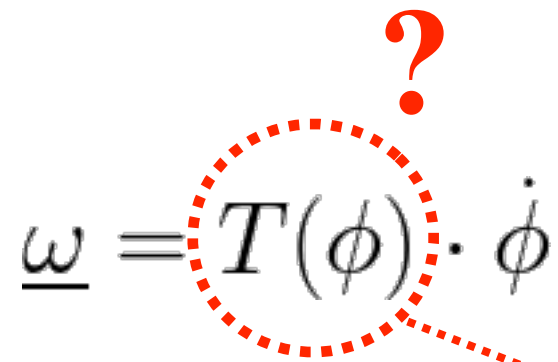
Determination of the Transformation: $\dot{\phi} \rightarrow \underline{\omega}$

$$\dot{\phi}(\underline{q}) = \frac{\partial \phi}{\partial \underline{q}} \dot{\underline{q}} \neq \underline{\omega}$$

$$\underline{\omega} = T(\phi) \cdot \dot{\phi}$$

Determination of the Transformation: $\dot{\phi} \rightarrow \underline{\omega}$

$$\dot{\phi}(\underline{q}) = \frac{\partial \phi}{\partial \underline{q}} \dot{\underline{q}} \neq \underline{\omega}$$

$$\underline{\omega} = T(\phi) \cdot \dot{\phi}$$


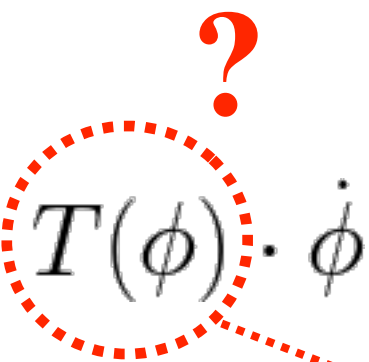
$$\underline{\omega} = T_{\text{EUL}}(\phi) \cdot \dot{\phi}$$

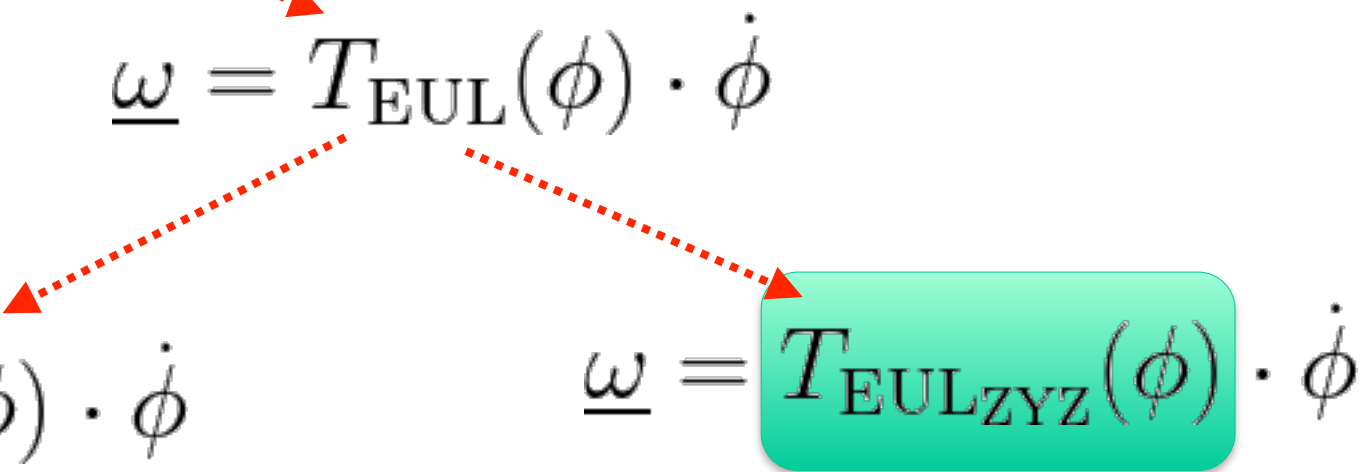

$$\underline{\omega} = T_{\text{RPY}}(\phi) \cdot \dot{\phi}$$

$$\underline{\omega} = T_{\text{EUL}_{\text{ZYZ}}}(\phi) \cdot \dot{\phi}$$

Determination of the Transformation: $\dot{\phi} \rightarrow \underline{\omega}$

$$\dot{\phi}(\underline{q}) = \frac{\partial \phi}{\partial \underline{q}} \dot{\underline{q}} \neq \underline{\omega}$$

$$\underline{\omega} = T(\phi) \cdot \dot{\phi}$$


$$\underline{\omega} = T_{\text{EUL}}(\phi) \cdot \dot{\phi}$$


$$\underline{\omega} = T_{\text{RPY}}(\phi) \cdot \dot{\phi}$$

$$\underline{\omega} = T_{\text{EULZYZ}}(\phi) \cdot \dot{\phi}$$

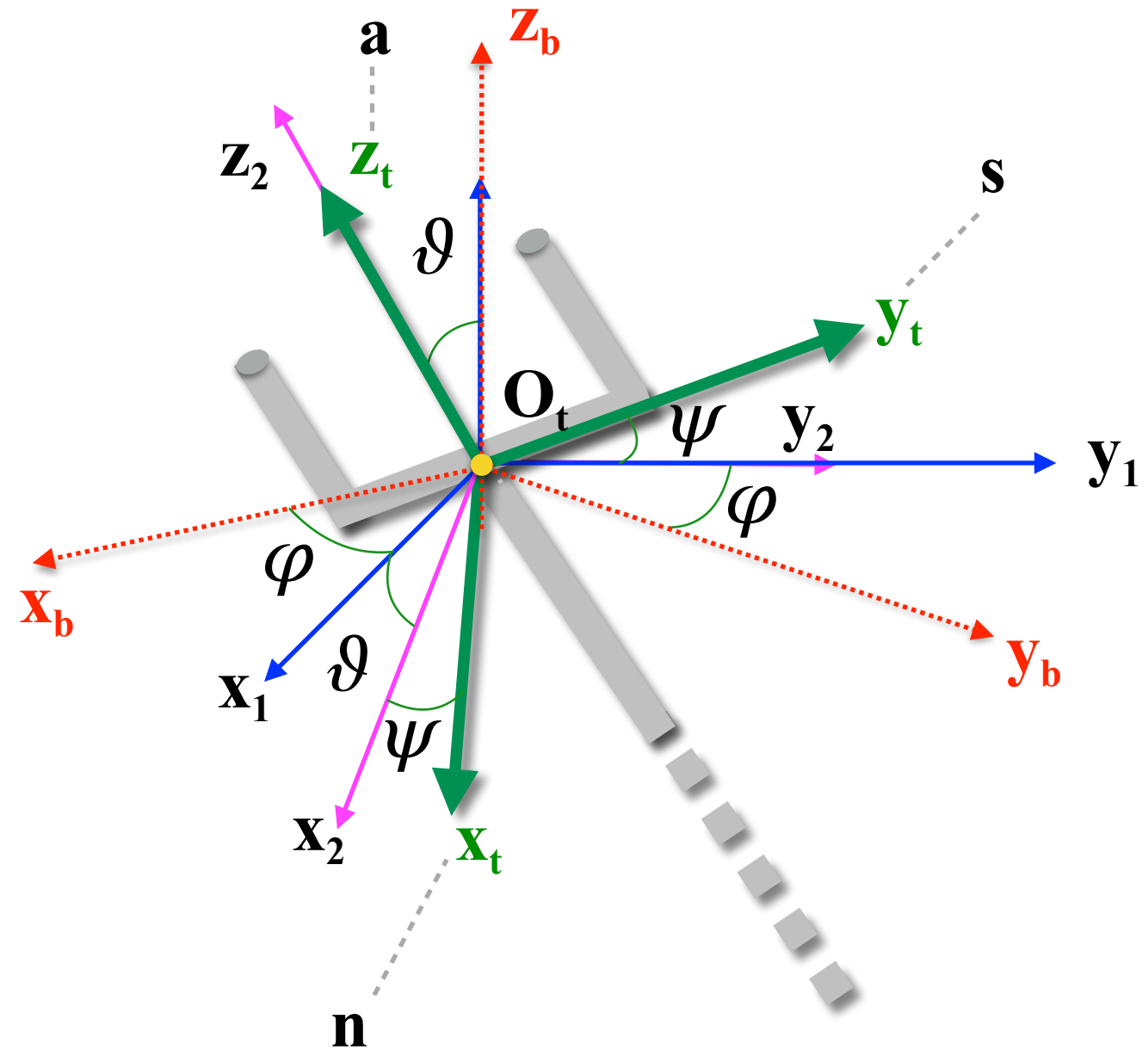
Determination of the Transformation

Matrix $T_{\text{EUL}_{\text{ZYZ}}}(\phi)$

$$\underline{\omega} = T_{\text{EUL}_{\text{ZYZ}}}(\phi) \cdot \dot{\phi}$$

Diagram illustrating the dimensions of the terms in the equation:

- $\underline{\omega}$ is a 3×1 vector (indicated by a pink arrow).
- $T_{\text{EUL}_{\text{ZYZ}}}(\phi)$ is a 3×3 matrix (indicated by a pink arrow and a dashed pink arc).
- $\dot{\phi}$ is a 3×1 vector (indicated by a pink arrow).



Determination of the Transformation

Matrix $T_{\text{EUL}_{\text{ZYZ}}}(\phi)$

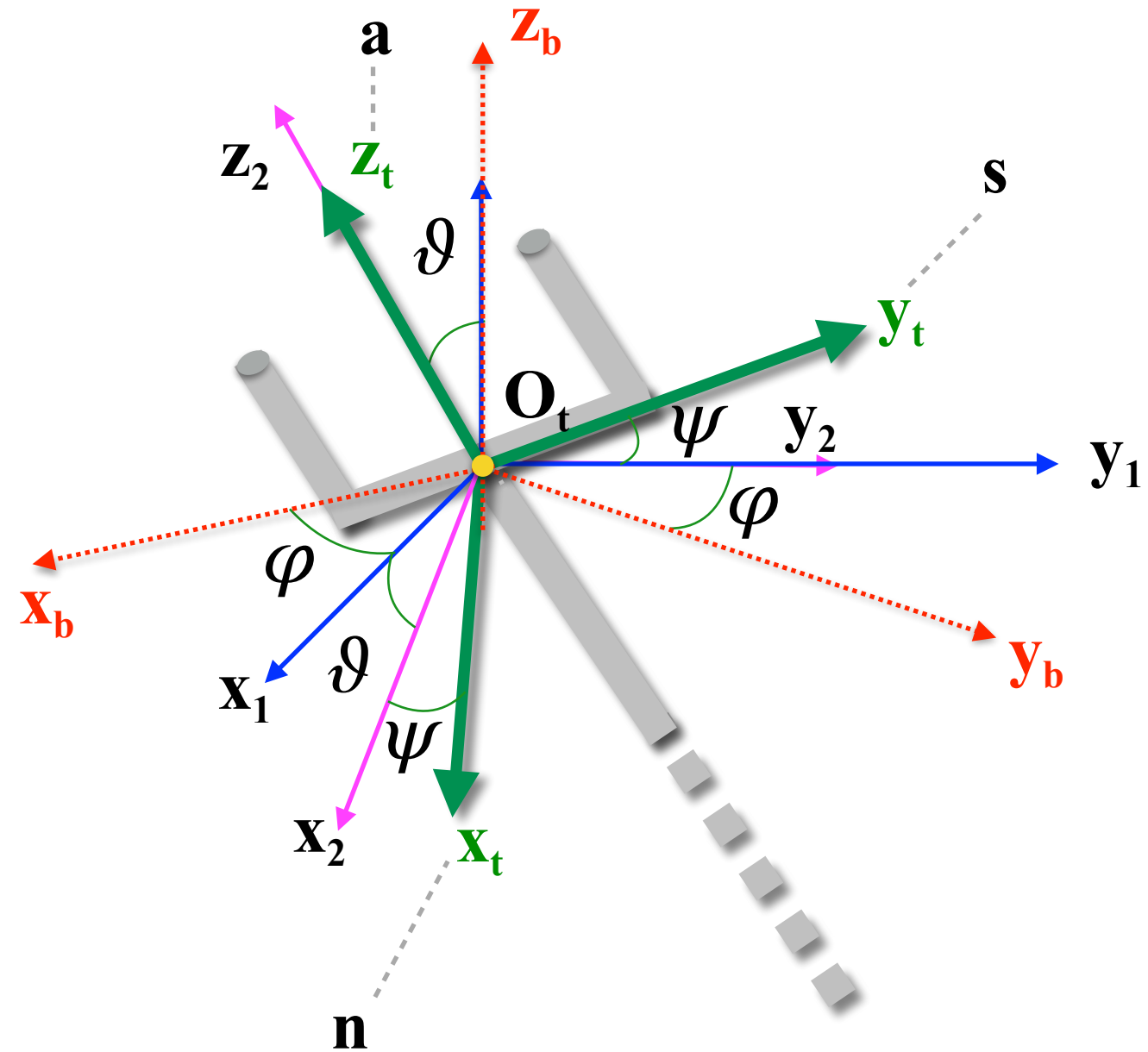
$$\underline{\omega} = T_{\text{EUL}_{\text{ZYZ}}}(\phi) \cdot \dot{\phi}$$

$\downarrow$ 3×1 $\downarrow$ 3×3 $\downarrow$ 3×1

$\downarrow$

?

$$\underline{\omega}(t) |_{\dot{\phi}} = \begin{bmatrix} T_{\dot{\phi}_x} \\ T_{\dot{\phi}_y} \\ T_{\dot{\phi}_z} \end{bmatrix} \dot{\phi}(t)$$



Determination of the Transformation

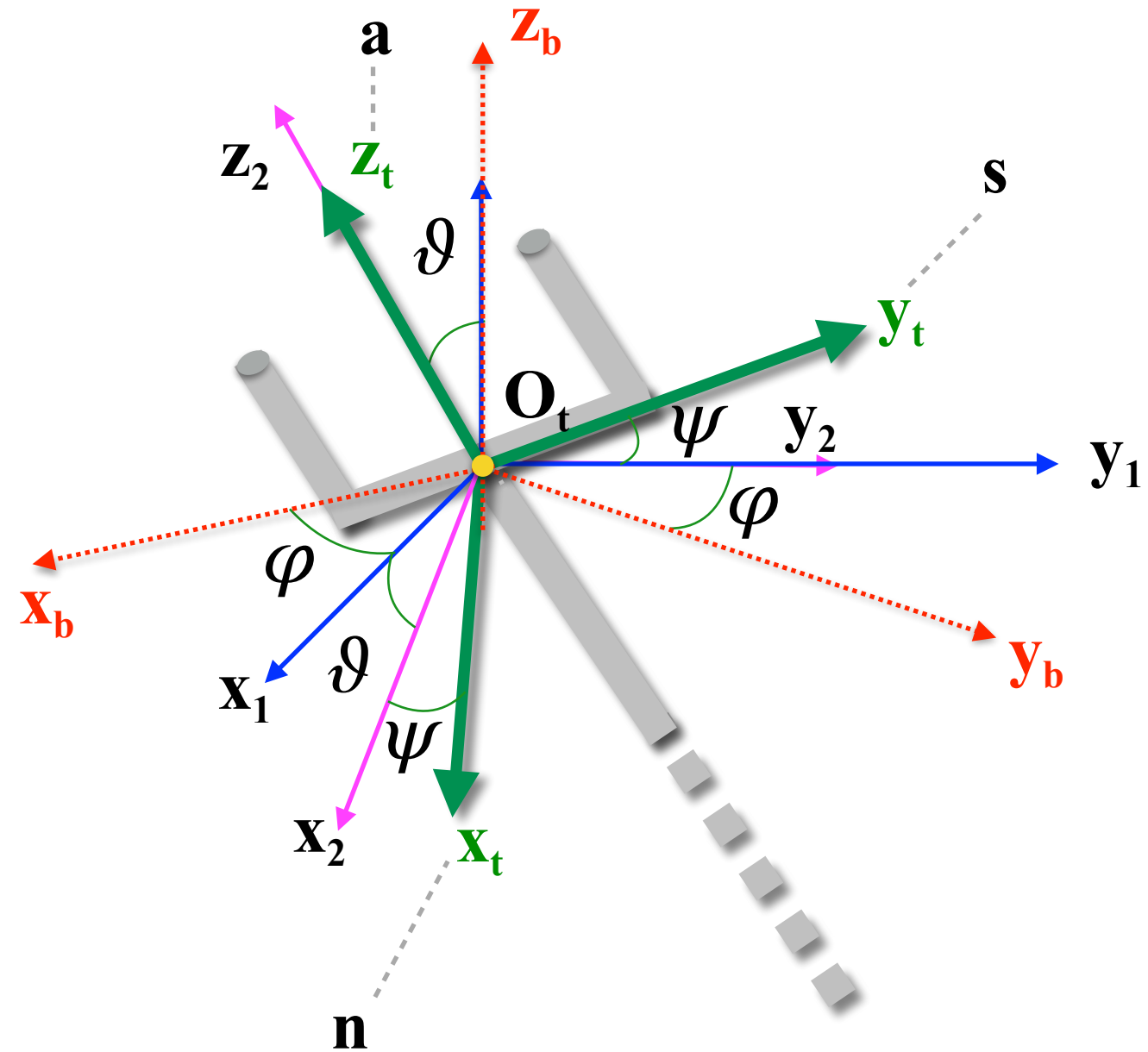
Matrix $T_{\text{EUL}_{\text{ZYZ}}}(\phi)$

$$\underline{\omega} = T_{\text{EUL}_{\text{ZYZ}}}(\phi) \cdot \dot{\phi}$$

$\downarrow$ 3×1 $\downarrow$ 3×3 $\downarrow$ 3×1

$\omega(t) |_{\dot{\vartheta}} = \begin{bmatrix} T_{\dot{\vartheta}_x} \\ T_{\dot{\vartheta}_y} \\ T_{\dot{\vartheta}_z} \end{bmatrix} \dot{\vartheta}(t)$

?



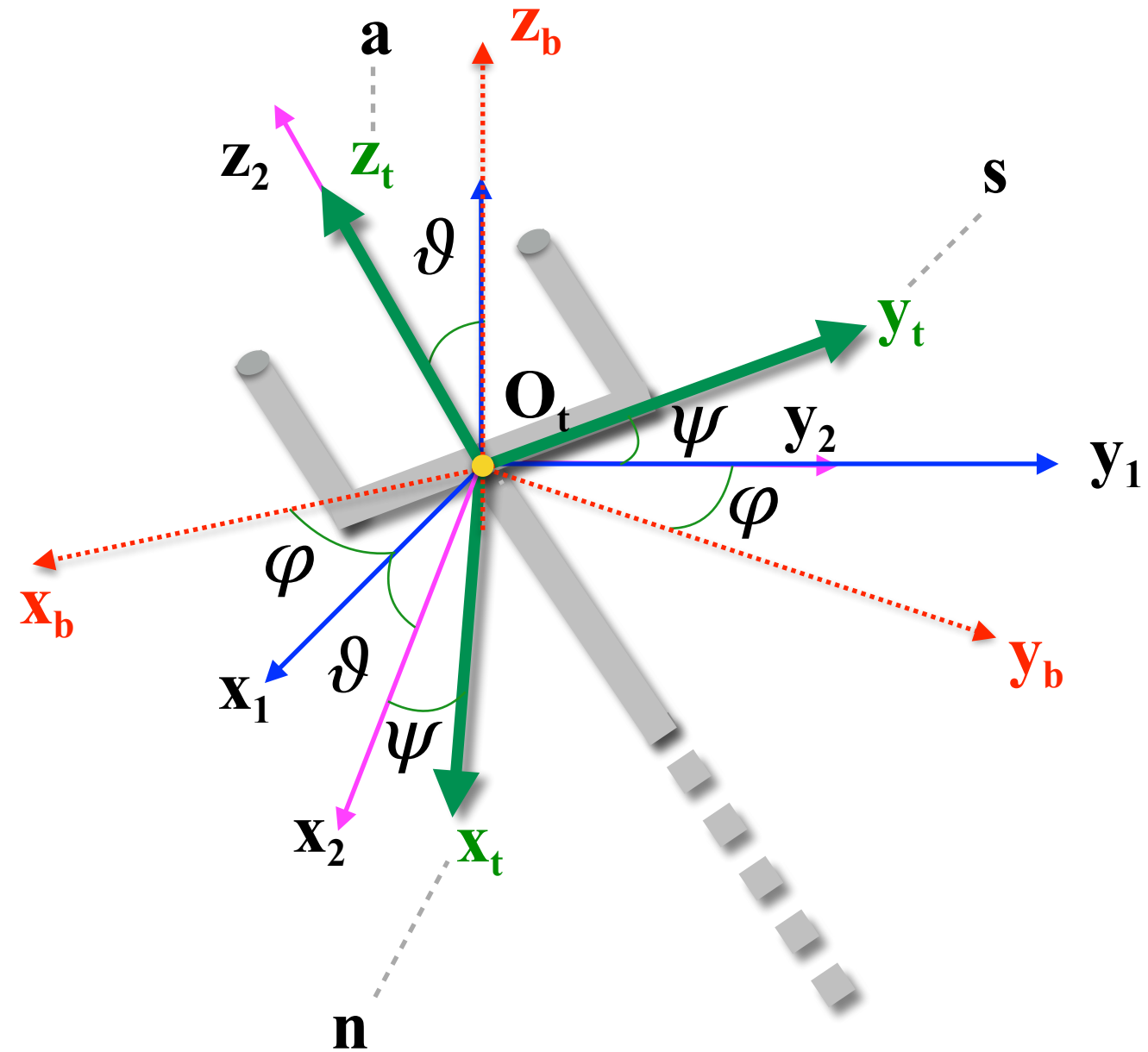
Determination of the Transformation

Matrix $T_{\text{EUL}_{\text{ZYZ}}}(\phi)$

$$\underline{\omega} = T_{\text{EUL}_{\text{ZYZ}}}(\phi) \cdot \dot{\phi}$$

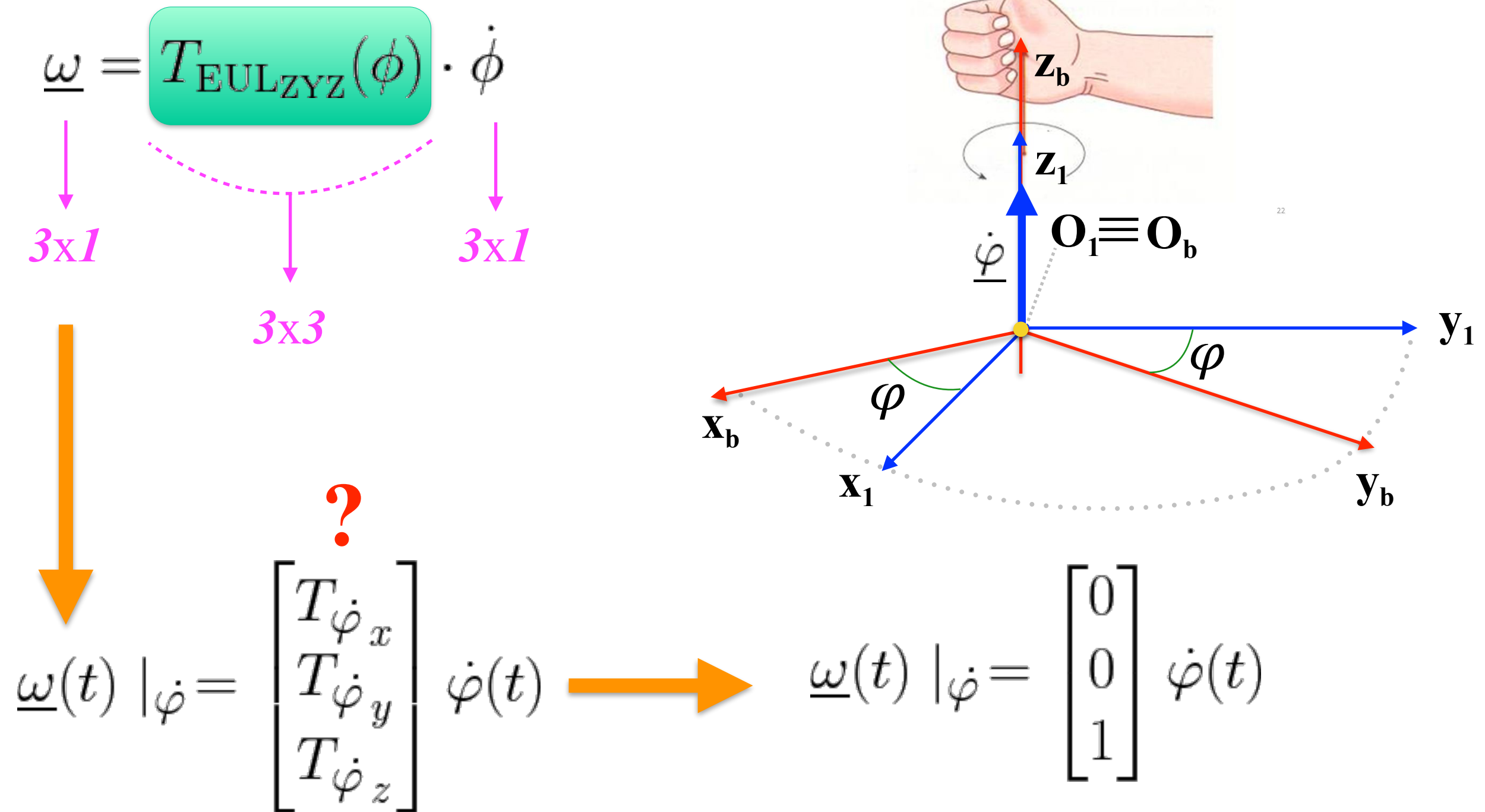
$\downarrow$ 3×1 $\downarrow$ 3×3 $\downarrow$ 3×1

$\omega(t) |_{\dot{\psi}} = \begin{bmatrix} T_{\dot{\psi}_x} \\ T_{\dot{\psi}_y} \\ T_{\dot{\psi}_z} \end{bmatrix} \dot{\psi}(t)$



Determination of the Transformation

Matrix $T_{\text{EUL}_{\text{ZYZ}}}(\phi)$



Determination of the Transformation

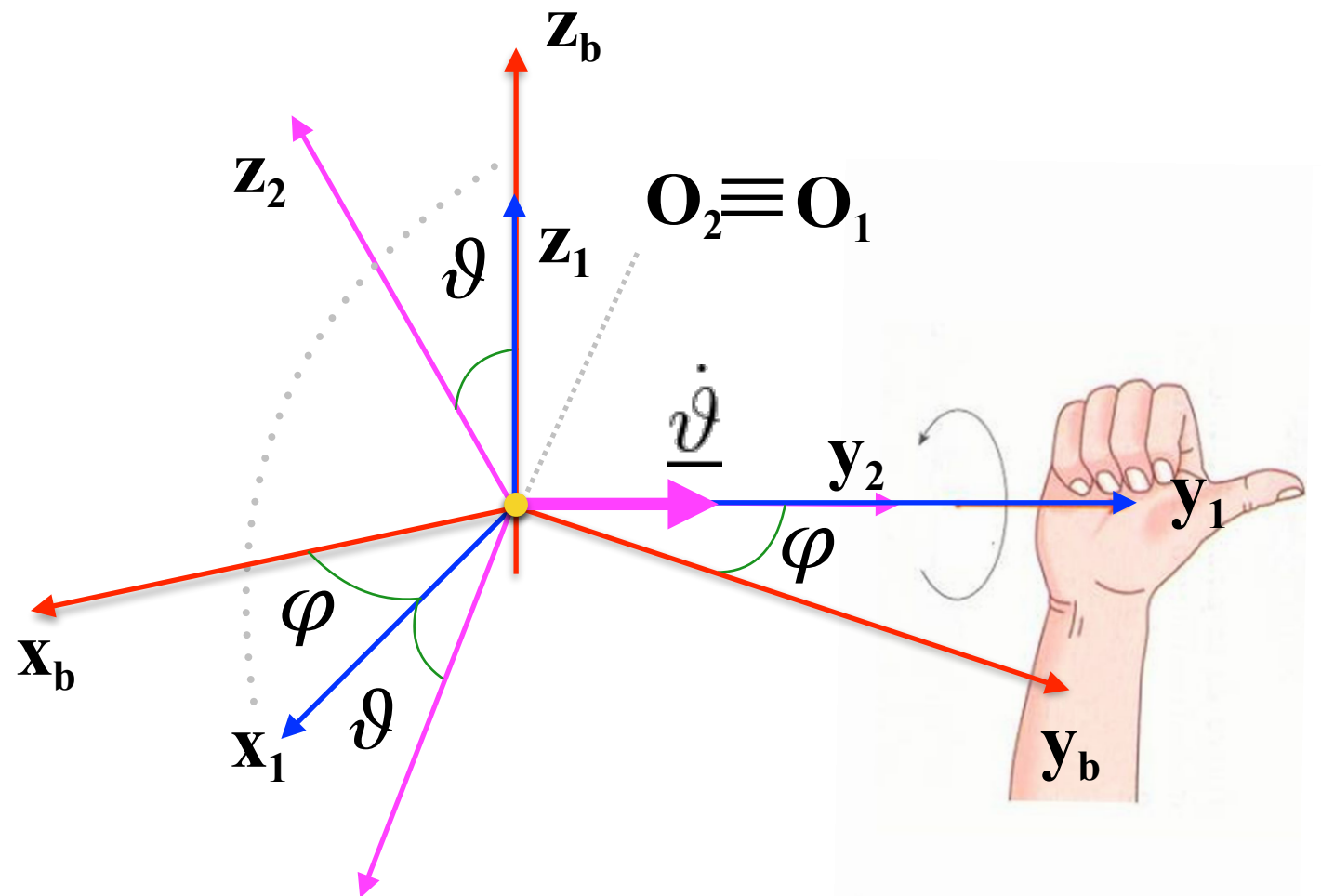
Matrix $T_{\text{EUL}_{\text{ZYZ}}}(\phi)$

$$\underline{\omega} = T_{\text{EUL}_{\text{ZYZ}}}(\phi) \cdot \dot{\phi}$$

$\downarrow$ 3×1 $\downarrow$ 3×3 $\downarrow$ 3×1

$\downarrow$

$$\underline{\omega}(t) |_{\dot{\vartheta}} = \begin{bmatrix} T_{\dot{\vartheta}_x} \\ T_{\dot{\vartheta}_y} \\ T_{\dot{\vartheta}_z} \end{bmatrix} \dot{\vartheta}(t) \longrightarrow \underline{\omega}(t) |_{\dot{\vartheta}} = \begin{bmatrix} -\sin(\varphi(t)) \\ \cos(\varphi(t)) \\ 0 \end{bmatrix} \dot{\vartheta}(t)$$



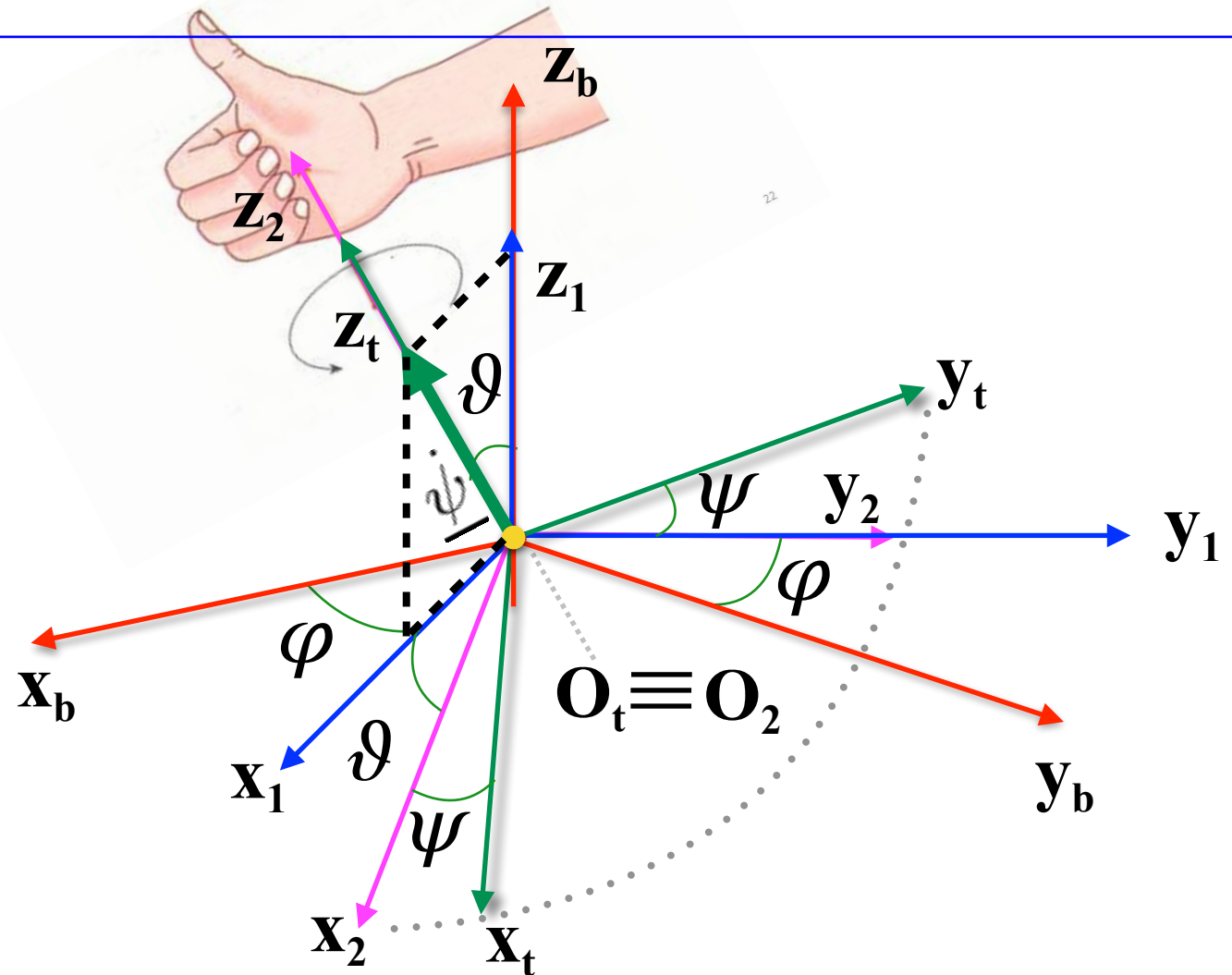
Determination of the Transformation

Matrix $T_{\text{EUL}_{\text{ZYZ}}}(\phi)$

$$\underline{\omega} = T_{\text{EUL}_{\text{ZYZ}}}(\phi) \cdot \dot{\phi}$$

$\downarrow$ 3×1 $\downarrow$ 3×3 $\downarrow$ 3×1

$\underline{\omega}(t) |_{\dot{\psi}} = \begin{bmatrix} T_{\dot{\psi}_x} \\ T_{\dot{\psi}_y} \\ T_{\dot{\psi}_z} \end{bmatrix} \dot{\psi}(t) \longrightarrow \underline{\omega}(t) |_{\dot{\psi}} = \begin{bmatrix} \sin(\vartheta(t)) \cos(\varphi(t)) \\ \sin(\vartheta(t)) \sin(\varphi(t)) \\ \cos(\vartheta(t)) \end{bmatrix} \dot{\psi}(t)$

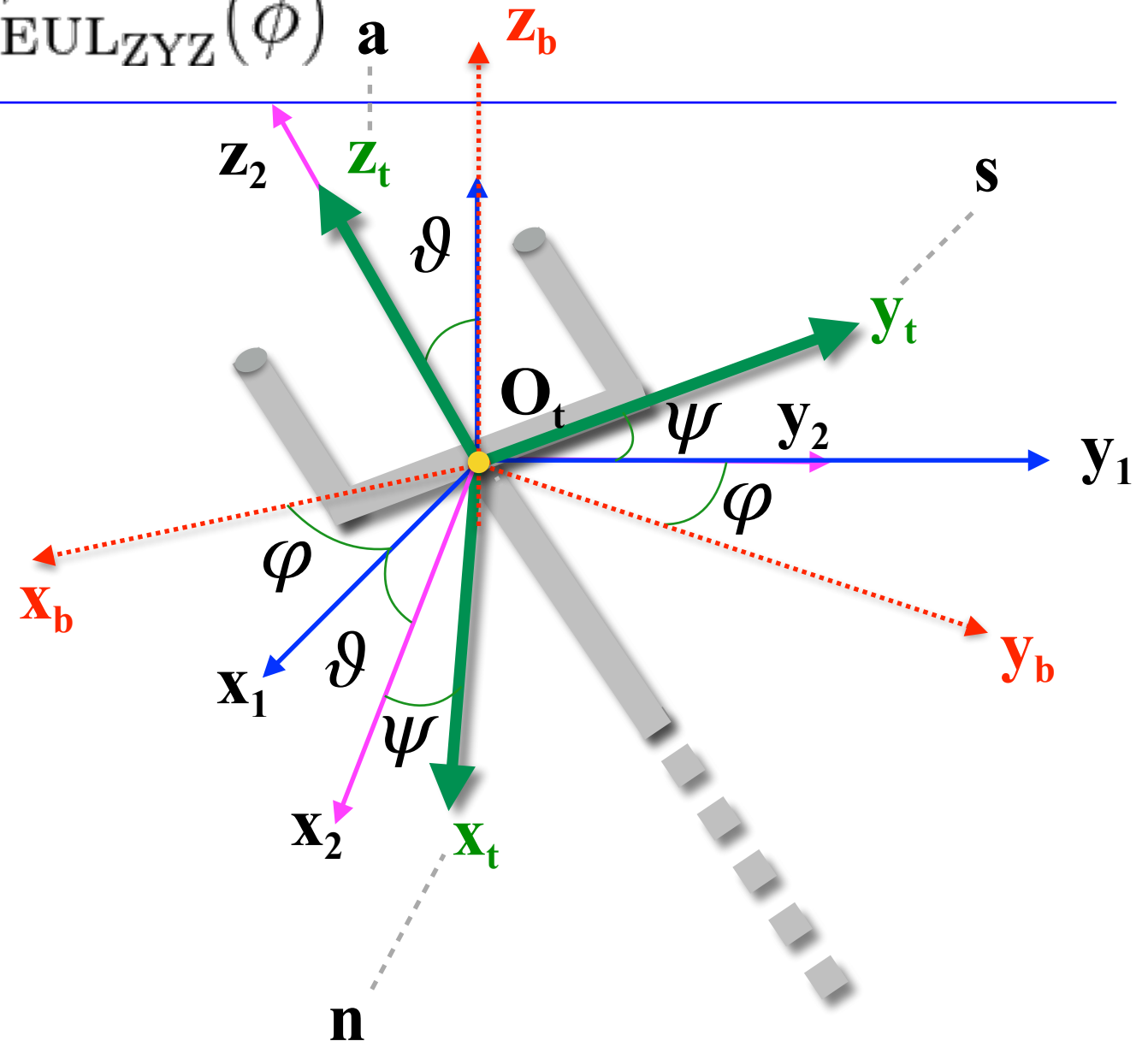


Determination of the Transformation

Matrix $T_{\text{EUL}_{\text{ZYZ}}}(\phi)$

$$\underline{\omega} = T_{\text{EUL}_{\text{ZYZ}}}(\phi) \cdot \dot{\phi}$$

$\downarrow$ 3×1 $\downarrow$ 3×3 $\downarrow$ 3×1



$$T_{\text{EUL}_{\text{ZYZ}}}(\phi) = \begin{bmatrix} 0 & -\sin(\varphi(t)) & \sin(\vartheta(t))\cos(\varphi(t)) \\ 0 & \cos(\varphi(t)) & \sin(\vartheta(t))\sin(\varphi(t)) \\ 1 & 0 & \cos(\vartheta(t)) \end{bmatrix}$$

Determination of the Transformation Matrix $T_A(\phi)$

$$\underline{\omega} = T_{\text{EULZYZ}}(\phi) \cdot \dot{\phi}$$
$$T_{\text{EULZYZ}}(\phi) = \begin{bmatrix} 0 & -\sin(\varphi(t)) & \sin(\vartheta(t))\cos(\varphi(t)) \\ 0 & \cos(\varphi(t)) & \sin(\vartheta(t))\sin(\varphi(t)) \\ 1 & 0 & \cos(\vartheta(t)) \end{bmatrix}$$

$$\underline{v}(t) = \begin{bmatrix} I & \underline{0} \\ \underline{0} & T(\phi) \end{bmatrix} \dot{\underline{x}}(t) = T_A(\phi) J_A(\underline{q}(t)) \dot{\underline{q}}(t)$$

Determination of the Transformation Matrix $T_A(\phi)$

$$\underline{\omega} = T_{\text{EULZYZ}}(\phi) \cdot \dot{\phi}$$
$$T_{\text{EULZYZ}}(\phi) = \begin{bmatrix} 0 & -\sin(\varphi(t)) & \sin(\vartheta(t))\cos(\varphi(t)) \\ 0 & \cos(\varphi(t)) & \sin(\vartheta(t))\sin(\varphi(t)) \\ 1 & 0 & \cos(\vartheta(t)) \end{bmatrix}$$

$$\underline{v}(t) = \begin{bmatrix} I & \underline{0} \\ \underline{0} & T(\phi) \end{bmatrix} \underline{\dot{x}}(t) = T_A(\phi) J_A(\underline{q}(t)) \underline{\dot{q}}(t)$$

Determination of the Transformation Matrix $T_A(\phi)$

$$\underline{\omega} = T_{\text{EULZYZ}}(\phi) \cdot \dot{\phi}$$

$$T_{\text{EULZYZ}}(\phi) = \begin{bmatrix} 0 & -\sin(\varphi(t)) & \sin(\vartheta(t))\cos(\varphi(t)) \\ 0 & \cos(\varphi(t)) & \sin(\vartheta(t))\sin(\varphi(t)) \\ 1 & 0 & \cos(\vartheta(t)) \end{bmatrix}$$

$$\underline{v}(t) = \begin{bmatrix} I & \underline{0} \\ \underline{0} & T(\phi) \end{bmatrix} \underline{\dot{x}}(t) = T_A(\phi) J_A(\underline{q}(t)) \underline{\dot{q}}(t)$$

$$J(\underline{q}(t)) = T_A(\phi(t)) J_A(\underline{q}(t))$$